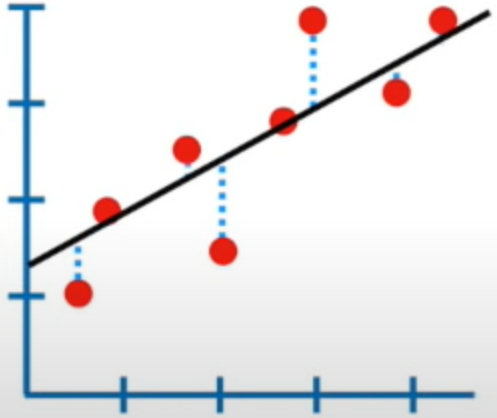
basics of **Least Squares** and **Linear Regression**, so if you're not already down with that, check out the Quest…

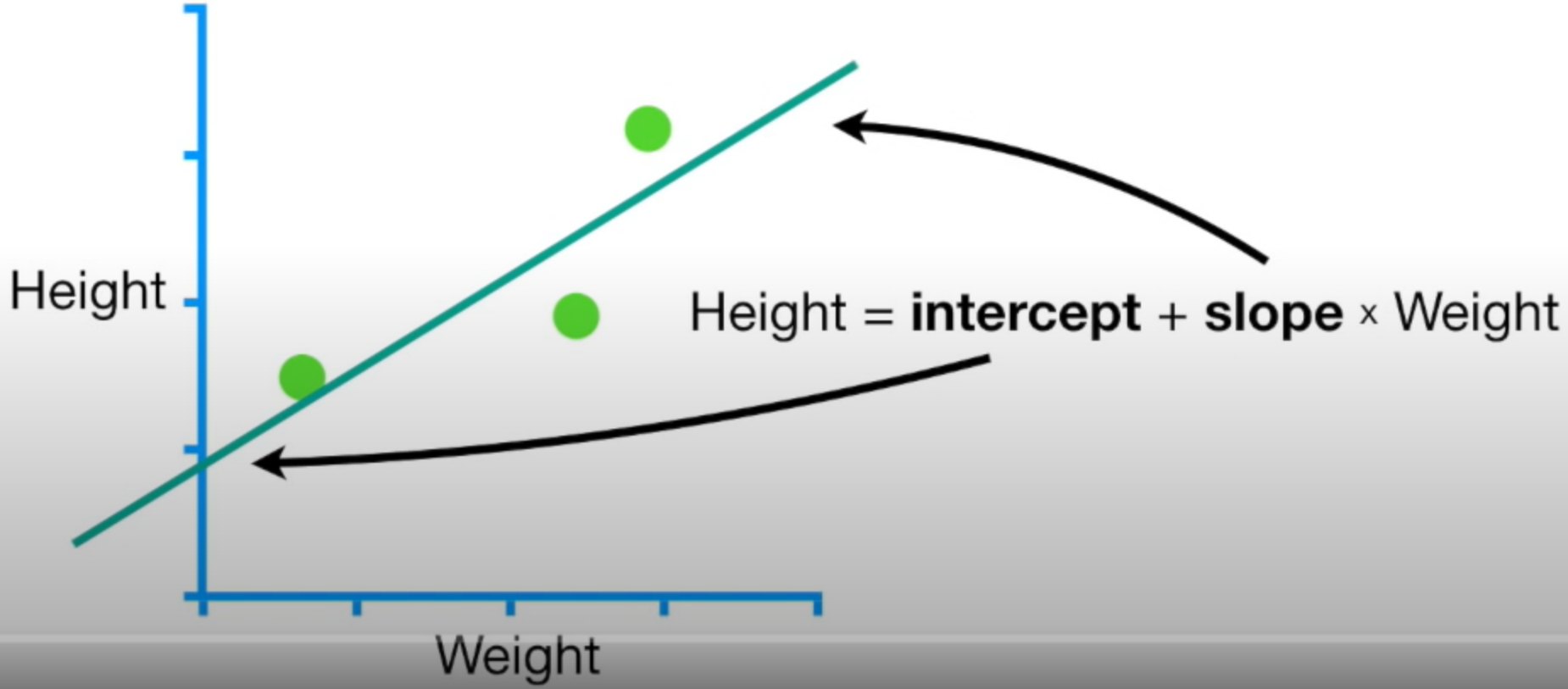
Fitting a Line to Data…

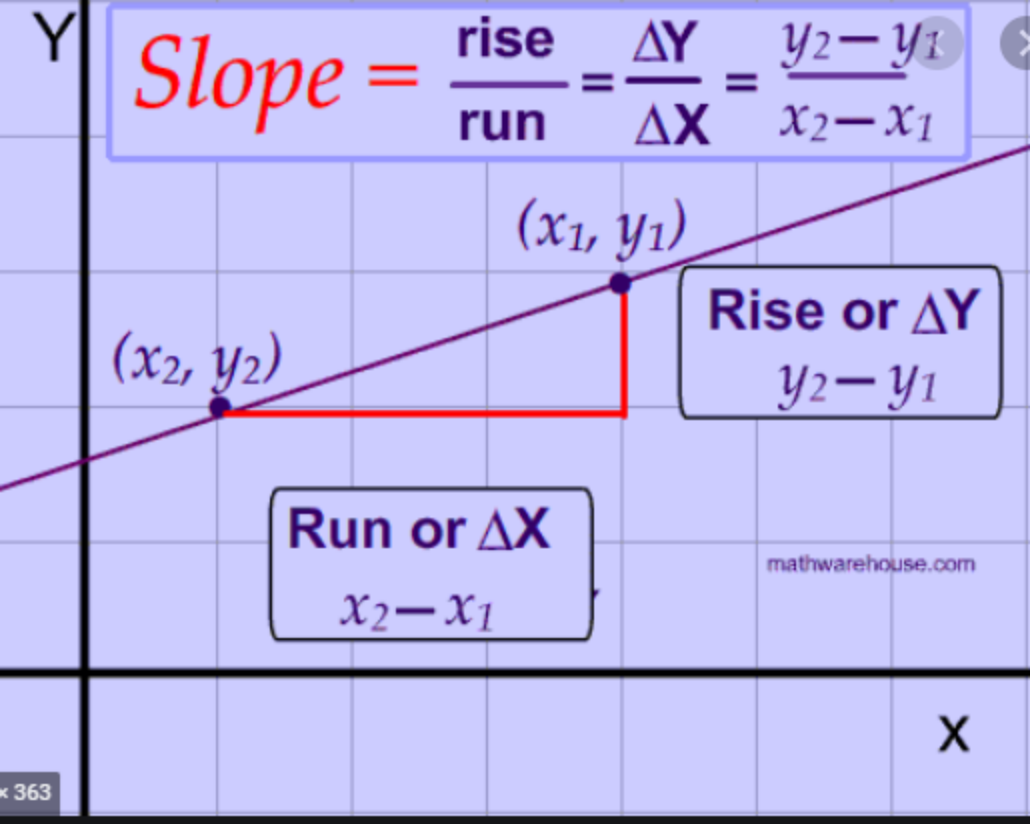


aka Linear Regression
aka Least Squares

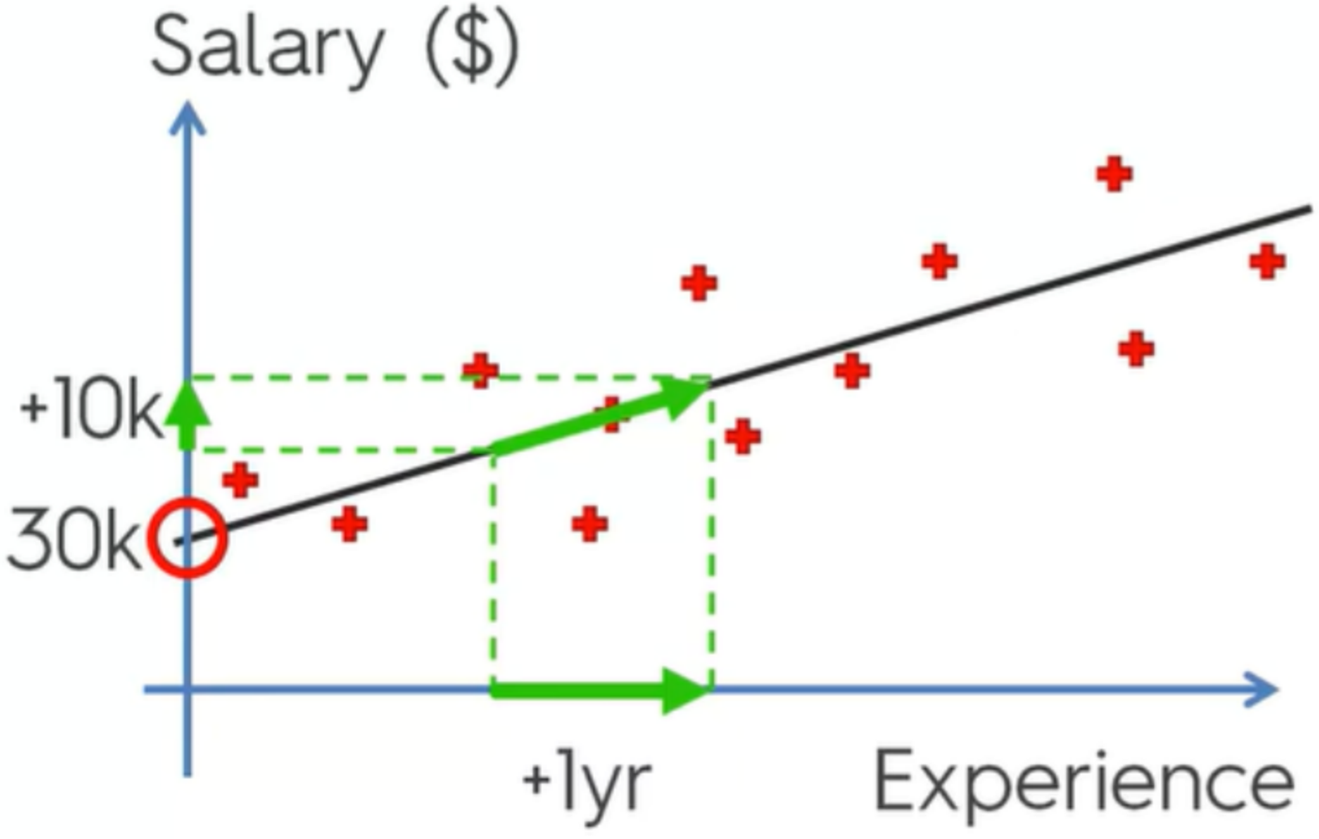
In Statistics, Machine Learning and other Data Science fields, we optimize a lot of stuff.

When we fit a line with **Linear Regression**, we optimize the **Intercept** and **Slope**.




Slope = rise/run = ΔY/ΔX = $y_2 - y_1$ / $x_2 - x_1$
Y
X
(x_1, y_1)
(x_2, y_2)
Rise or ΔY
$y_2 - y_1$
Run or ΔX
$x_2 - x_1$
mathwarehouse.com

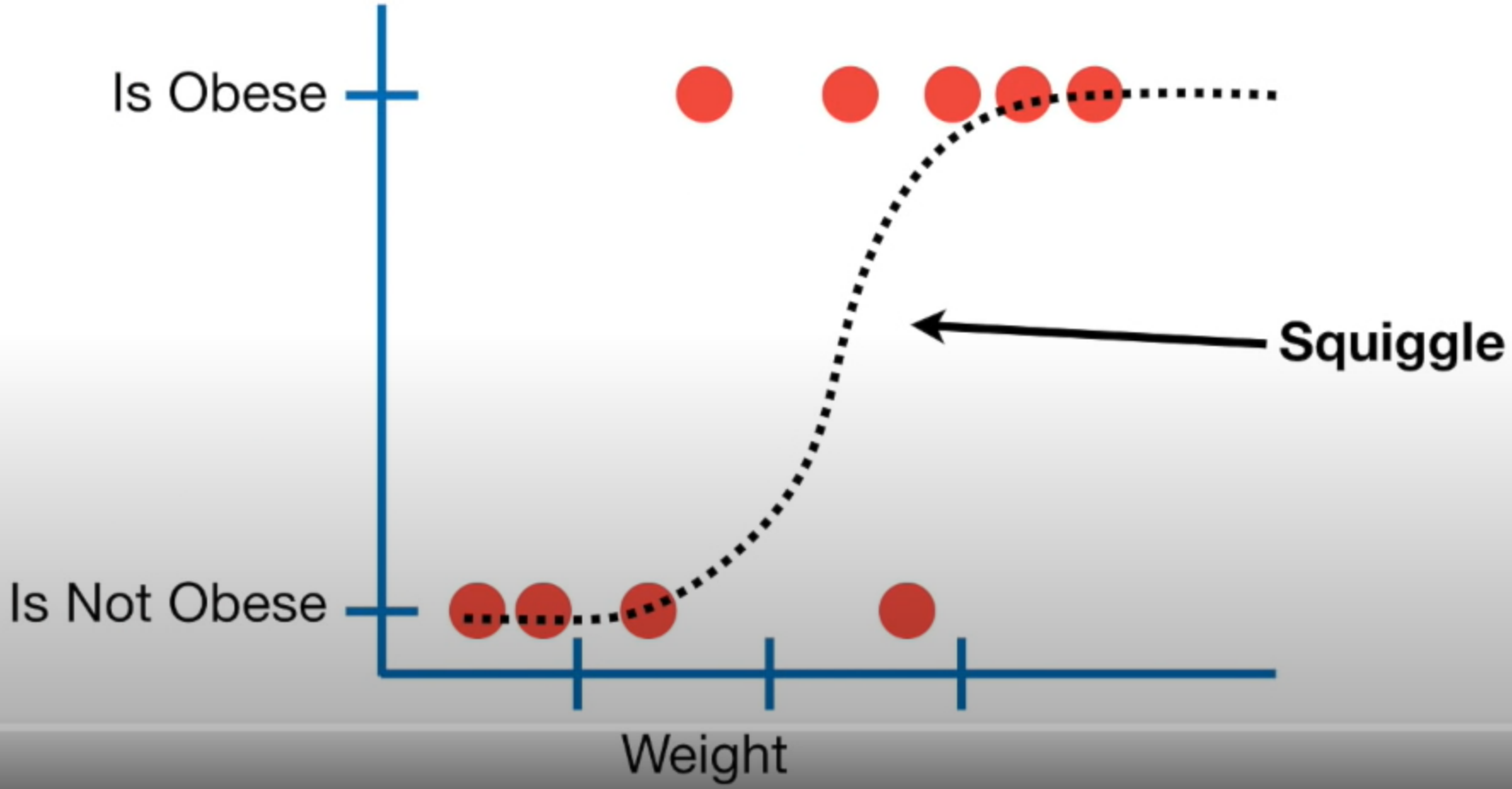
Simple Linear Regression:



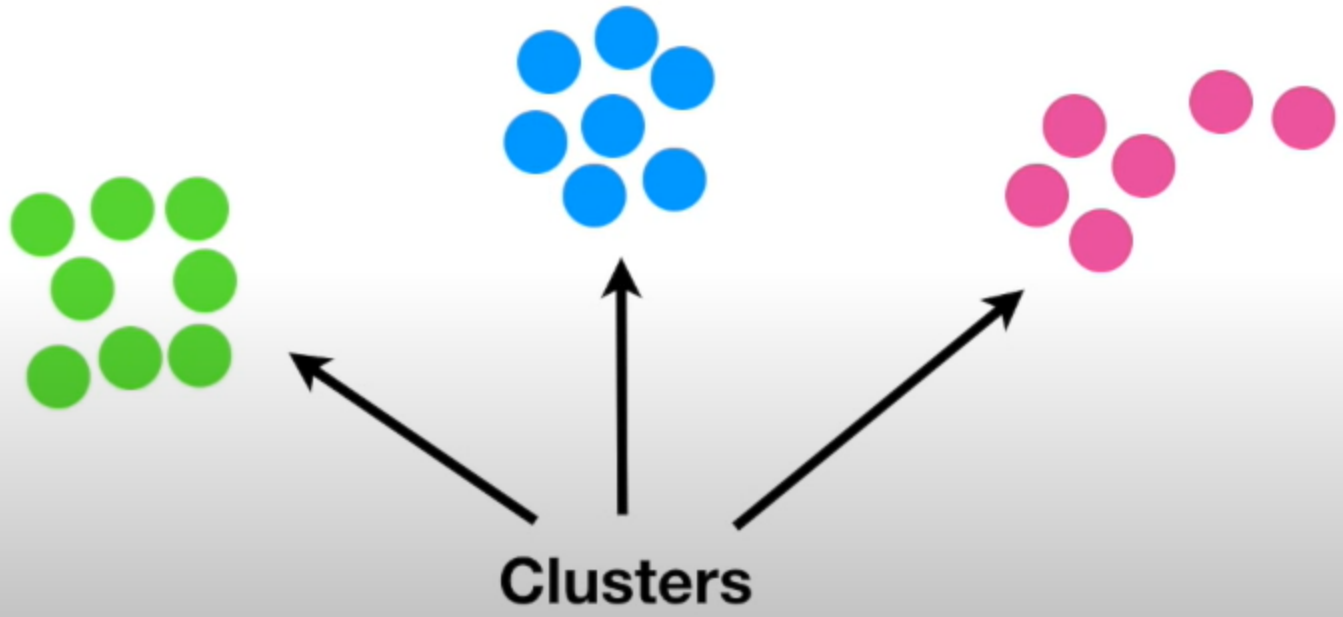
$$y = b_0 + b_1 * x$$

Salary = b_0 + b_1 *Experience

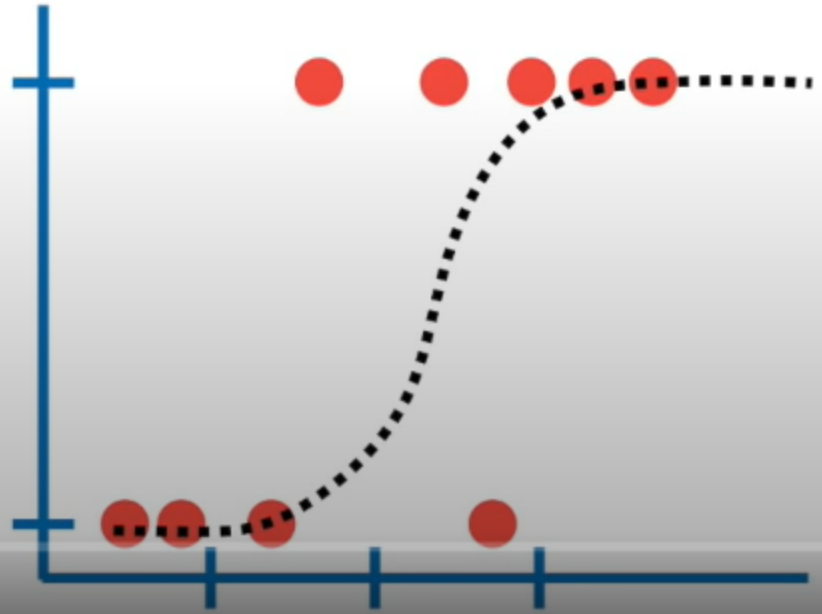
When we use **Logistic Regression**,
we optimize a squiggle.

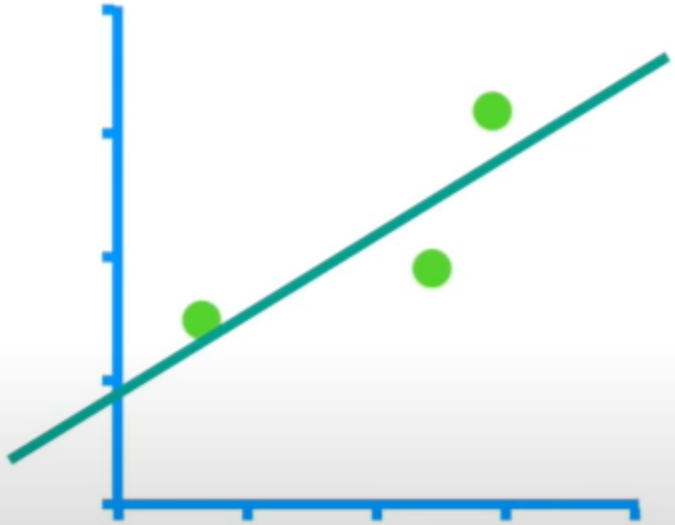


And when we use **t-SNE**, we optimize clusters.

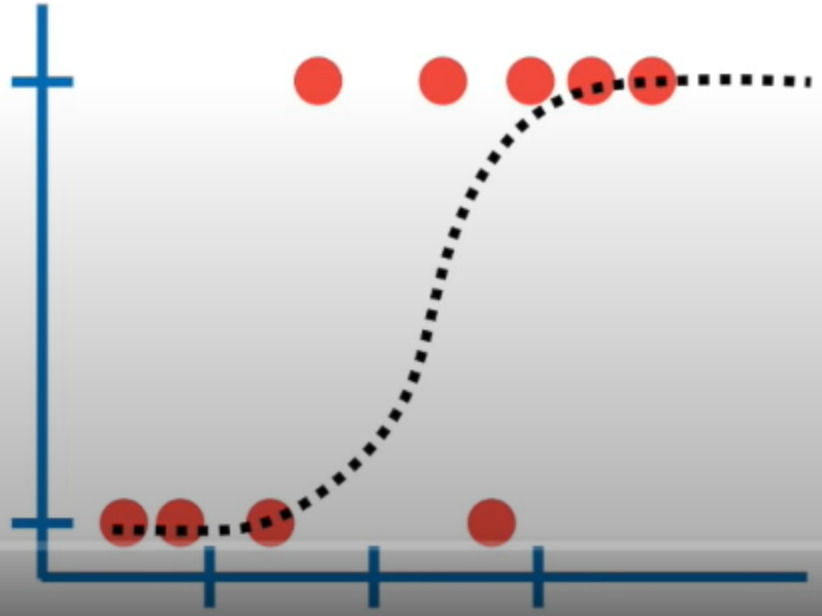


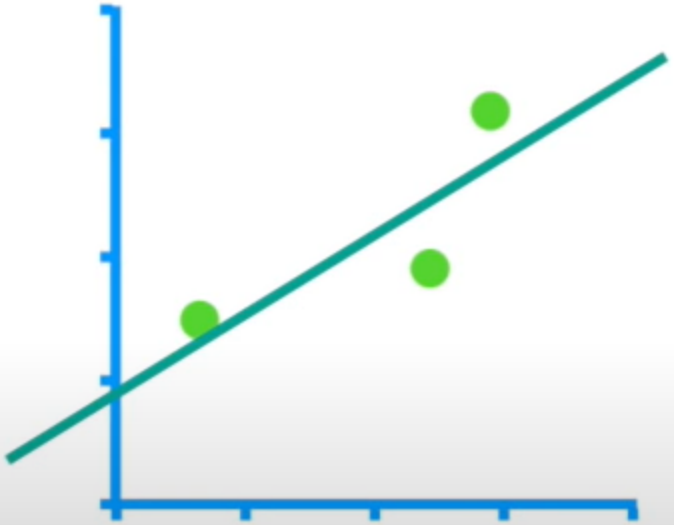
These are just a few examples of the stuff we optimize, there are tons more.



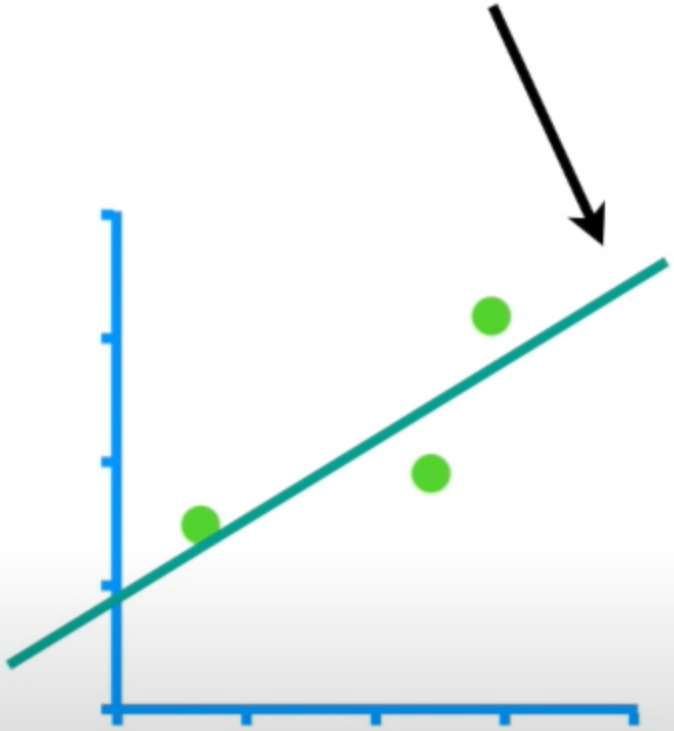


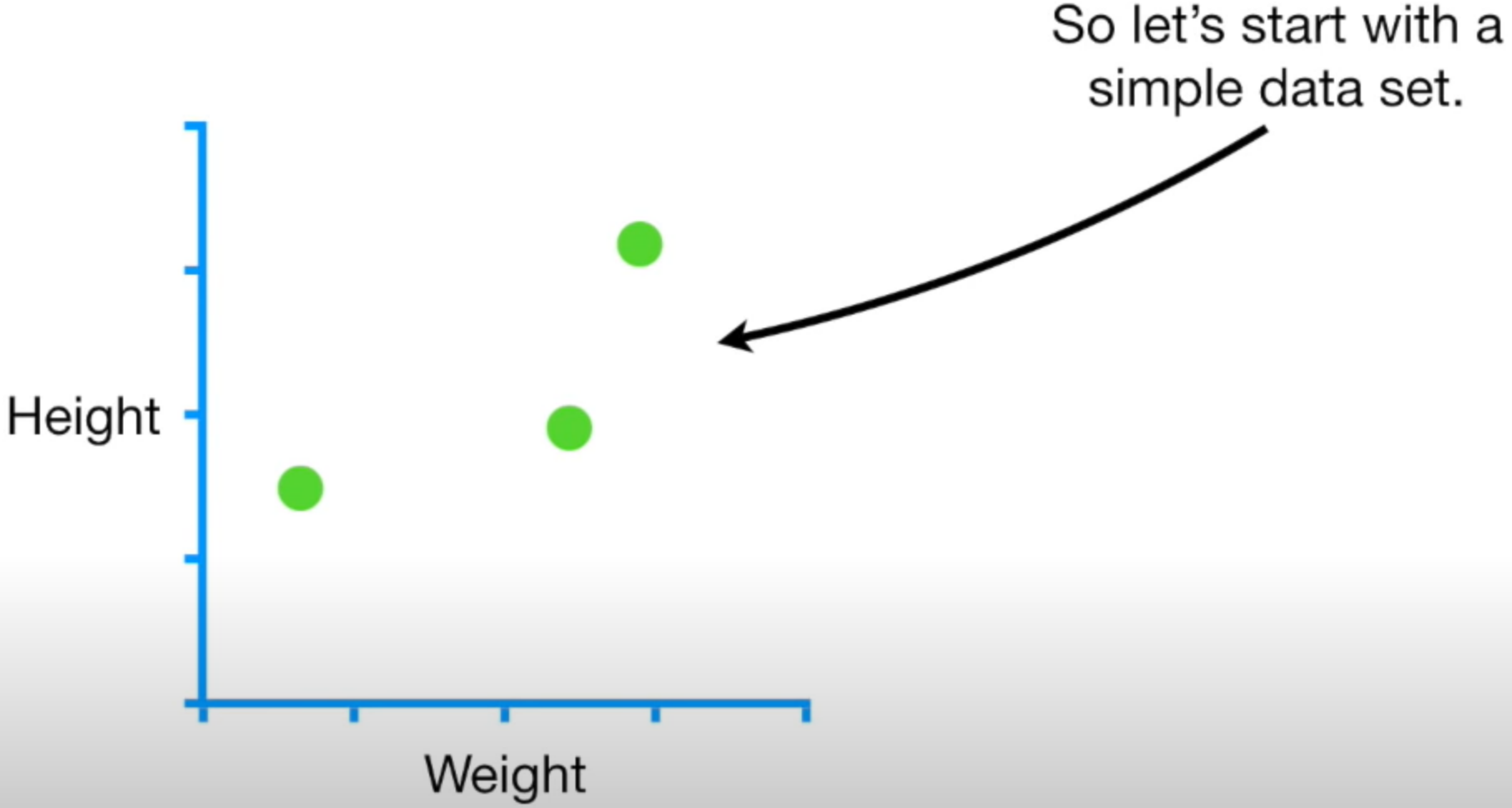
The cool thing is that **Gradient Descent** can optimize all these things and much more.

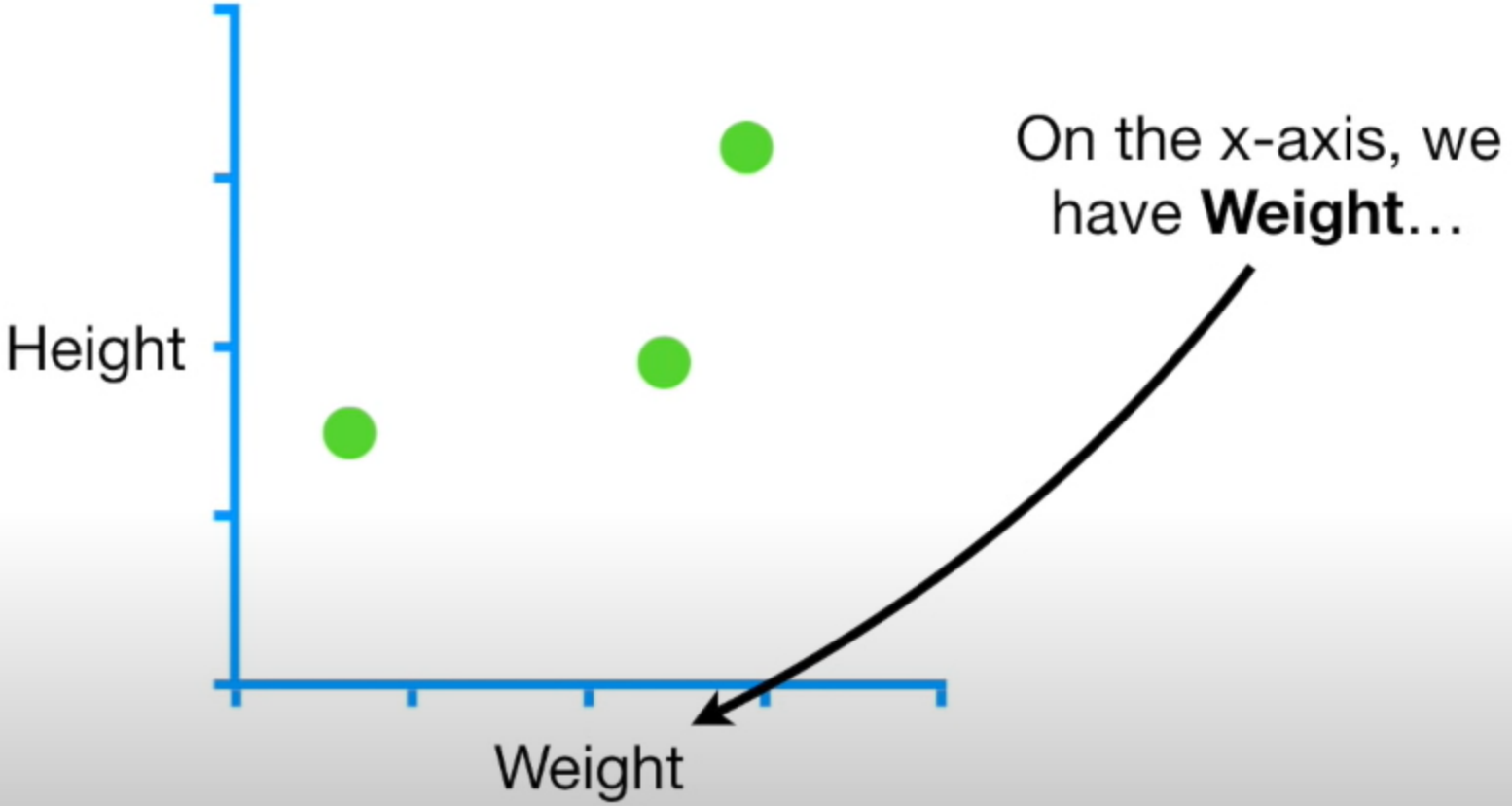


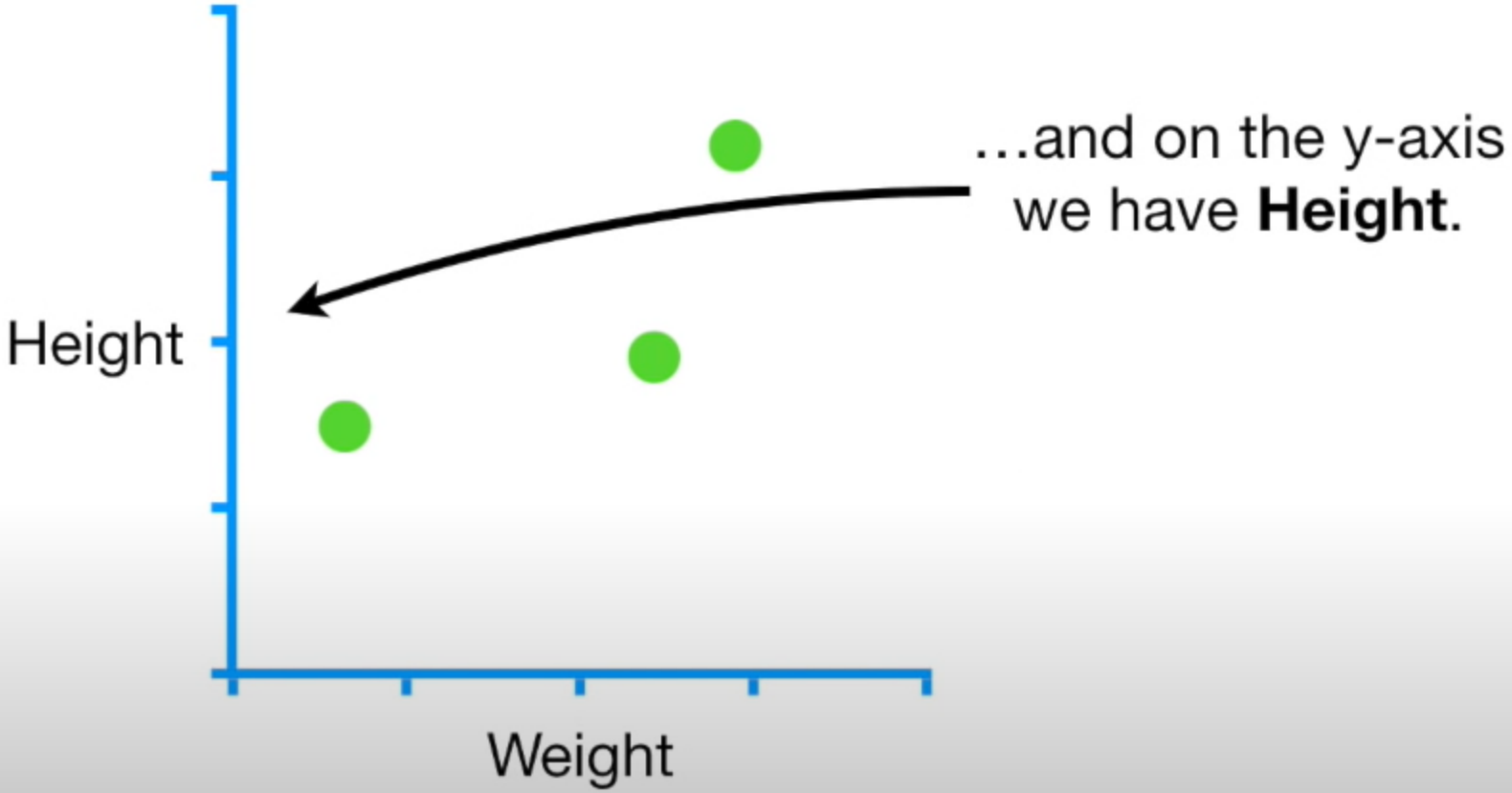


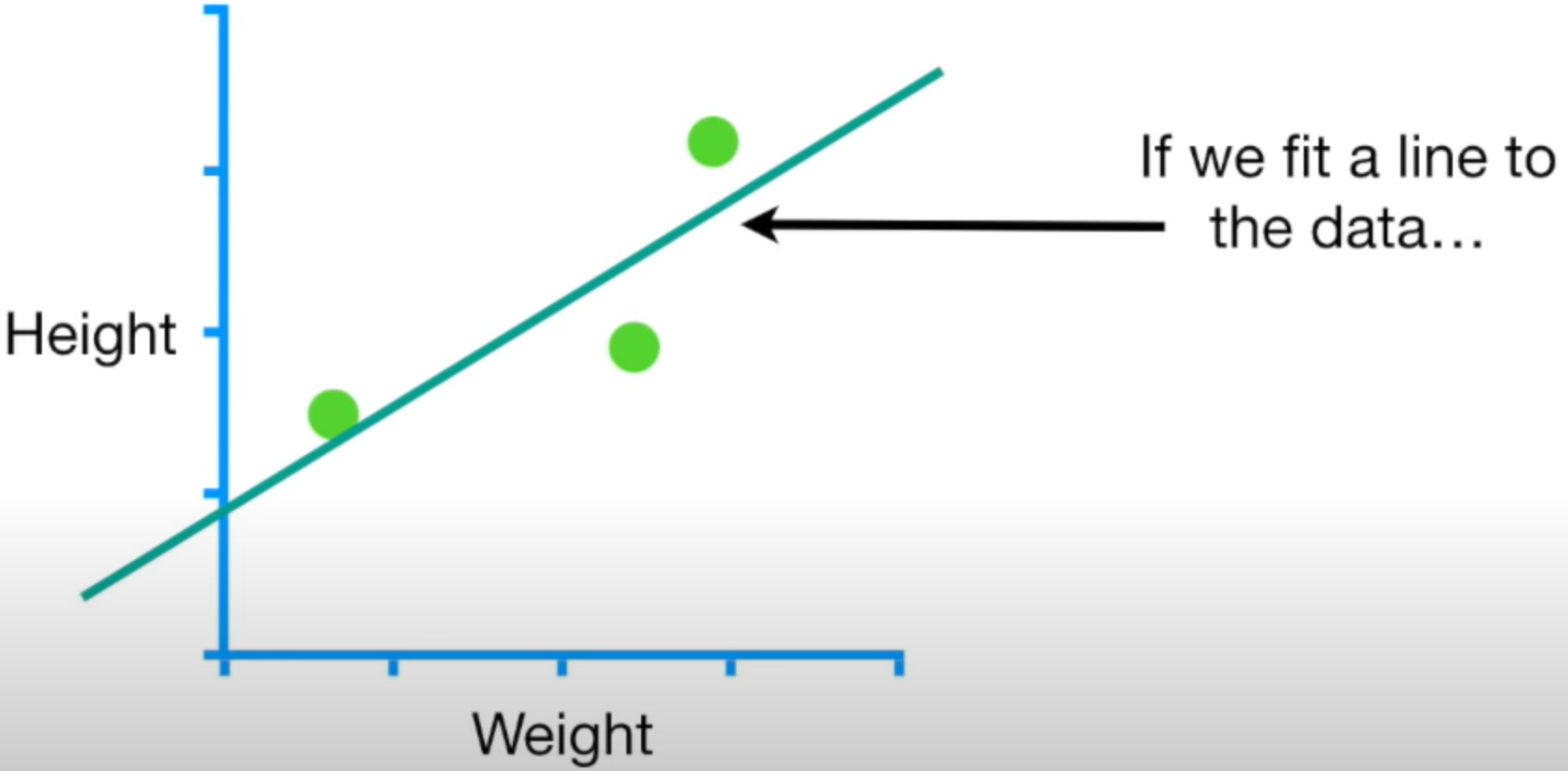
So if we learn how to optimize this line using **Gradient Descent**…

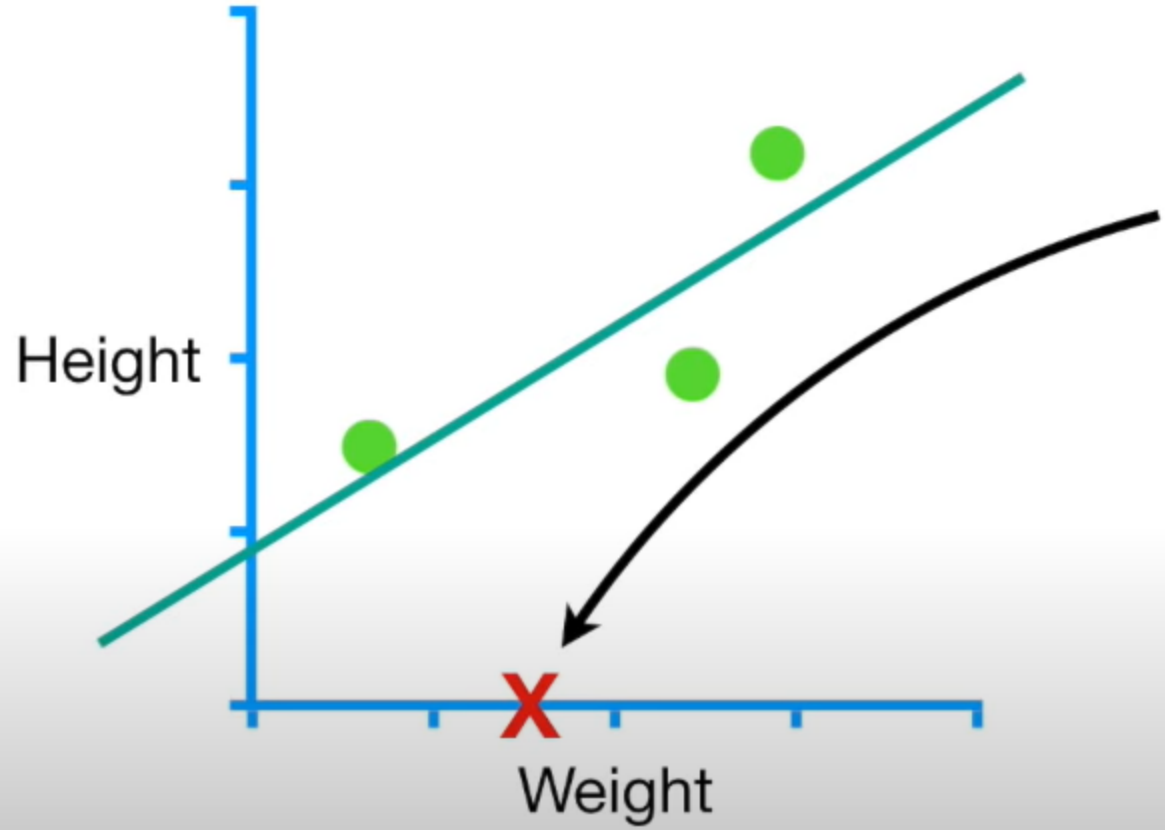



So let's start with a simple data set.
Height
Weight

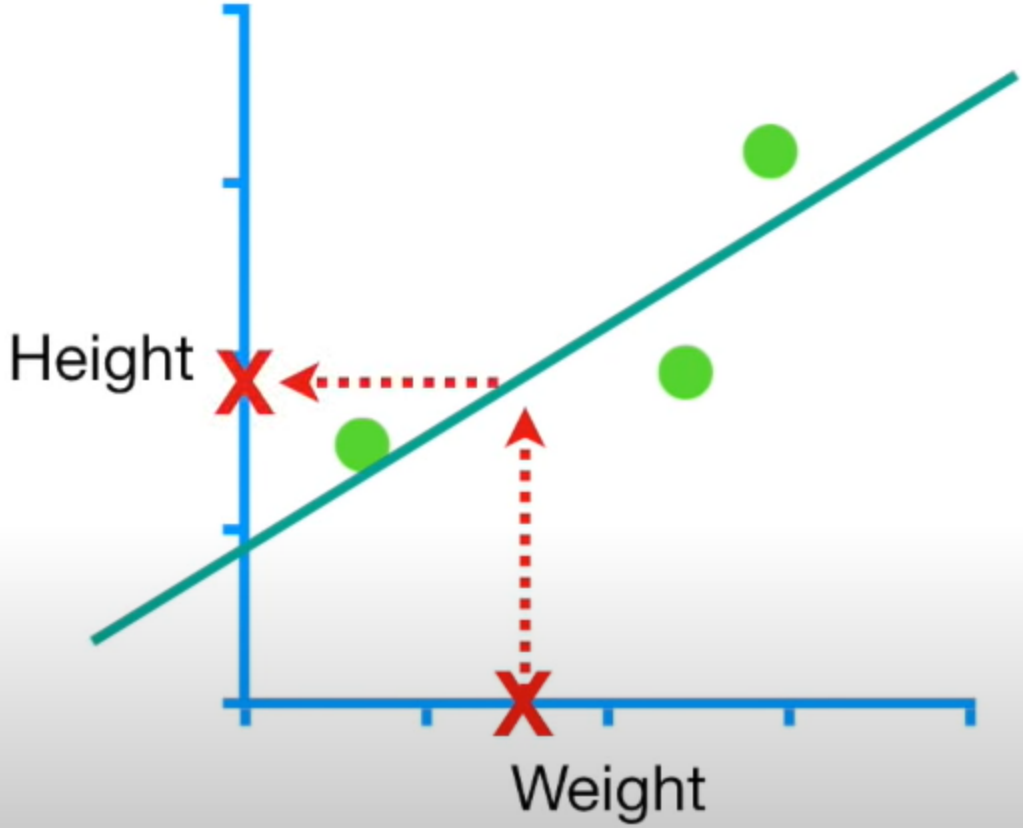

On the x-axis, we have **Weight**…
Height
Weight


…and on the y-axis
we have **Height**.
Height
Weight

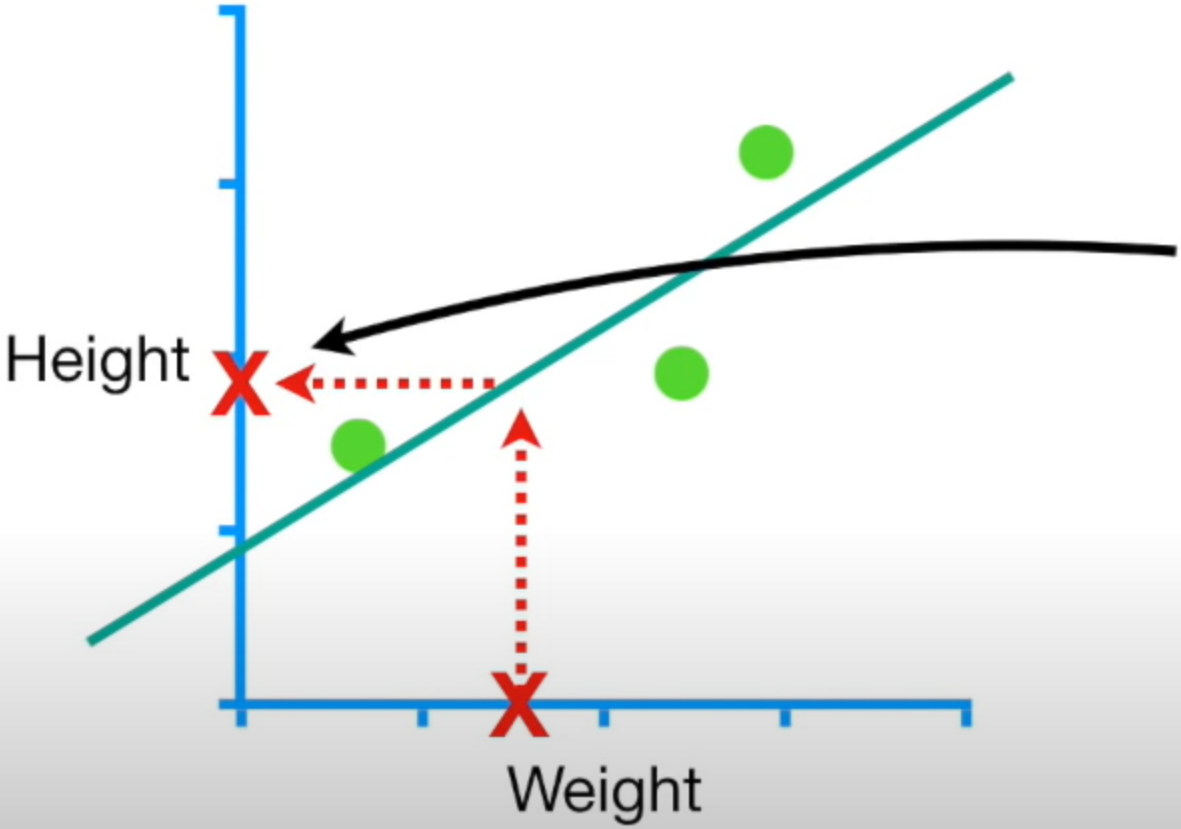

If we fit a line to
the data…
Height
Weight



…and someone tells us
that they weigh **1.5**…

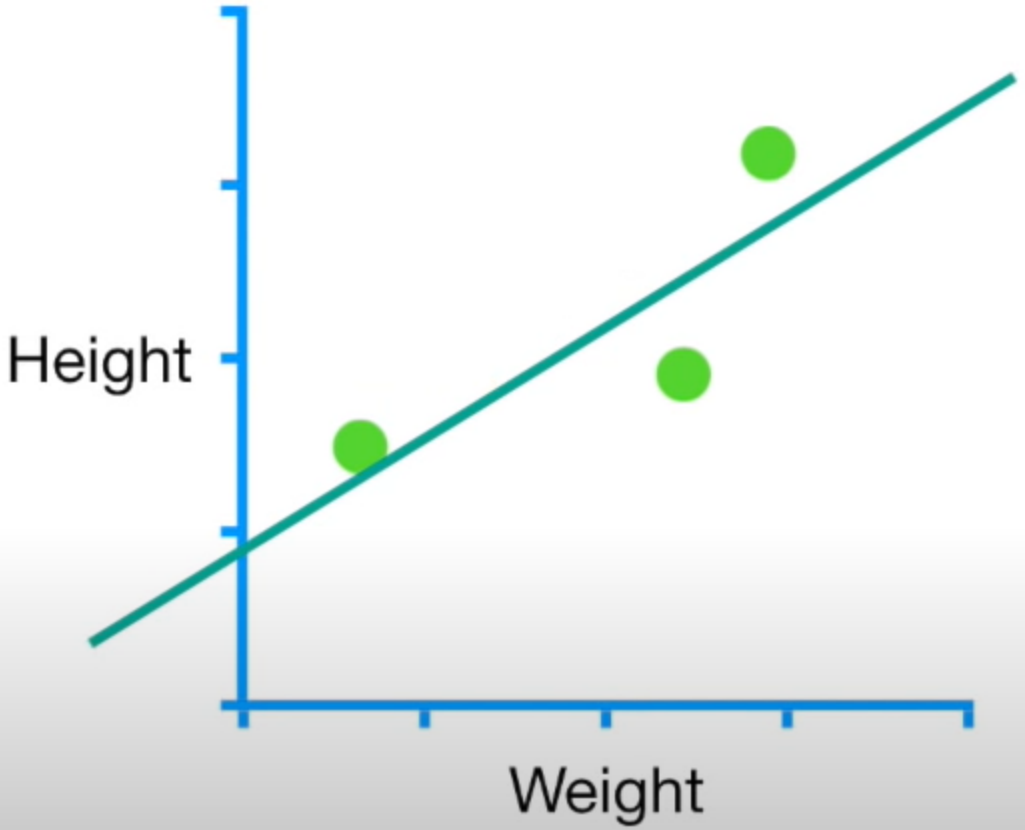


…we can use the line to predict that they will be **1.9** tall.



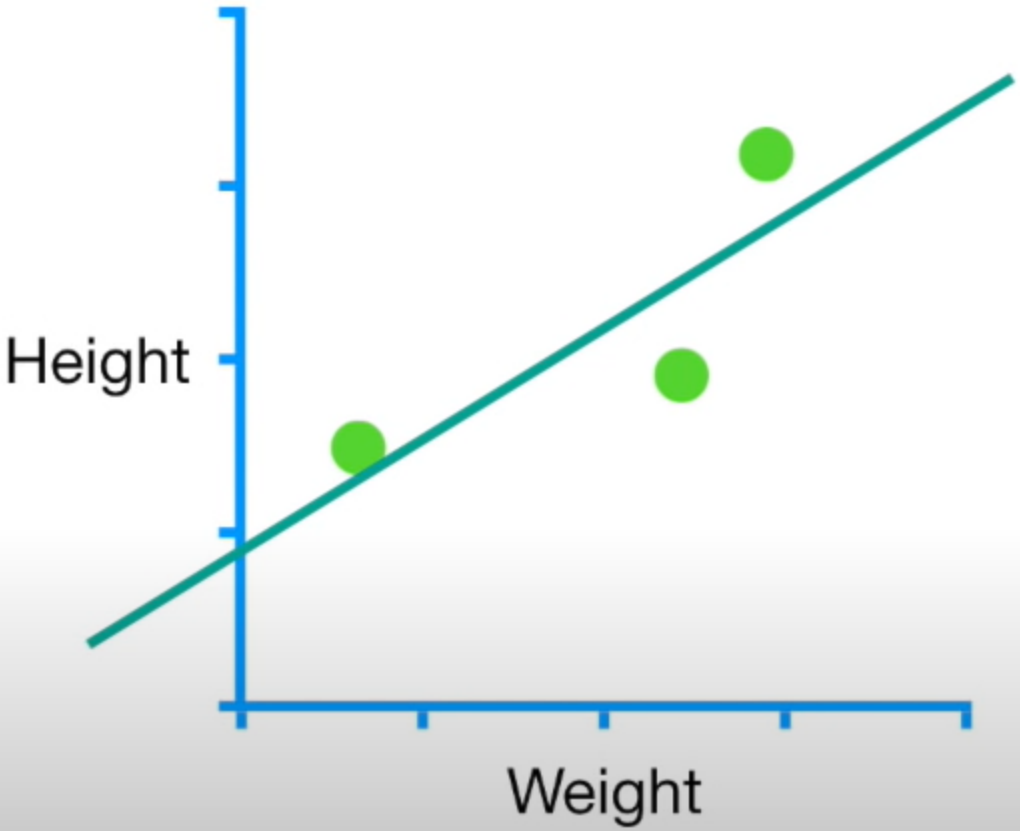
…we can use the line to predict that they will be **1.9** tall.

Predicted Height = intercept + slope × **Weight**



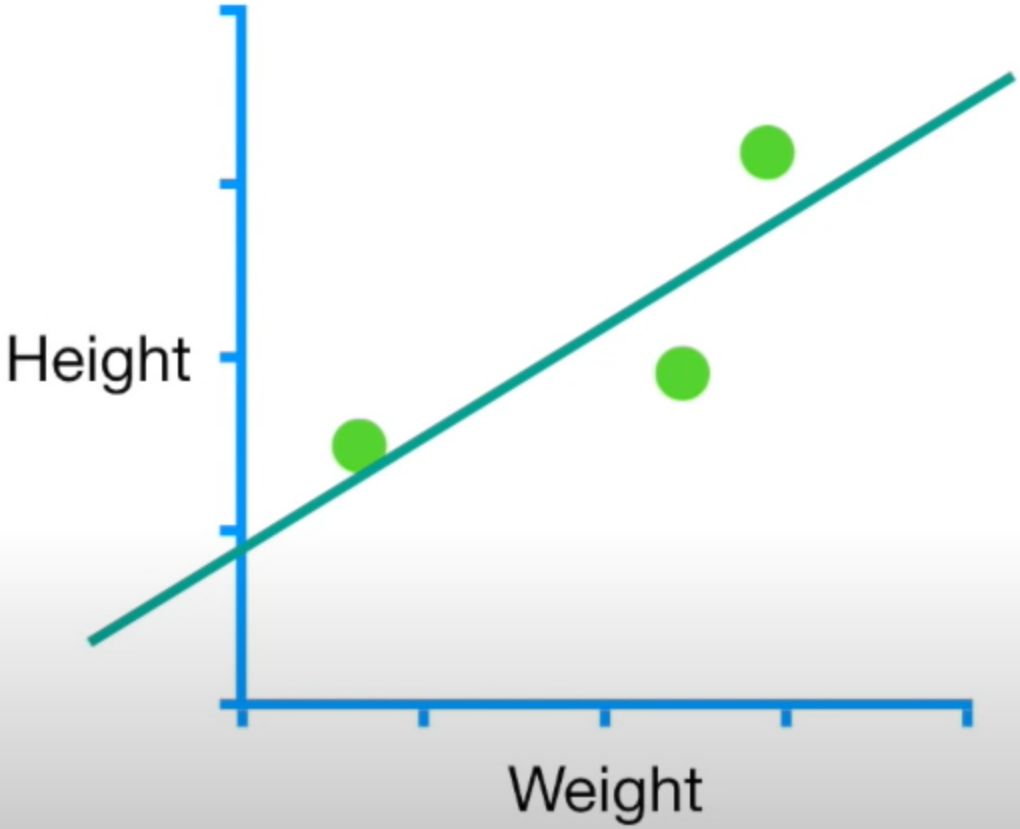
So let's learn how **Gradient Descent** can fit a line to data by finding the optimal values for the **Intercept** and the **Slope**.

Predicted Height = intercept + slope × **Weight**



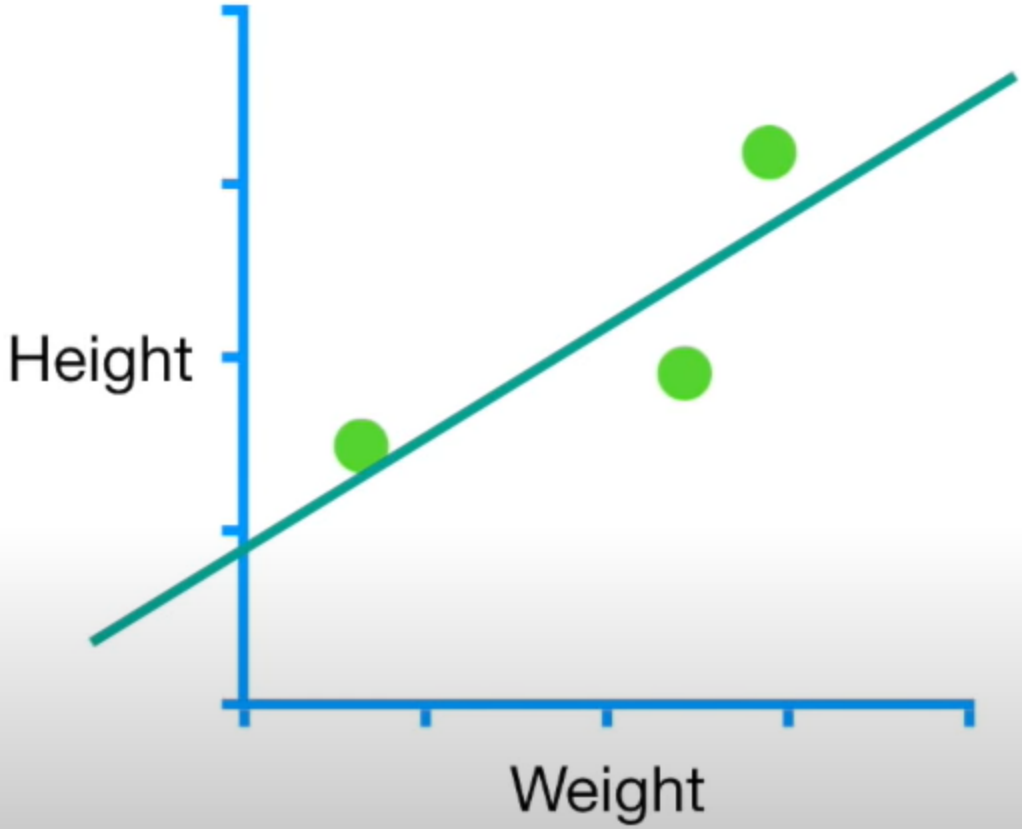
Actually, we'll start by using **Gradient Descent** to find the **Intercept**.

Predicted Height = intercept + slope × **Weight**



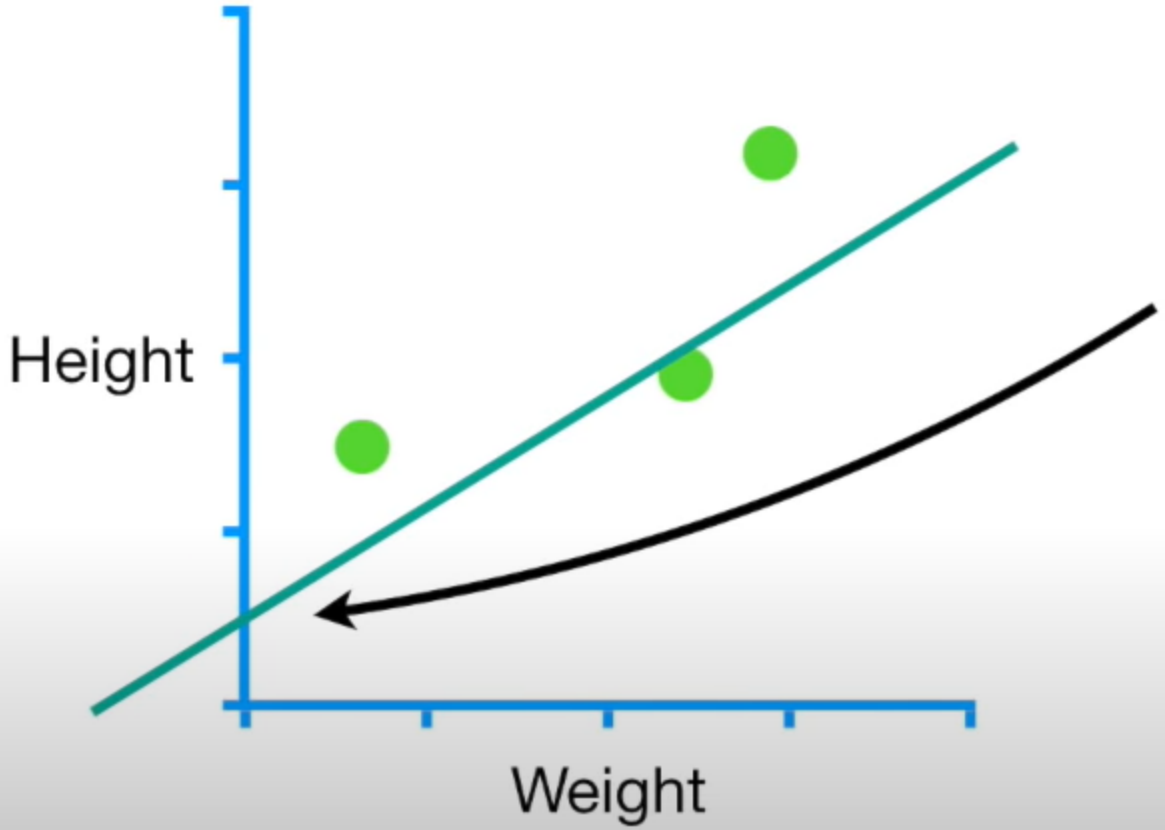
Then, once we understand how **Gradient Descent** works, we'll use it to solve for the **Intercept** *and* the **Slope**.

Predicted Height = intercept + slope × **Weight**



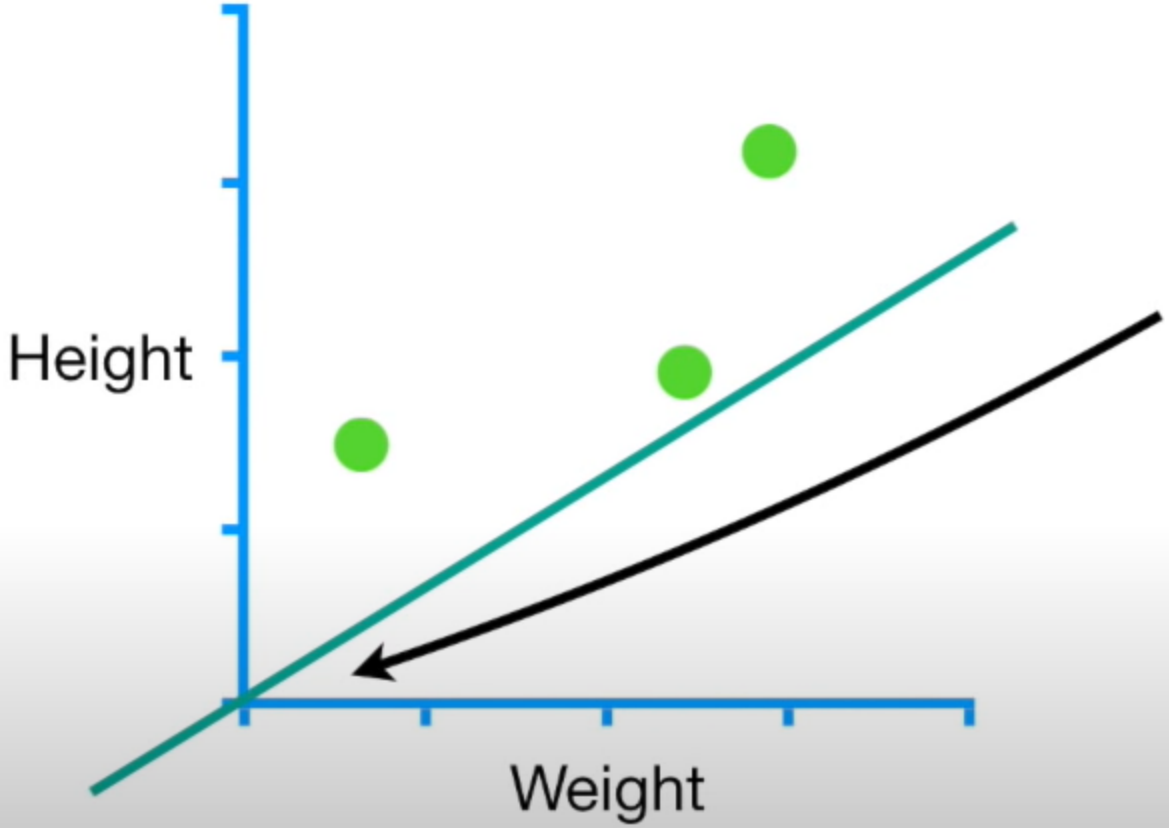
So for now, let's just plug in the **Least Squares** estimate for the **Slope**, **0.64**.

Predicted Height = intercept + 0.64 x **Weight**



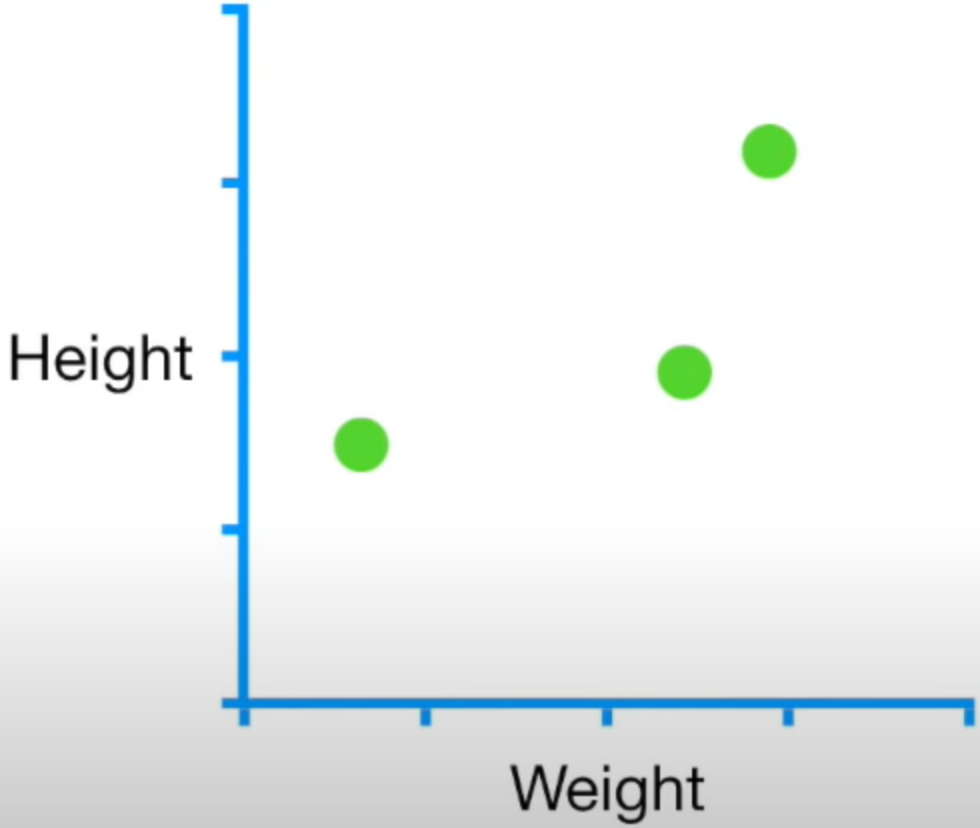
…and we'll use **Gradient Descent** to find the the optimal value for the Intercept.

Predicted Height = intercept + 0.64 × **Weight**



…and we'll use **Gradient Descent** to find the the optimal value for the Intercept.

Predicted Height = intercept + 0.64 × **Weight**

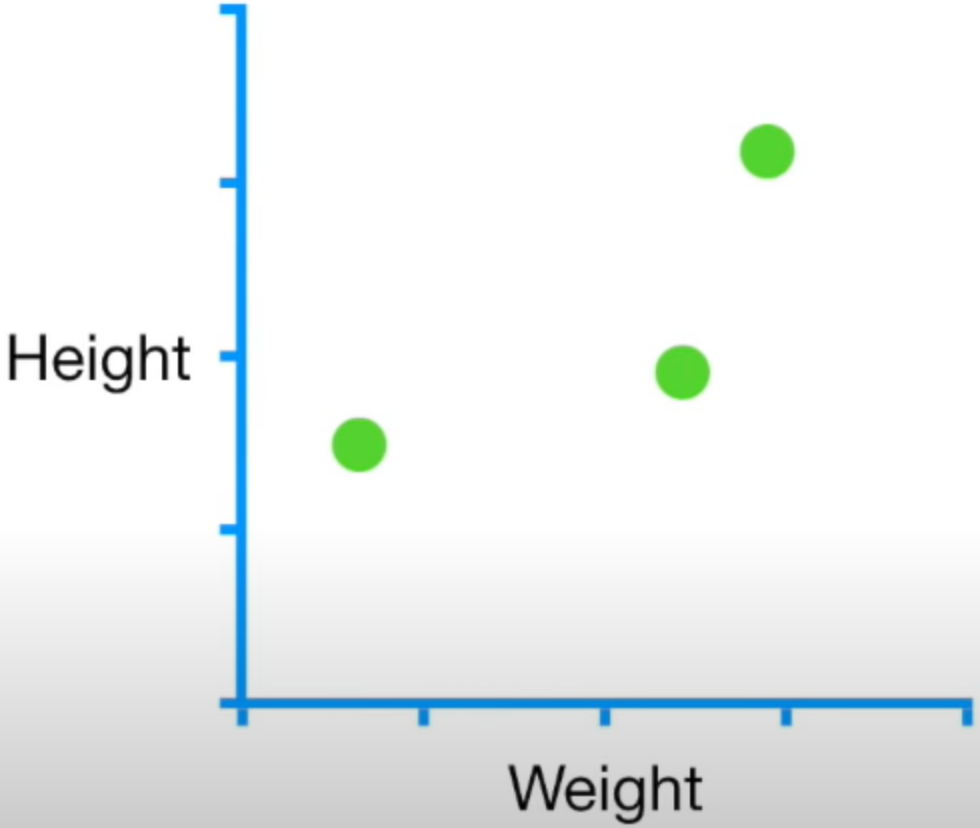


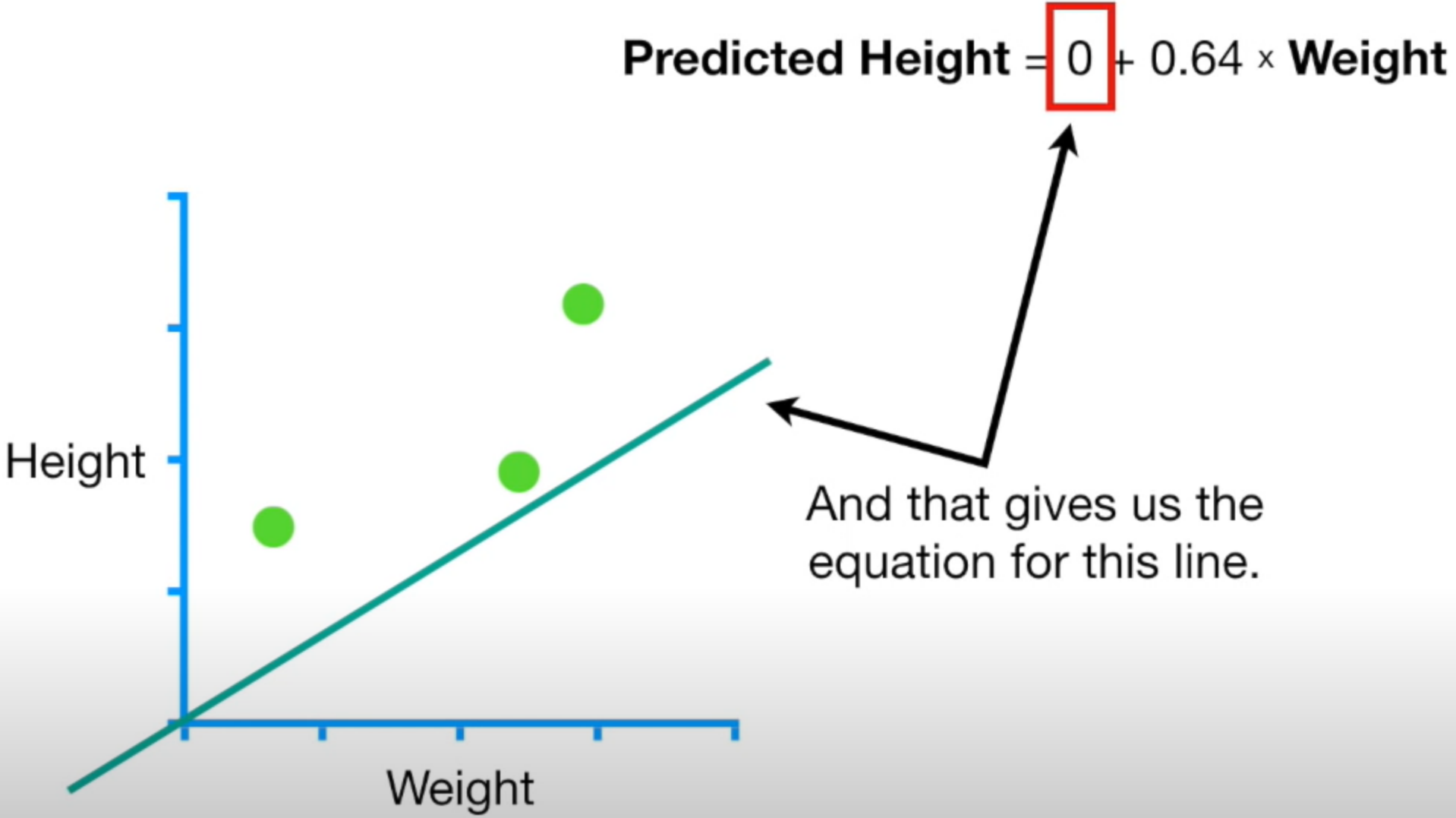
The first thing we do is pick a random value for the **Intercept**.

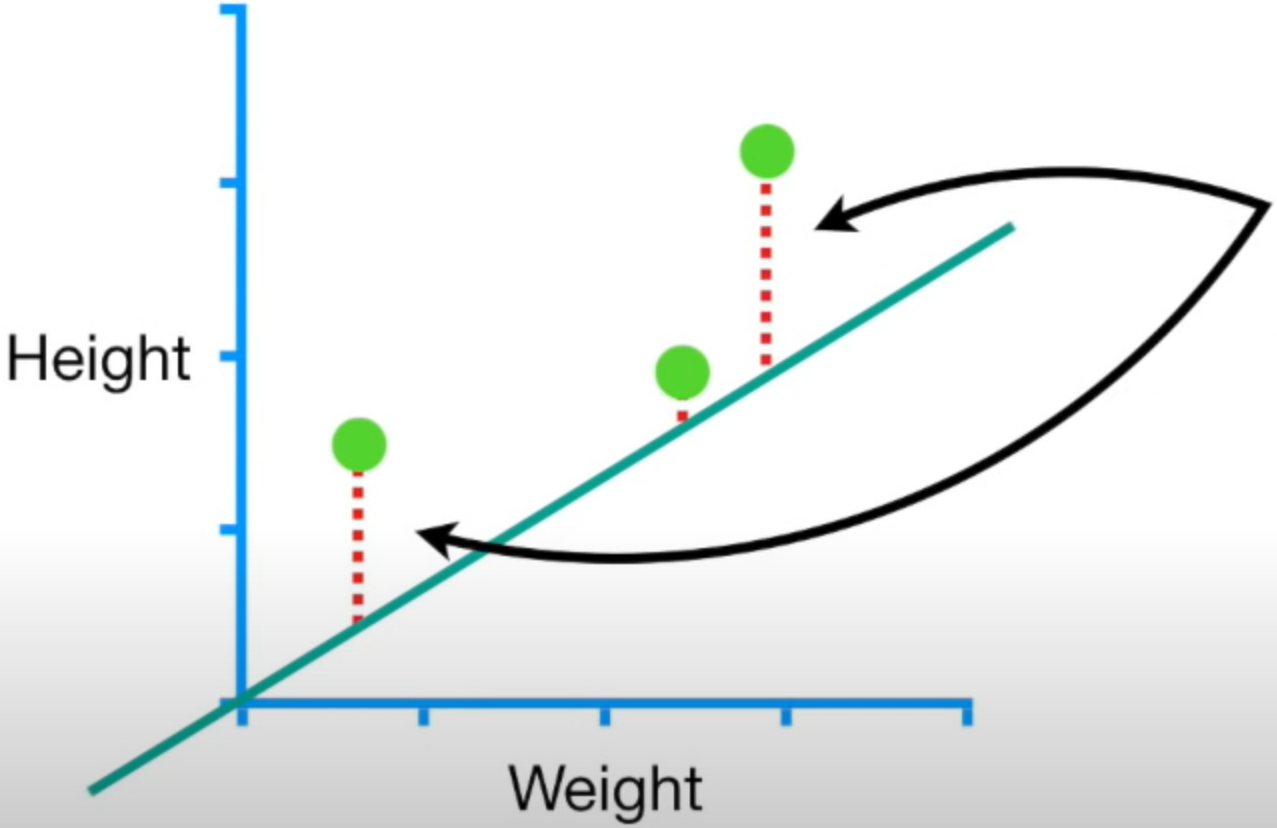
This is just an initial guess that gives **Gradient Descen**t something to improve upon.

Predicted Height = 0 + 0.64 × **Weight**

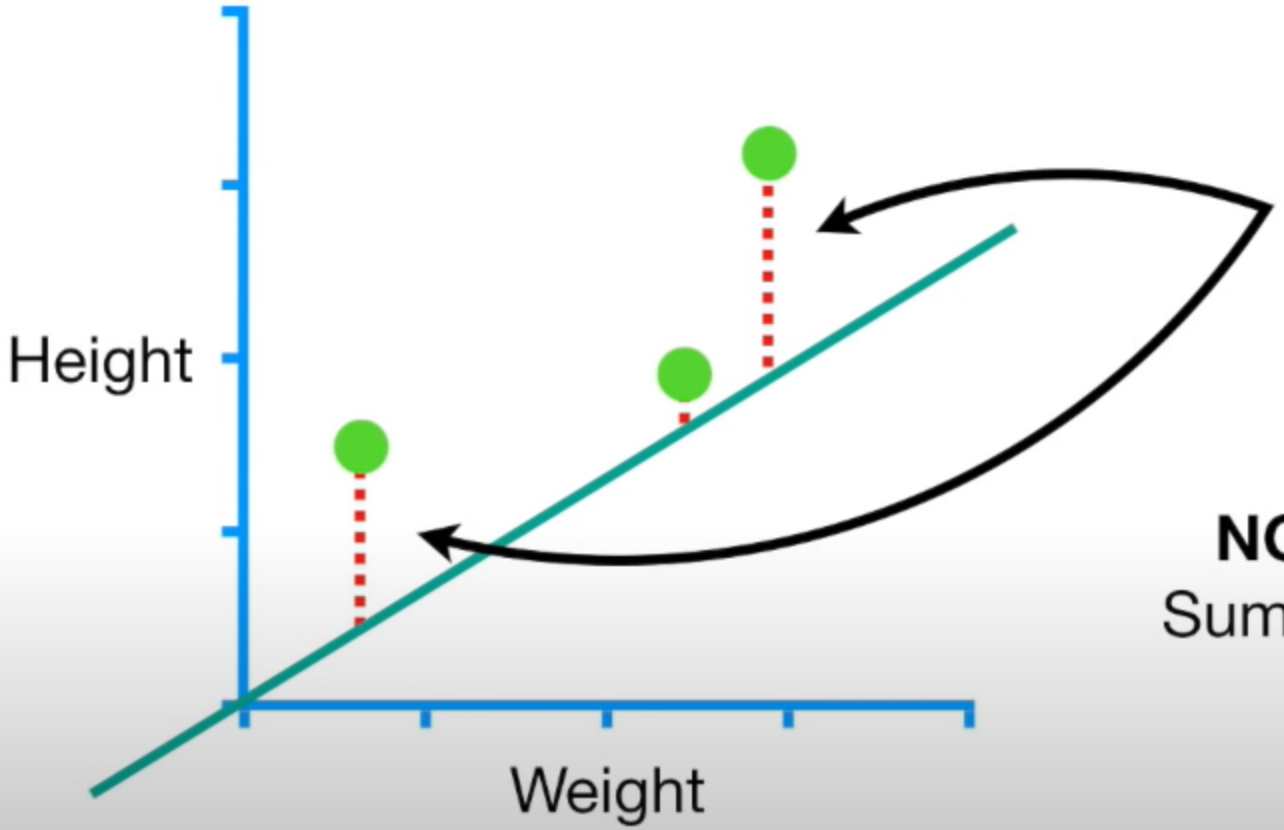
In this case, we'll use **0**,
but any number will do.




Predicted Height = 0 + 0.64 x Weight
Height
Weight
And that gives us the equation for this line.

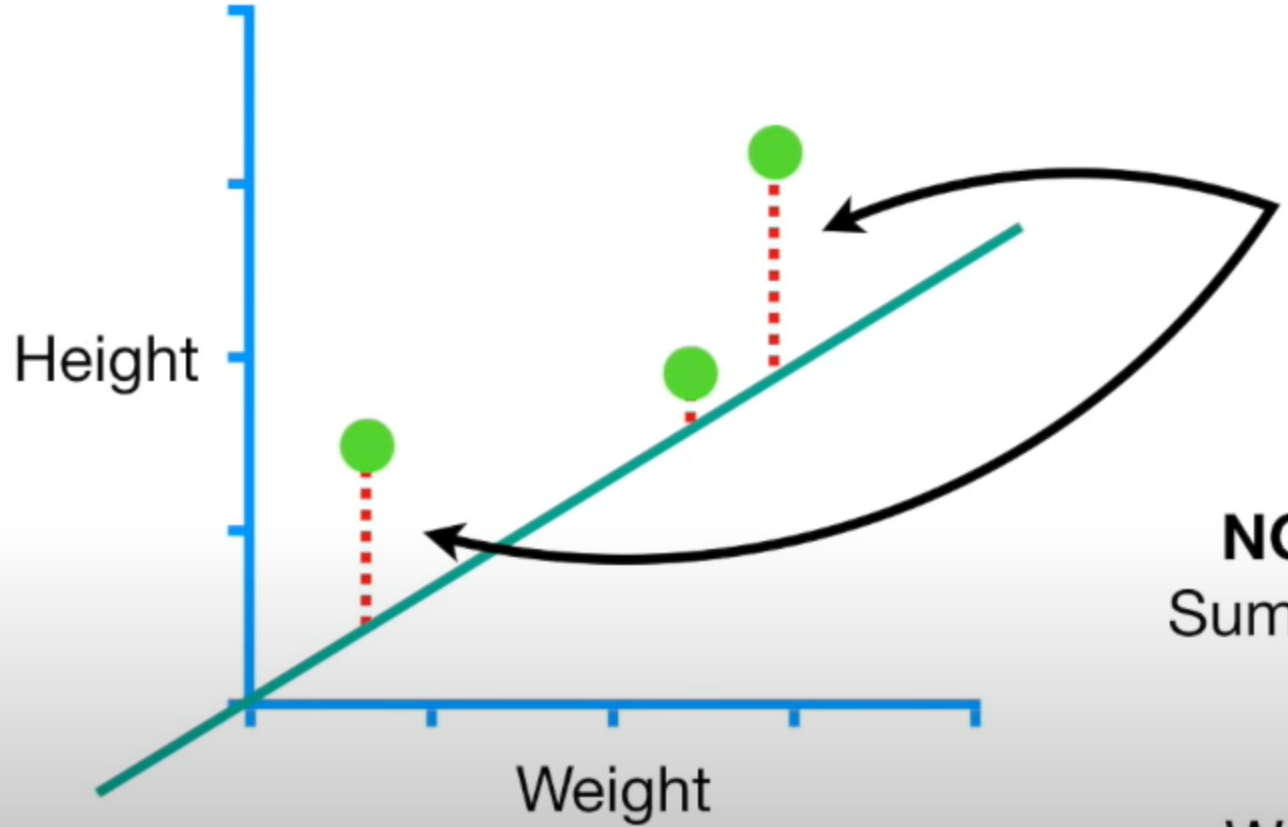


In this example, we will evaluate how well this line fits the data with the **Sum of the Squared Residuals**.



In this example, we will evaluate how well this line fits the data with the **Sum of the Squared Residuals**.

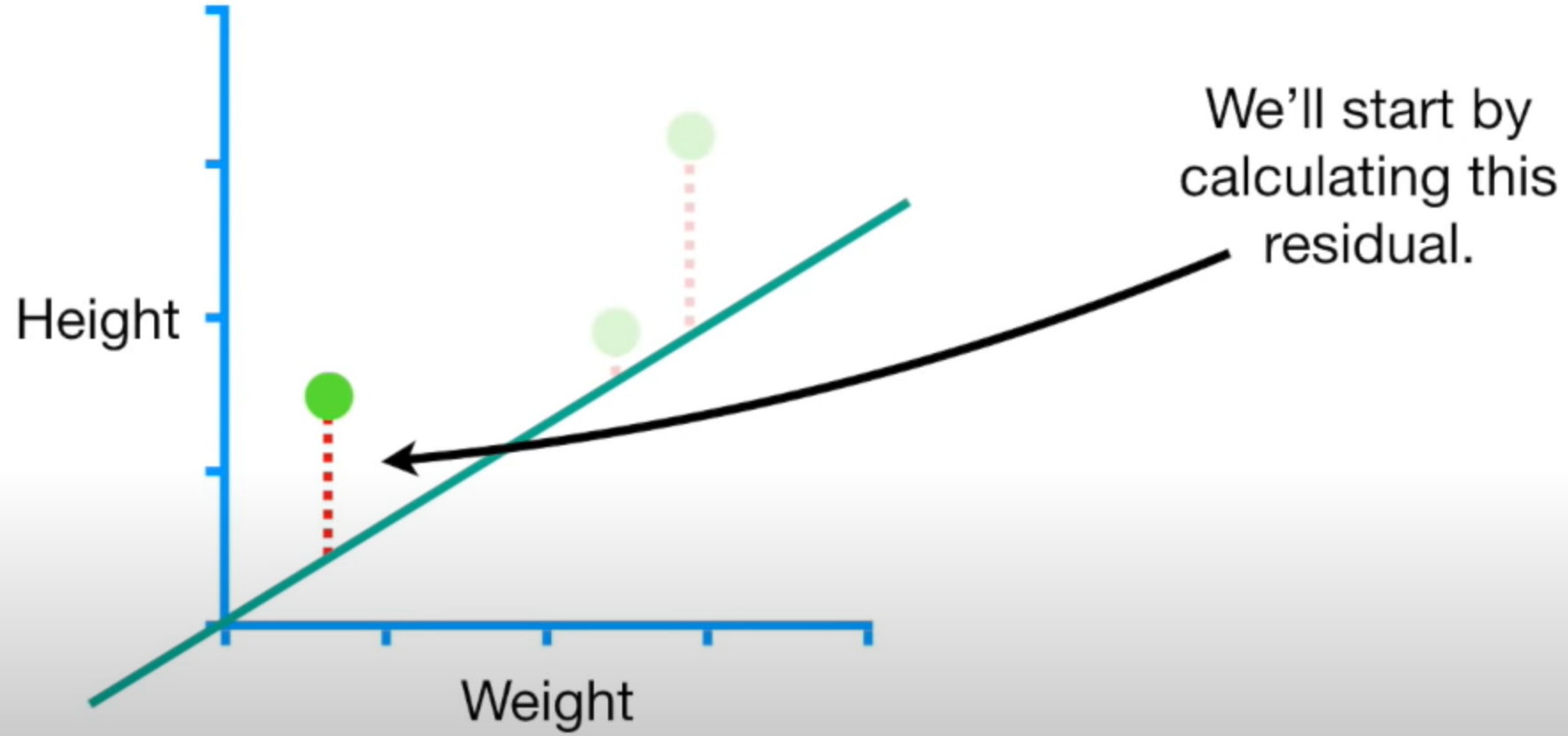
NOTE: In Machine Learning lingo, The Sum of the Squared Residuals is a type of **Loss Function**.

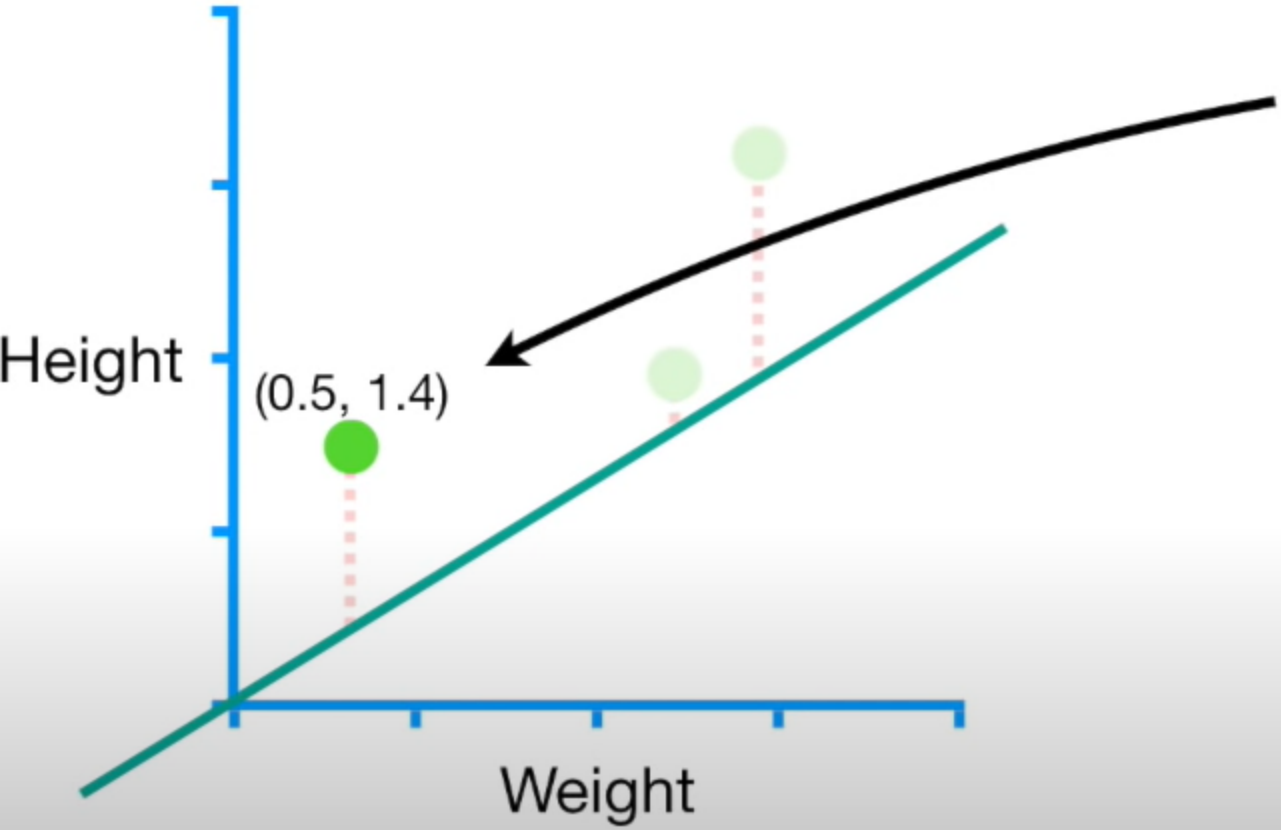


In this example, we will evaluate how well this line fits the data with the **Sum of the Squared Residuals**.

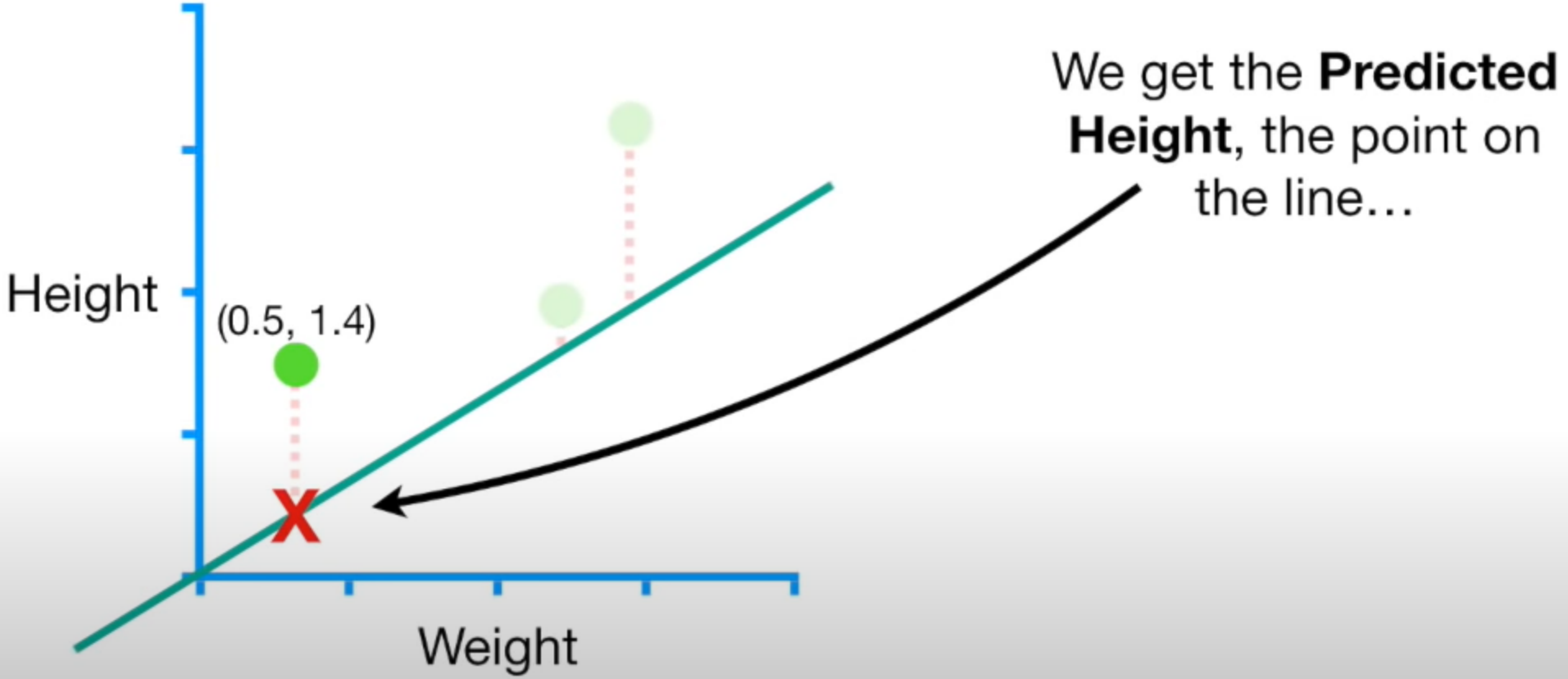
NOTE: In Machine Learning lingo, The Sum of the Squared Residuals is a type of **Loss Function**.

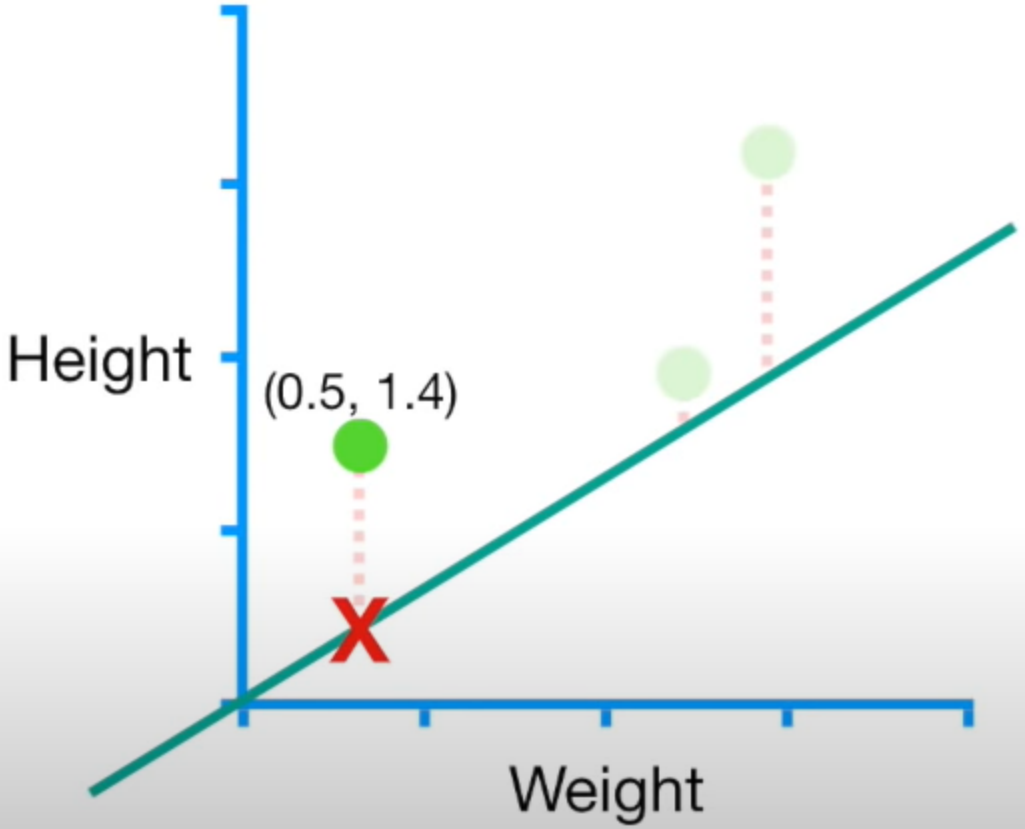
We'll talk more about **Loss Functions** towards the end of the video.


We'll start by calculating this residual.
Height
Weight



This datapoint represents a person with **Weight 0.5** and **Height 1.4**.

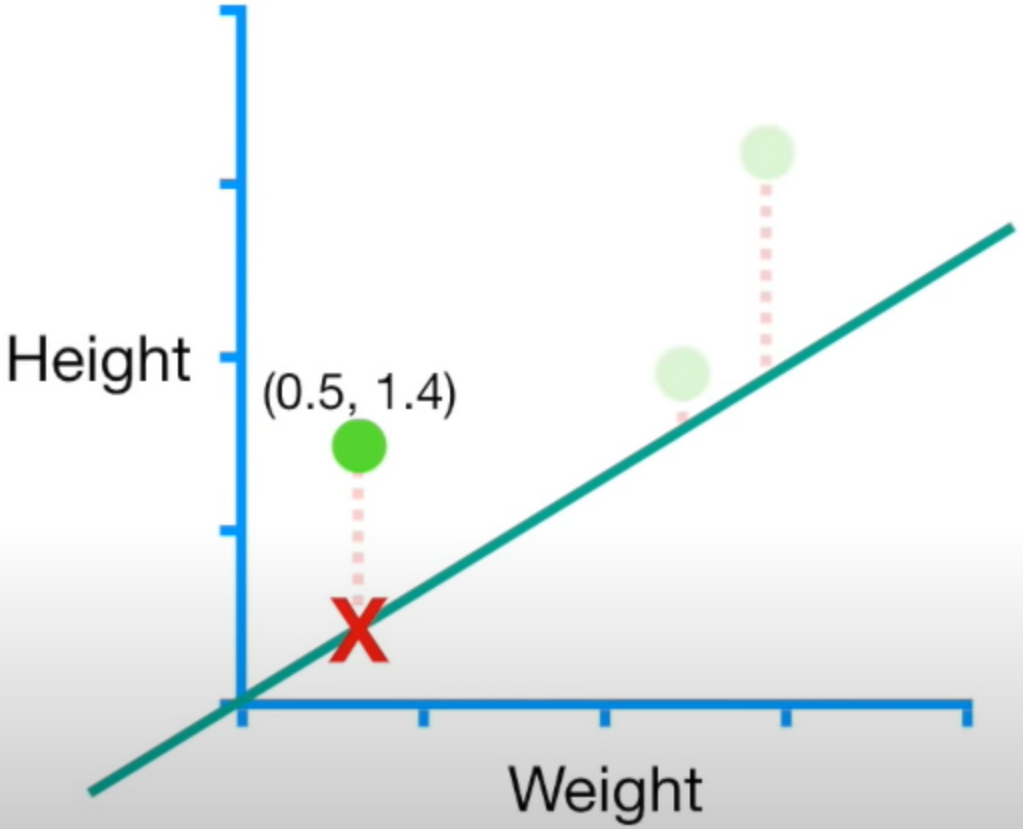

Height
(0.5, 1.4)
Weight
We get the **Predicted Height**, the point on the line…



We get the **Predicted Height**, the point on the line…

…by plugging **Weight = 0.5** into the equation for the line…

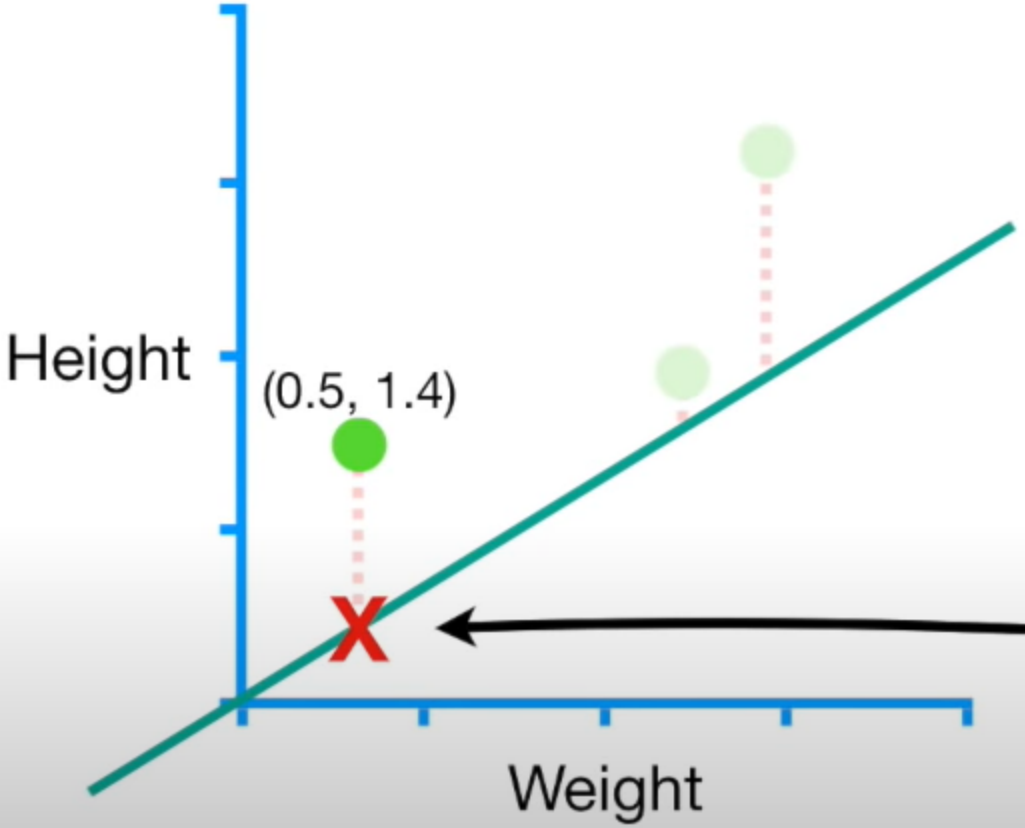
Predicted Height = 0 + 0.64 × **Weight**



We get the **Predicted Height**, the point on the line…

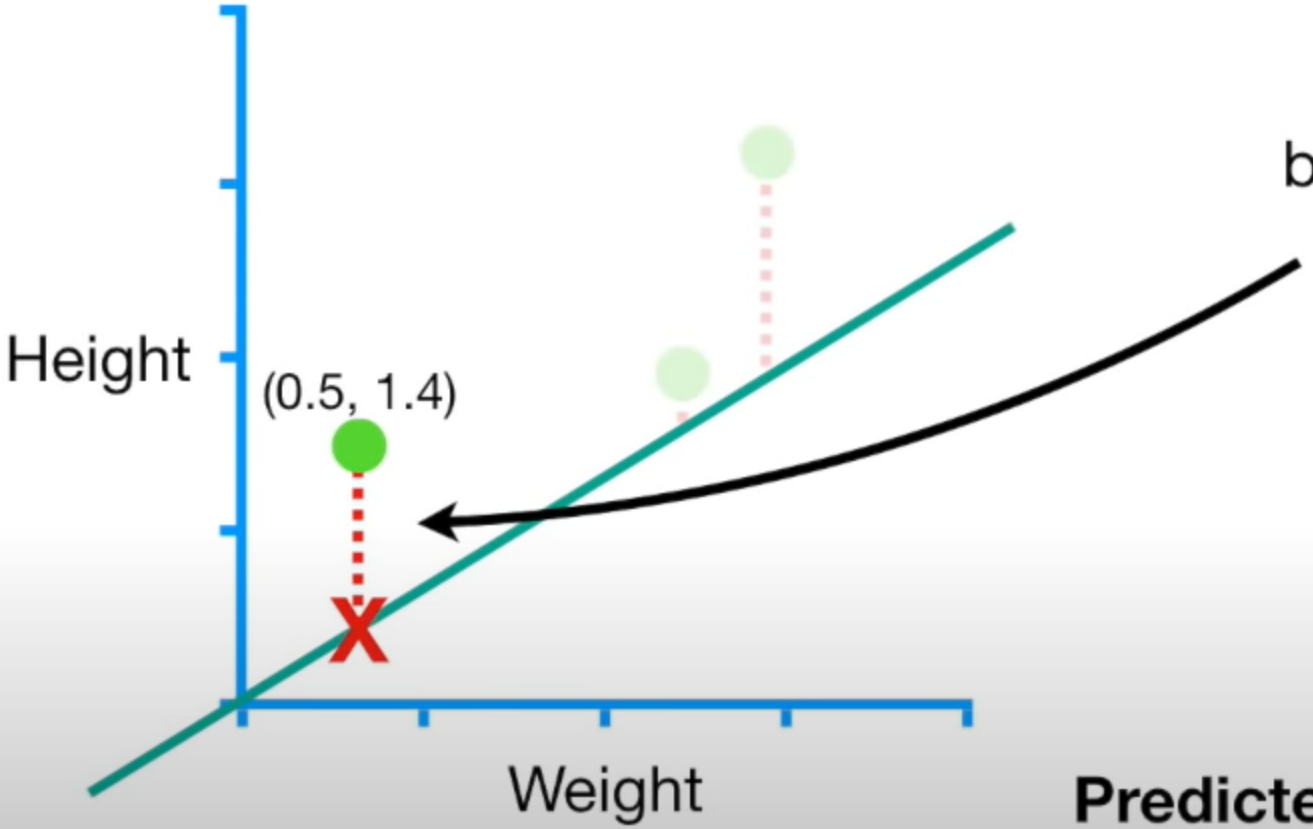
…by plugging **Weight = 0.5** into the equation for the line…

Predicted Height = 0 + 0.64 × **0.5**



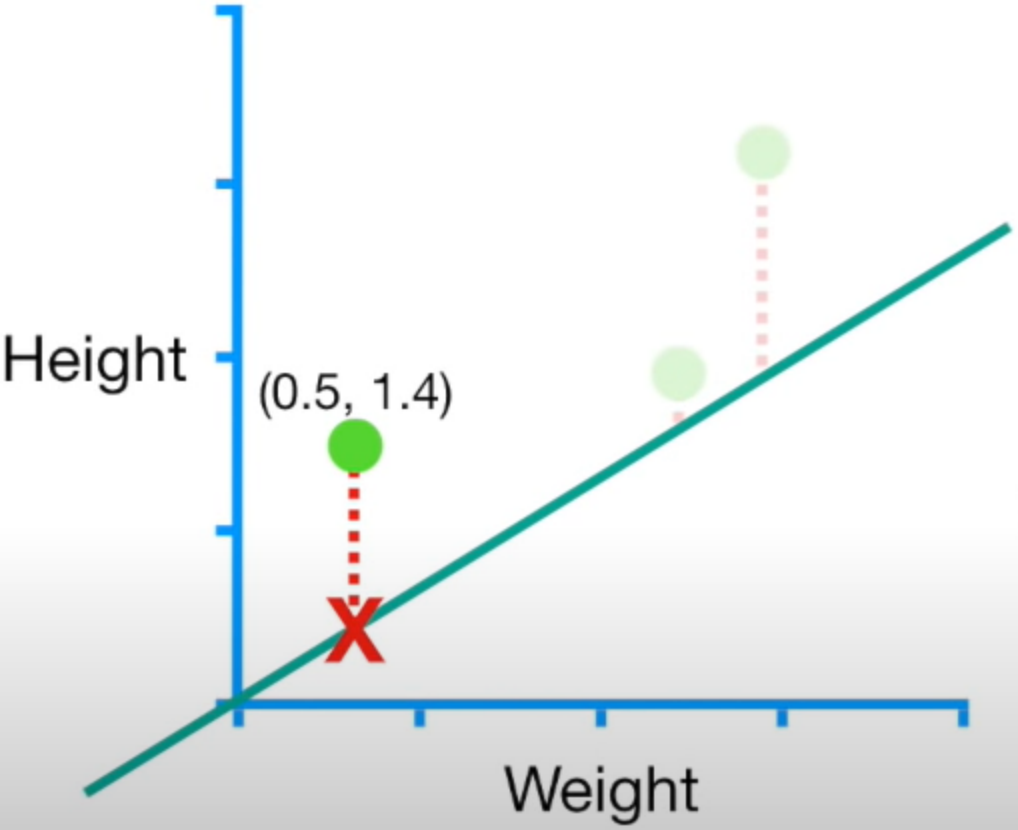
…and the **Predicted Height** is **0.32**.

Predicted Height = 0 + 0.64 × **0.5** = **0.32**



The residual is the difference between the **Observed Height**, and the **Predicted Height**…

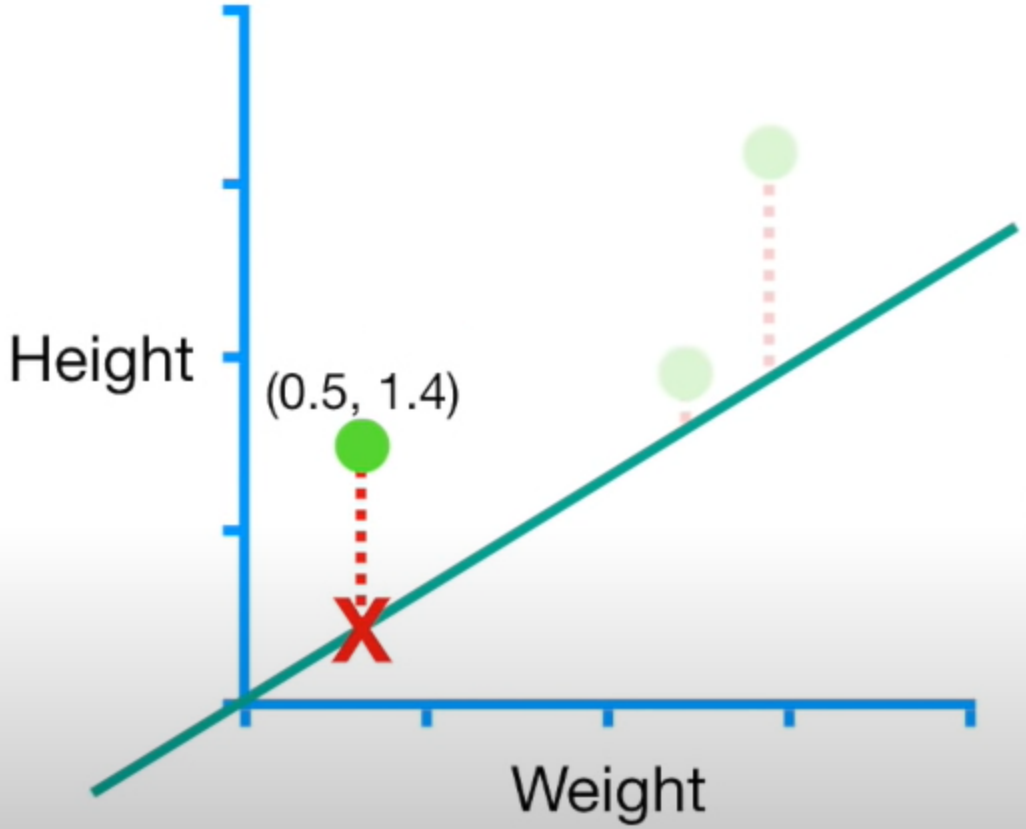
Predicted Height = 0 + 0.64 × **0.5** = **0.32**



The residual is the difference between the **Observed Height**, and the **Predicted Height**…

Residual = **Observed Height** - **Predicted Height**

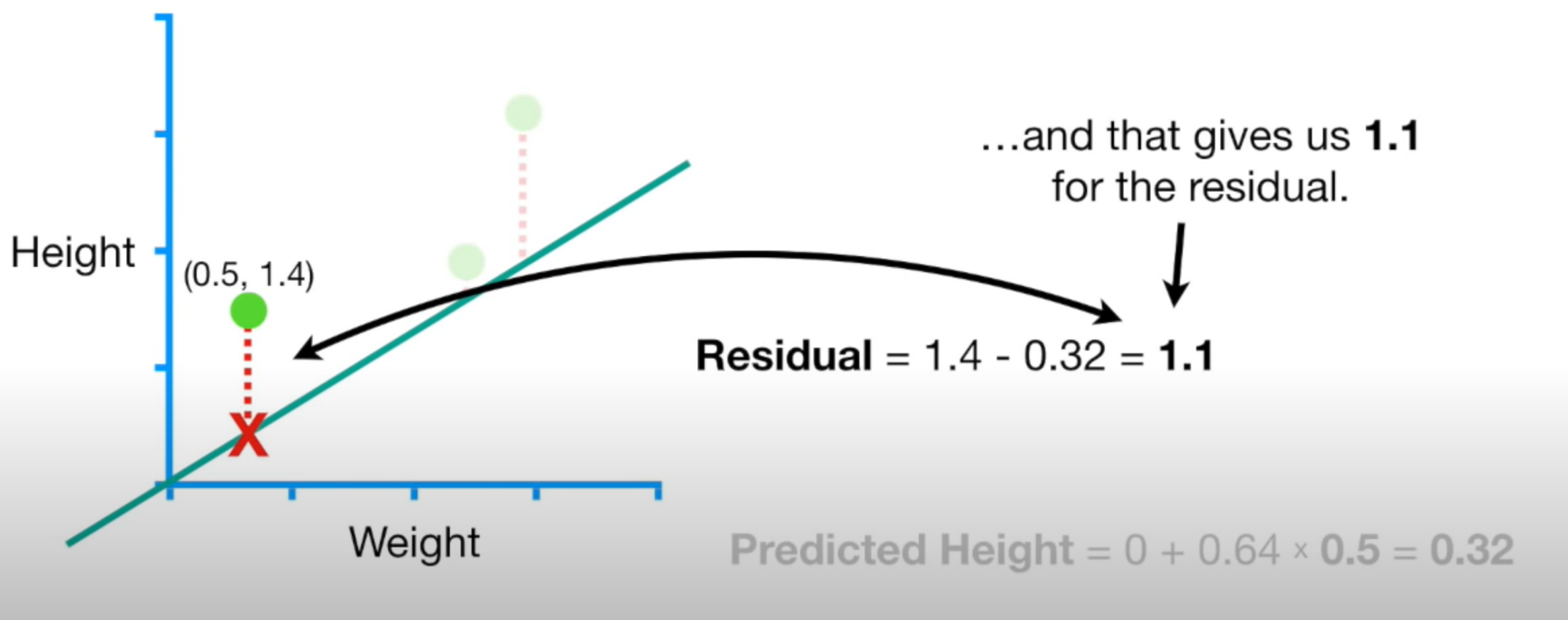
Predicted Height = 0 + 0.64 × **0.5** = **0.32**



…so we calculate the difference between **1.4** (the **Observed Height**)… …and **0.32** (the **Predicted Height**)…

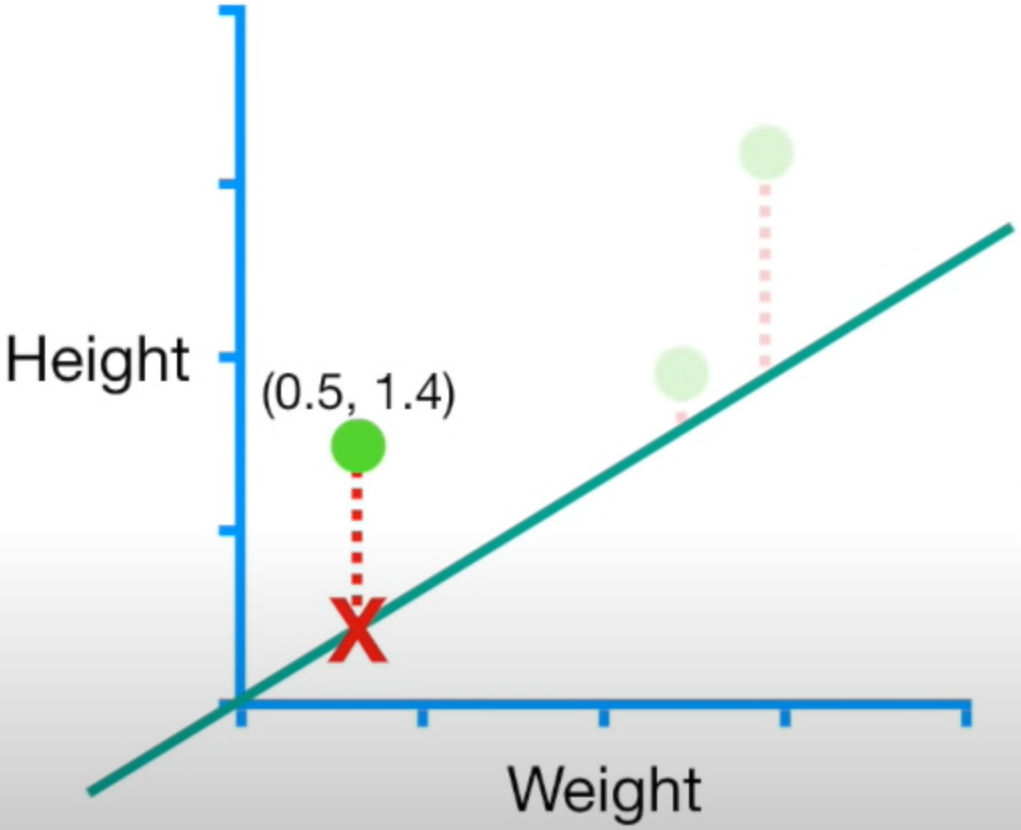
Residual = 1.4 - 0.32

Predicted Height = 0 + 0.64 × **0.5** = **0.32**


Height
(0.5, 1.4)
X
Weight
…and that gives us **1.1** for the residual.
Residual = 1.4 - 0.32 = **1.1**
Predicted Height = 0 + 0.64 × **0.5** = **0.32**

Sum of squared residuals =

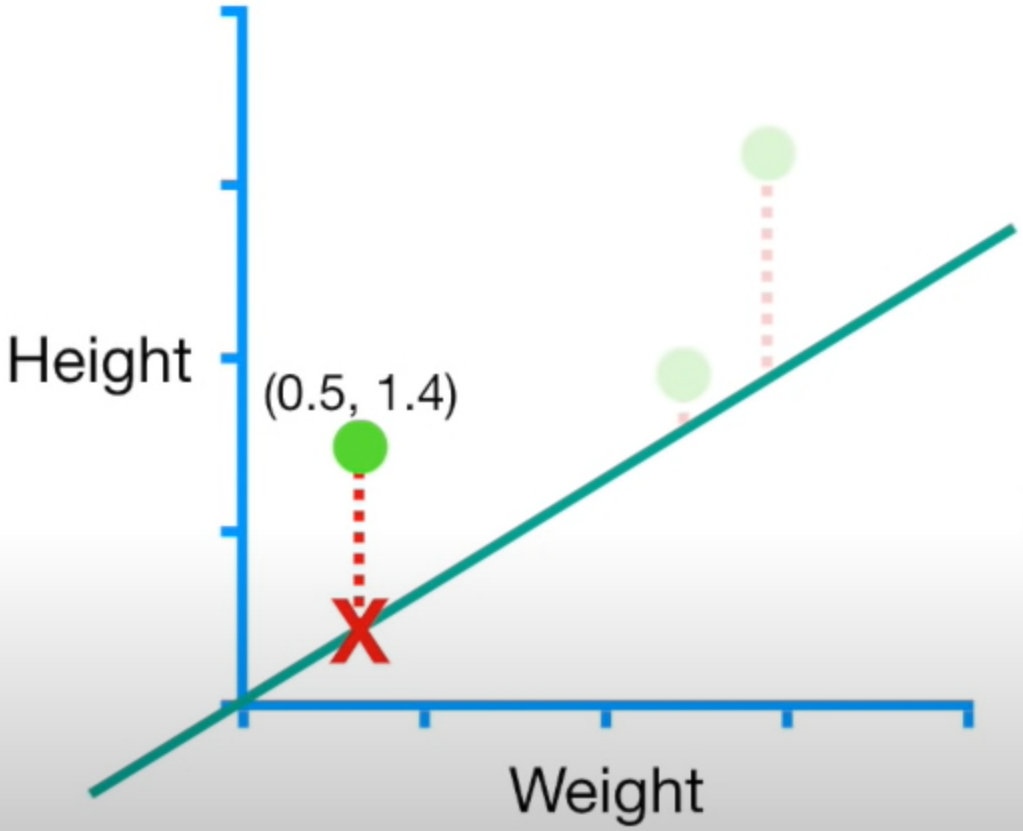
We'll keep track of the Sum of the Squared Residuals up here.



Residual = 1.4 - 0.32 = **1.1**

Predicted Height = 0 + 0.64 × **0.5** = **0.32**

Sum of squared residuals = 1.1^2

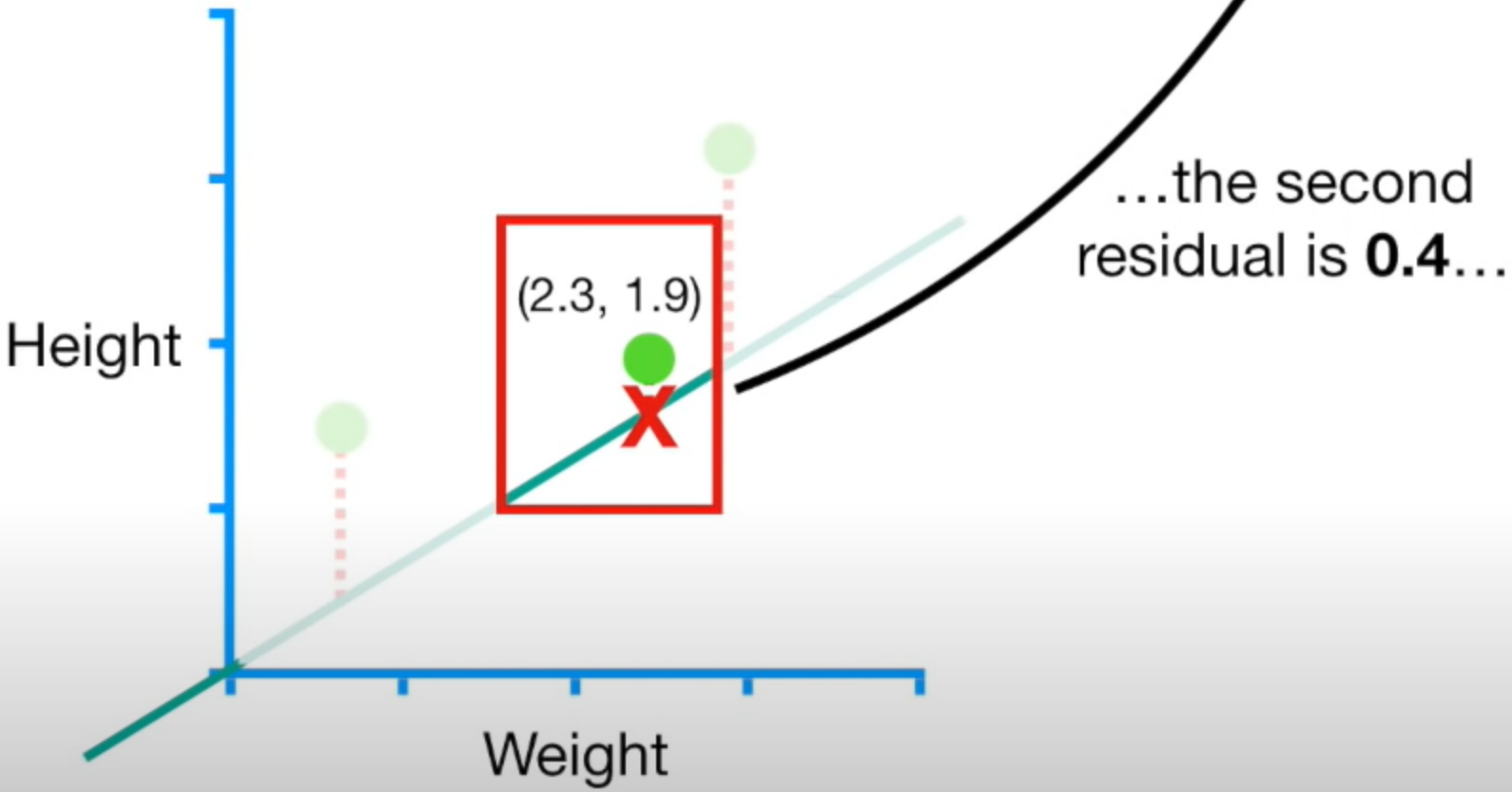


Here's the square of the first residual…

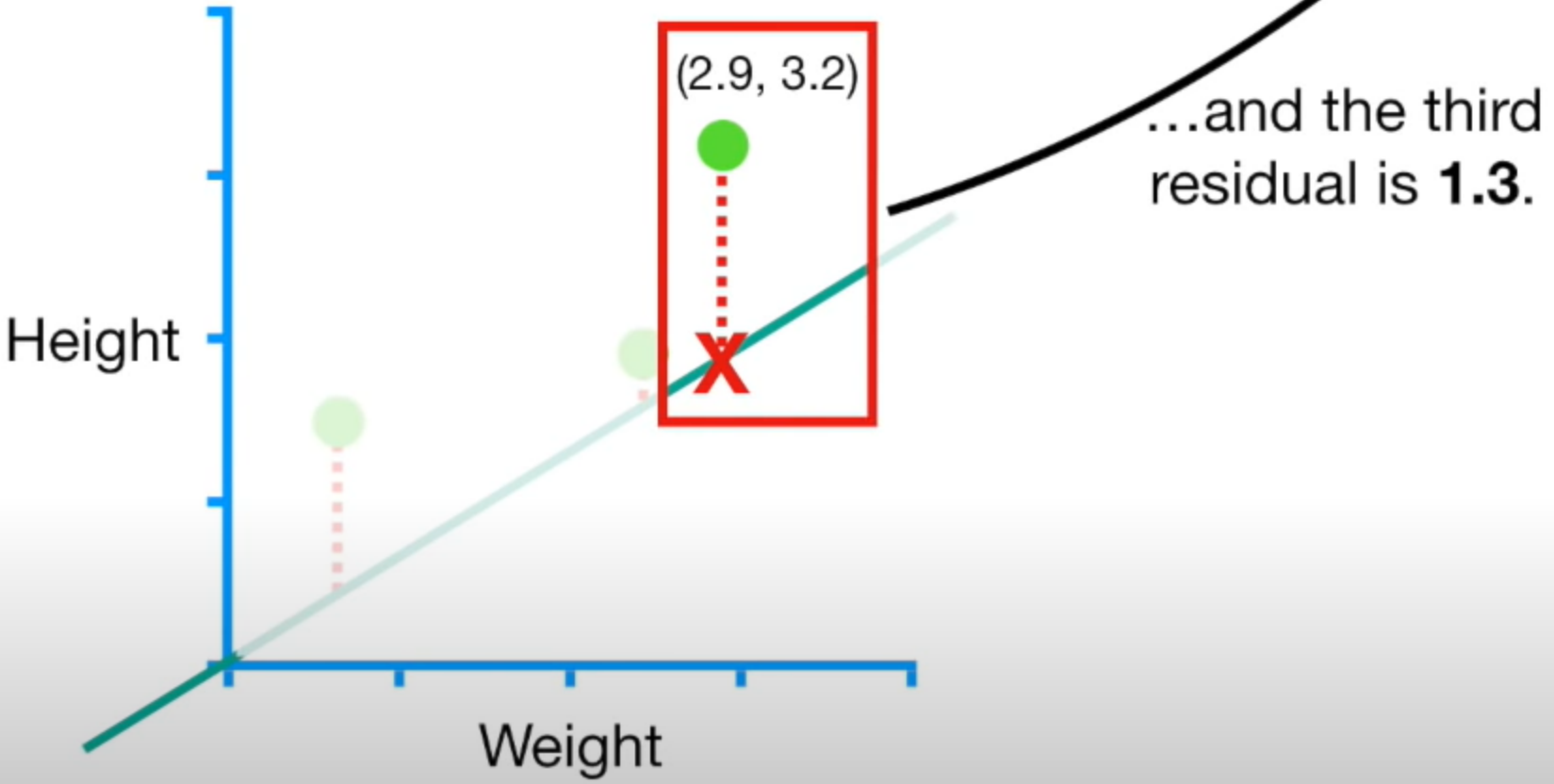
Residual = 1.4 - 0.32 = **1.1**

Predicted Height = 0 + 0.64 × **0.5** = **0.32**

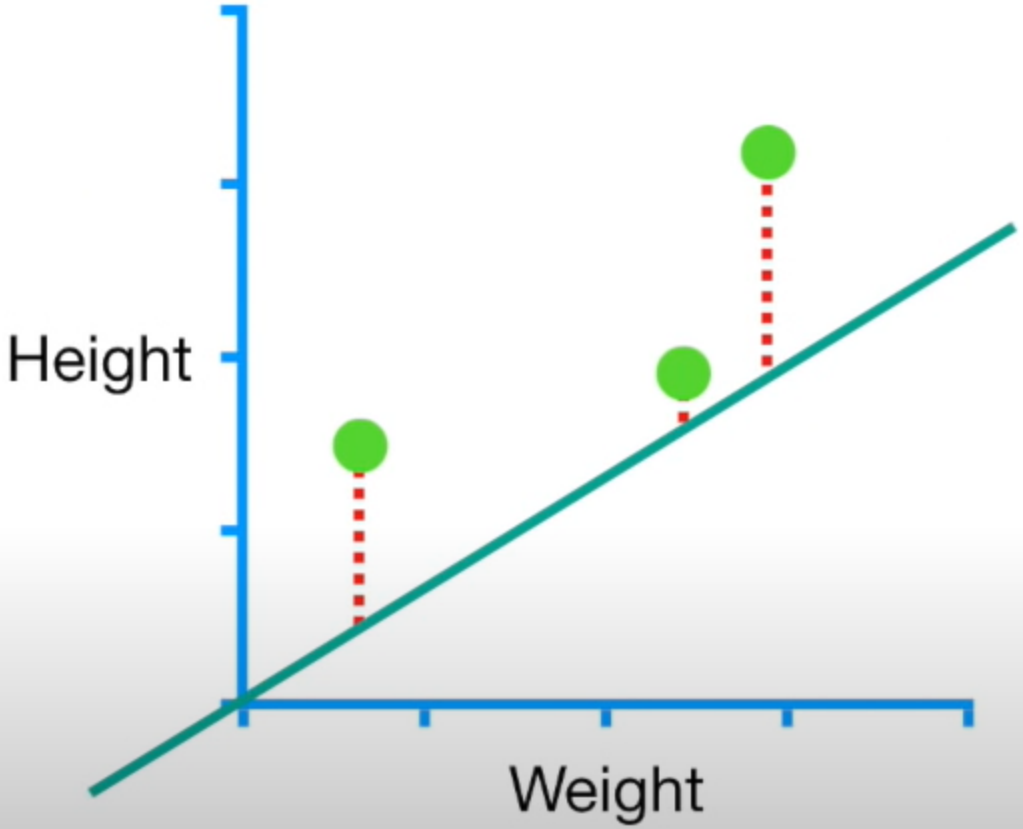
Sum of squared residuals = $1.1^2 + 0.4^2$



Sum of squared residuals = $1.1^2 + 0.4^2 + 1.3^2$

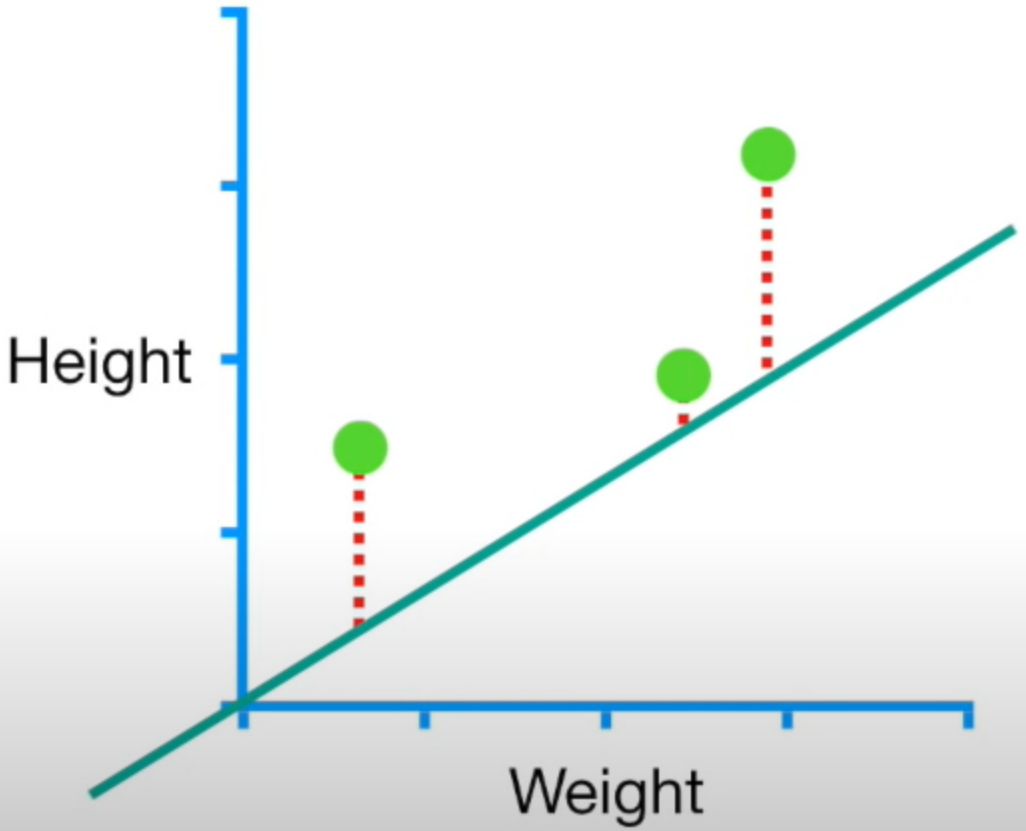


Sum of squared residuals = $1.1^2 + 0.4^2 + 1.3^2 = 3.1$

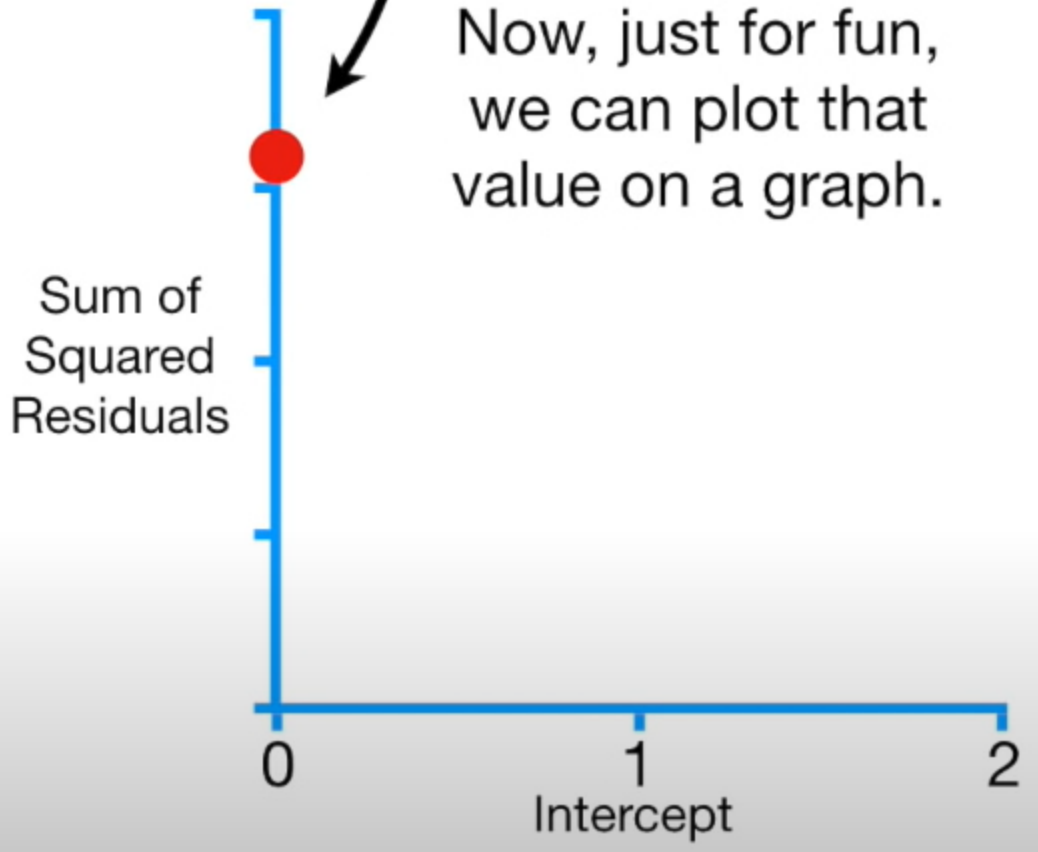


In the end, **3.1** is the Sum of the Squared Residuals.

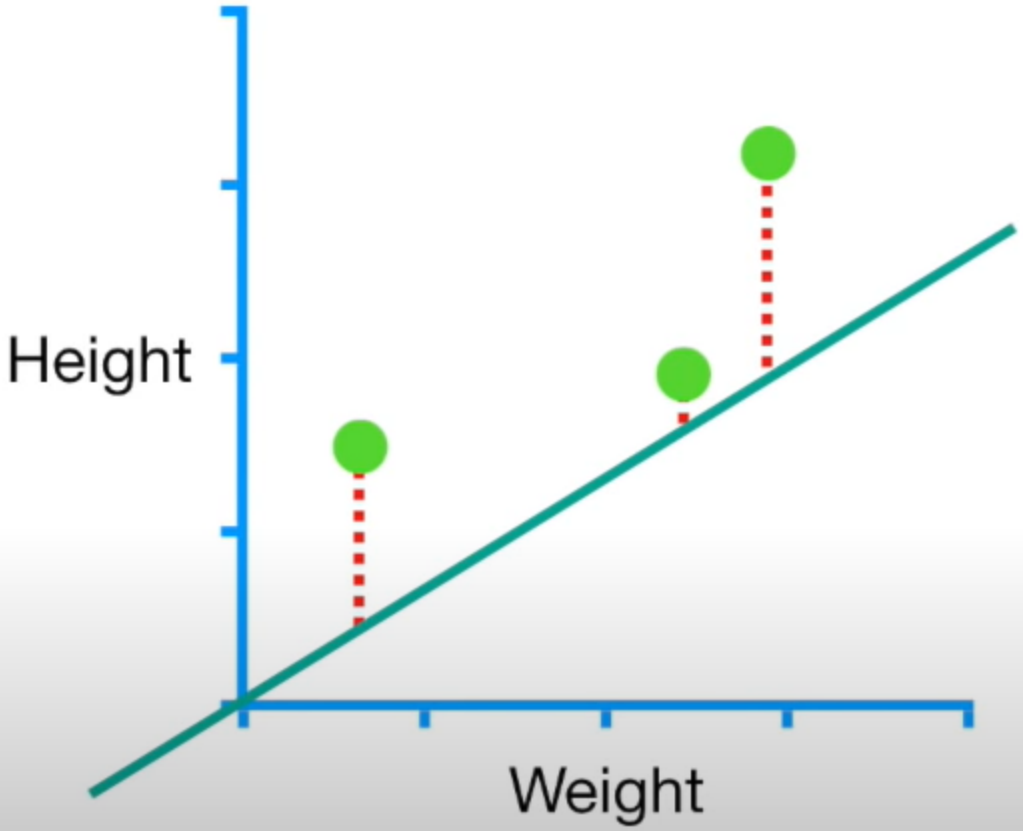
Sum of squared residuals = $1.1^2 + 0.4^2 + 1.3^2 = 3.1$

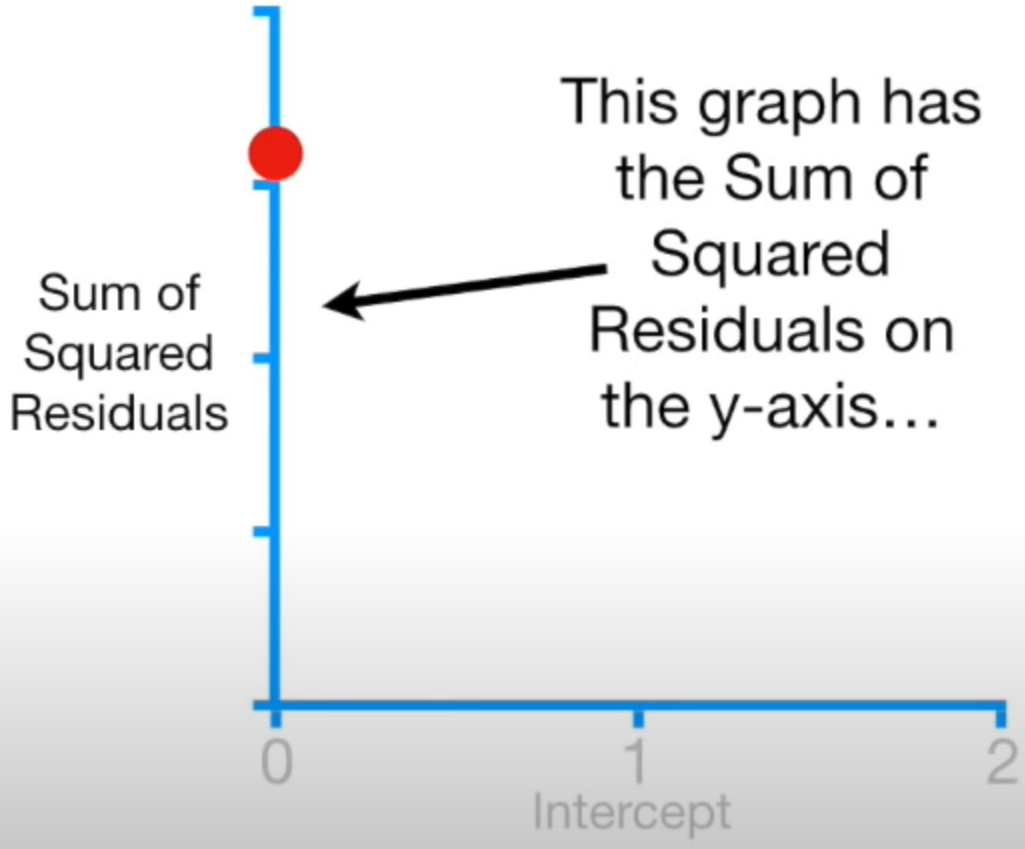


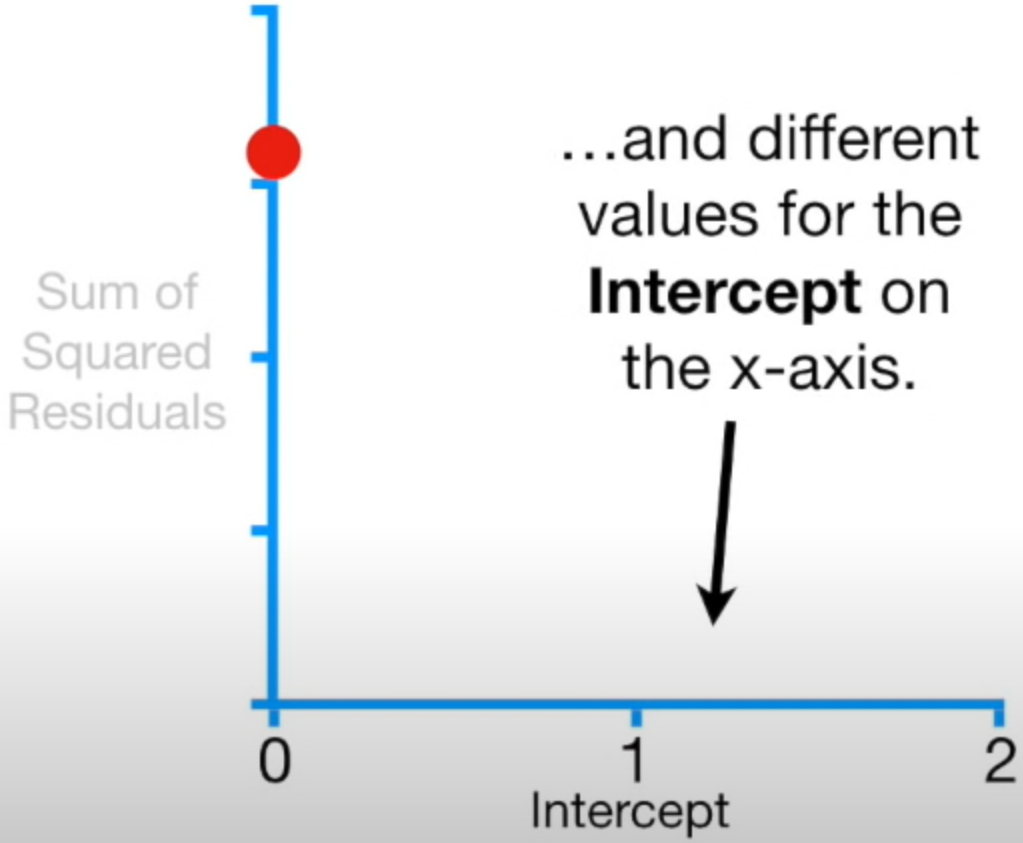
Now, just for fun, we can plot that value on a graph.

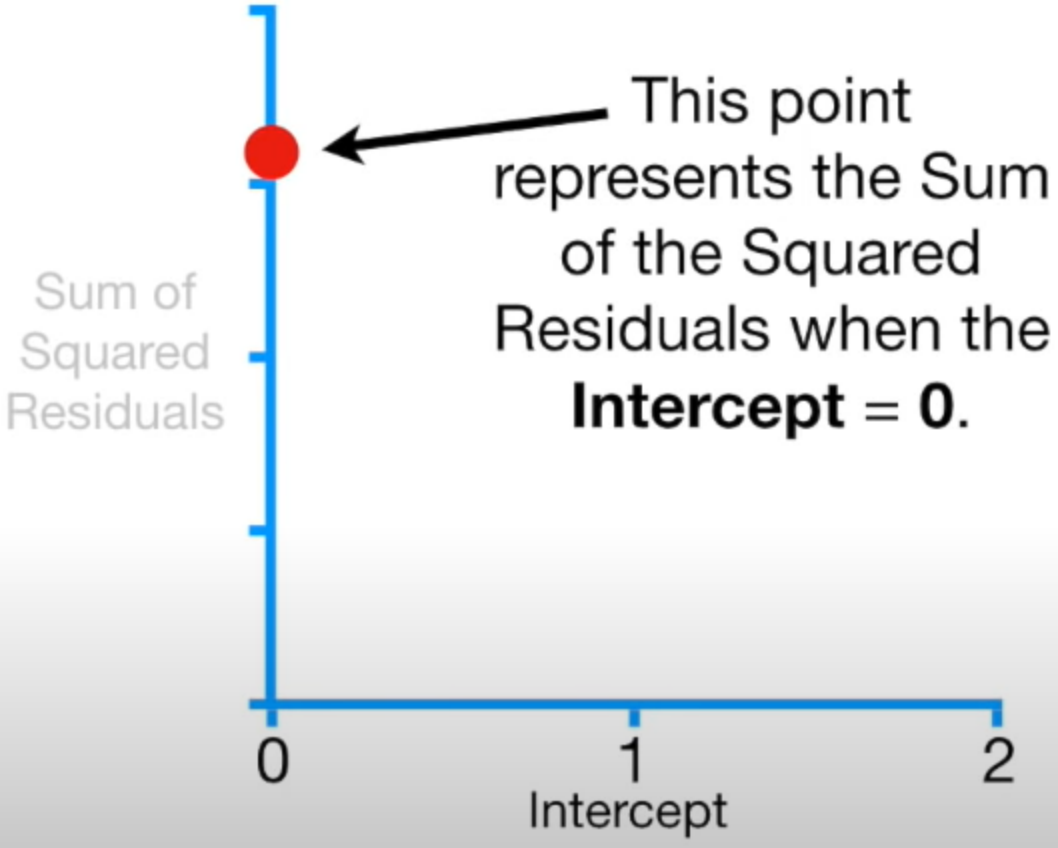


Sum of squared residuals = $1.1^2 + 0.4^2 + 1.3^2 = 3.1$

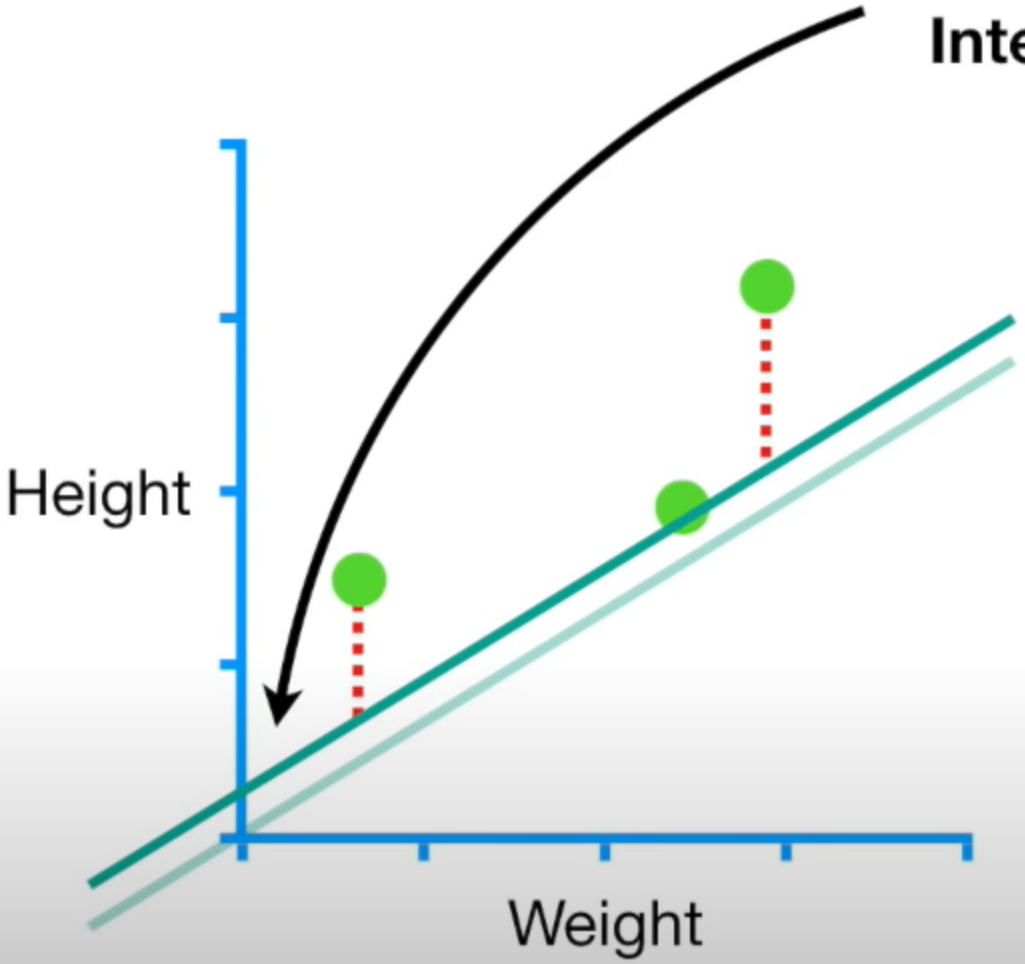




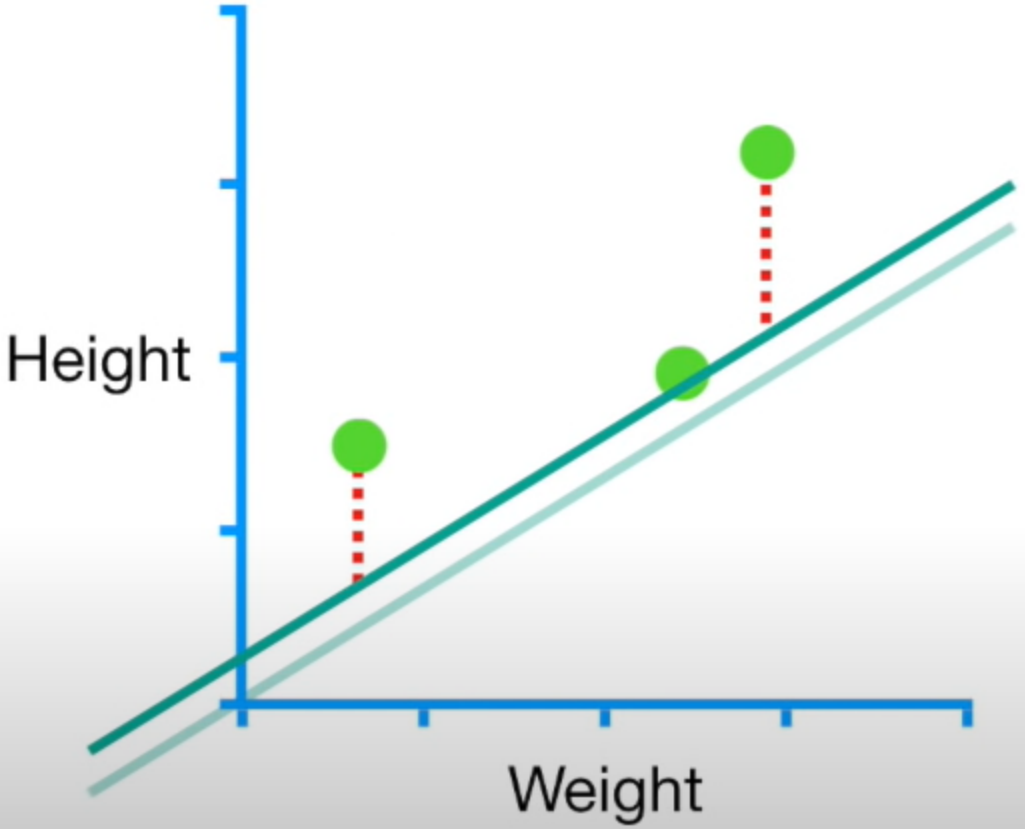

Sum of Squared Residuals
…and different values for the **Intercept** on the x-axis.
0
1
2
Intercept

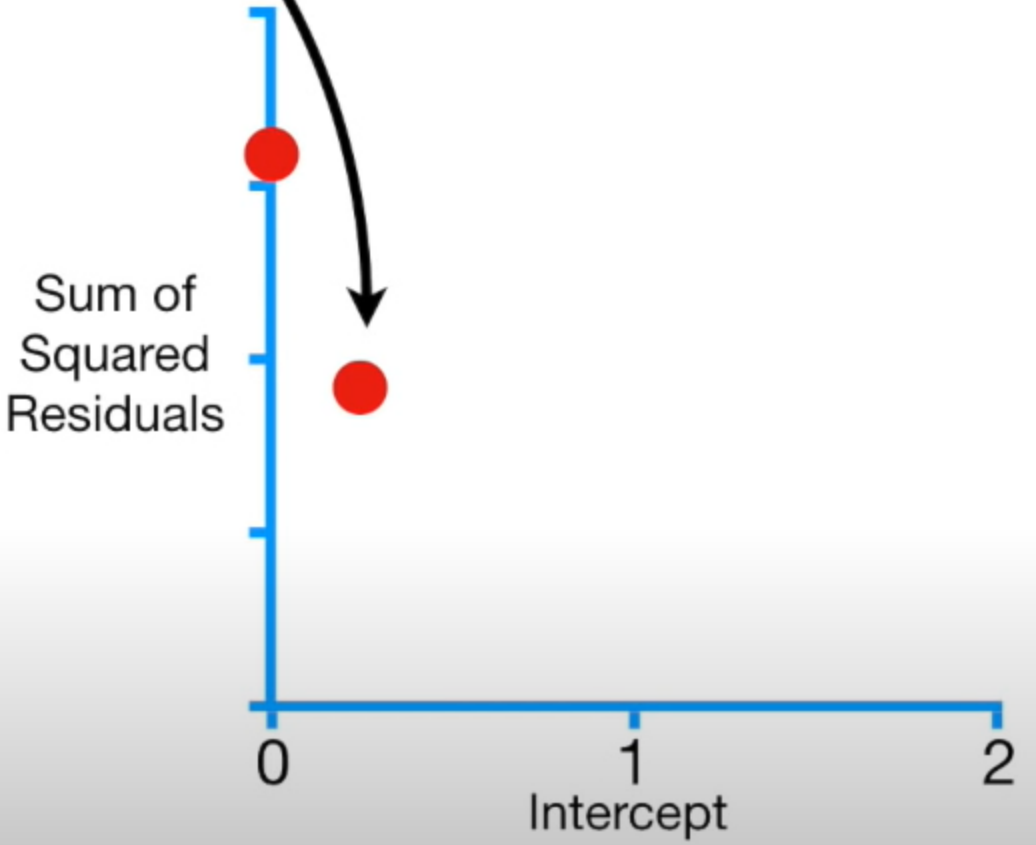

This point represents the Sum of the Squared Residuals when the **Intercept = 0**.
Sum of Squared Residuals
0
1
2
Intercept

However, if the
Intercept = **0.25**…

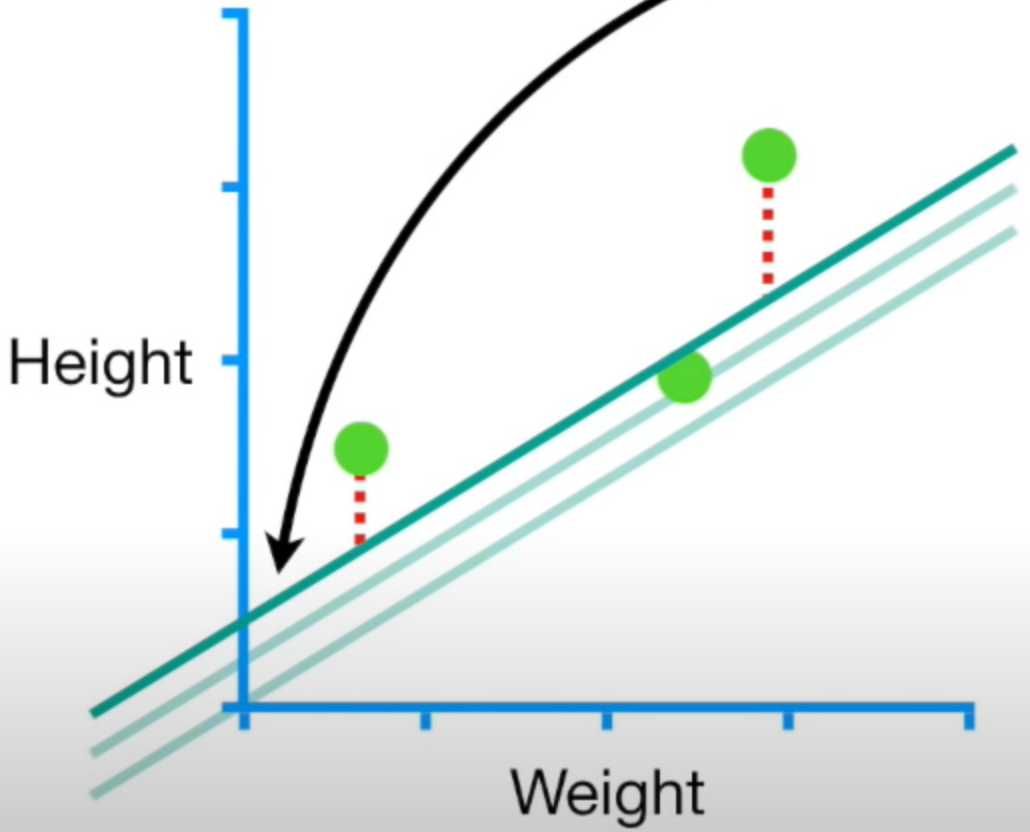


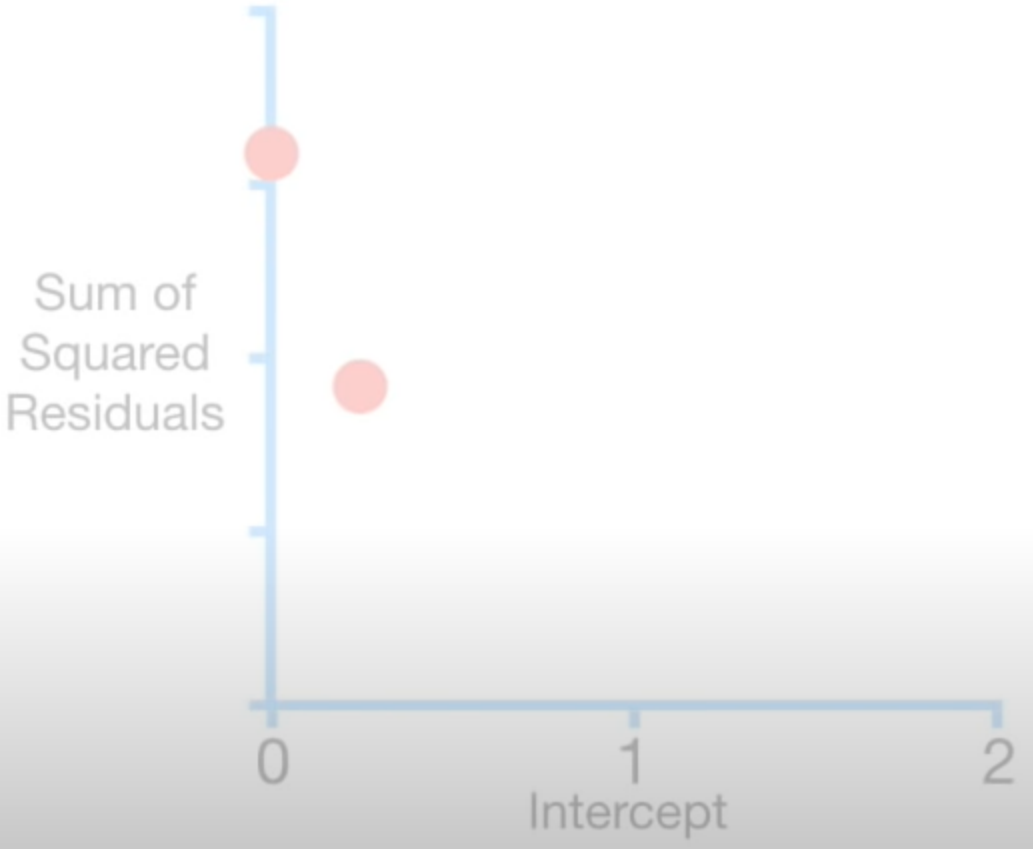
…then we would get this point on the graph.

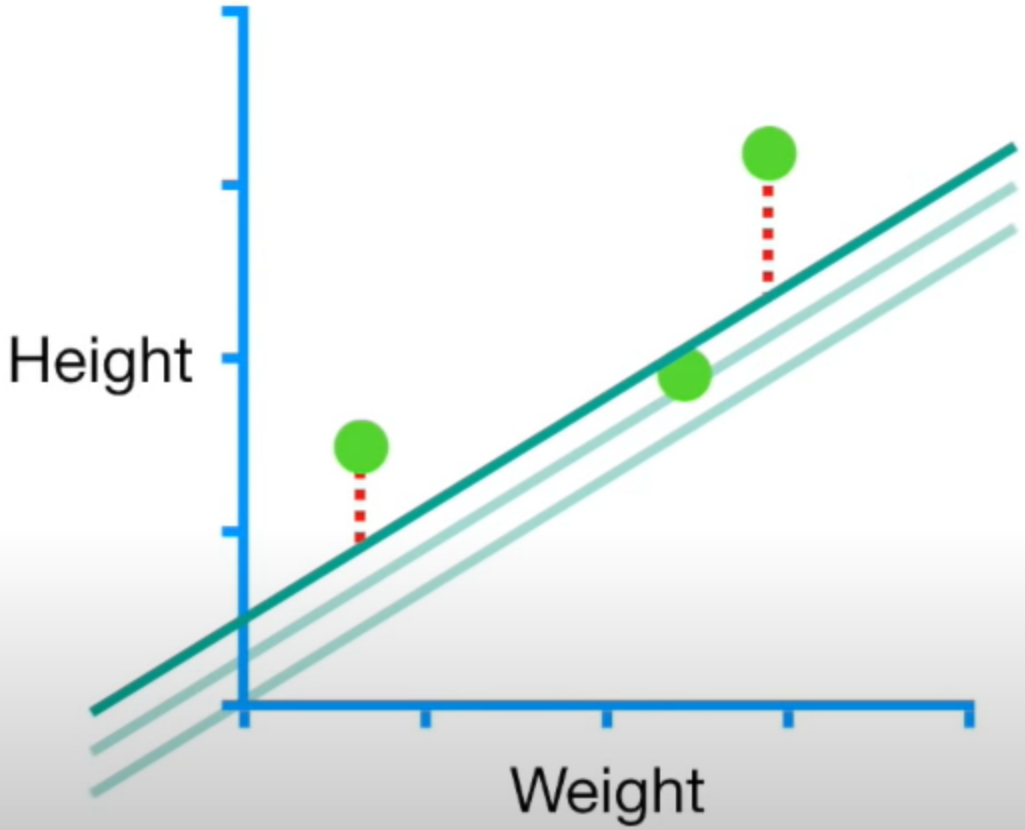




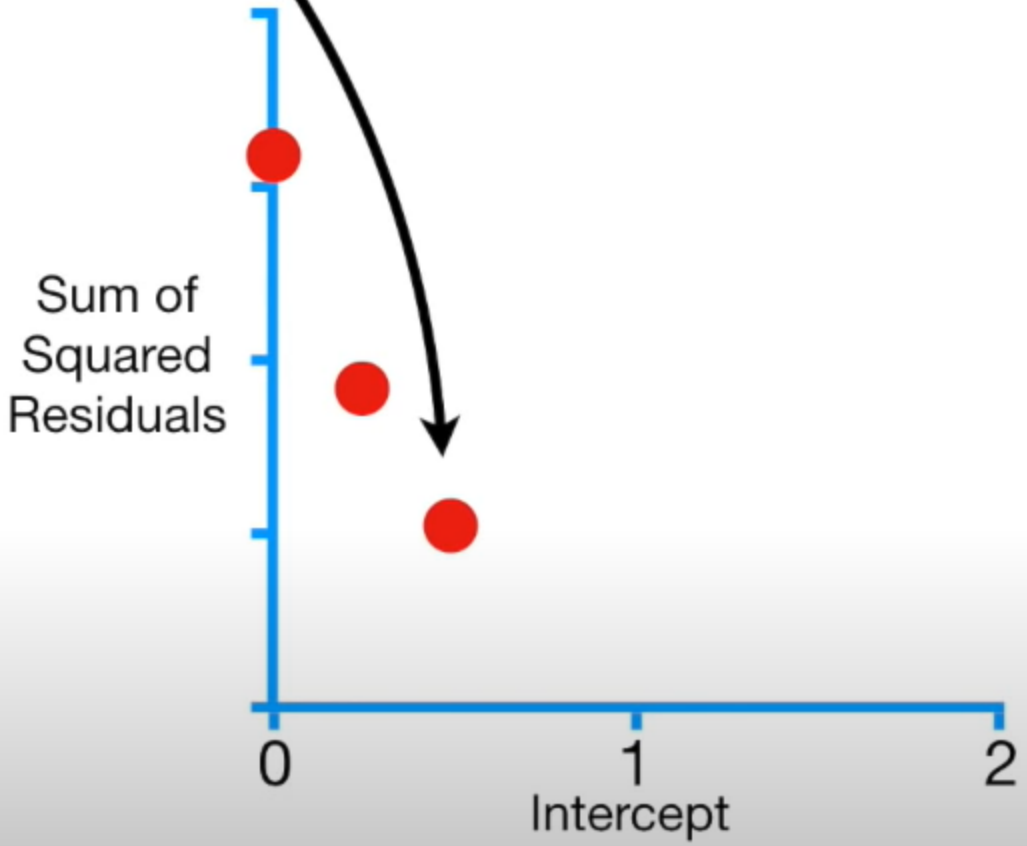
And if the
Intercept = 0.5…



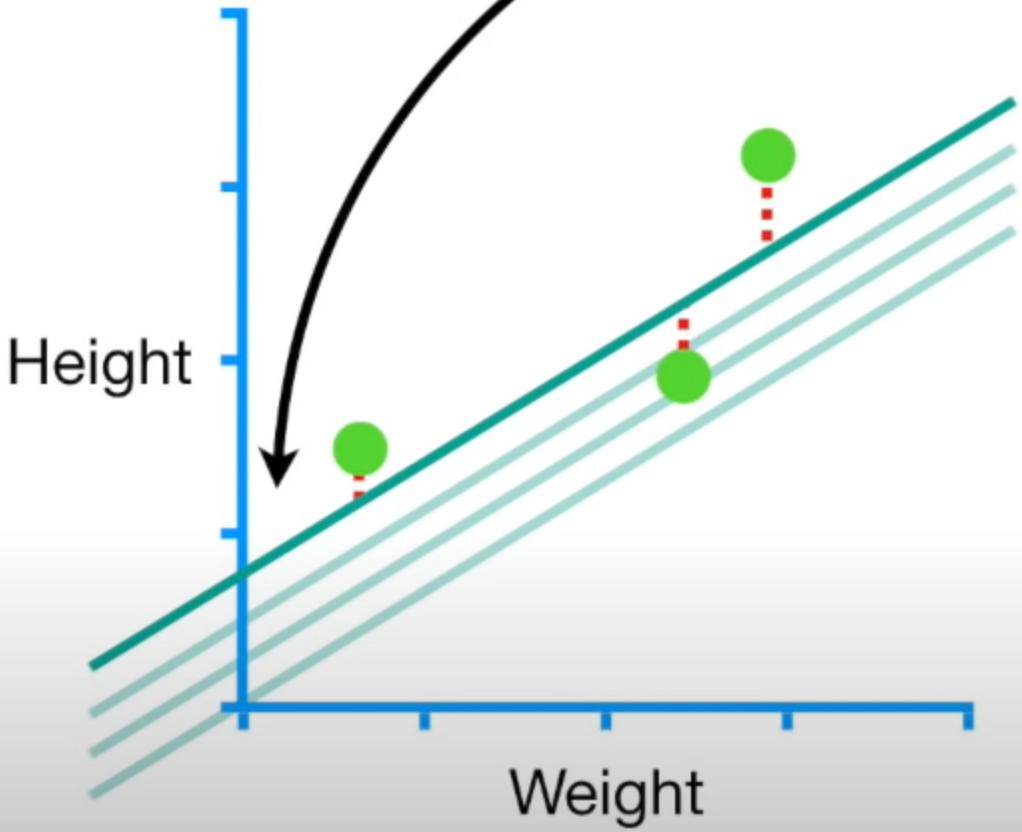


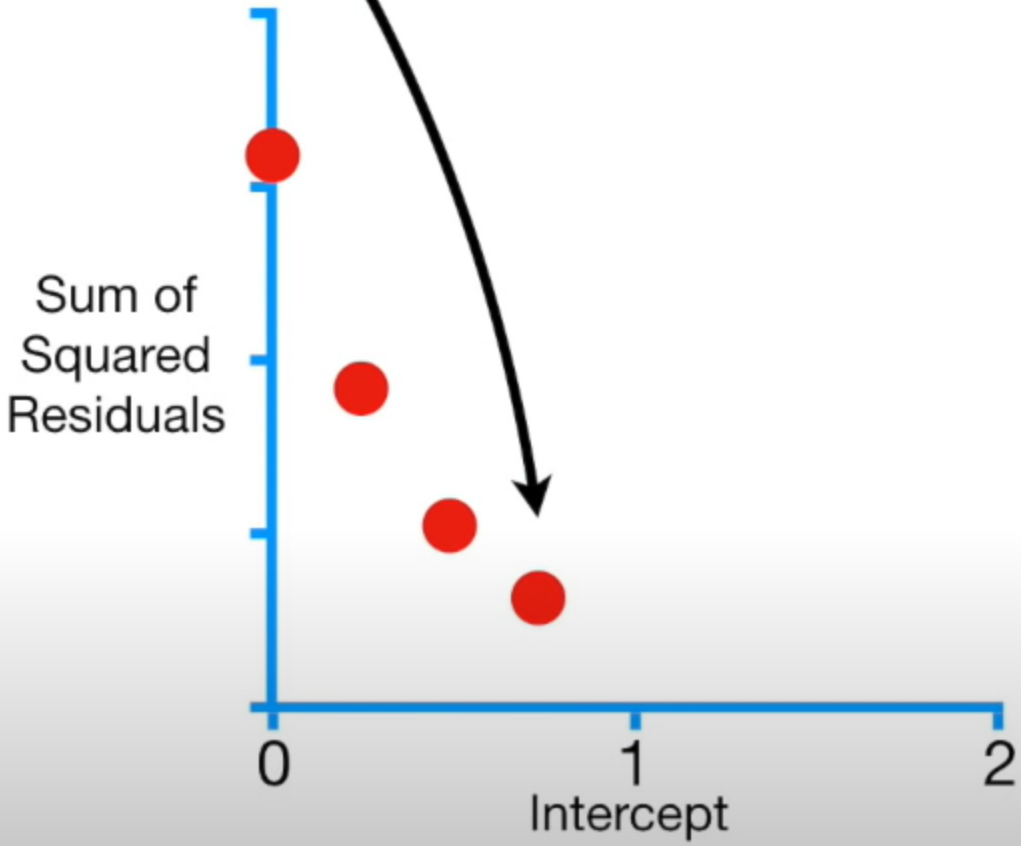


…then we would get this point.

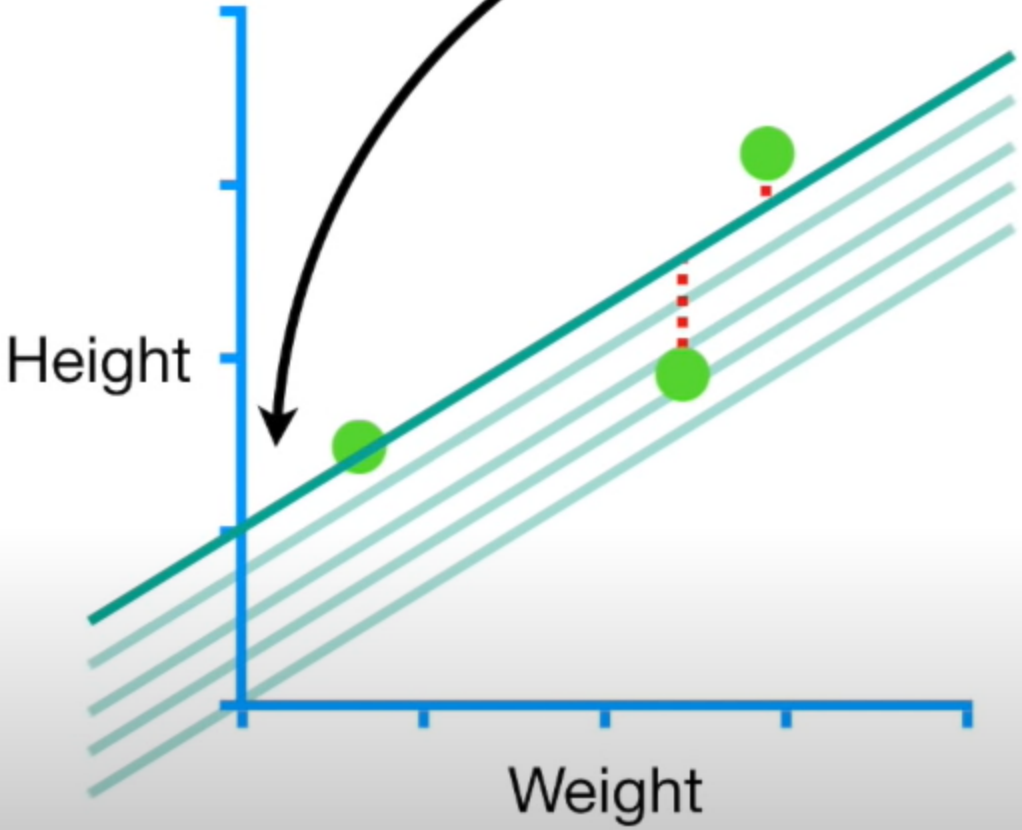


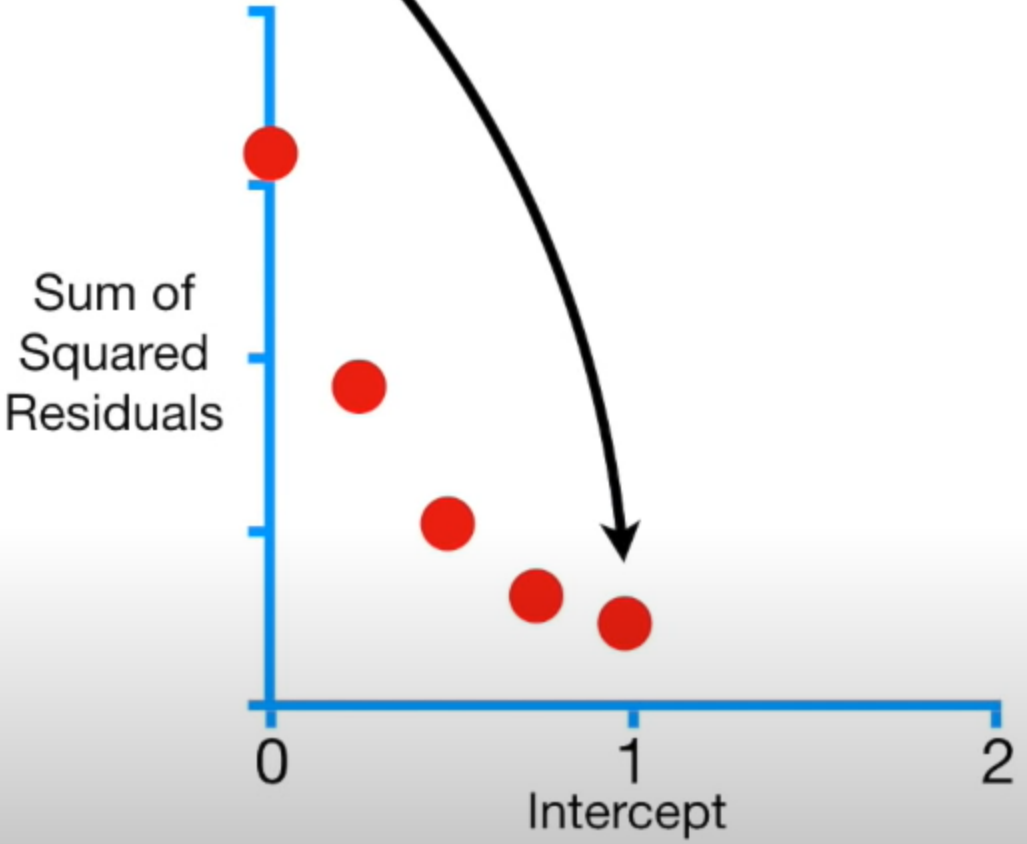
And for increasing values for the **Intercept**, we get these points.



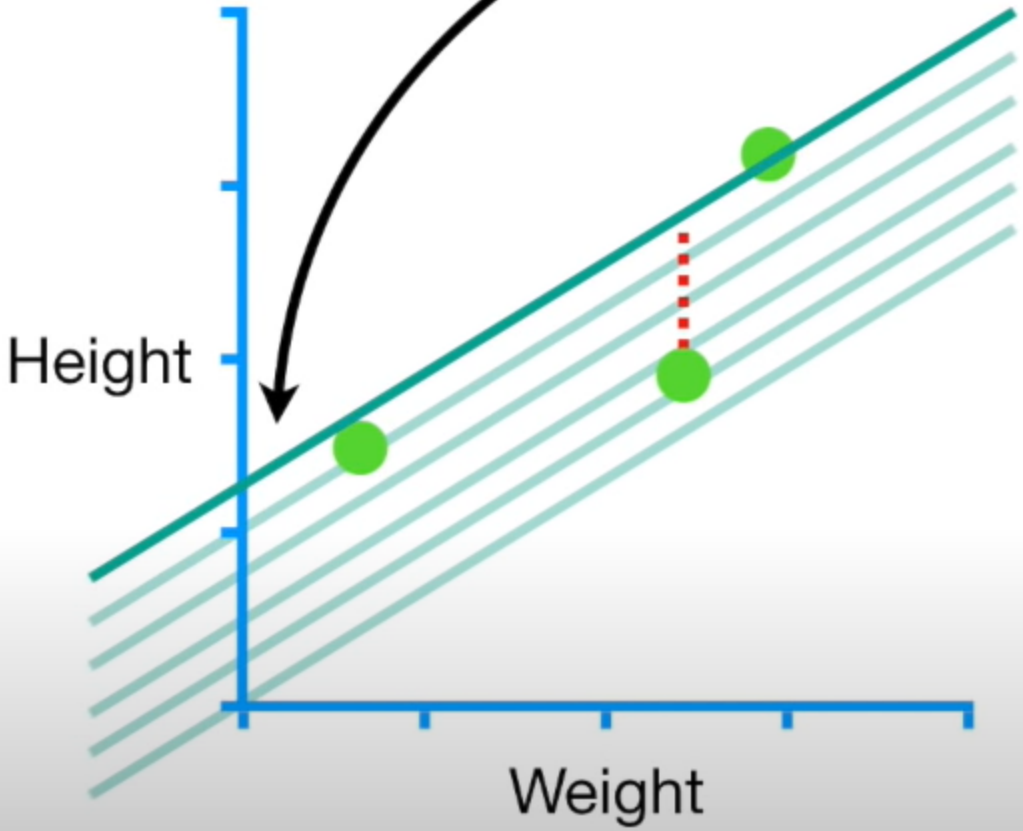


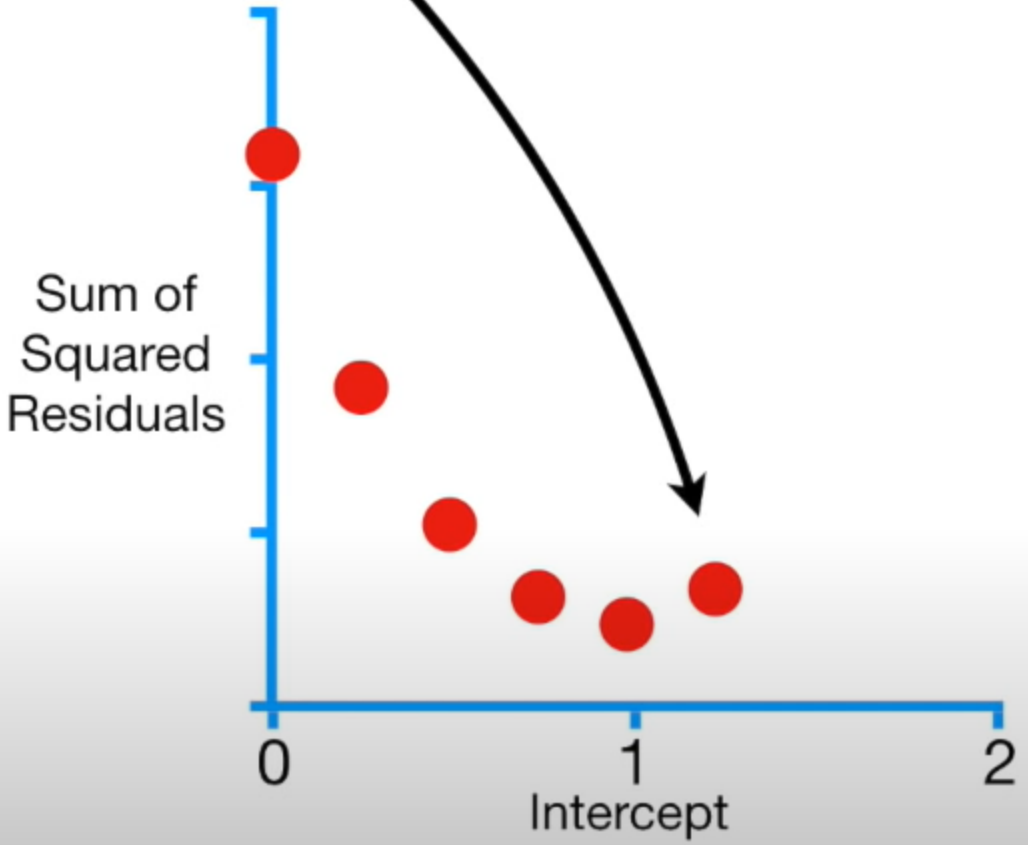
And for increasing values for the **Intercept**, we get these points.



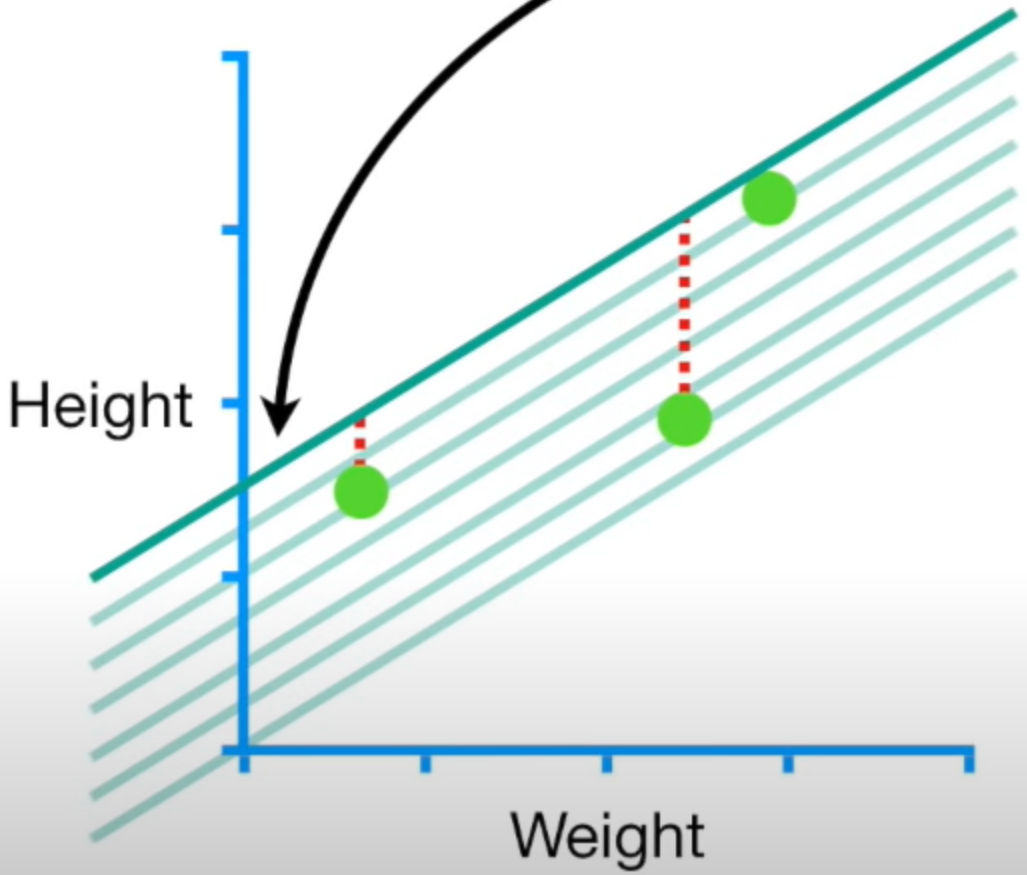


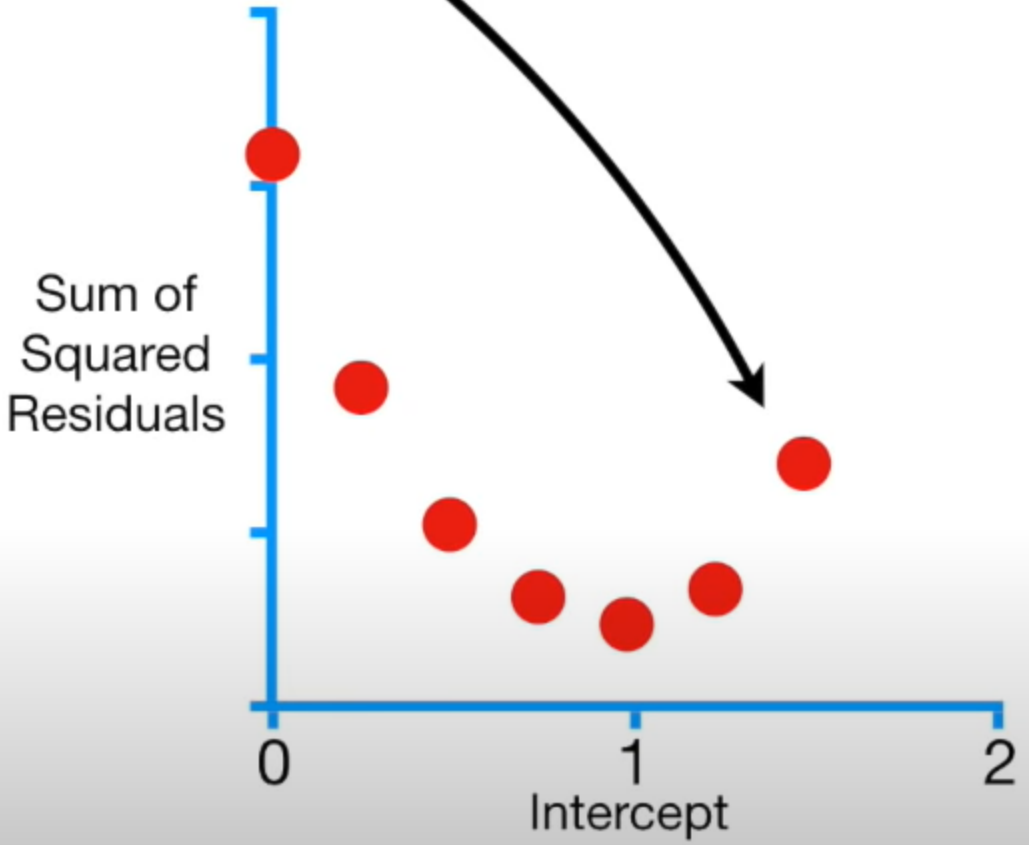
And for increasing values for the **Intercept**, we get these points.



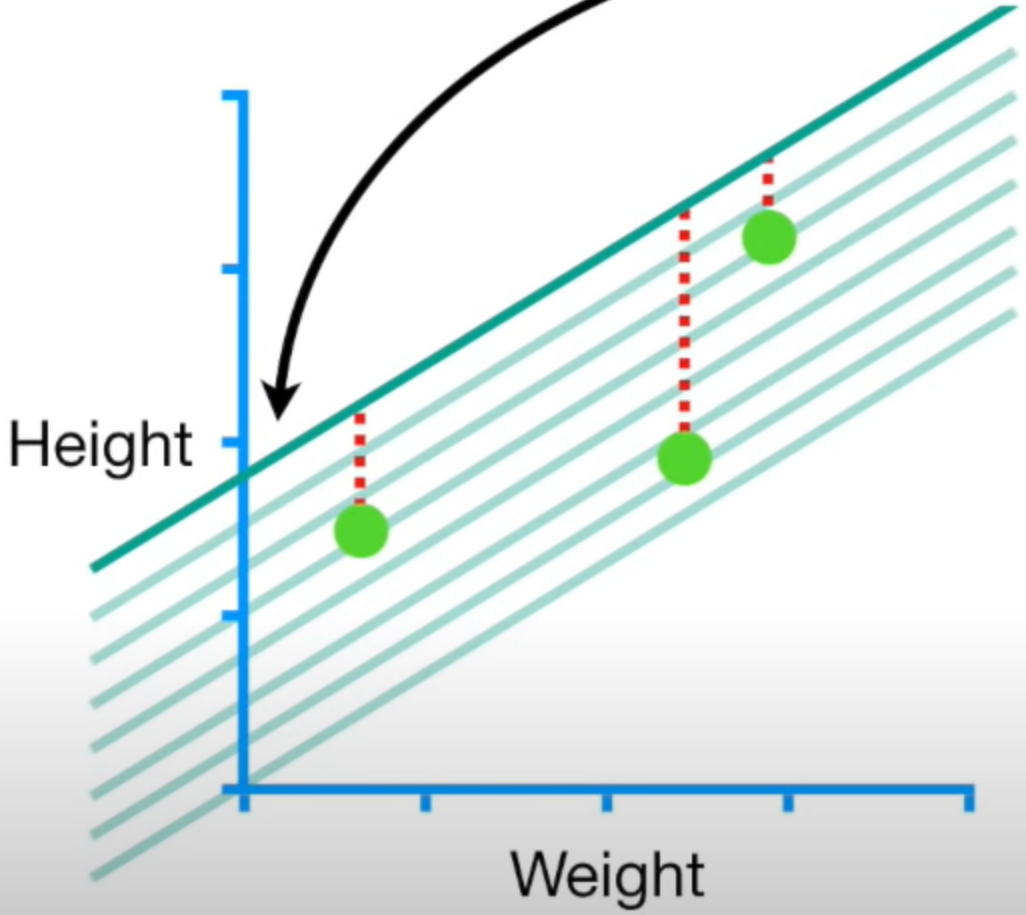


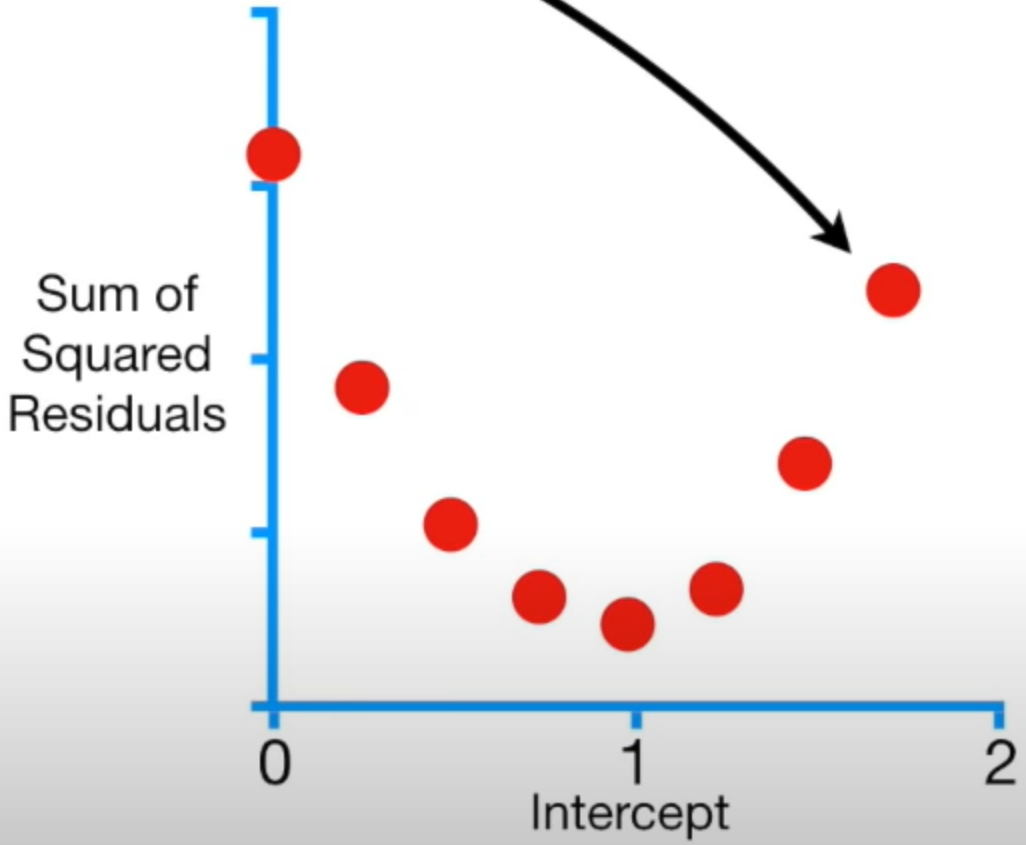
And for increasing values for the **Intercept**, we get these points.



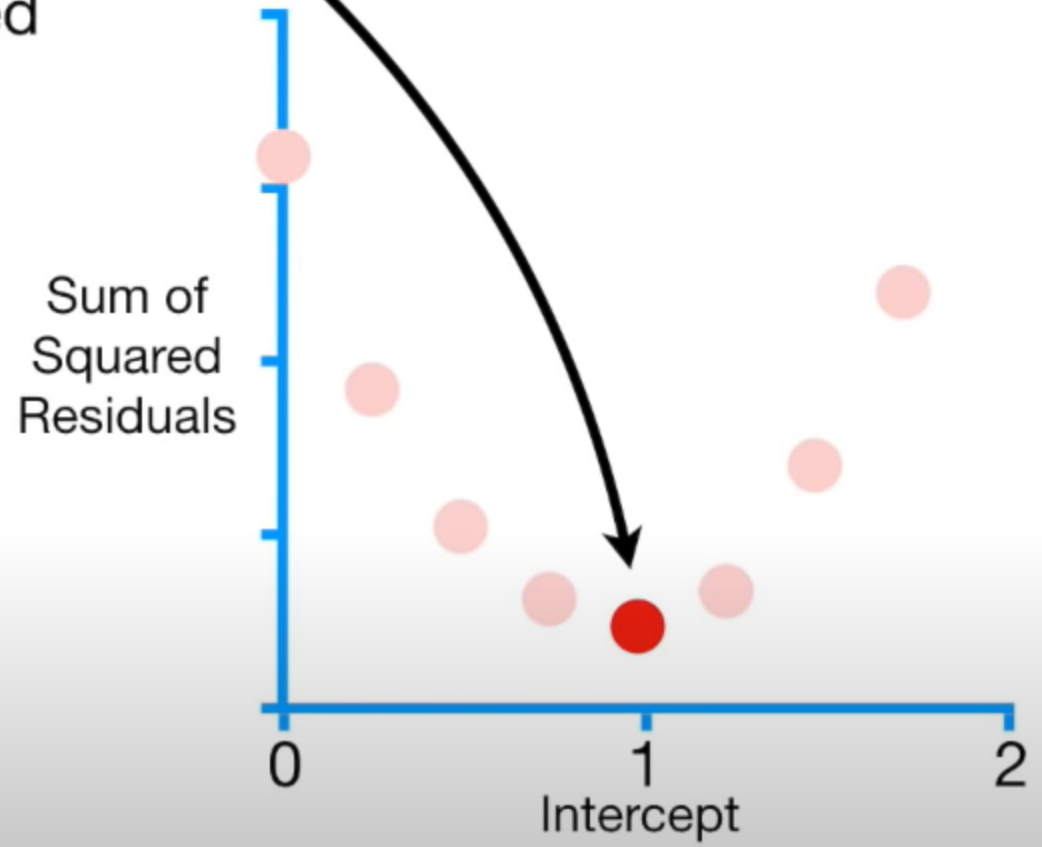


And for increasing values for the **Intercept**, we get these points.

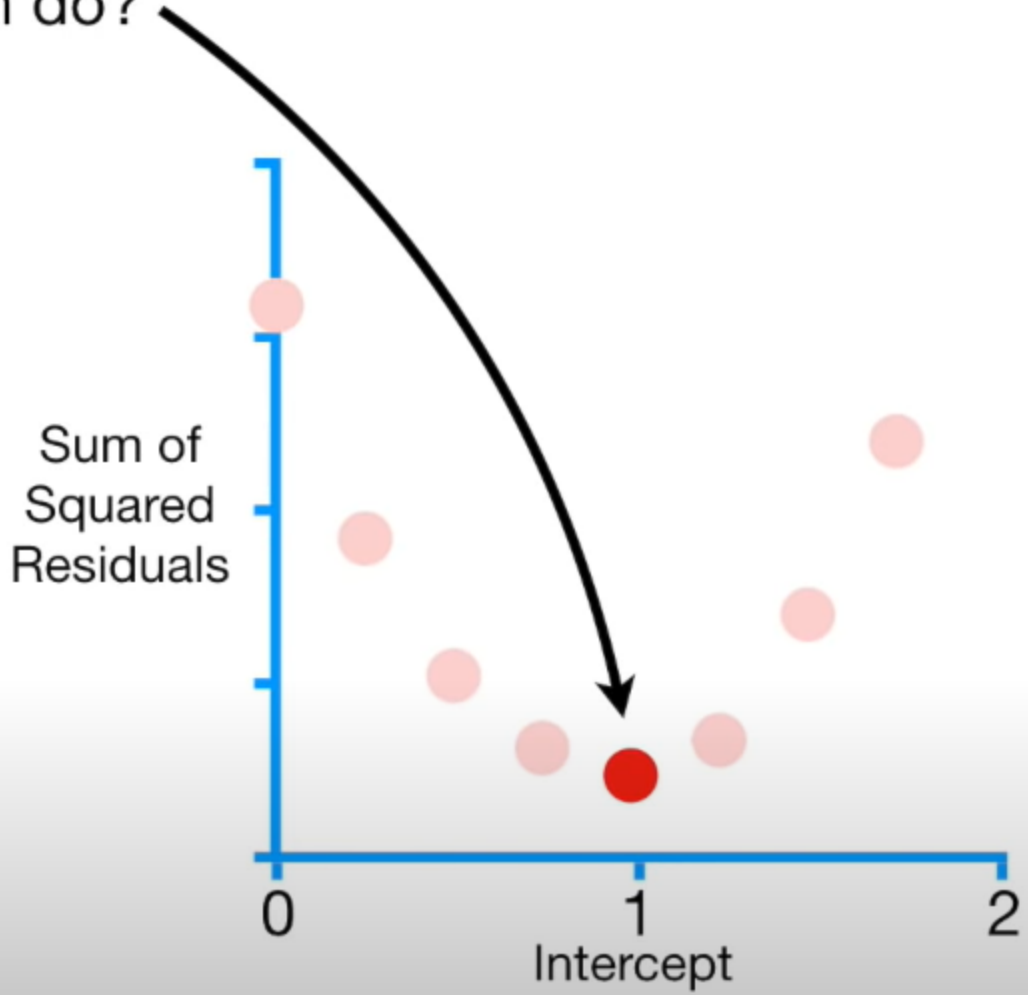




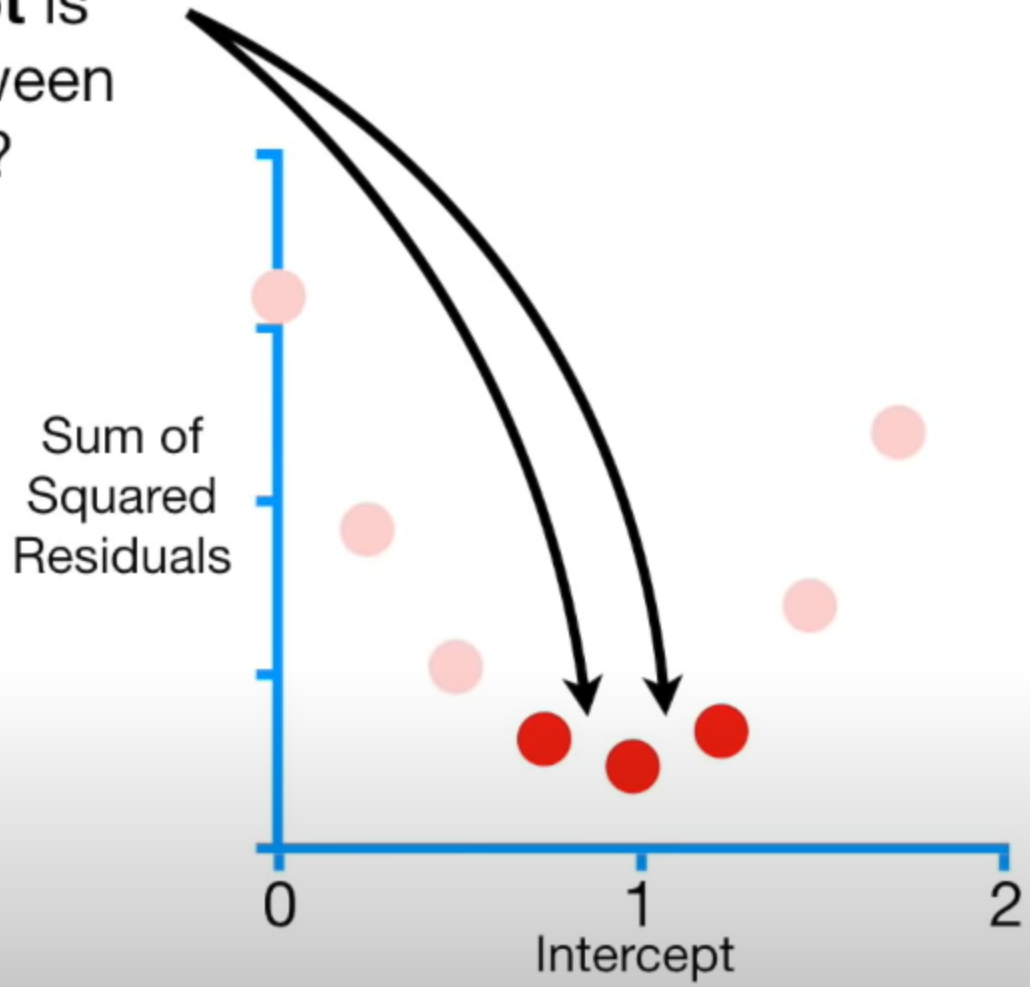
Of the points that we calculated for the graph, this one has the lowest Sum of Squared Residuals…



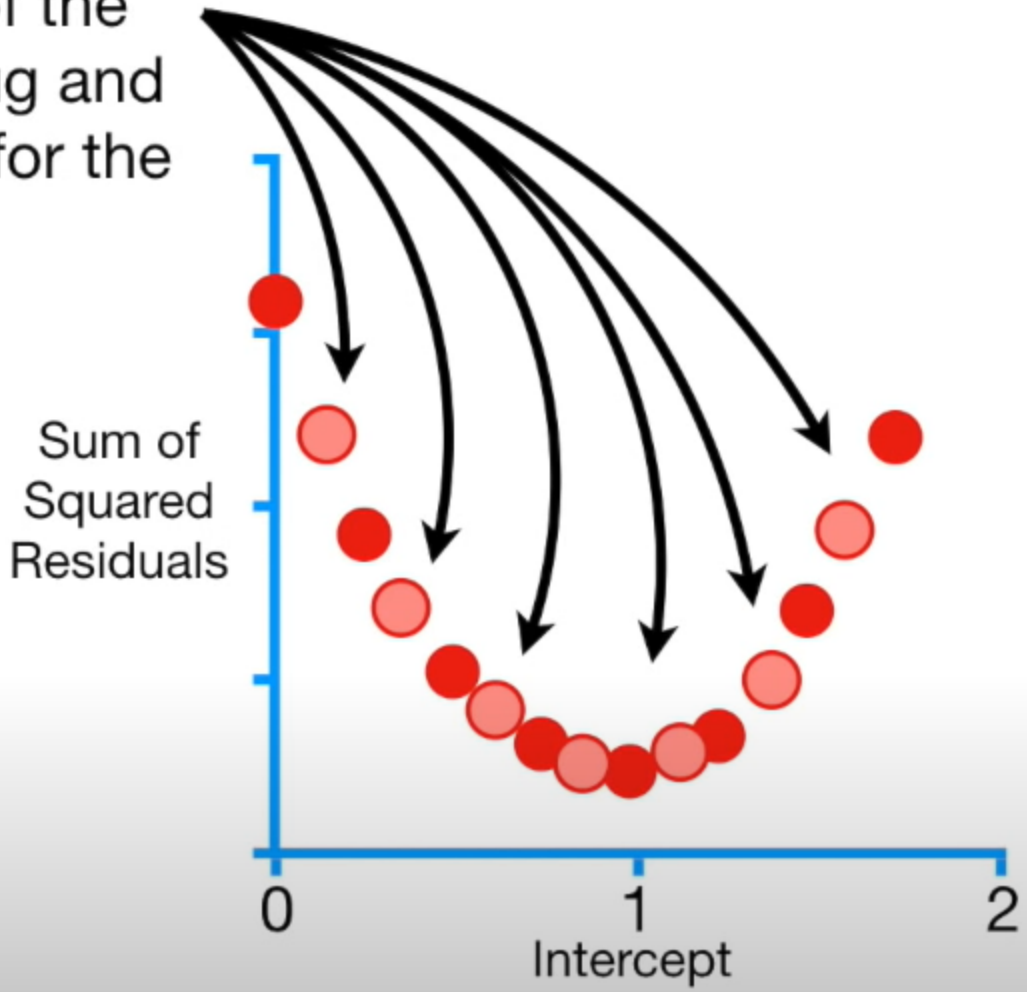
…but is it the best we can do?



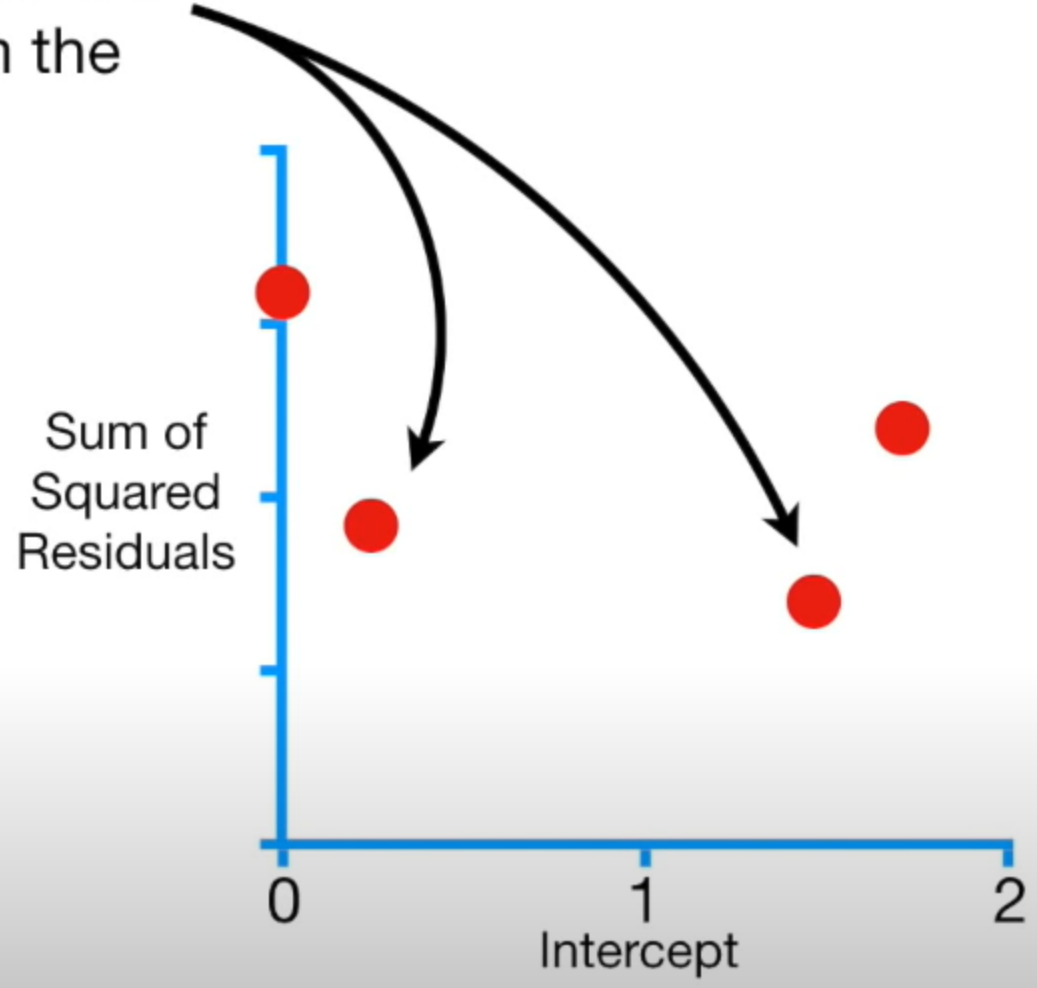
What if the best value for the **Intercept** is somewhere between these values?



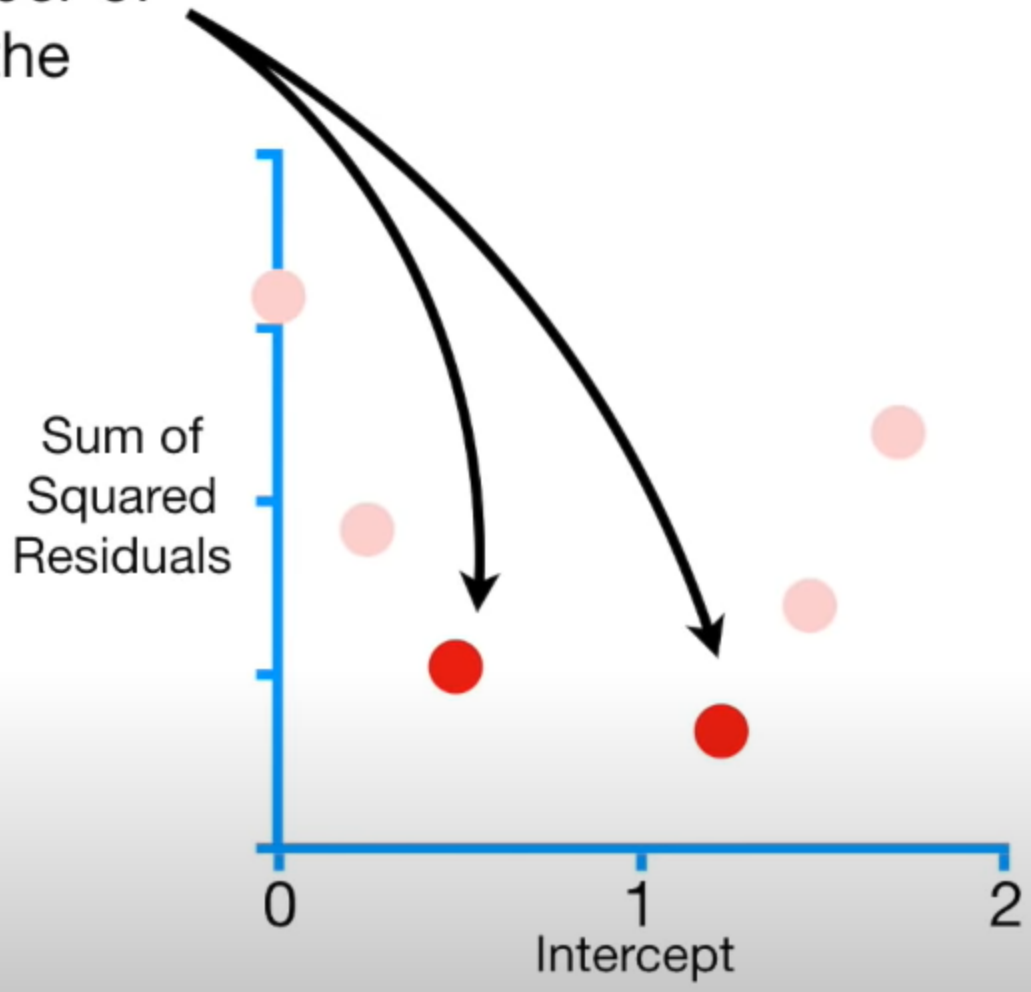
A slow and painful method for finding the minimal Sum of the Squared Residuals is to plug and chug a bunch more values for the **Intercept**.



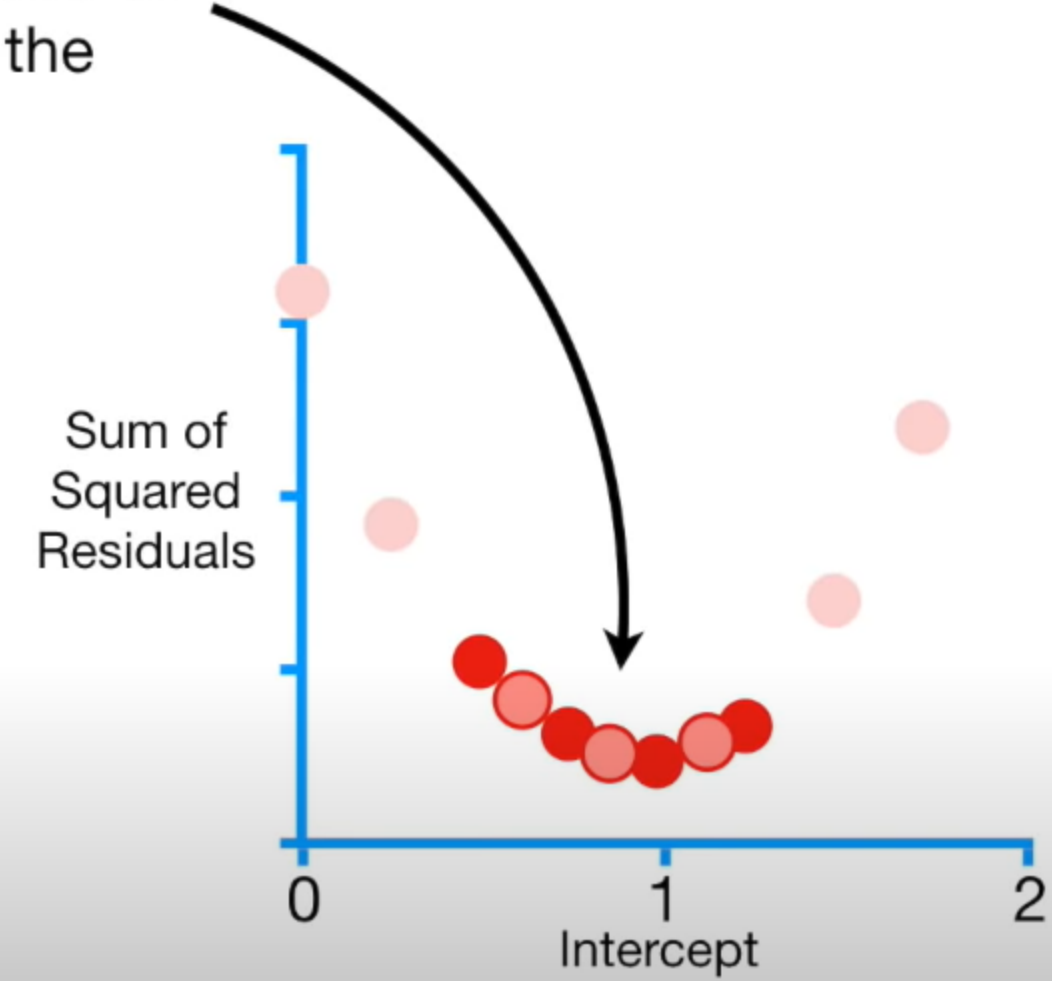
Gradient Descent only does a few calculations far from the optimal solution…



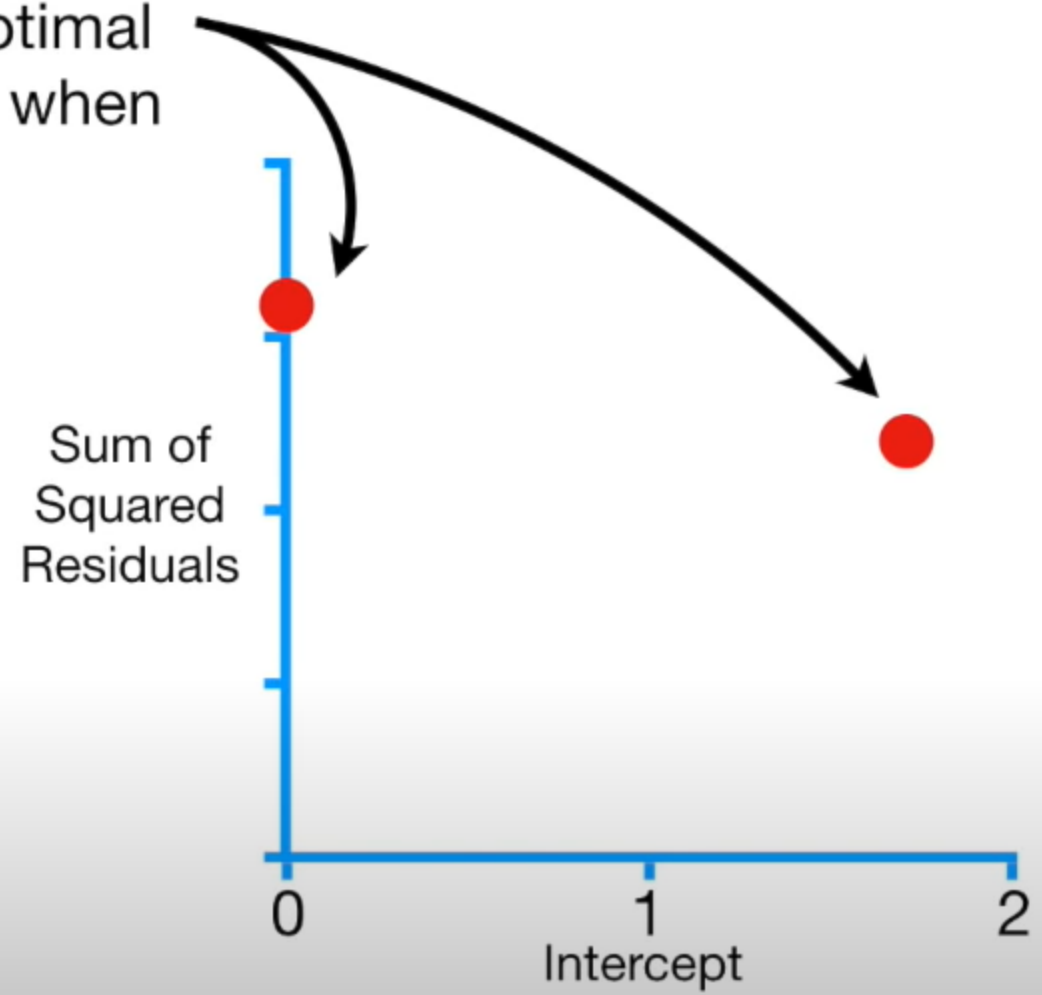
…and increases the number of calculations closer to the optimal value.



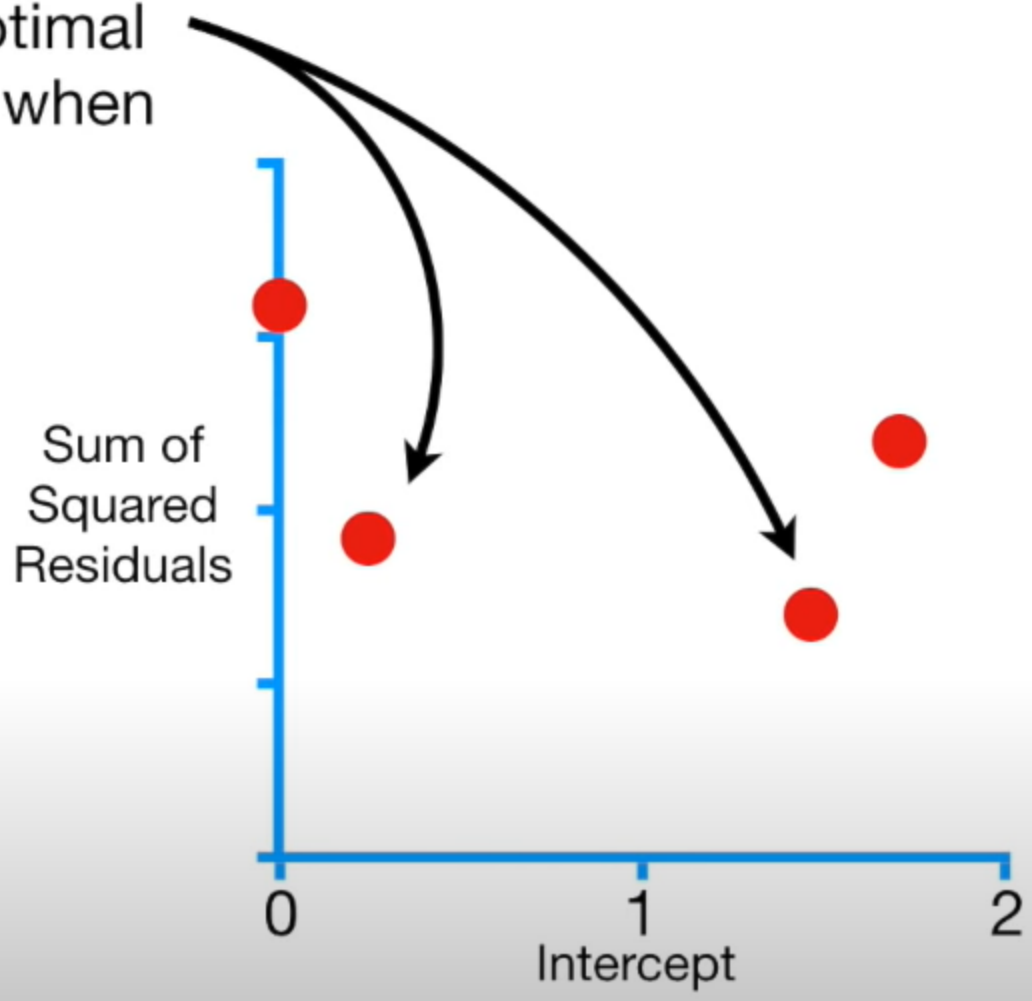
…and increases the number of calculations closer to the optimal value.



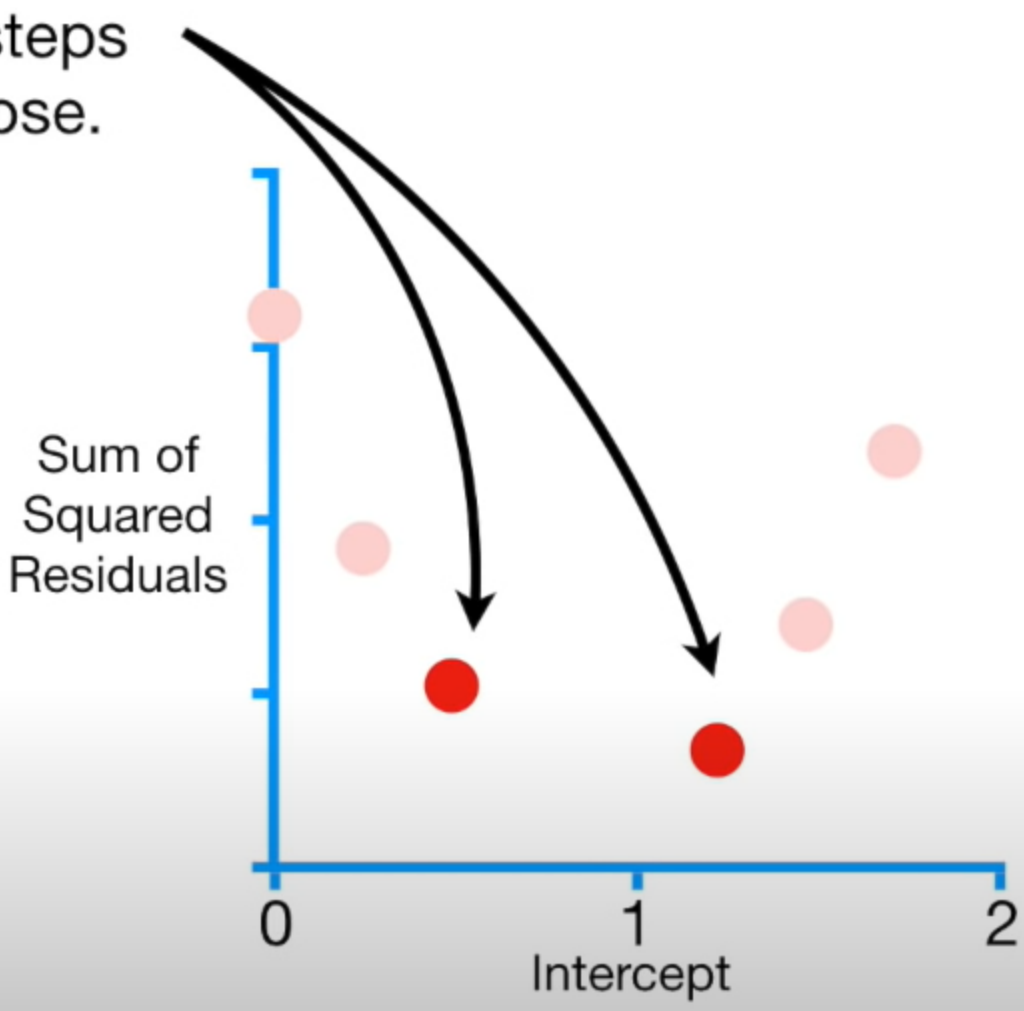
In other words, **Gradient Descent** identifies the optimal value by taking big steps when it is far away…



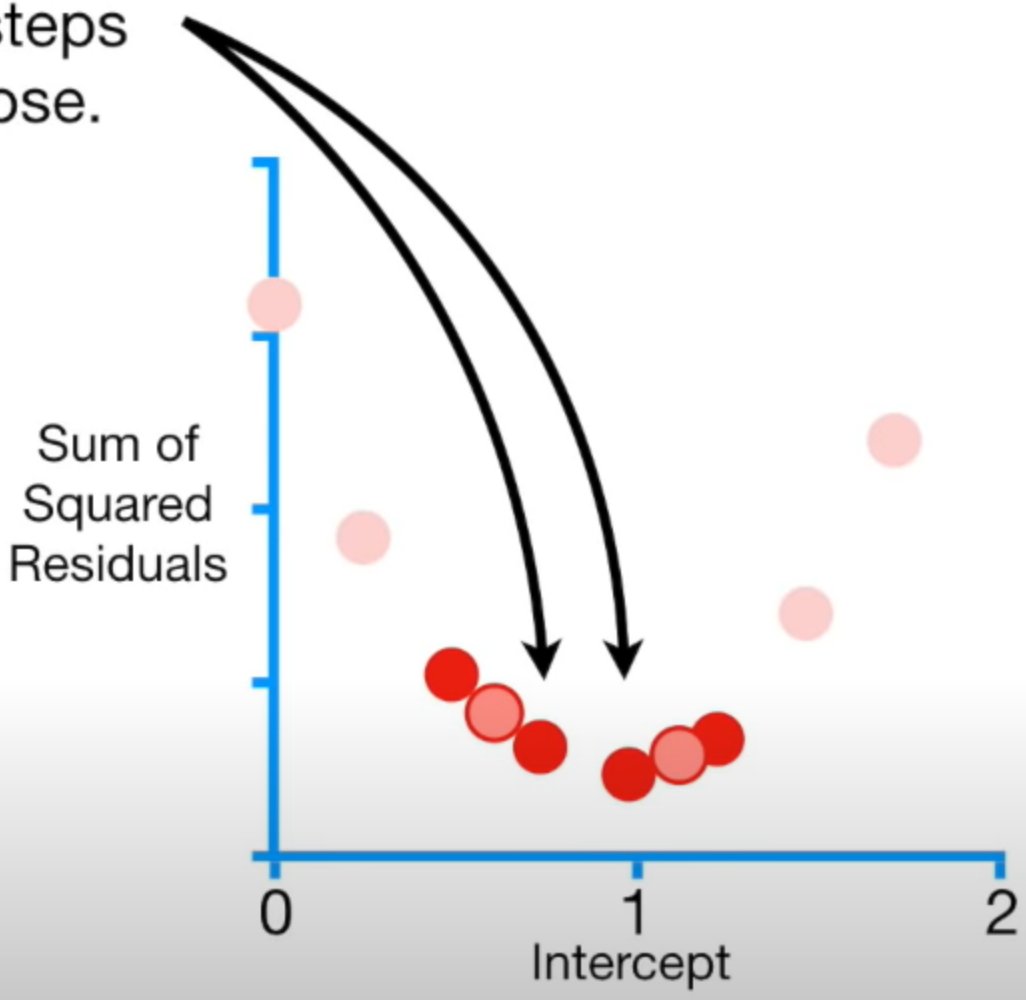
In other words, **Gradient Descent** identifies the optimal value by taking big steps when it is far away…



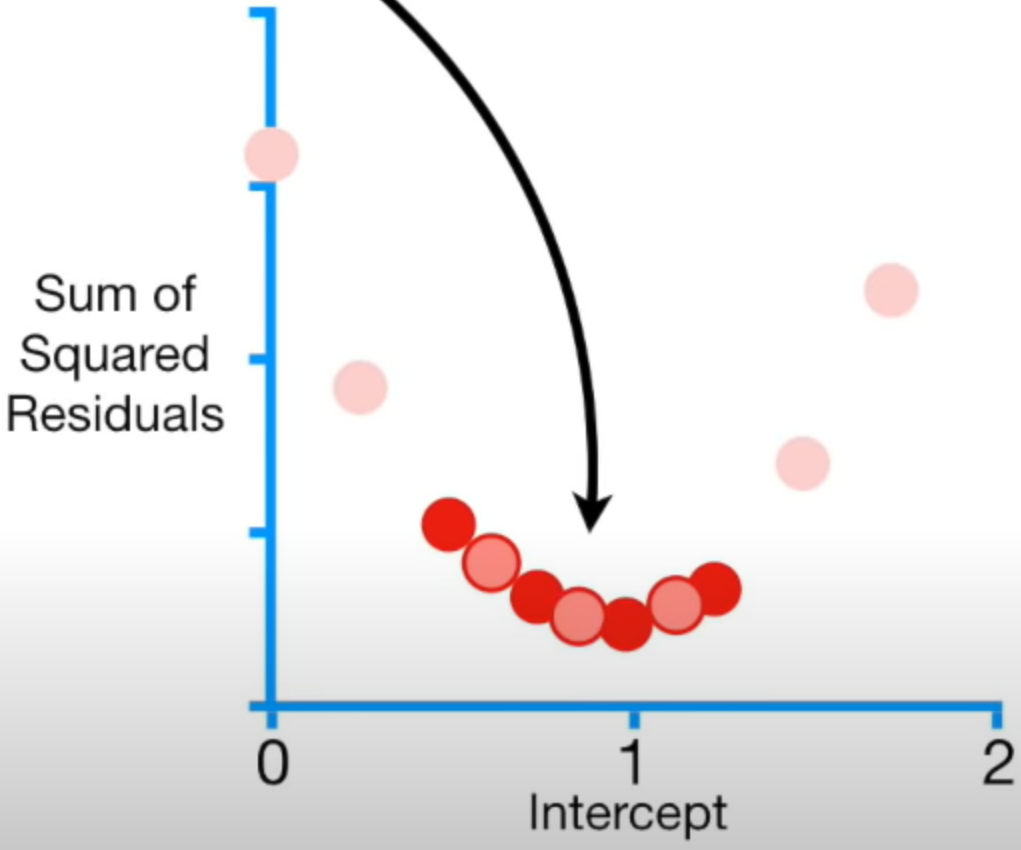
…and baby steps
when it is close.



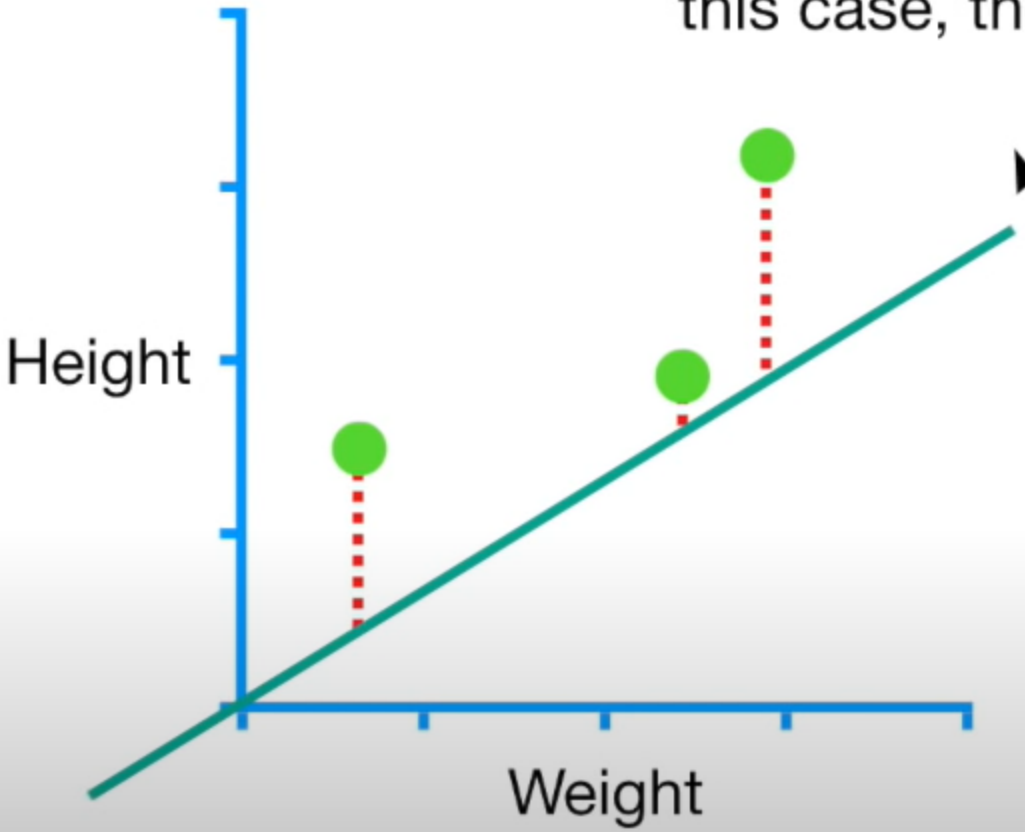
…and baby steps
when it is close.



…and baby steps
when it is close.



So let's get back to using **Gradient Descent** to find the optimal value for the **Intercept**, starting from a random value. In this case, the random value was **0**.

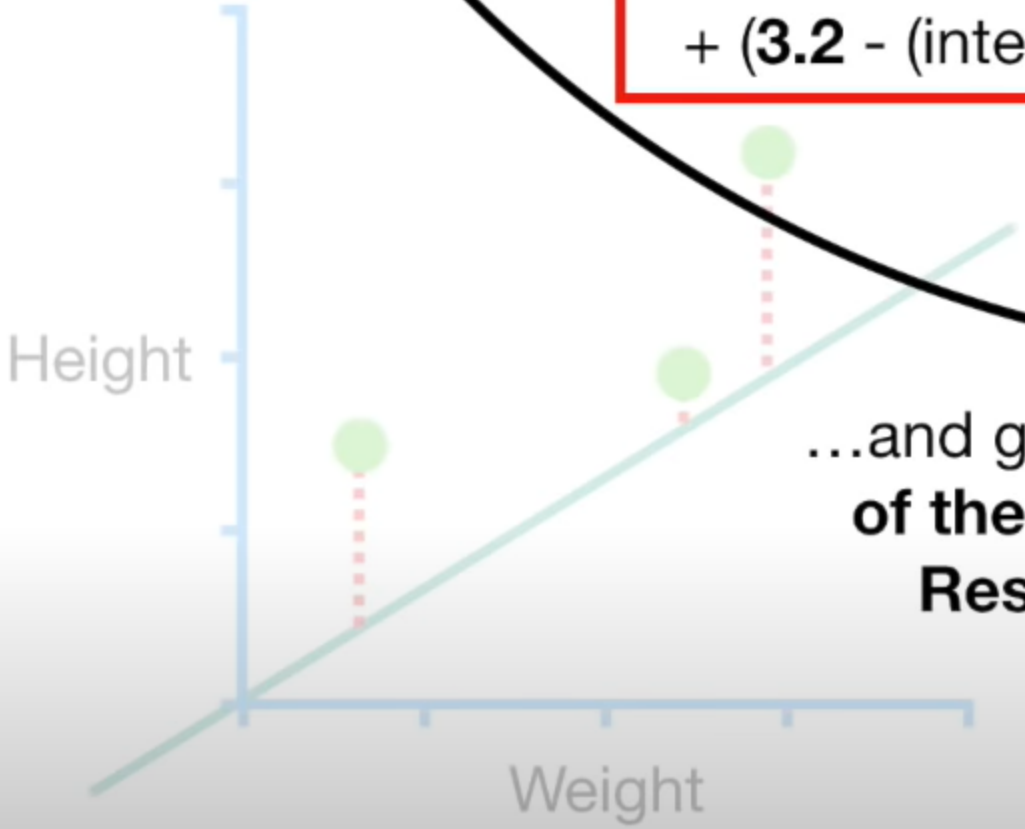


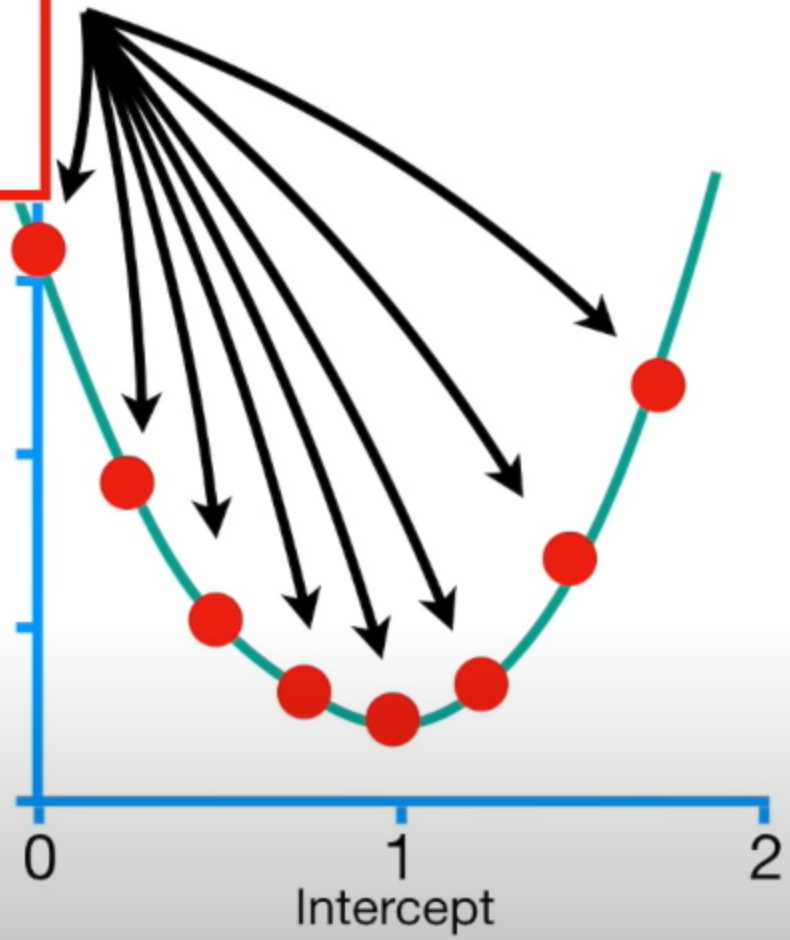
Sum of squared residuals = (**1.4** - (intercept + 0.64 × **0.5**))^2

+ (**1.9** - (intercept + 0.64 × **2.3**))^2

+ (**3.2** - (intercept + 0.64 × **2.9**))^2

…and get the **Sum of the Squared Residuals**.



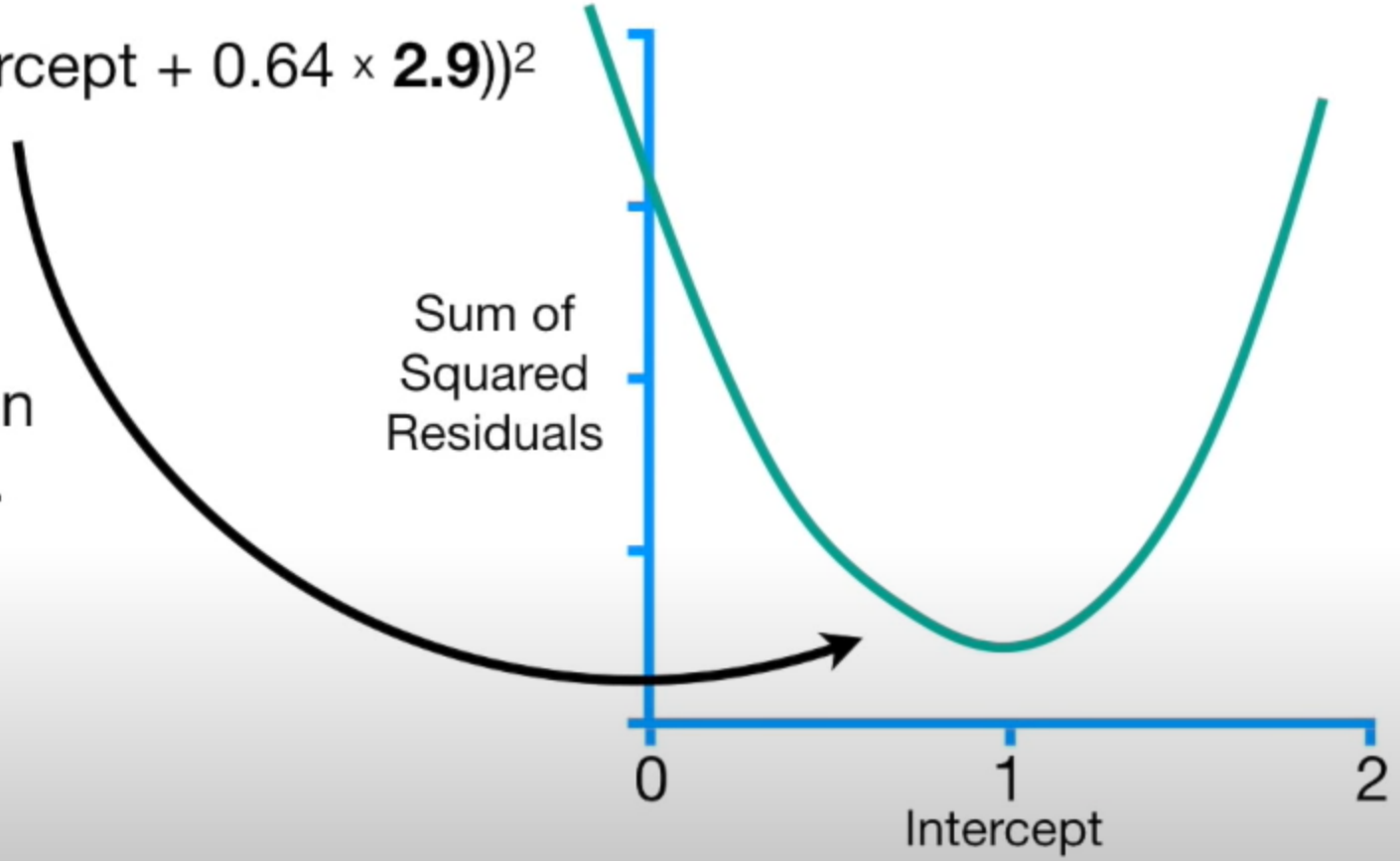


Sum of squared residuals = (**1.4** - (intercept + 0.64 × **0.5**))^2

+ (**1.9** - (intercept + 0.64 × **2.3**))^2

+ (**3.2** - (intercept + 0.64 × **2.9**))^2

Thus, we now have an equation for this curve…

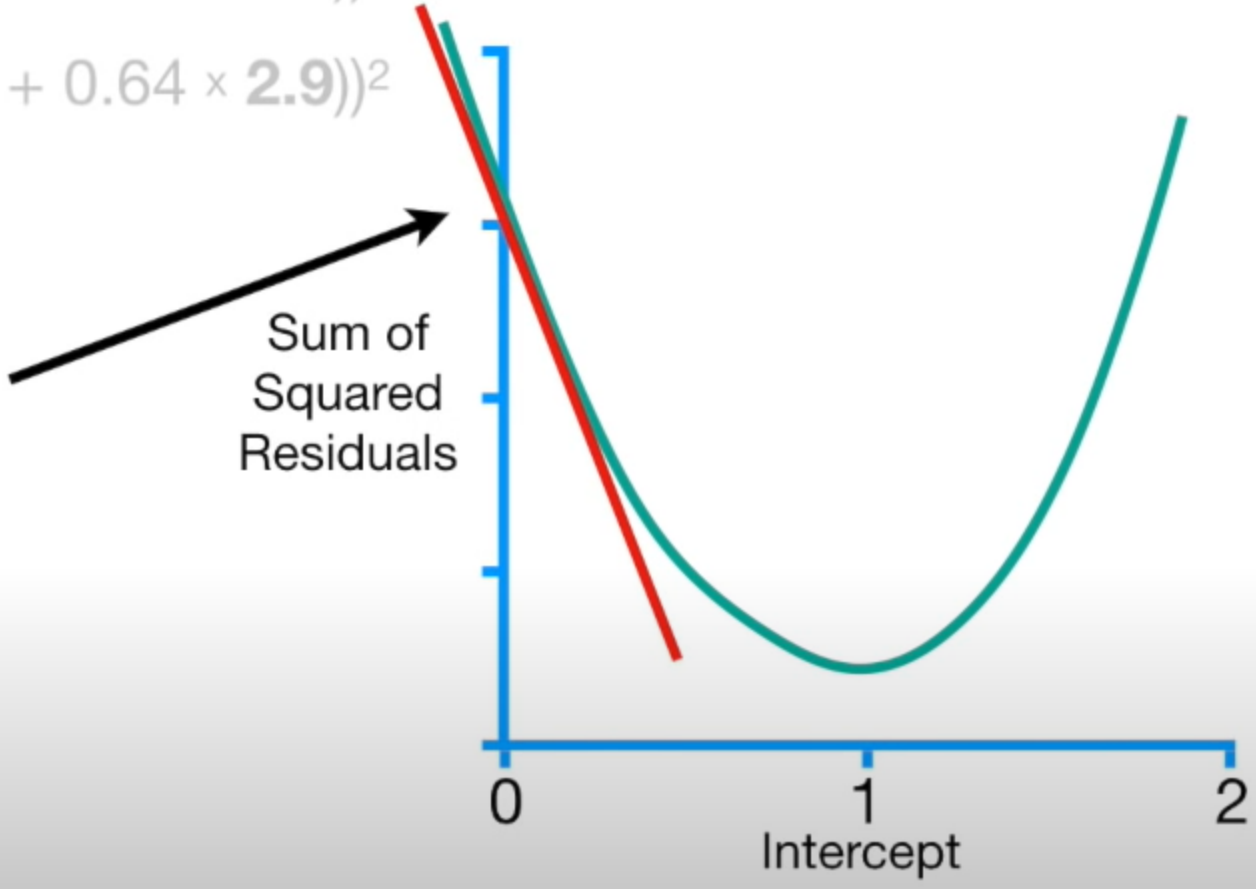


Sum of squared residuals = (**1.4** - (intercept + 0.64 × **0.5**))^2

+ (**1.9** - (intercept + 0.64 × **2.3**))^2

+ (**3.2** - (intercept + 0.64 × **2.9**))^2

…and we can take the derivative of this function and determine the slope at any value for the **Intercept**.

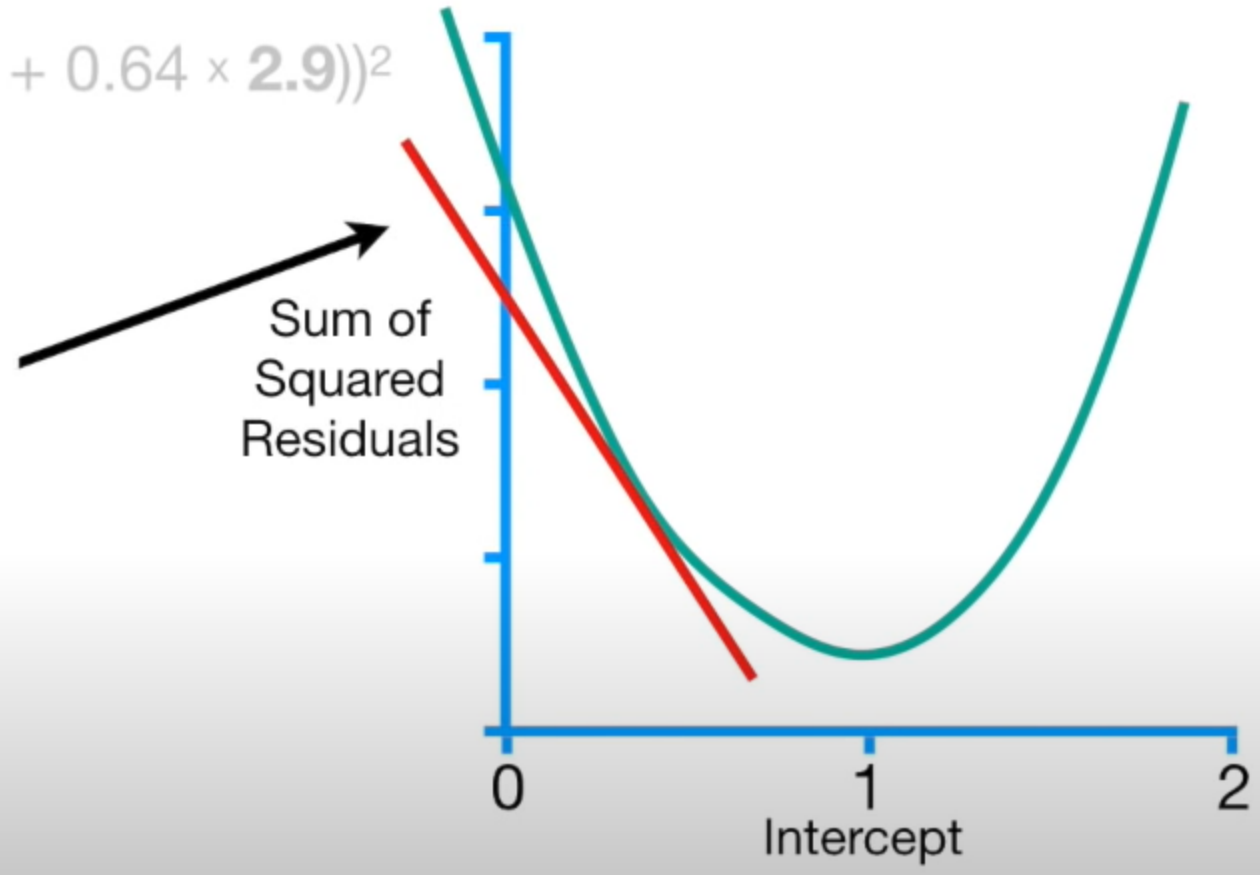


Sum of squared residuals = (**1.4** - (intercept + 0.64 × **0.5**))^2

+ (**1.9** - (intercept + 0.64 × **2.3**))^2

+ (**3.2** - (intercept + 0.64 × **2.9**))^2

…and we can take the derivative of this function and determine the slope at any value for the **Intercept**.

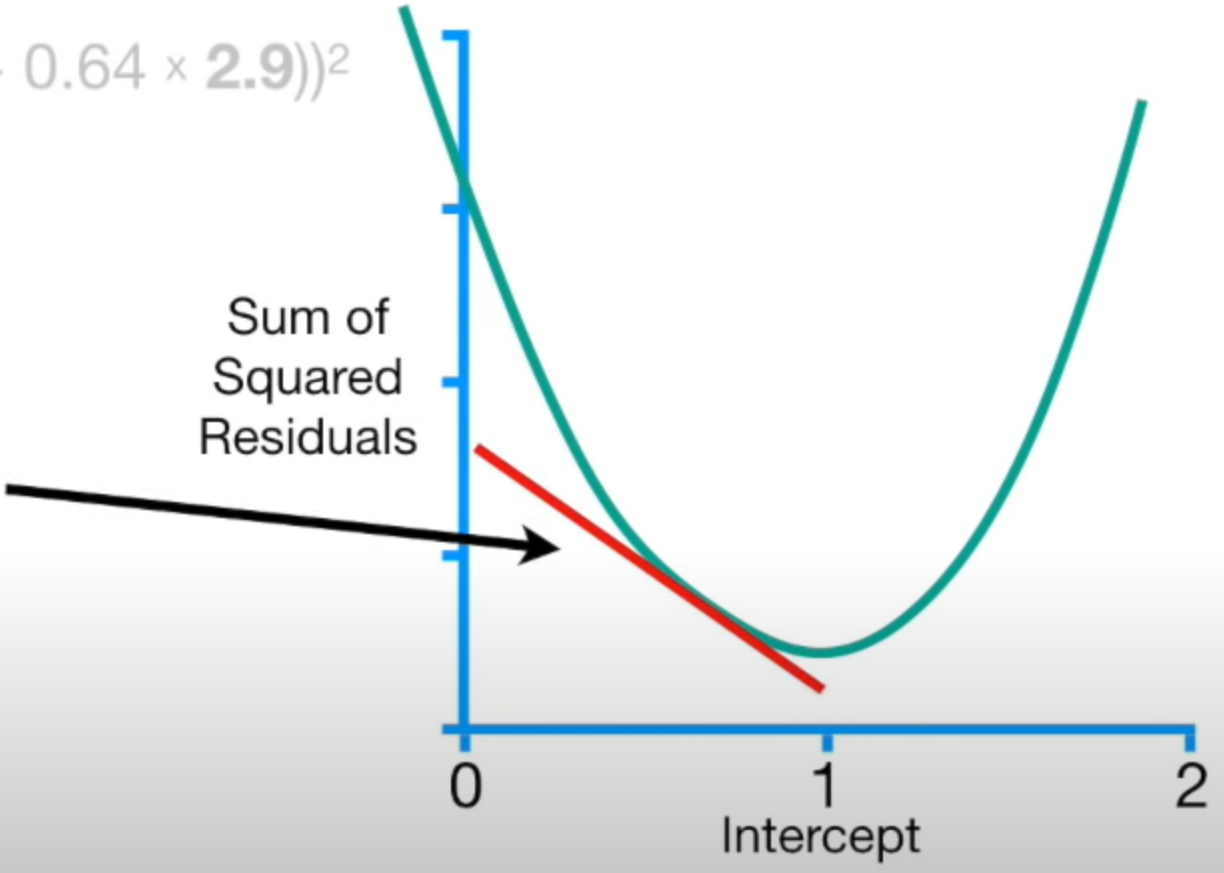


Sum of squared residuals = (**1.4** - (intercept + 0.64 × **0.5**))^2

+ (**1.9** - (intercept + 0.64 × **2.3**))^2

+ (**3.2** - (intercept + 0.64 × **2.9**))^2

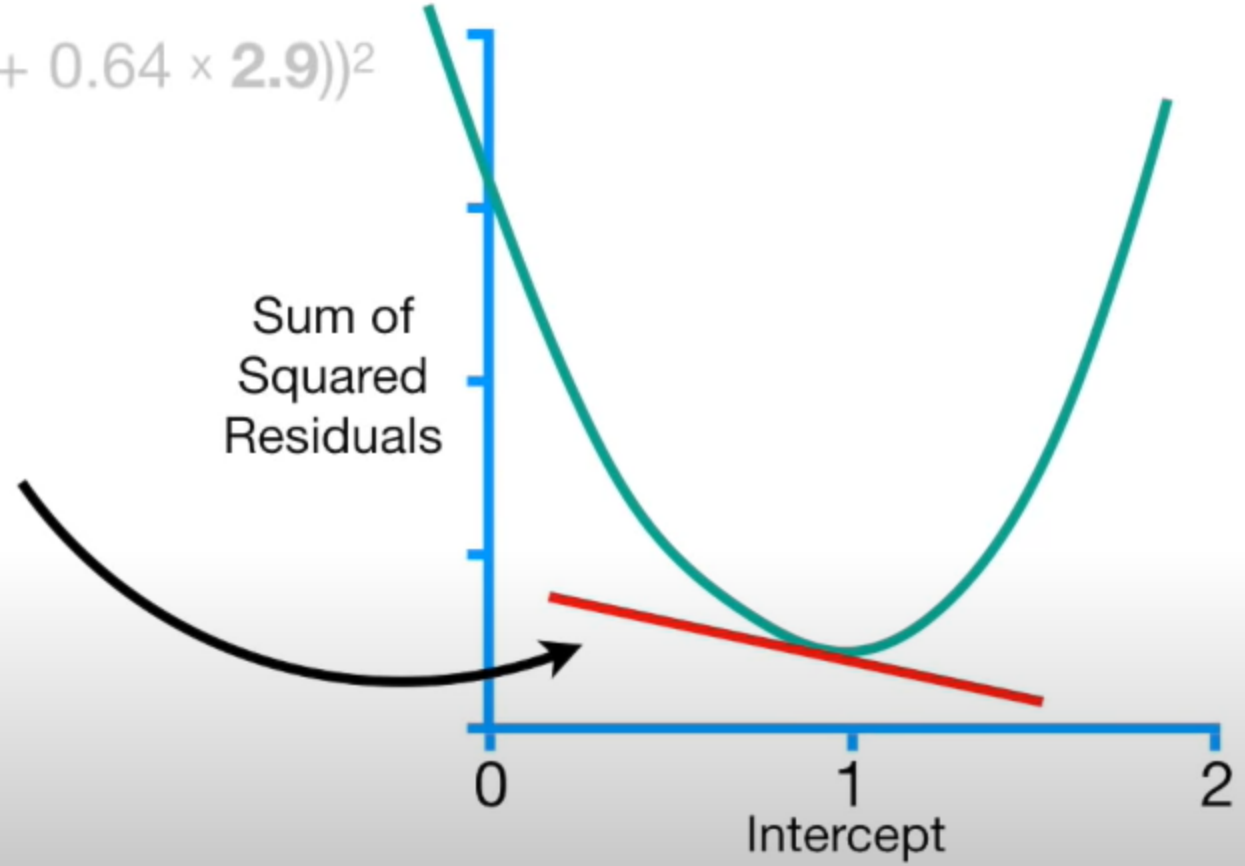
…and we can take the derivative of this function and determine the slope at any value for the **Intercept**.



$+ (\mathbf{1.9} - (\text{intercept} + 0.64 \times \mathbf{2.3}))^2$

$+ (\mathbf{3.2} - (\text{intercept} + 0.64 \times \mathbf{2.9}))^2$

…and we can take the derivative of this function and determine the slope at any value for the **Intercept**.

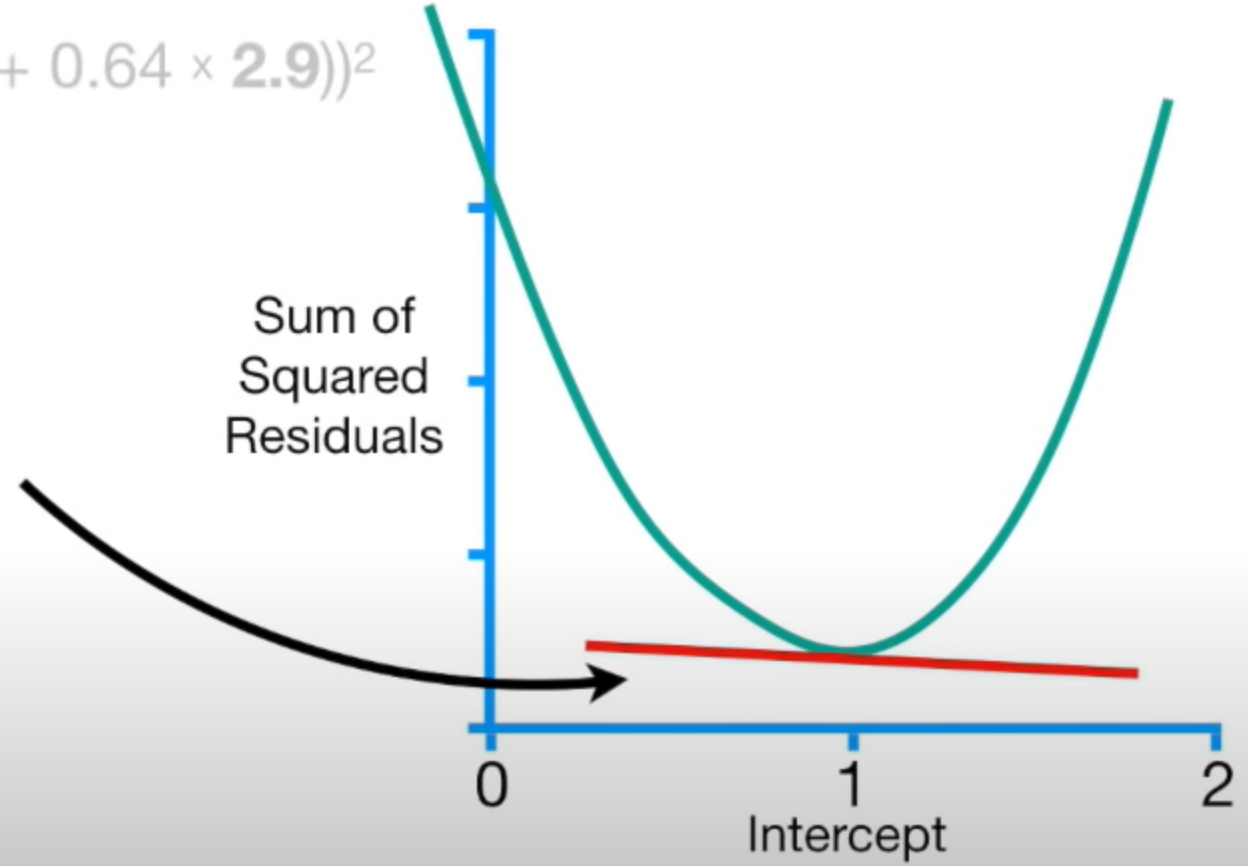


Sum of squared residuals = (**1.4** - (intercept + 0.64 × **0.5**))^2

+ (**1.9** - (intercept + 0.64 × **2.3**))^2

+ (**3.2** - (intercept + 0.64 × **2.9**))^2

…and we can take the derivative of this function and determine the slope at any value for the **Intercept**.

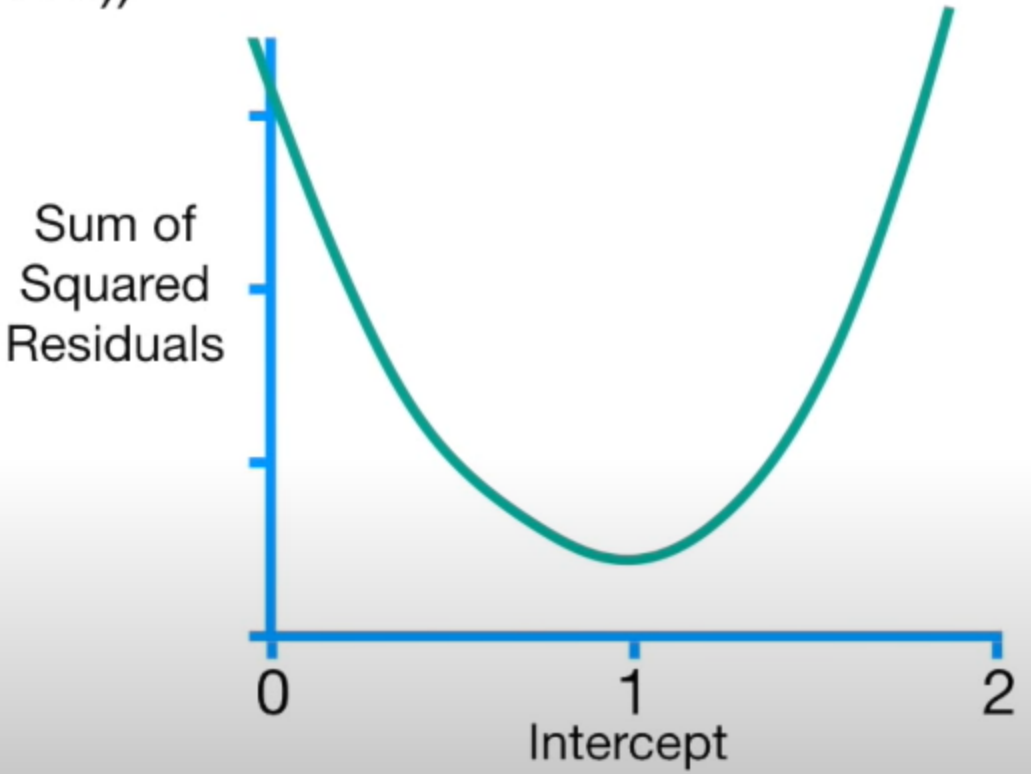


Sum of squared residuals = (**1.4** - (intercept + 0.64 × **0.5**))^2

+ (**1.9** - (intercept + 0.64 × **2.3**))^2

+ (**3.2** - (intercept + 0.64 × **2.9**))^2

So let's take the derivative of the Sum of the Squared Residuals with respect to the **Intercept**.



Sum of squared residuals = $(\mathbf{1.4} - (\text{intercept} + 0.64 \times \mathbf{0.5}))^2$

$+ (\mathbf{1.9} - (\text{intercept} + 0.64 \times \mathbf{2.3}))^2$

$+ (\mathbf{3.2} - (\text{intercept} + 0.64 \times \mathbf{2.9}))^2$

…the derivative of the first part…

$$\frac{d}{d\ intercept} \text{Sum of squared residuals} = \frac{d}{d\ intercept} (\mathbf{1.4} - (\text{intercept} + 0.64 \times \mathbf{0.5}))^2$$

Sum of squared residuals = $(\mathbf{1.4} - (\text{intercept} + 0.64 \times \mathbf{0.5}))^2$

$+ (\mathbf{1.9} - (\text{intercept} + 0.64 \times \mathbf{2.3}))^2$

$+ (\mathbf{3.2} - (\text{intercept} + 0.64 \times \mathbf{2.9}))^2$

…plus the derivative of the second part…

$$\frac{d}{d\ intercept} \text{Sum of squared residuals} = \frac{d}{d\ intercept} (\mathbf{1.4} - (\text{intercept} + 0.64 \times \mathbf{0.5}))^2$$

$$+ \frac{d}{d\ intercept} (\mathbf{1.9} - (\text{intercept} + 0.64 \times \mathbf{2.3}))^2$$

Sum of squared residuals = $(\mathbf{1.4} - (\text{intercept} + 0.64 \times \mathbf{0.5}))^2$

$+ (\mathbf{1.9} - (\text{intercept} + 0.64 \times \mathbf{2.3}))^2$

$+ (\mathbf{3.2} - (\text{intercept} + 0.64 \times \mathbf{2.9}))^2$

…plus the derivative of the third part.

$$\frac{d}{d\,\text{intercept}} \text{Sum of squared residuals} = \frac{d}{d\,\text{intercept}} (\mathbf{1.4} - (\text{intercept} + 0.64 \times \mathbf{0.5}))^2$$

$$+ \frac{d}{d\,\text{intercept}} (\mathbf{1.9} - (\text{intercept} + 0.64 \times \mathbf{2.3}))^2$$

$$+ \frac{d}{d\,\text{intercept}} (\mathbf{3.2} - (\text{intercept} + 0.64 \times \mathbf{2.9}))^2$$

Sum of squared residuals = $(\mathbf{1.4} - (\text{intercept} + 0.64 \times \mathbf{0.5}))^2$

$+ (\mathbf{1.9} - (\text{intercept} + 0.64 \times \mathbf{2.3}))^2$

$+ (\mathbf{3.2} - (\text{intercept} + 0.64 \times \mathbf{2.9}))^2$

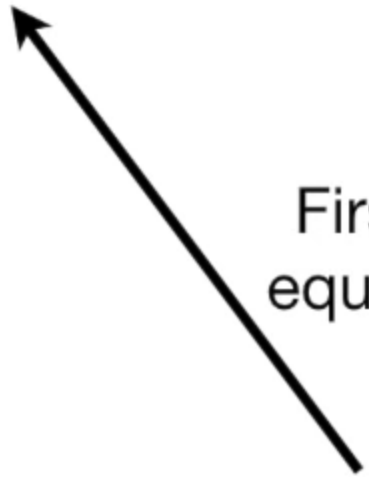
Let's start by taking the derivative of the first part.

$$\frac{d}{d\ intercept} \text{Sum of squared residuals} = \frac{d}{d\ intercept} (\mathbf{1.4} - (\text{intercept} + 0.64 \times \mathbf{0.5}))^2$$

$$+ \frac{d}{d\ intercept} (\mathbf{1.9} - (\text{intercept} + 0.64 \times \mathbf{2.3}))^2$$

$$+ \frac{d}{d\ intercept} (\mathbf{3.2} - (\text{intercept} + 0.64 \times \mathbf{2.9}))^2$$

$$\frac{d}{d\ intercept}(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5}))^2 =$$



First, we'll move this part of the equation up here so that we have room to work.

$$\frac{d}{d\ intercept}\text{Sum of squared residuals} = \frac{d}{d\ intercept}(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5}))^2$$

$$+ \frac{d}{d\ intercept}(\mathbf{1.9} - (intercept + 0.64 \times \mathbf{2.3}))^2$$

$$+ \frac{d}{d\ intercept}(\mathbf{3.2} - (intercept + 0.64 \times \mathbf{2.9}))^2$$

$$\frac{d}{d\,intercept}(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5}))^2 =$$

To take the derivative of this, we need to apply…

$$\frac{d}{d\,intercept}\text{Sum of squared residuals} = \frac{d}{d\,intercept}(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5}))^2$$

$$+ \frac{d}{d\,intercept}(\mathbf{1.9} - (intercept + 0.64 \times \mathbf{2.3}))^2$$

$$+ \frac{d}{d\,intercept}(\mathbf{3.2} - (intercept + 0.64 \times \mathbf{2.9}))^2$$

The Chain Rule

$$\frac{d}{d\,intercept}(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5}))^2 = 2(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5}))$$

So we start by moving the square to the front…

$$\frac{d}{d\,intercept}\text{Sum of squared residuals} = \frac{d}{d\,intercept}(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5}))^2$$

$$+ \frac{d}{d\,intercept}(\mathbf{1.9} - (intercept + 0.64 \times \mathbf{2.3}))^2$$

$$+ \frac{d}{d\,intercept}(\mathbf{3.2} - (intercept + 0.64 \times \mathbf{2.9}))^2$$

$$\frac{d}{d\,intercept}(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5})^2 = 2(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5})) \times -1$$

…and multiply that by the derivative of the stuff inside the parentheses.

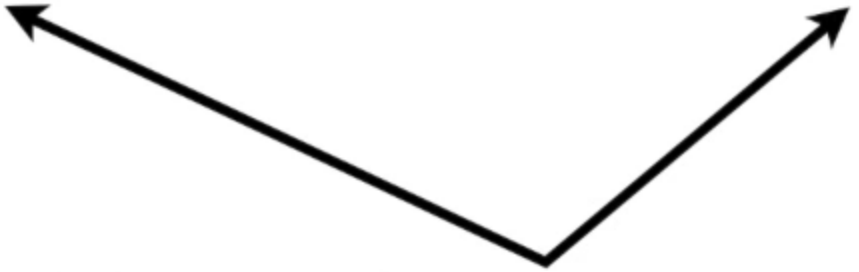
$$\frac{d}{d\ intercept}\boxed{(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5})^2} = 2(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5}))\boxed{\times -1}$$

$$\frac{d}{d\ intercept}\ \mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5})$$

$$\frac{d}{d\ intercept}\ \cancel{\mathbf{1.4}} + (-1)intercept - \cancel{0.64 \times \mathbf{0.5}} = -1$$

These parts don't contain a term for the **Intercept**, so they go away.

$$\frac{d}{d\,intercept}(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5}))^2 = 2(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5})) \times -1$$



…then we simplify by multiplying 2 by -1…

$$\frac{d}{d\,intercept}\text{Sum of squared residuals} = \frac{d}{d\,intercept}(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5}))^2$$

$$+ \frac{d}{d\,intercept}(\mathbf{1.9} - (intercept + 0.64 \times \mathbf{2.3}))^2$$

$$+ \frac{d}{d\,intercept}(\mathbf{3.2} - (intercept + 0.64 \times \mathbf{2.9}))^2$$

$$\frac{d}{d\ intercept}(\mathbf{1.4} - (\text{intercept} + 0.64 \times \mathbf{0.5}))^2 = 2(\mathbf{1.4} - (\text{intercept} + 0.64 \times \mathbf{0.5})) \times -1$$

$$= -2(\mathbf{1.4} - (\text{intercept} + 0.64 \times \mathbf{0.5}))$$

…then we simplify by multiplying 2 by -1…

$$\frac{d}{d\,intercept}(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5}))^2 = 2(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5})) \times -1$$

$$= -2(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5}))$$

…and this…

…is the derivative of the first part…

$$\frac{d}{d\,intercept}\text{Sum of squared residuals} = \frac{d}{d\,intercept}(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5}))^2$$

$$+ \frac{d}{d\,intercept}(\mathbf{1.9} - (intercept + 0.64 \times \mathbf{2.3}))^2$$

$$+ \frac{d}{d\,intercept}(\mathbf{3.2} - (intercept + 0.64 \times \mathbf{2.9}))^2$$

$$\frac{d}{d\,intercept}(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5}))^2 = 2(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5})) \times -1$$

$$= -2(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5}))$$

…so we plug it in.

$$\frac{d}{d\,intercept}\text{Sum of squared residuals} = -2(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5}))$$

$$+ \frac{d}{d\,intercept}(\mathbf{1.9} - (intercept + 0.64 \times \mathbf{2.3}))^2$$

$$+ \frac{d}{d\,intercept}(\mathbf{3.2} - (intercept + 0.64 \times \mathbf{2.9}))^2$$

Now we need to take the derivative of the next two parts.

$$\frac{d}{d\ intercept}\text{ Sum of squared residuals } = -2(\mathbf{1.4} - (\text{intercept} + 0.64 \times \mathbf{0.5})$$

$$+ \frac{d}{d\ intercept}(\mathbf{1.9} - (\text{intercept} + 0.64 \times \mathbf{2.3}))^2$$

$$+ \frac{d}{d\ intercept}(\mathbf{3.2} - (\text{intercept} + 0.64 \times \mathbf{2.9}))^2$$

$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

$$-2(\mathbf{1.4} - (\text{intercept} + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (\text{intercept} + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (\text{intercept} + 0.64 \times \mathbf{2.9}))$$

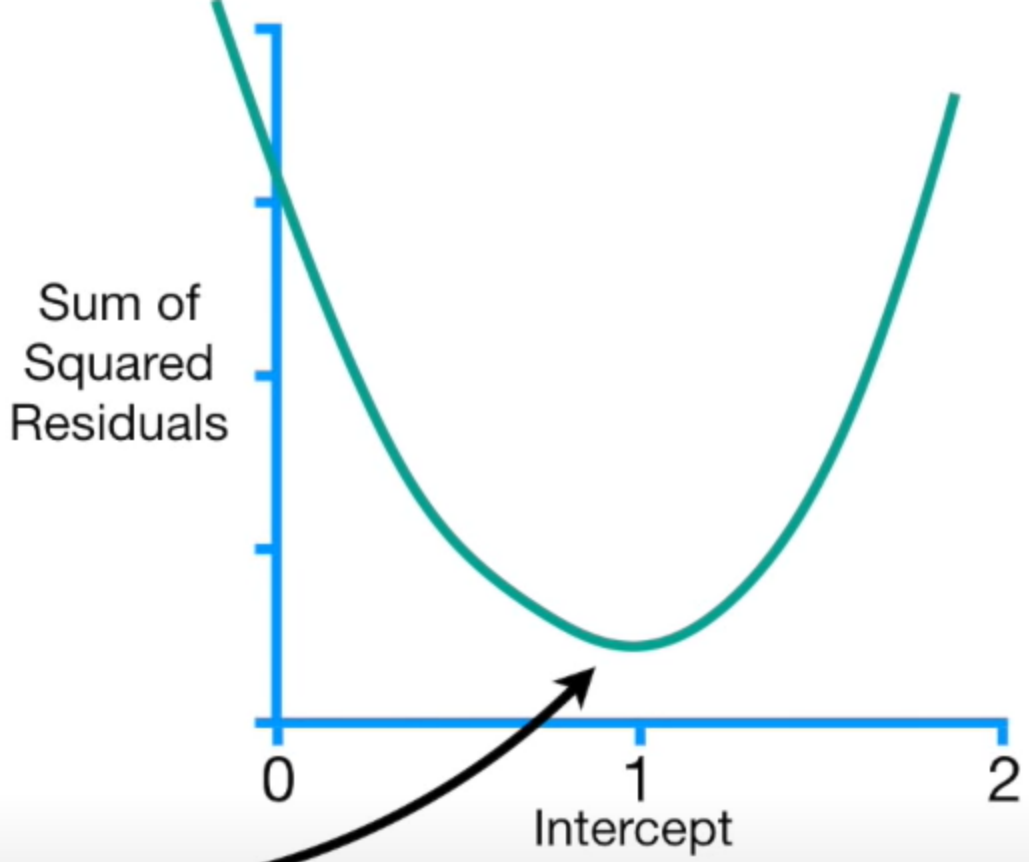
Let's move the derivative up here so that it's not taking up half of the screen.

$$\frac{d}{d\,intercept}$$ Sum of squared residuals =

$$-2(\mathbf{1.4} - (\text{intercept} + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (\text{intercept} + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (\text{intercept} + 0.64 \times \mathbf{2.9}))$$



Now that we have the derivative, **Gradient Descent** will use it to find where the Sum of Squared Residuals is lowest.

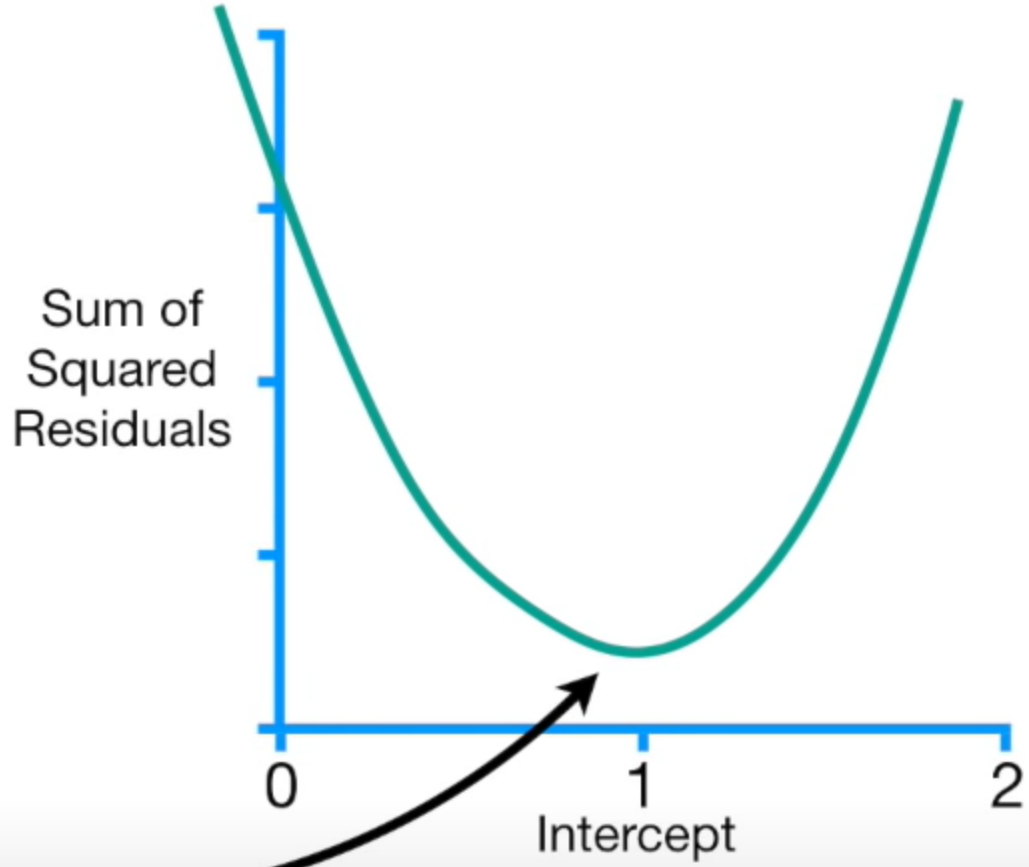
$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

$$-2(\mathbf{1.4} - (\text{intercept} + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (\text{intercept} + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (\text{intercept} + 0.64 \times \mathbf{2.9}))$$

NOTE: If we were using **Least Squares** to solve for the optimal value for the **Intercept**, we would simply find where the the slope of the curve = **0**.

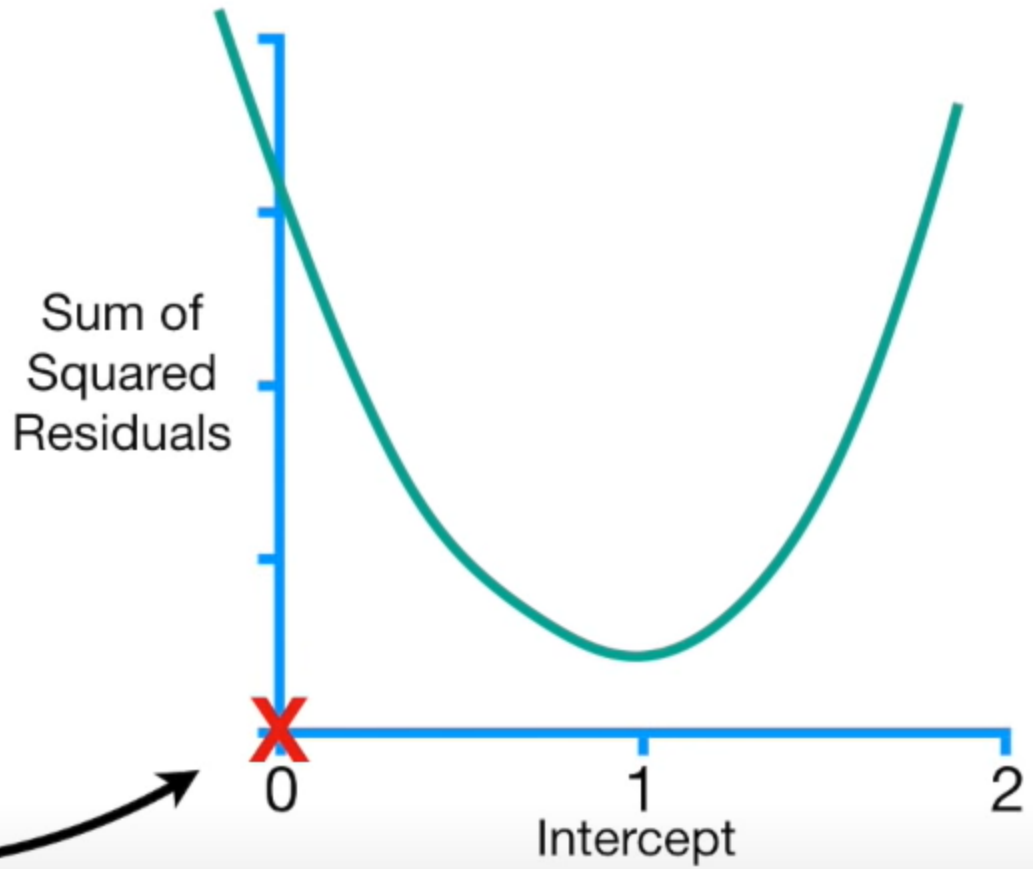


$$\frac{d}{d\,intercept}$$ Sum of squared residuals =

-2(**1.4** - (intercept + 0.64 × **0.5**)

+ -2(**1.9** - (intercept + 0.64 × **2.3**))

+ -2(**3.2** - (intercept + 0.64 × **2.9**))



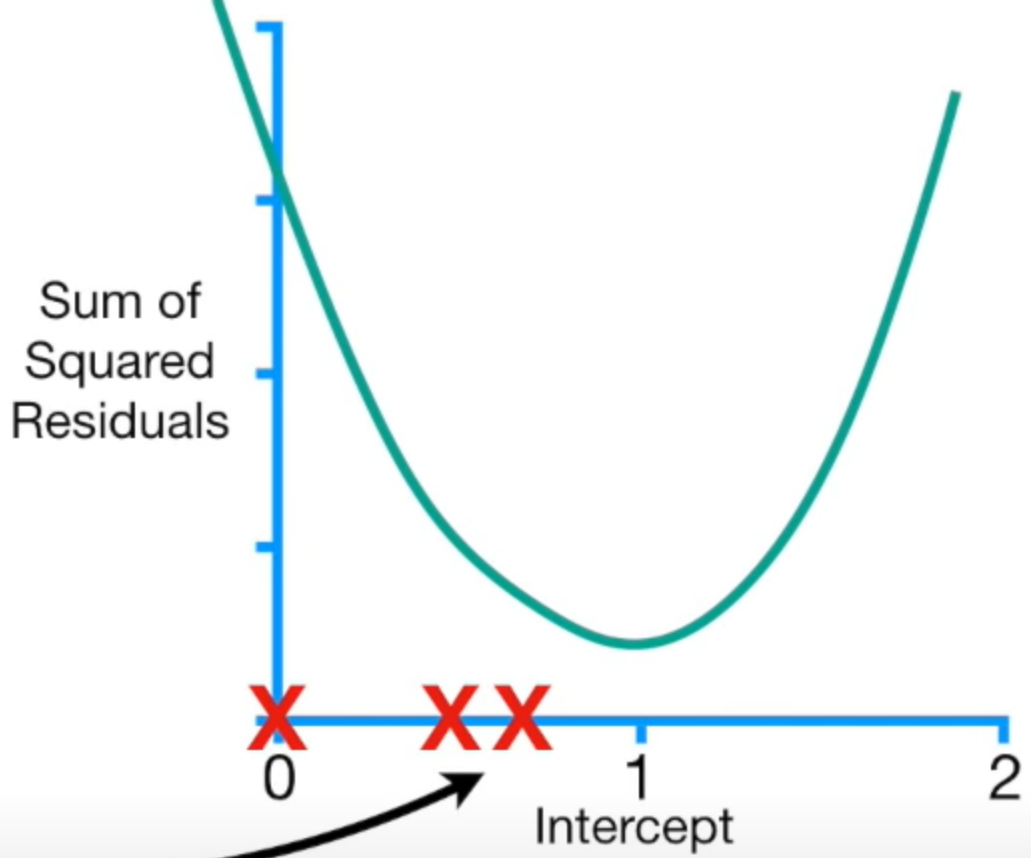
In contrast, **Gradient Descent** finds the minimum value by taking steps from an initial guess until it reaches the best value.

$$\frac{d}{d\,intercept}$$ Sum of squared residuals =

$$-2(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (intercept + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (intercept + 0.64 \times \mathbf{2.9}))$$



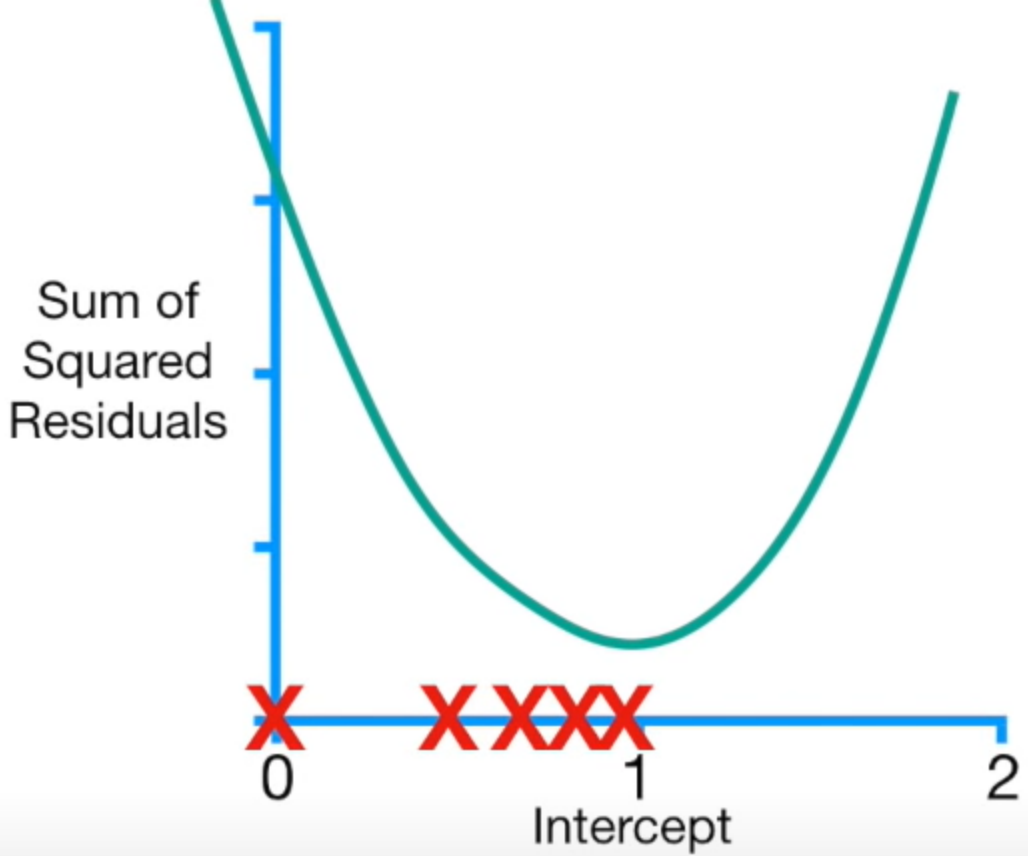
In contrast, **Gradient Descent** finds the minimum value by taking steps from an initial guess until it reaches the best value.

$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

$$-2(\mathbf{1.4} - (\text{intercept} + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (\text{intercept} + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (\text{intercept} + 0.64 \times \mathbf{2.9}))$$



This makes **Gradient Descent** very useful when it is not possible to solve for where the derivative = **0**, and this is why **Gradient Descent** can be used in so many different situations.

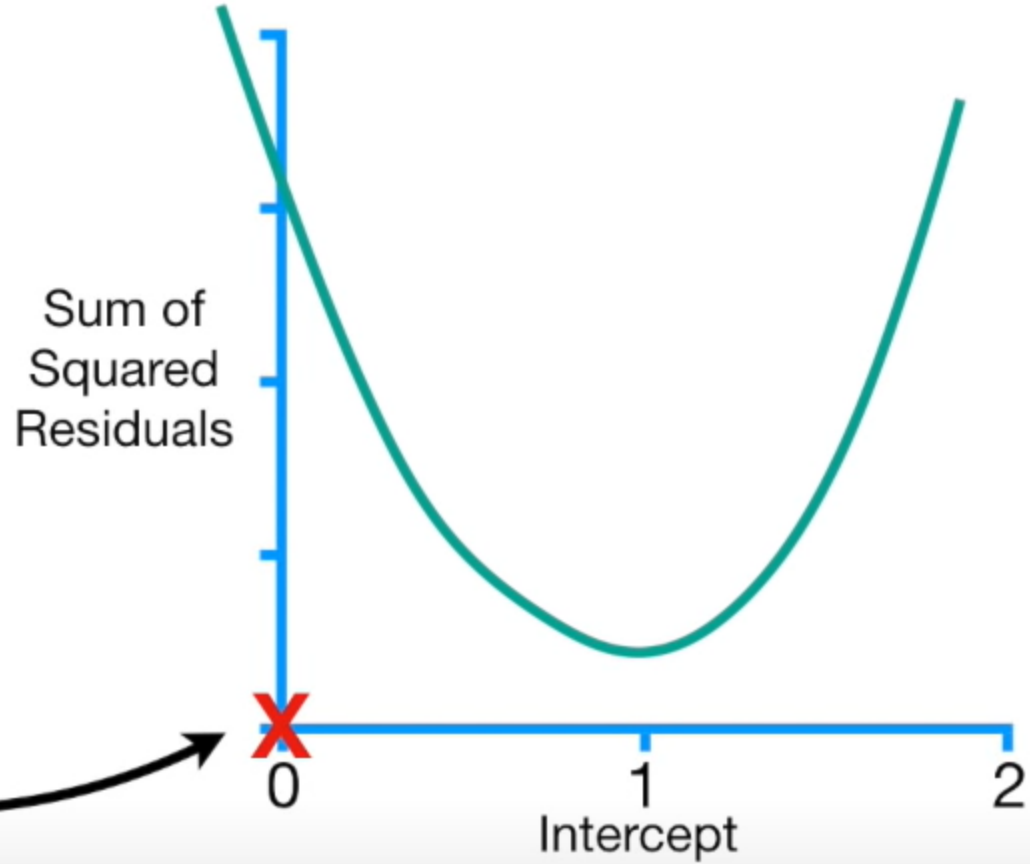
$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

$$-2(\mathbf{1.4} - (\text{intercept} + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (\text{intercept} + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (\text{intercept} + 0.64 \times \mathbf{2.9}))$$

Remember, we started by setting the **Intercept** to a random number. In this case, that was **0**.



$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

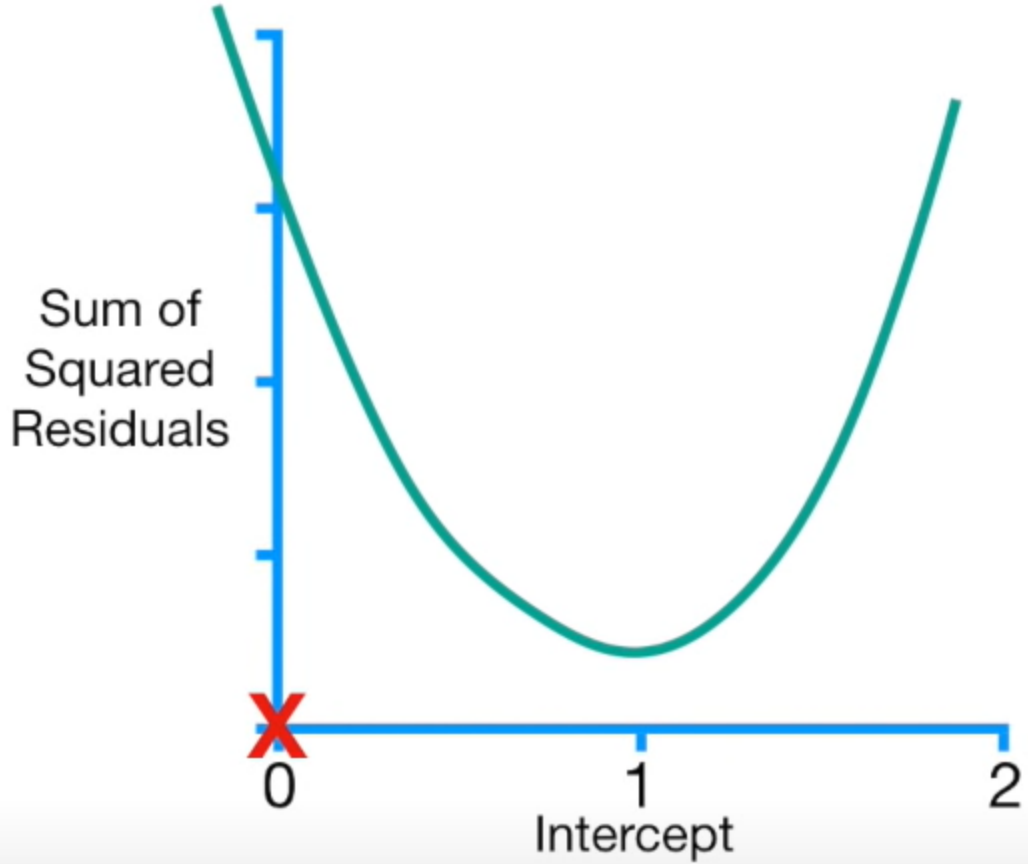
$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$

…and we get **-5.7**.



$$\frac{d}{d\,intercept}$$ Sum of squared residuals =

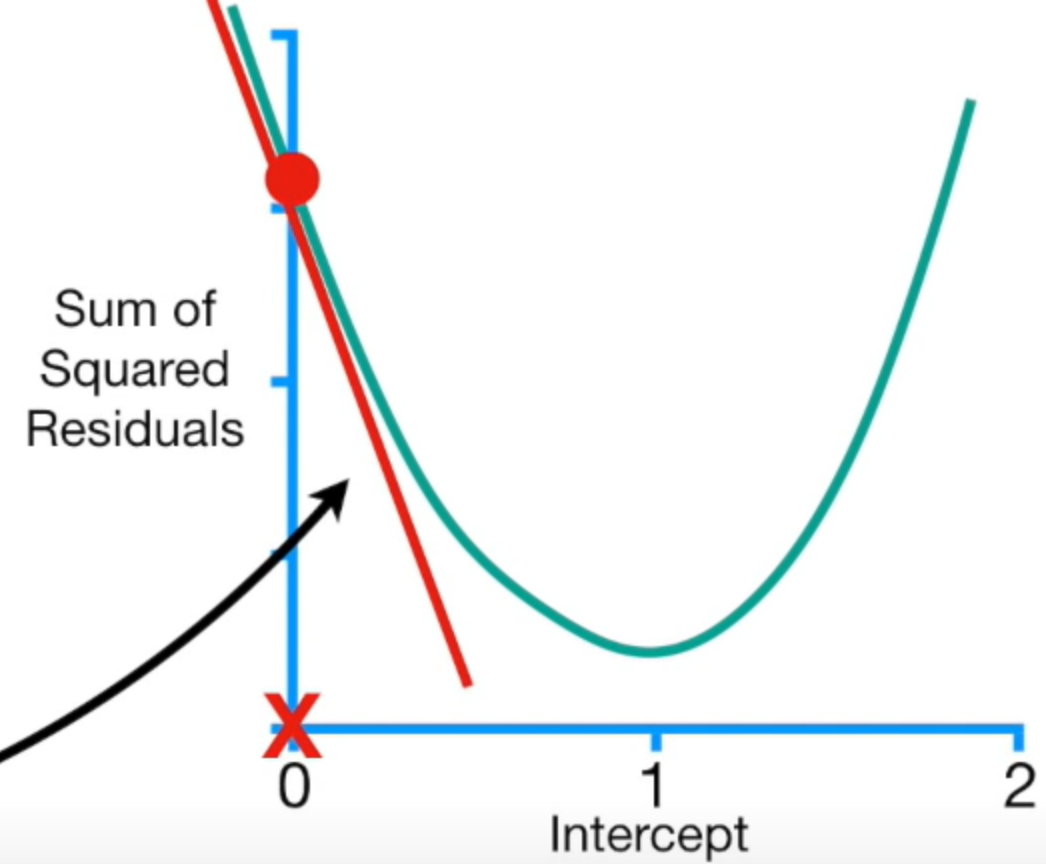
$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$

So when the **Intercept** = **0**,
the slope of the curve = **-5.7**.



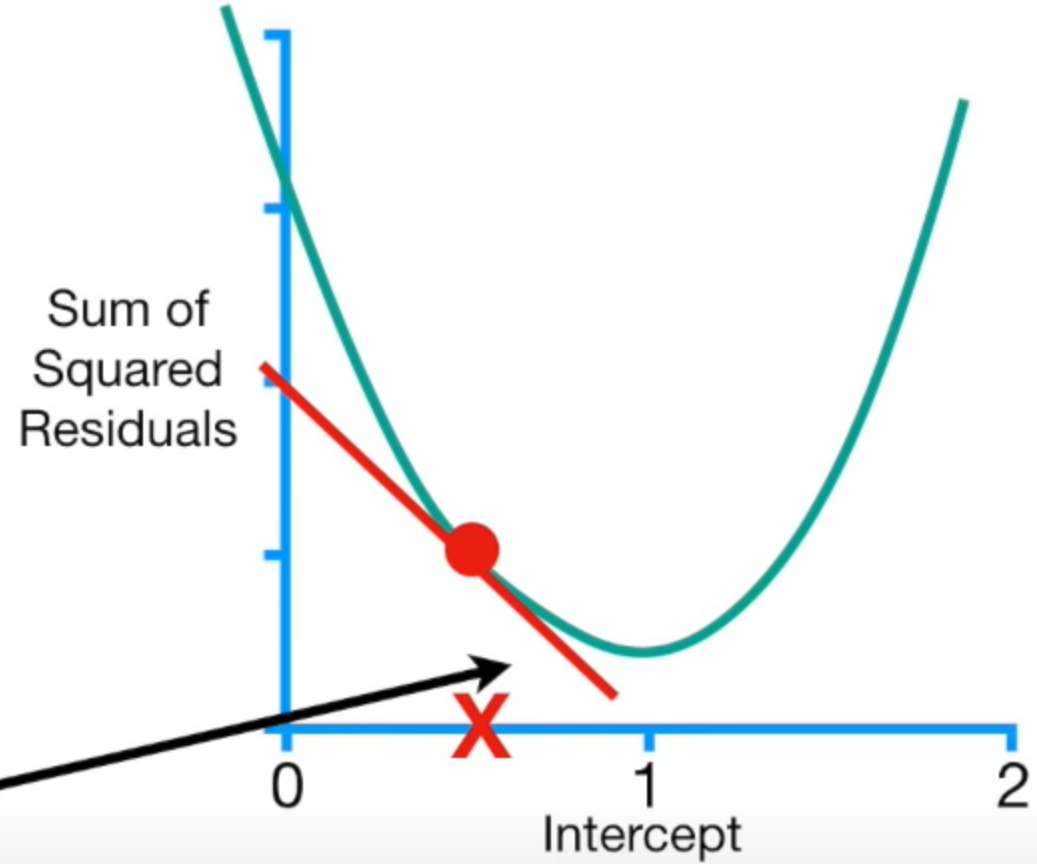
$\frac{d}{d\ intercept}$ Sum of squared residuals =

$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$



NOTE: The closer we get to the optimal value for the **Intercept**, the closer the slope of the curve gets to **0**.

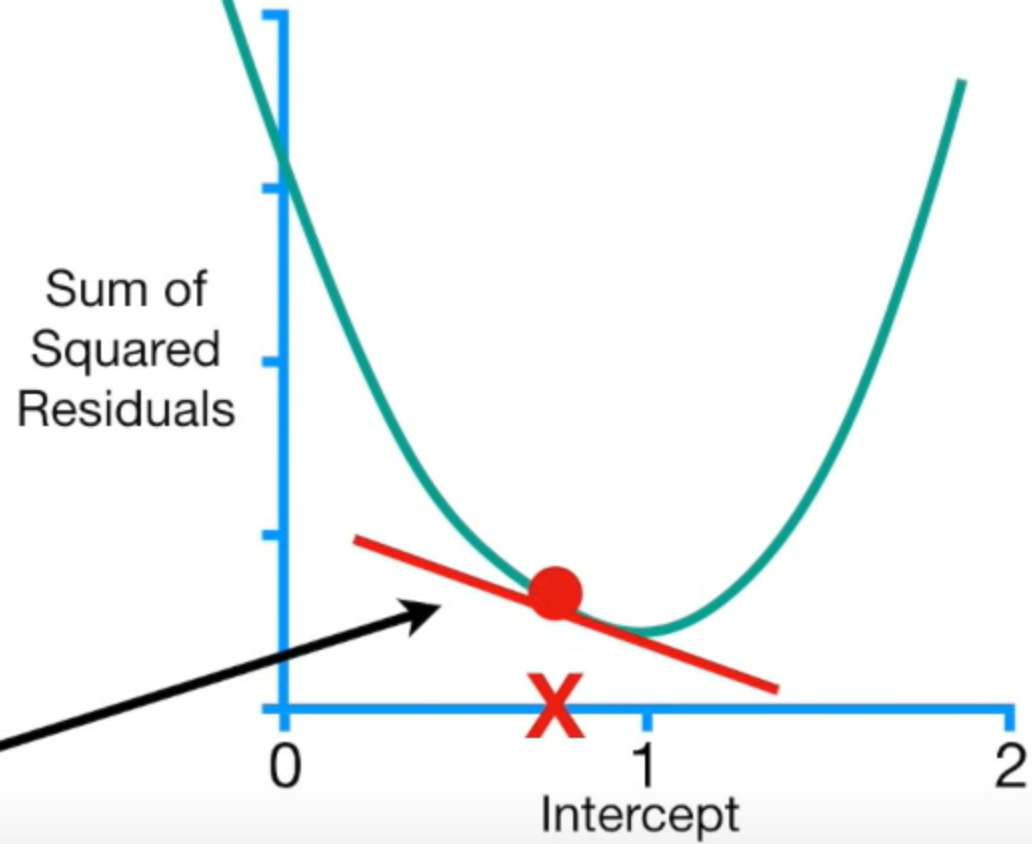
$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$



NOTE: The closer we get to the optimal value for the **Intercept**, the closer the slope of the curve gets to **0**.

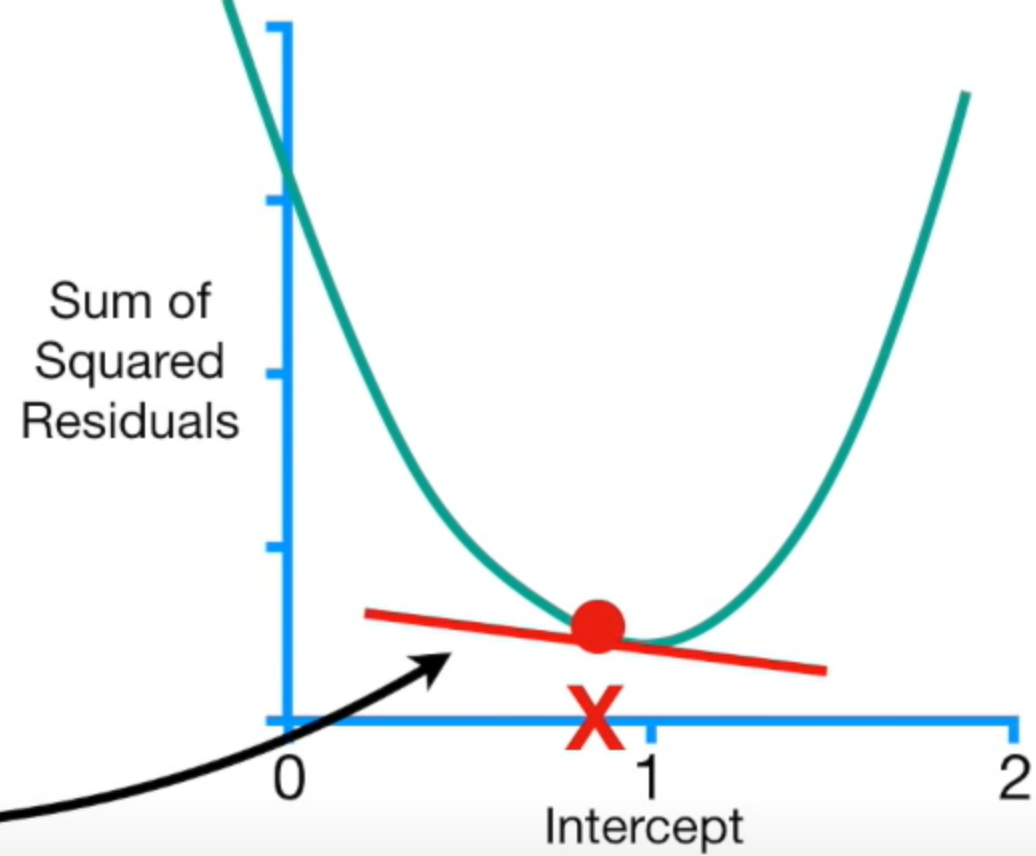
$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$



NOTE: The closer we get to the optimal value for the **Intercept**, the closer the slope of the curve gets to **0**.

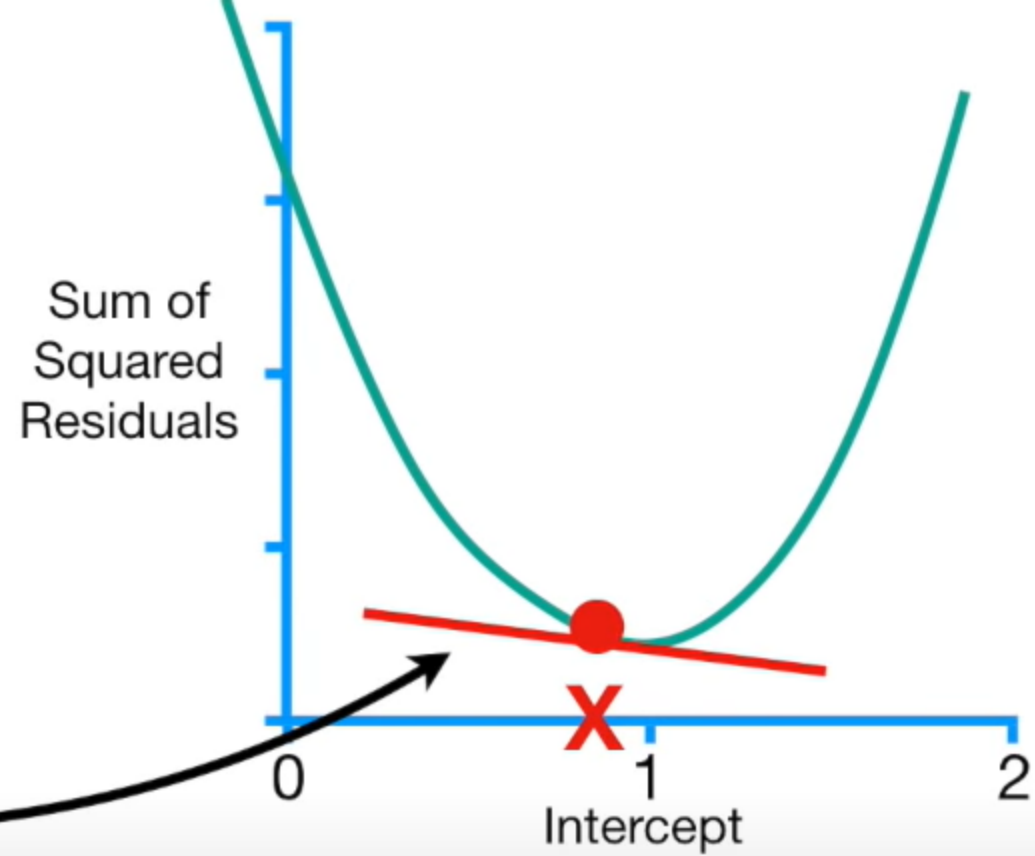
$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$



This means that when the slope of the curve is close to **0**…

$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

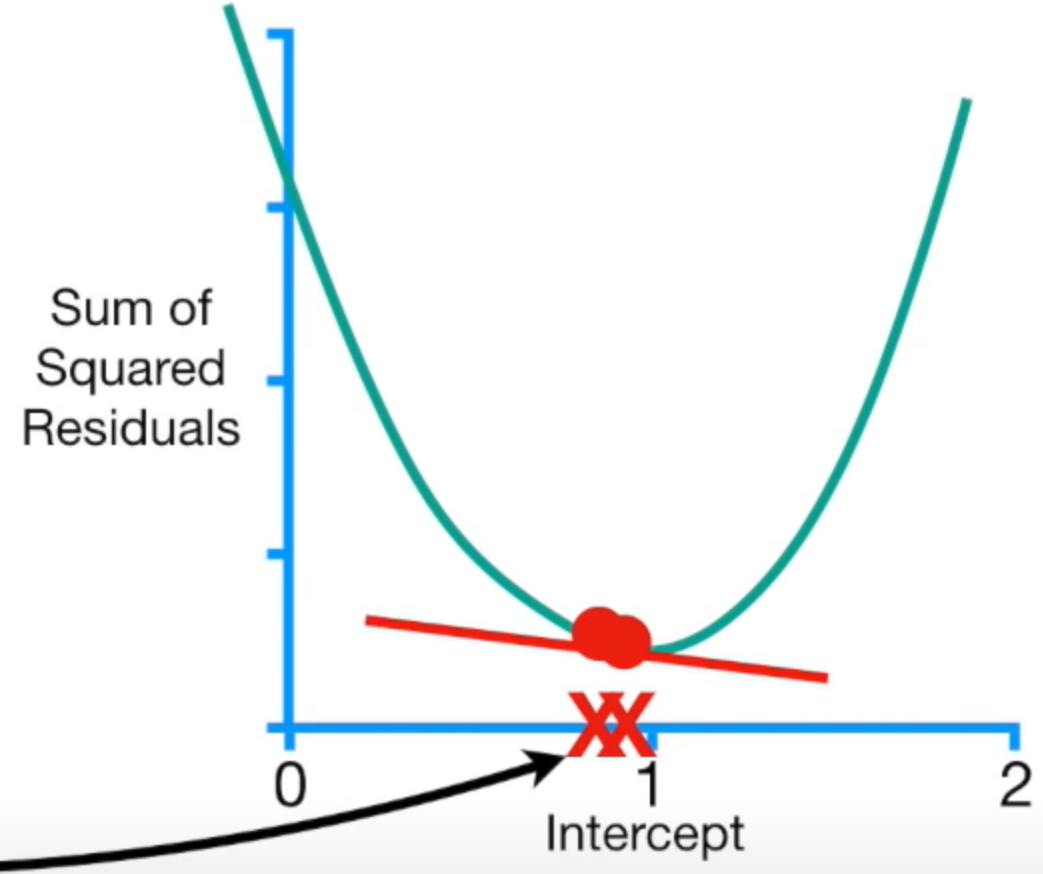
$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$

…then we should take baby steps, because we are close to the optimal value…



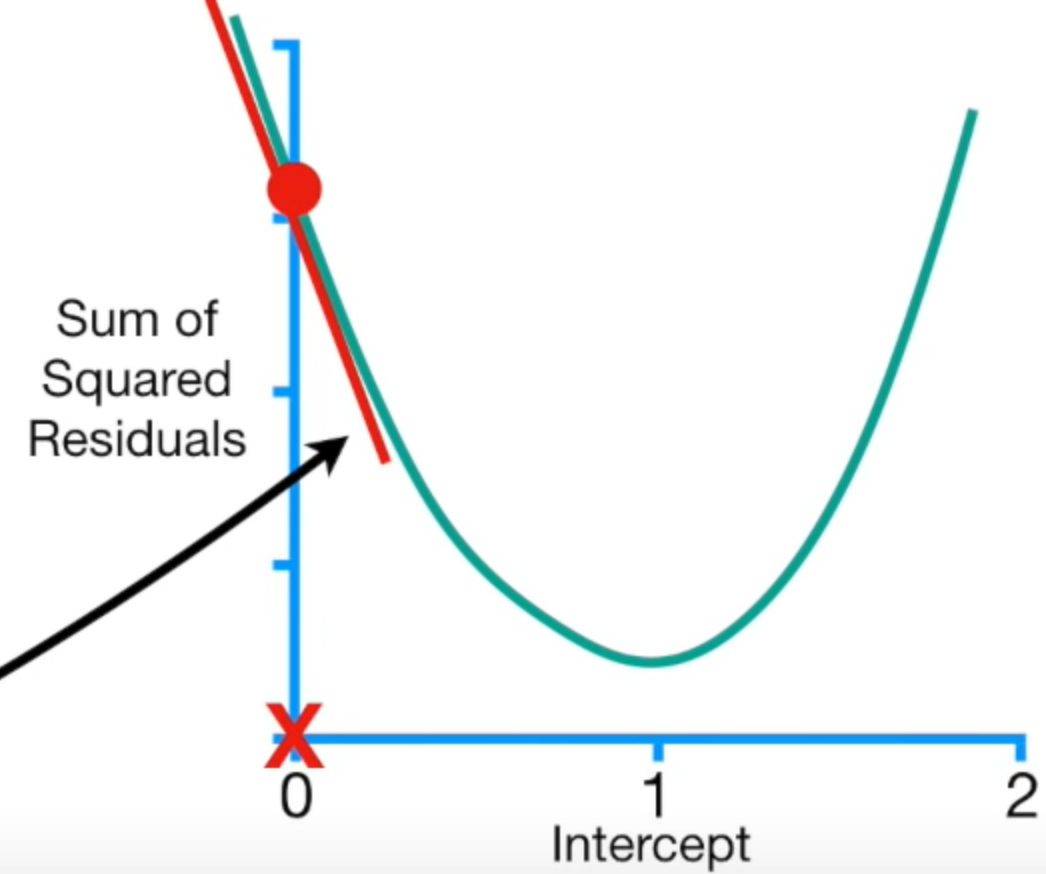
$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$



…and when the slope is far from **0**…

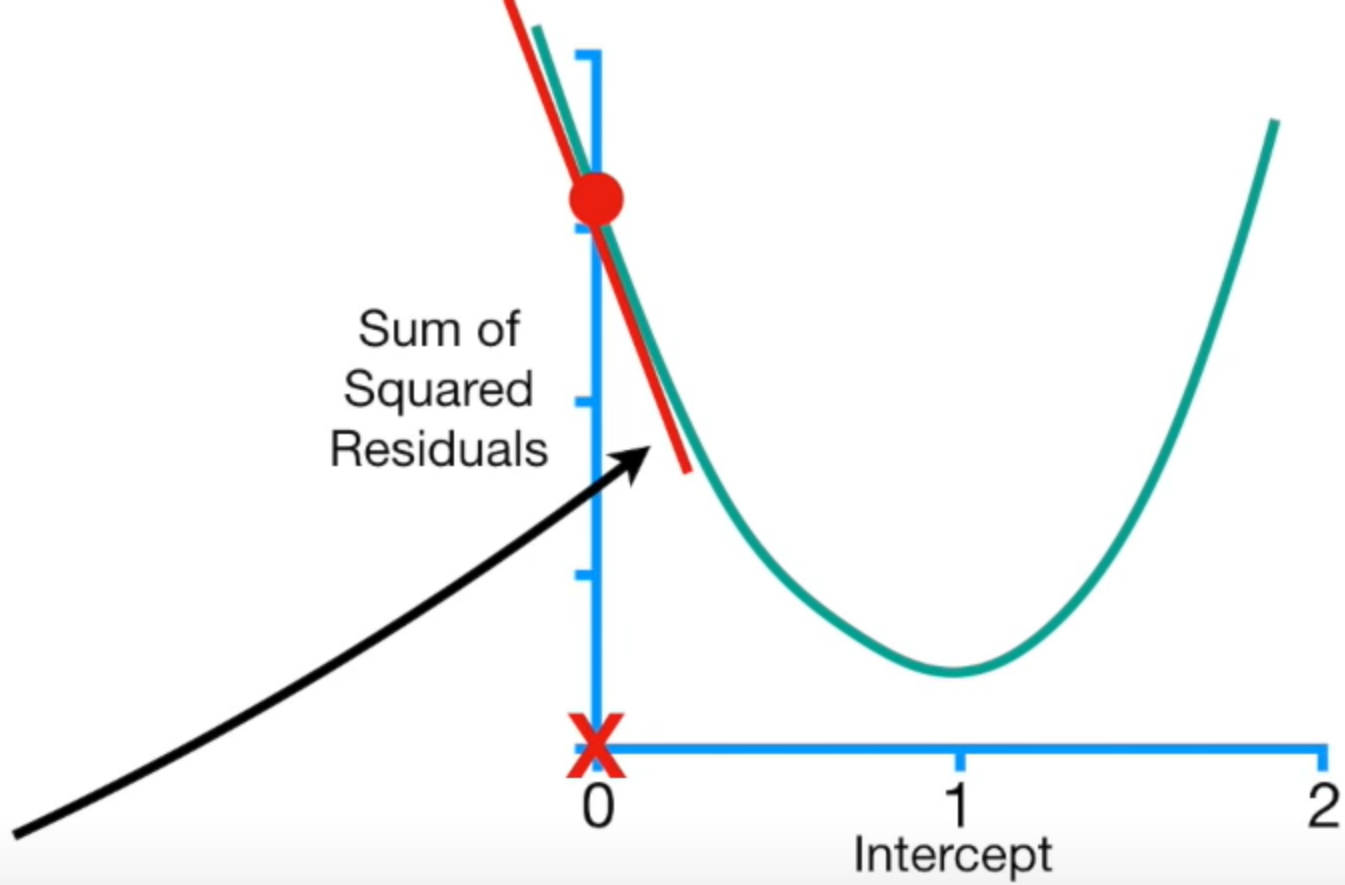
$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$



…and when the slope is far from **0**…

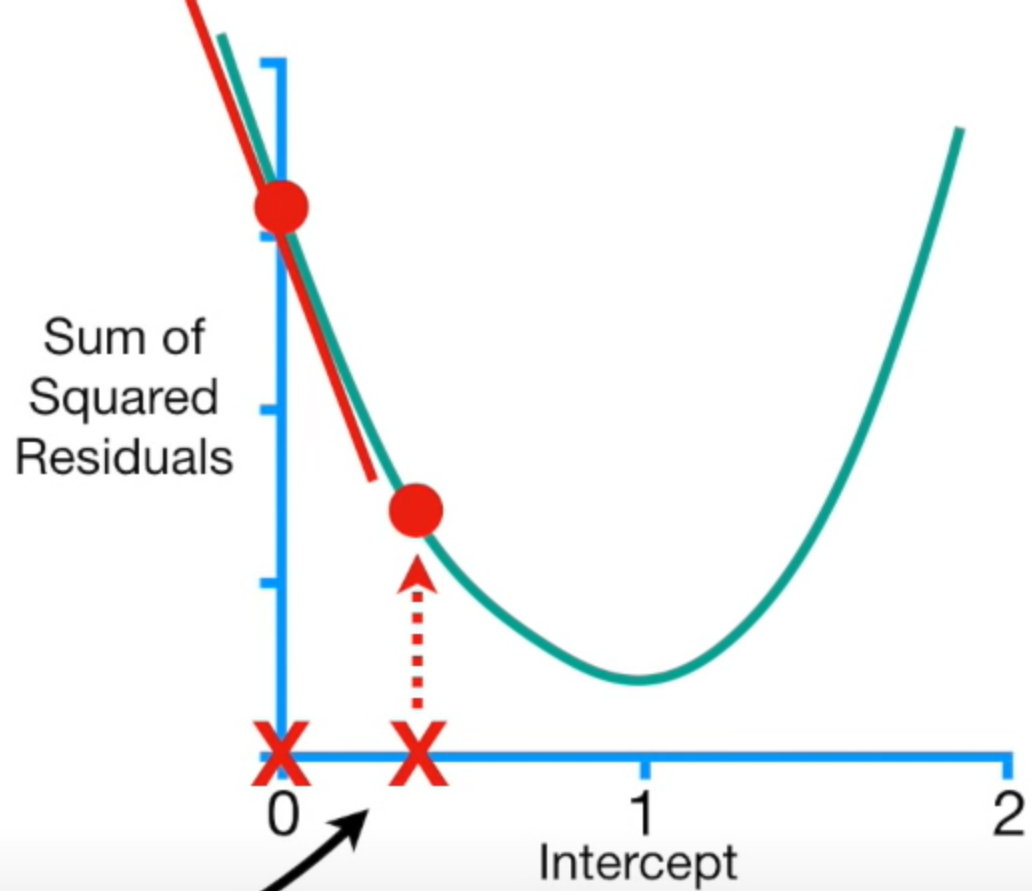
$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$



…then we should take big steps, because we are far from the optimal value.

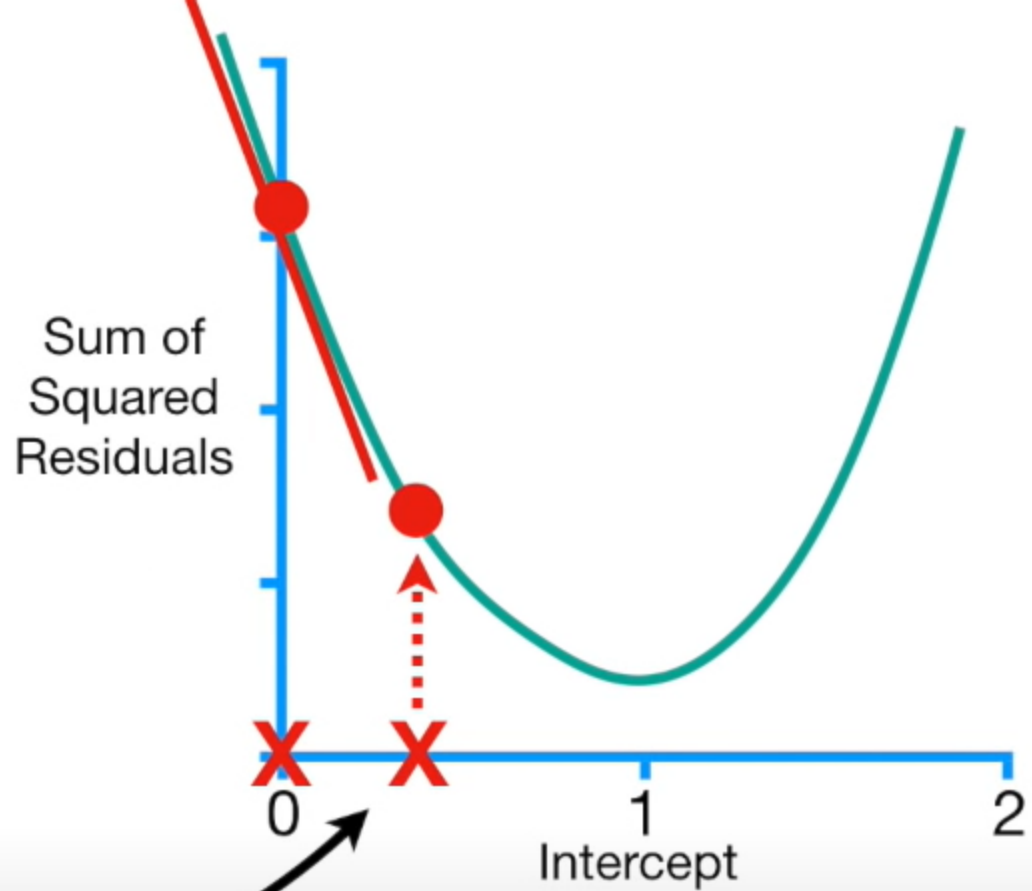
$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$



…then we should take big steps, because we are far from the optimal value.

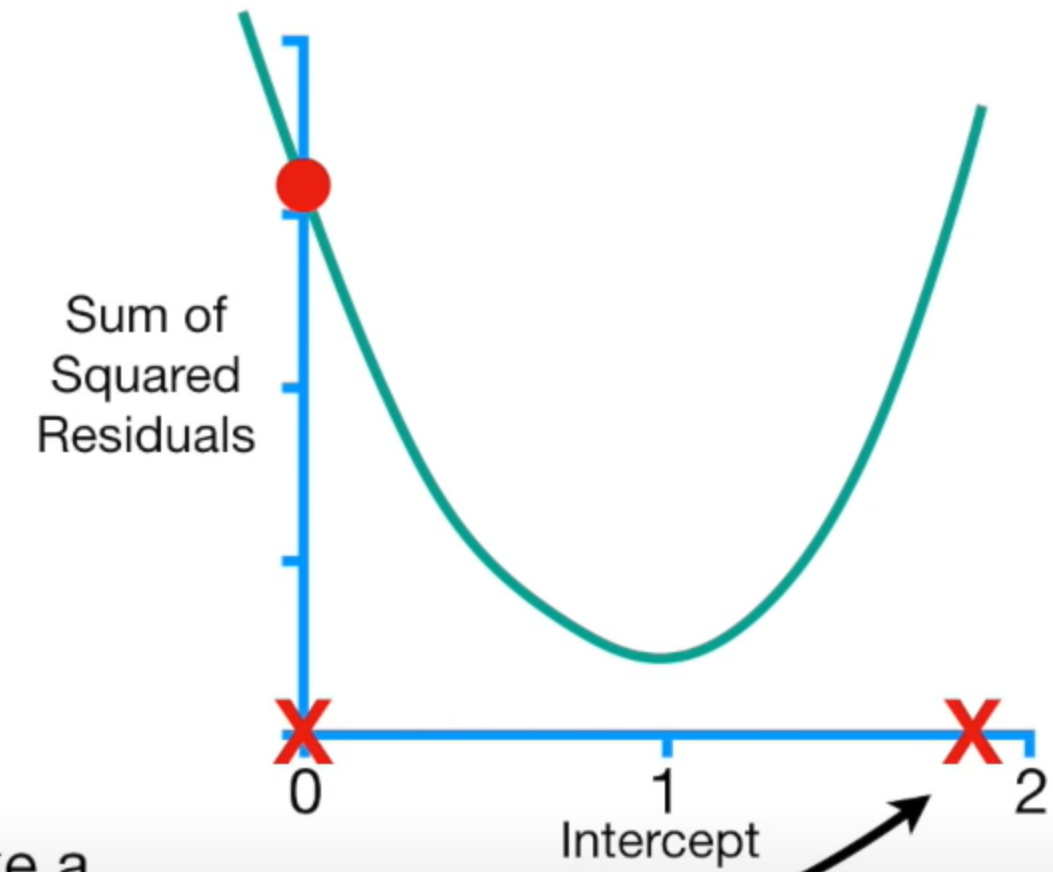
$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$



However, if we take a super huge step…

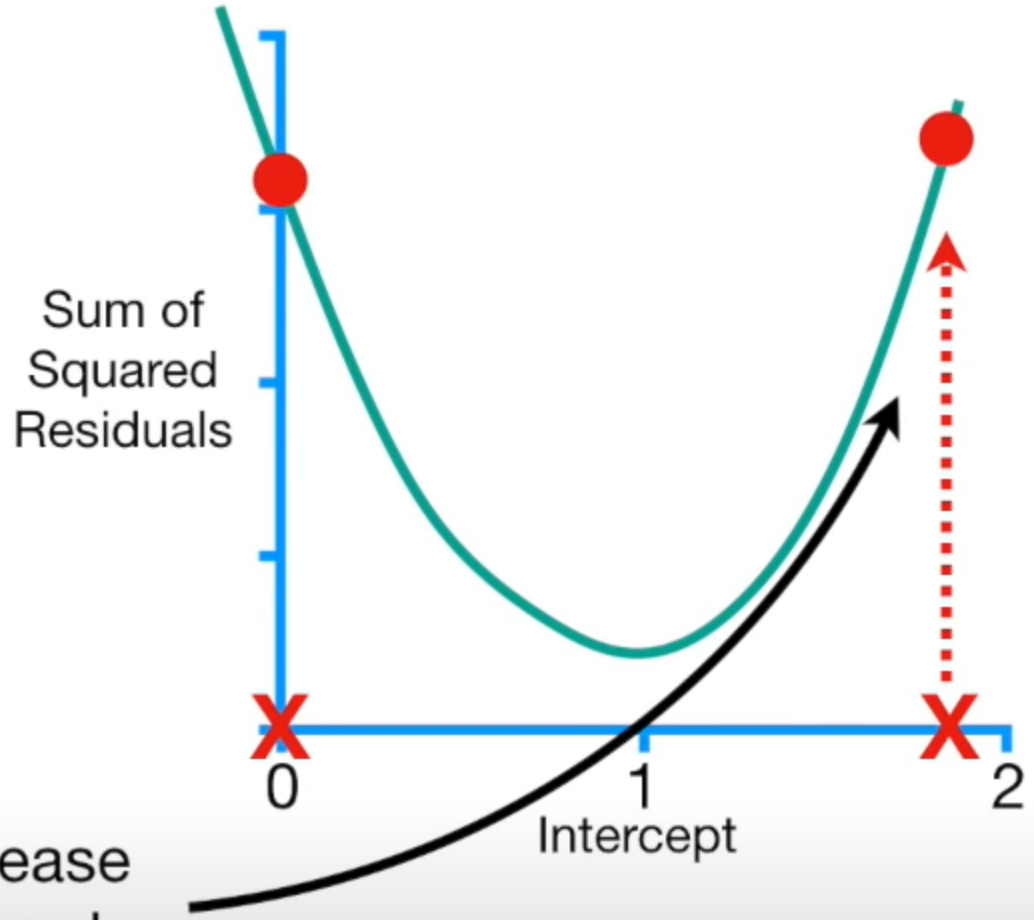
red residuals =

+ 0.64 × **0.5**)

+ 0.64 × **2.3**))

+ 0.64 × **2.9**))

-5.7



…then we would increase the Sum of the Squared Residuals!

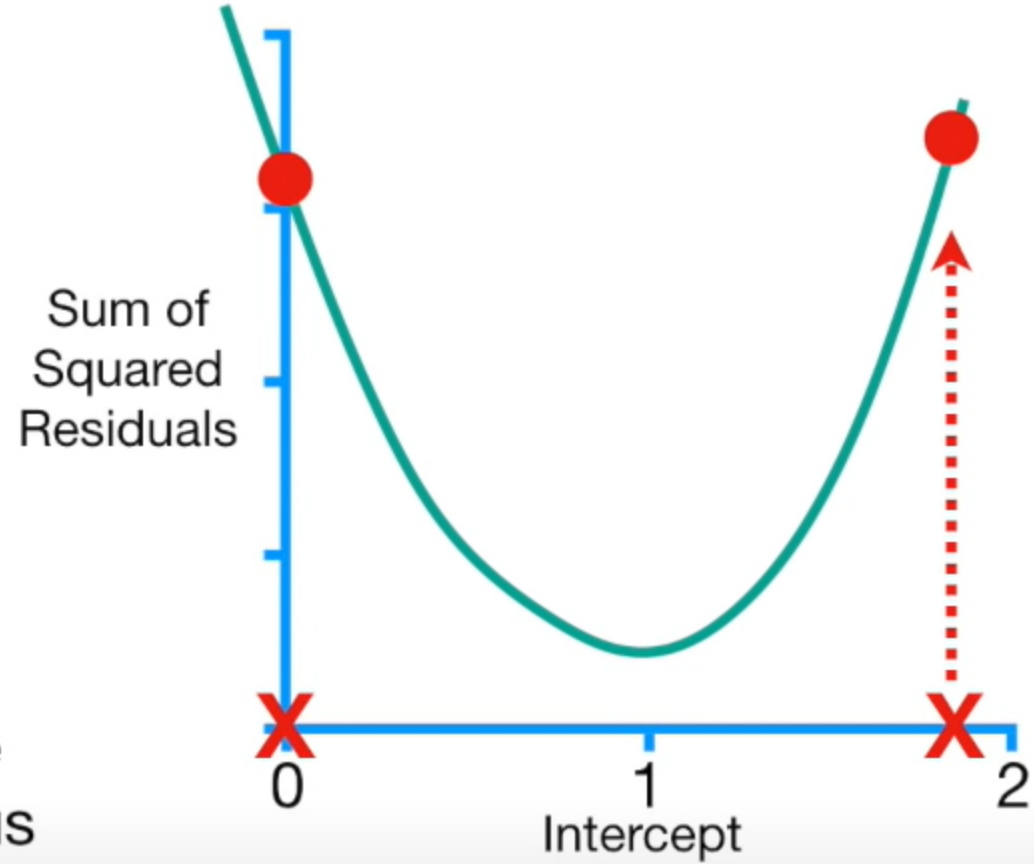
$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$



So the size of the step should be related to the slope, since it tells us if we should take a baby step or a big step, but we need to make sure the big step is not too big.

$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

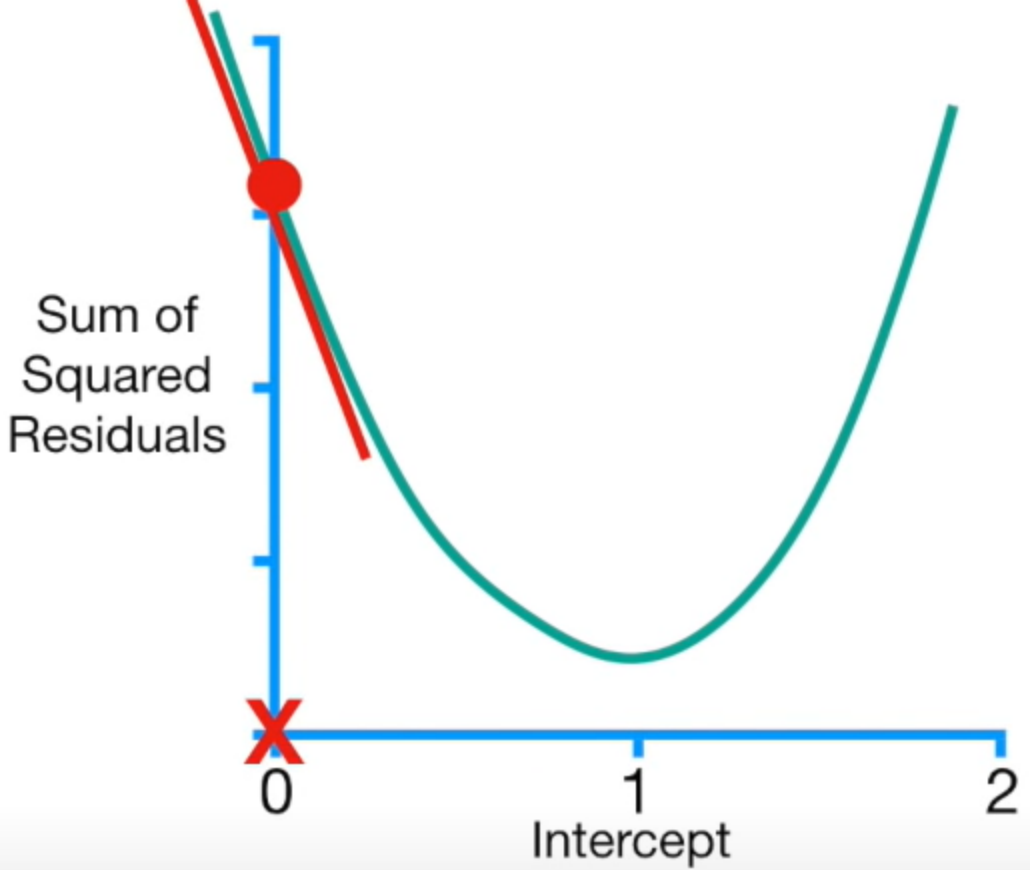
$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$

Step Size = -5.7

Gradient Descent determines the **Step Size** by multiplying the **slope**…



$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

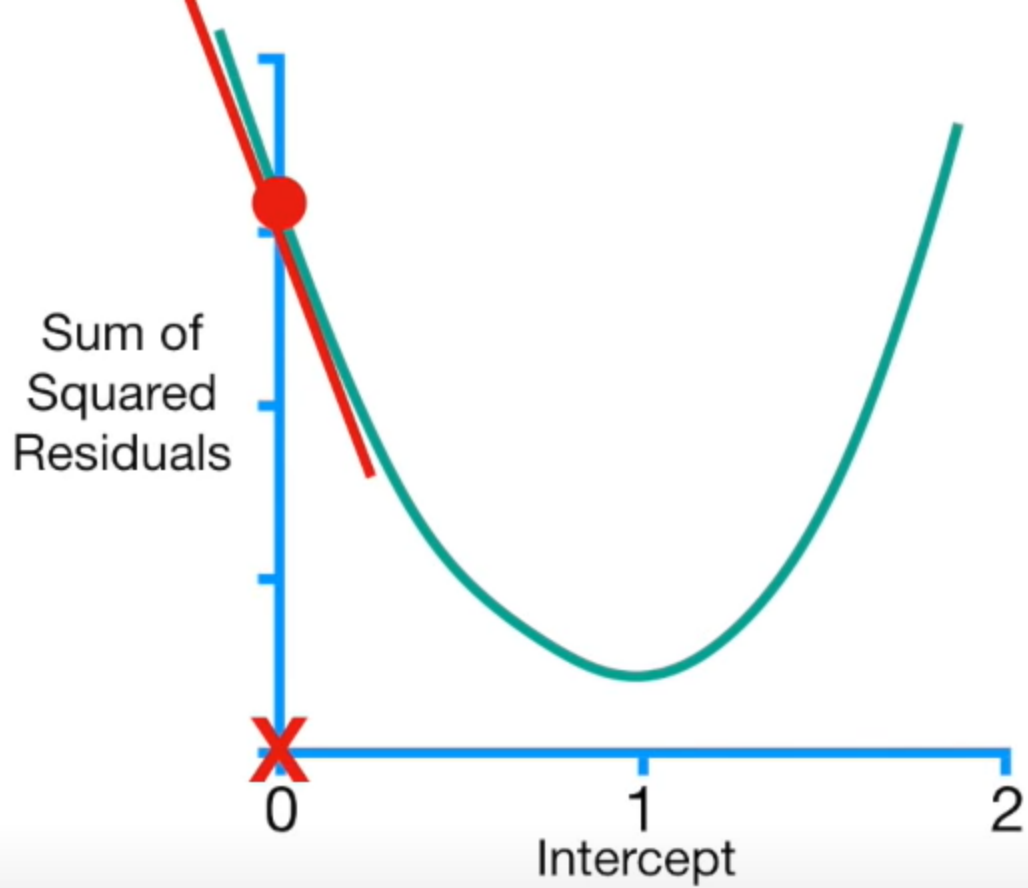
$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$

Step Size = -5.7 x 0.1

…by a small number called
The Learning Rate.



$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

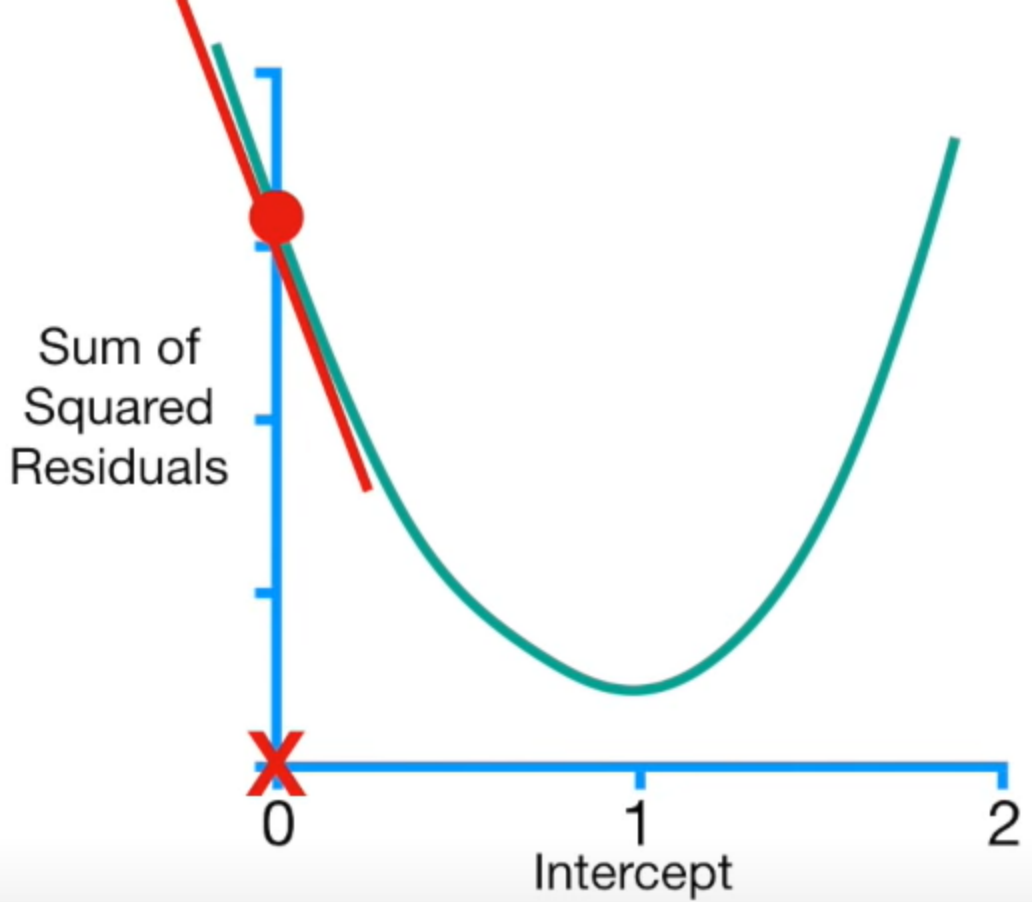
$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$

Step Size = -5.7 x 0.1

NOTE: We'll talk more about **Learning Rates** later.



$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

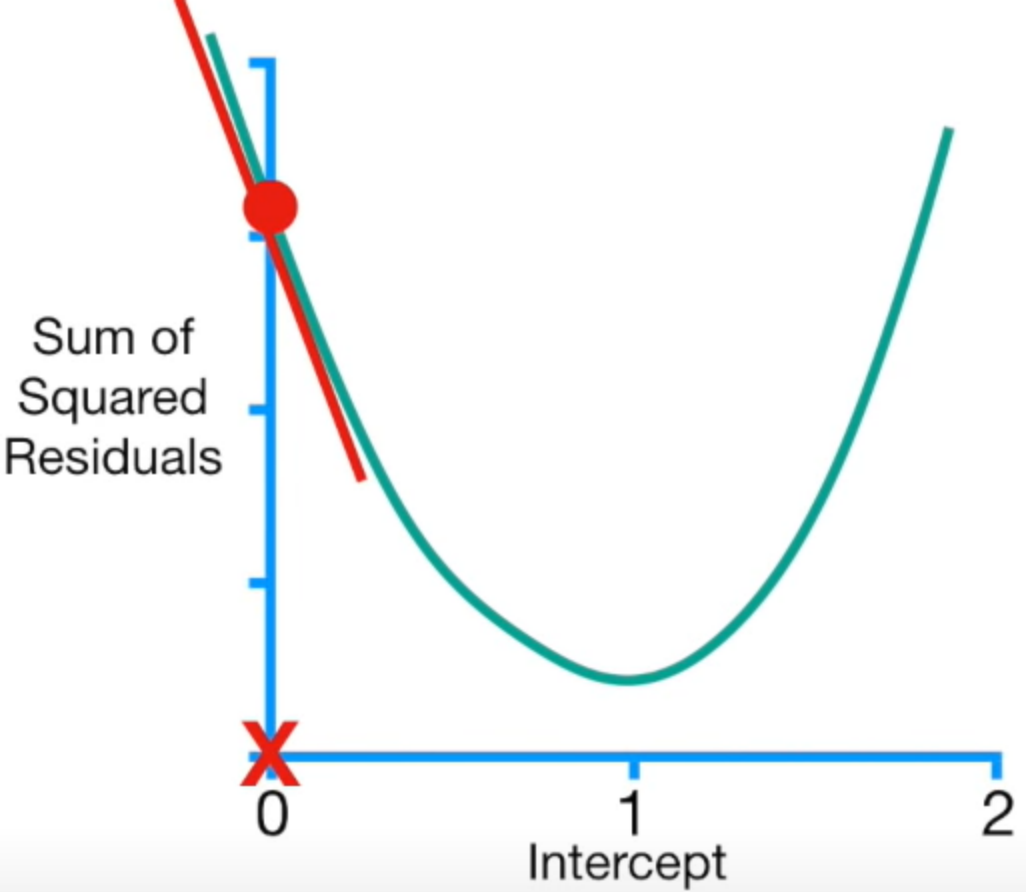
$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$

Step Size = -5.7 × 0.1 = -0.57

When the **Intercept** = **0**, the **Step Size** = **-0.57**.



$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

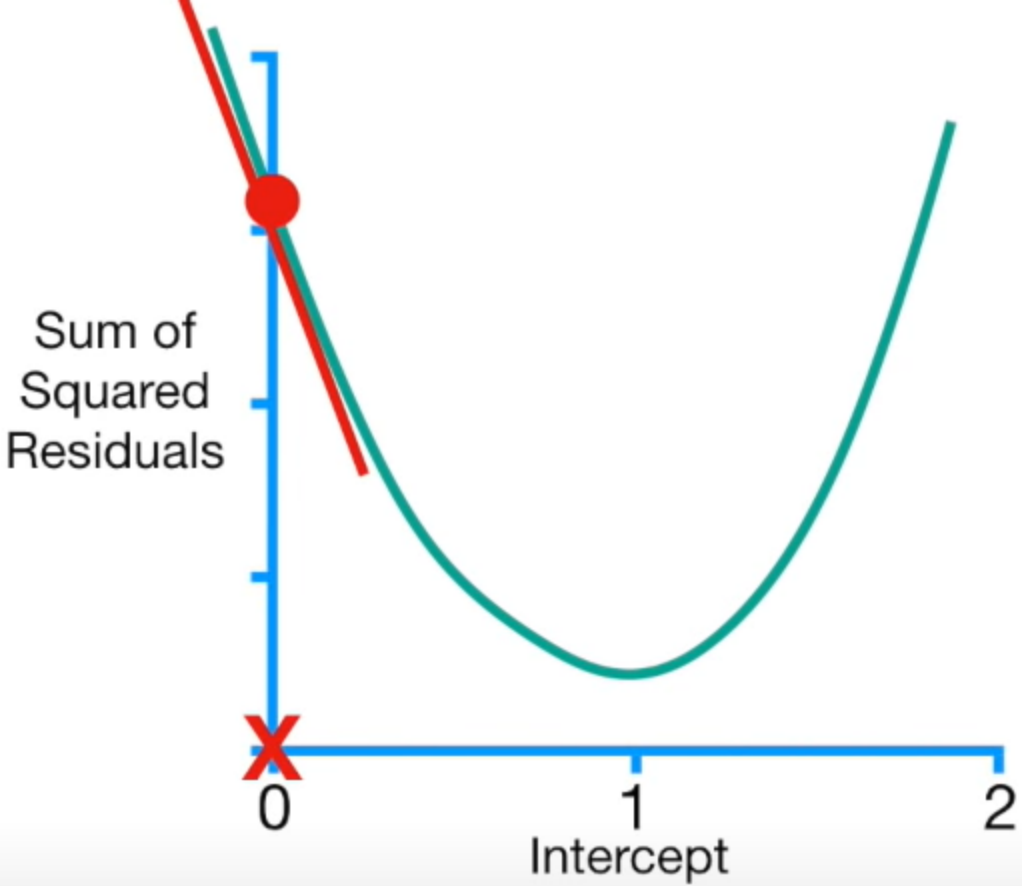
$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$

Step Size = -5.7 x 0.1 = -0.57



When the **Intercept** = **0**, the **Step Size** = **-0.57**.

$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

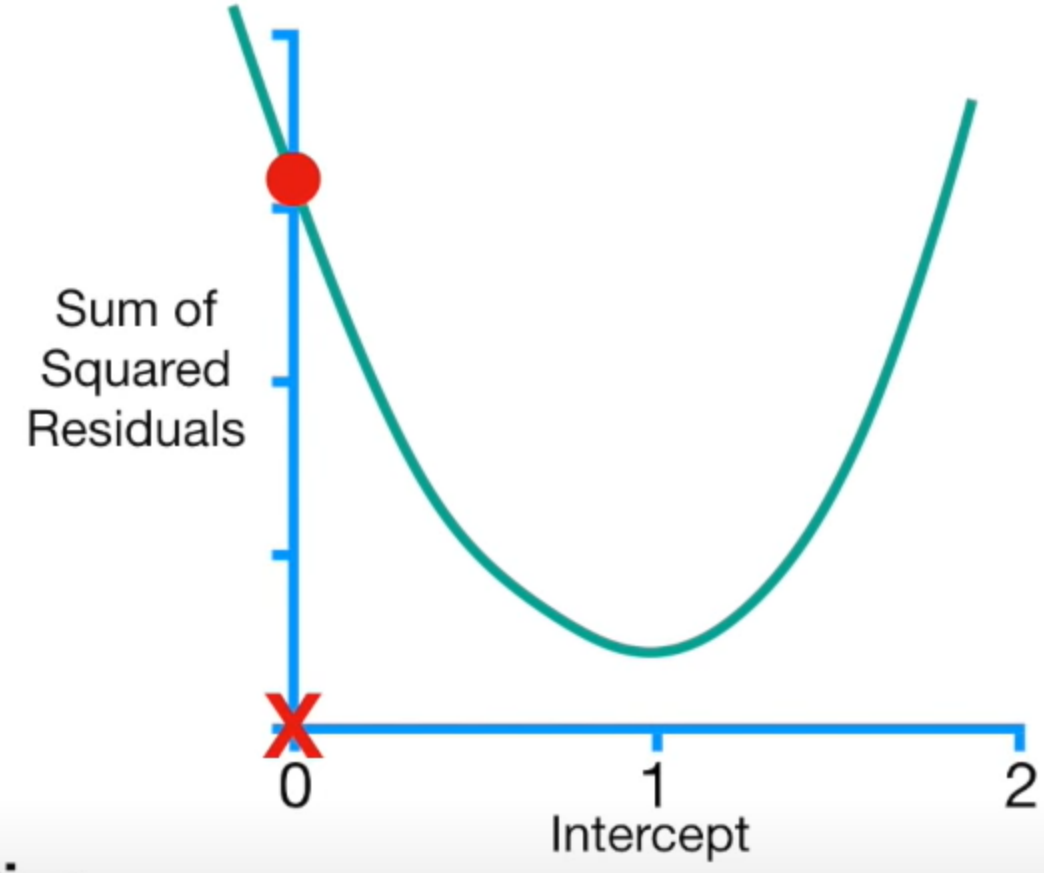
$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$

Step Size = -5.7 × 0.1 = -0.57

New Intercept = ←

With the **Step Size**, we can calculate a **New Intercept**.



$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

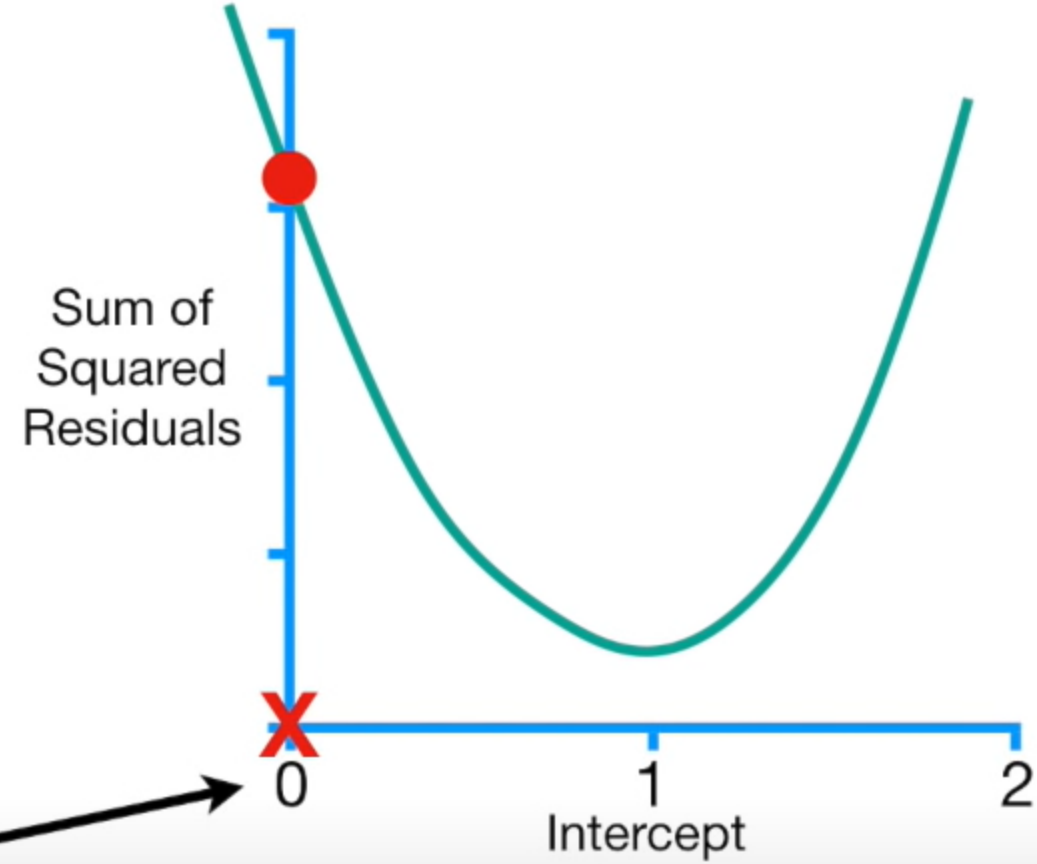
$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5}))$$

$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$

Step Size $= -5.7 \times 0.1 = -0.57$



New Intercept = **Old Intercept** - Step Size

The **New Intercept** is
the **Old Intercept**…

$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

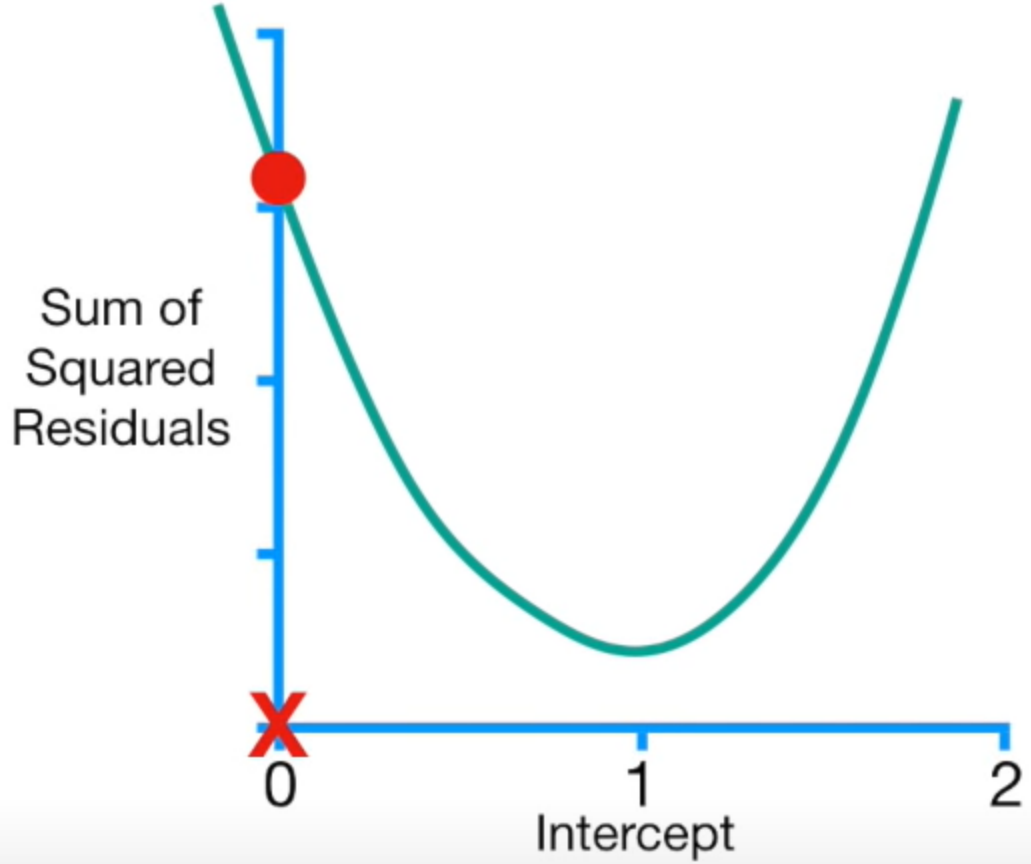
$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$

Step Size = -5.7 × 0.1 = -0.57



New Intercept = **Old Intercept** - **Step Size**

…minus the **Step Size**.

$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

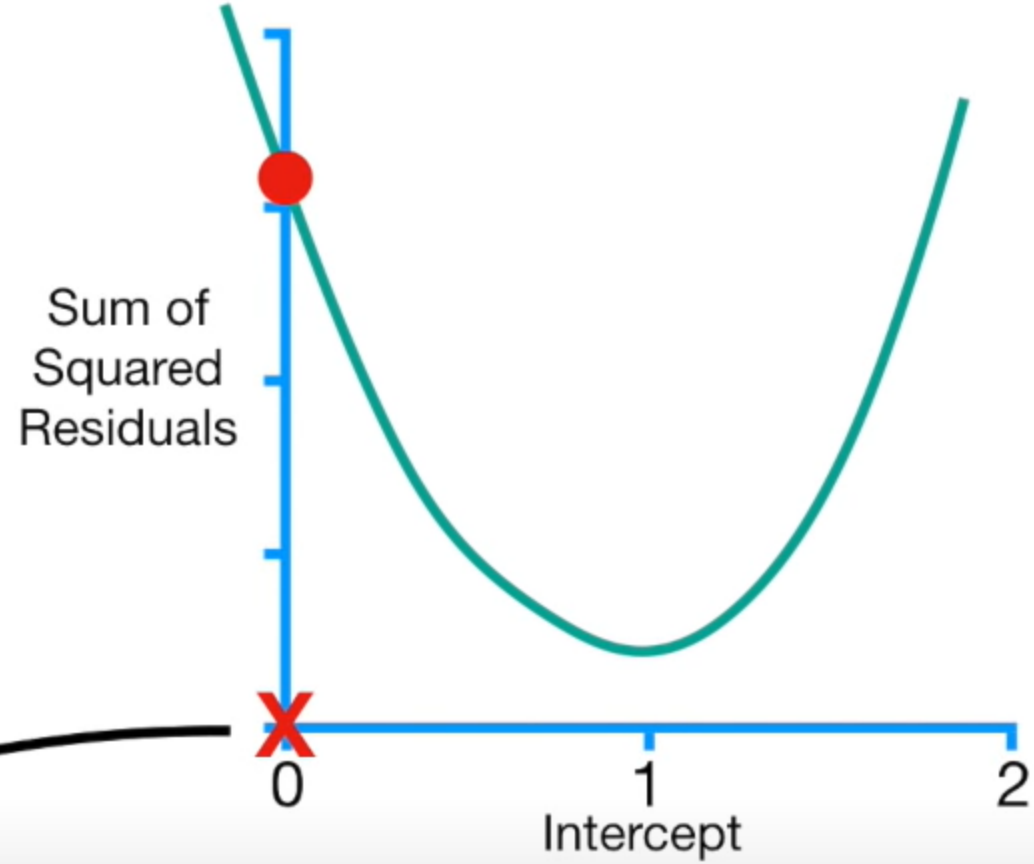
$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$

$$\textbf{Step Size} = -5.7 \times 0.1 = -0.57$$



New Intercept = **Old Intercept** - **Step Size**

So we plug in the
numbers…

$$\frac{d}{d\,intercept}$$ Sum of squared residuals =

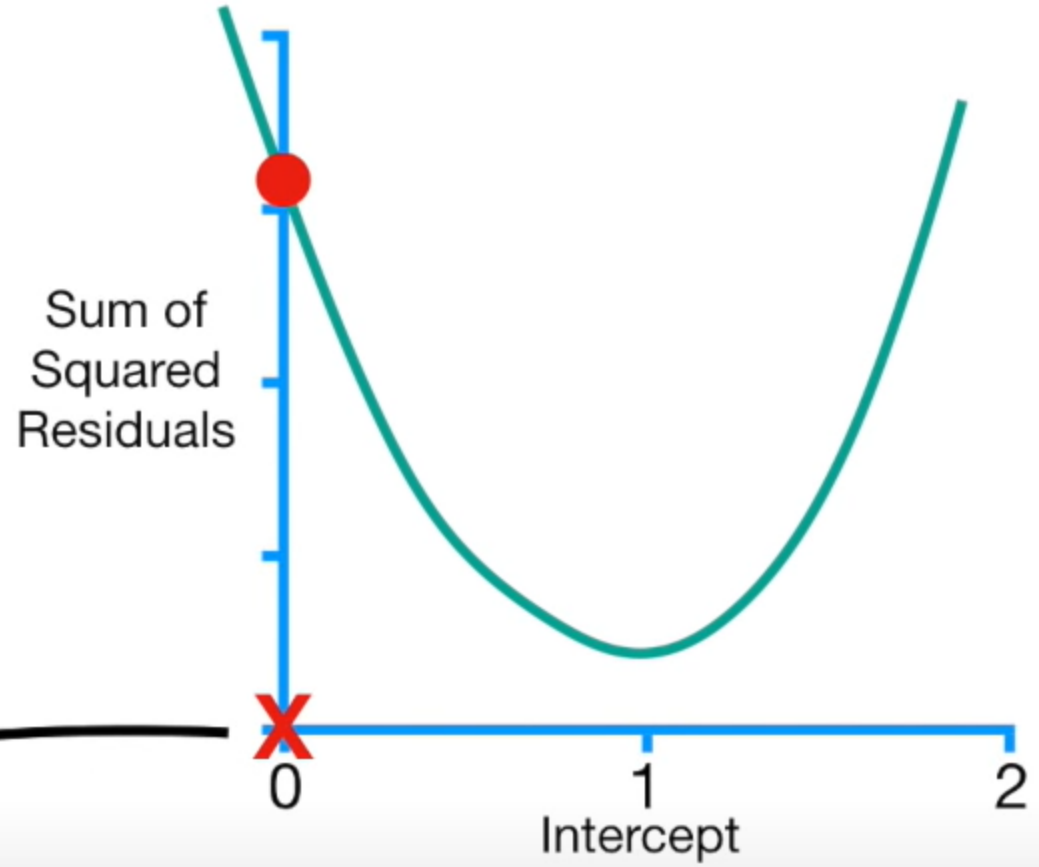
$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$

Step Size $= -5.7 \times 0.1 = -0.57$



New Intercept $= 0$ - **Step Size**

So we plug in the numbers…

$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

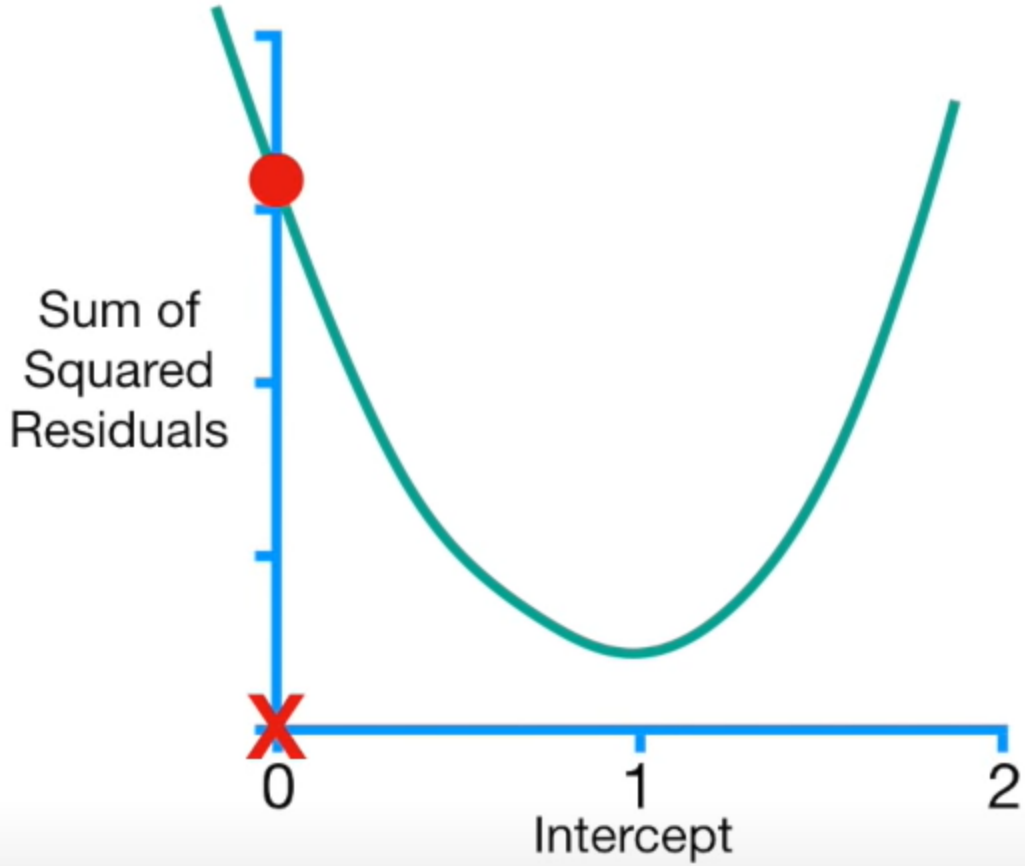
$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$

Step Size = -5.7 x 0.1 = -0.57

New Intercept = 0 - **Step Size**



So we plug in the numbers…

$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

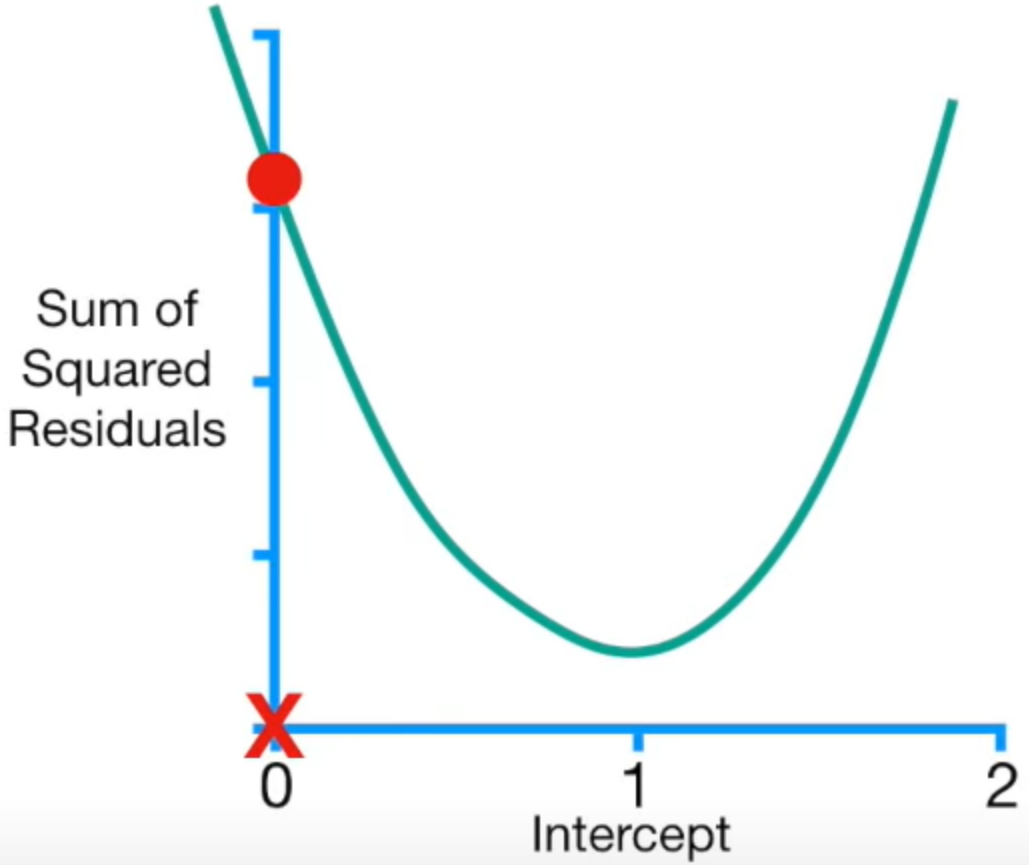
$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

$$= -5.7$$

Step Size = -5.7 × 0.1 = -0.57

New Intercept = 0 - (-0.57)



So we plug in the numbers

$$\frac{d}{d\,intercept}$$ Sum of squared residuals =

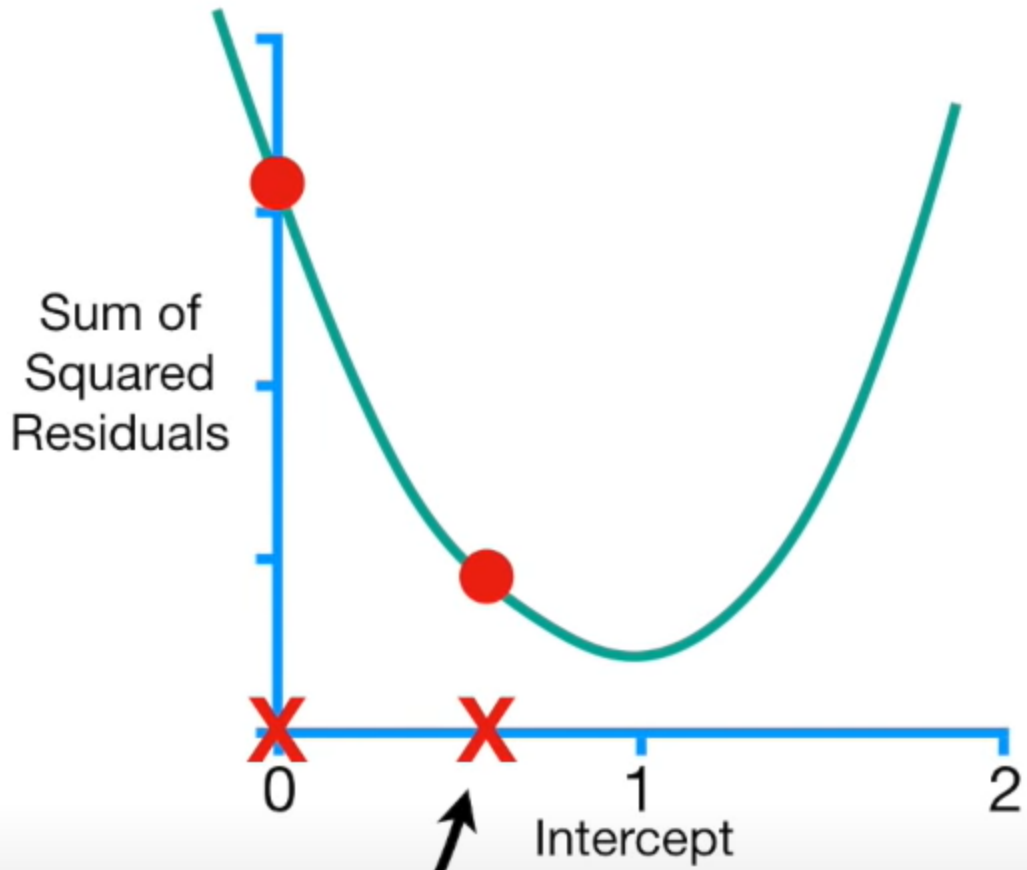
$$-2(\mathbf{1.4} - (0 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0 + 0.64 \times \mathbf{2.9}))$$

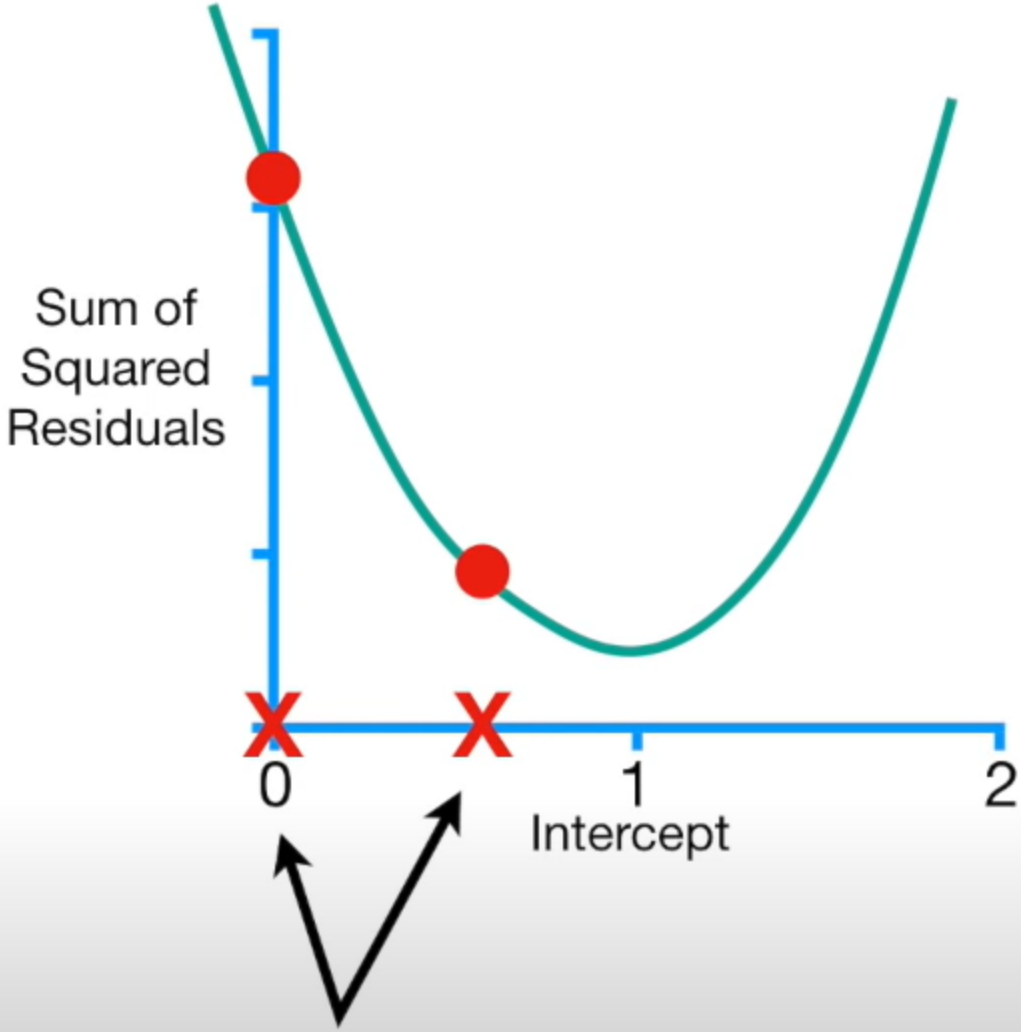
$$= -5.7$$

Step Size $= -5.7 \times 0.1 = -0.57$



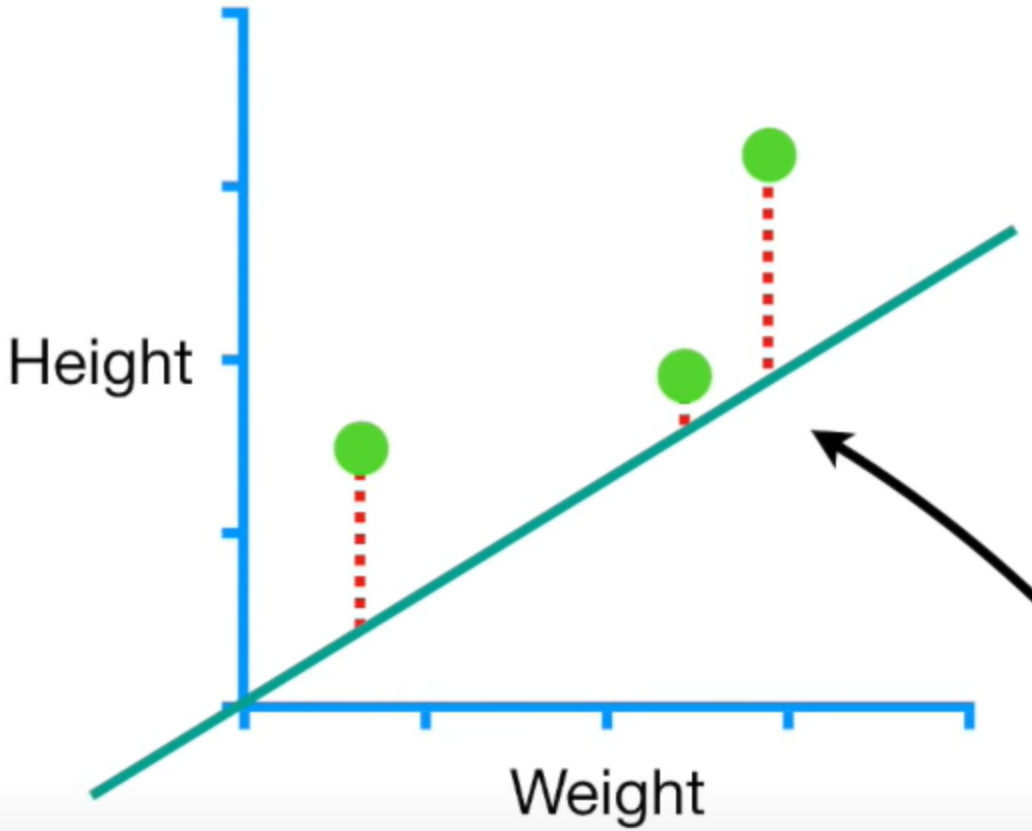
New Intercept $= 0 - (-0.57) =$ **0.57**

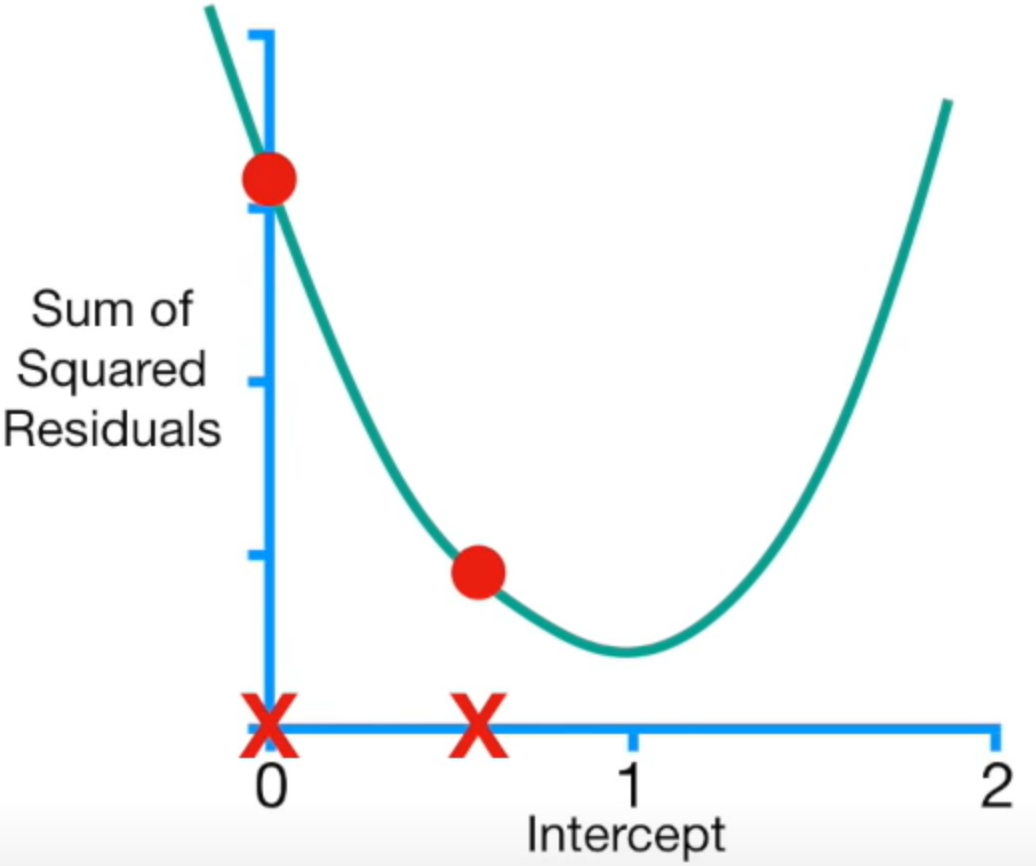
…and the the **New Intercept** = **0.57**.



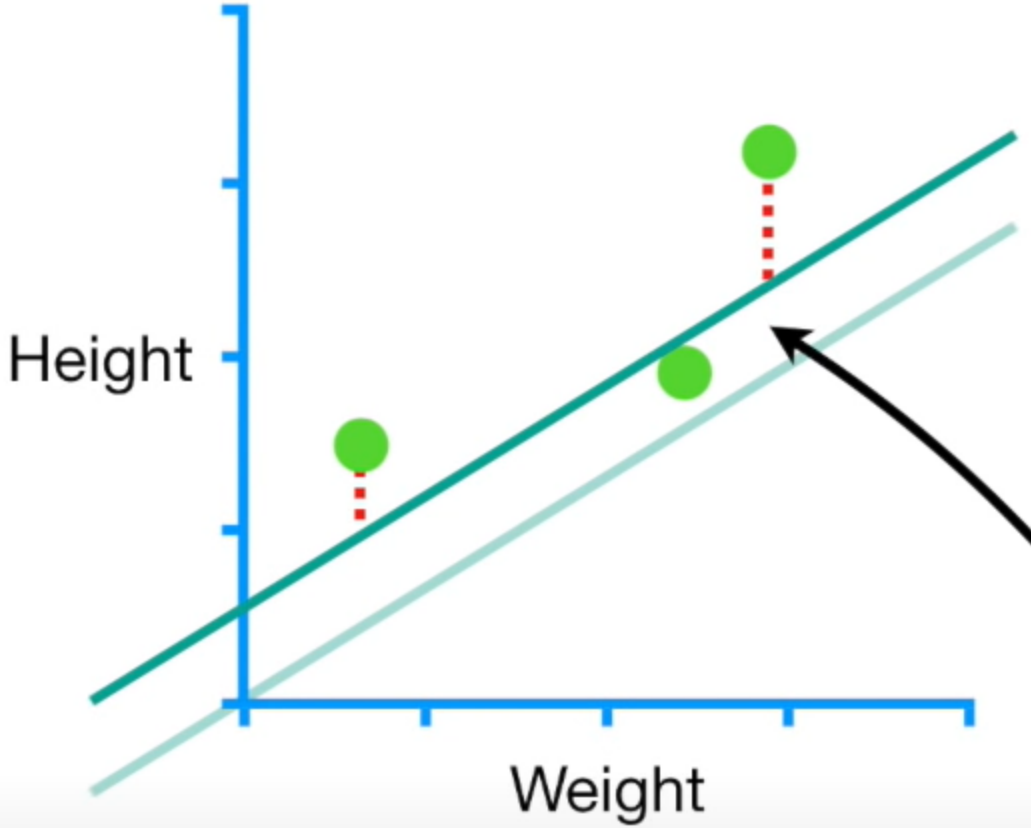
!!

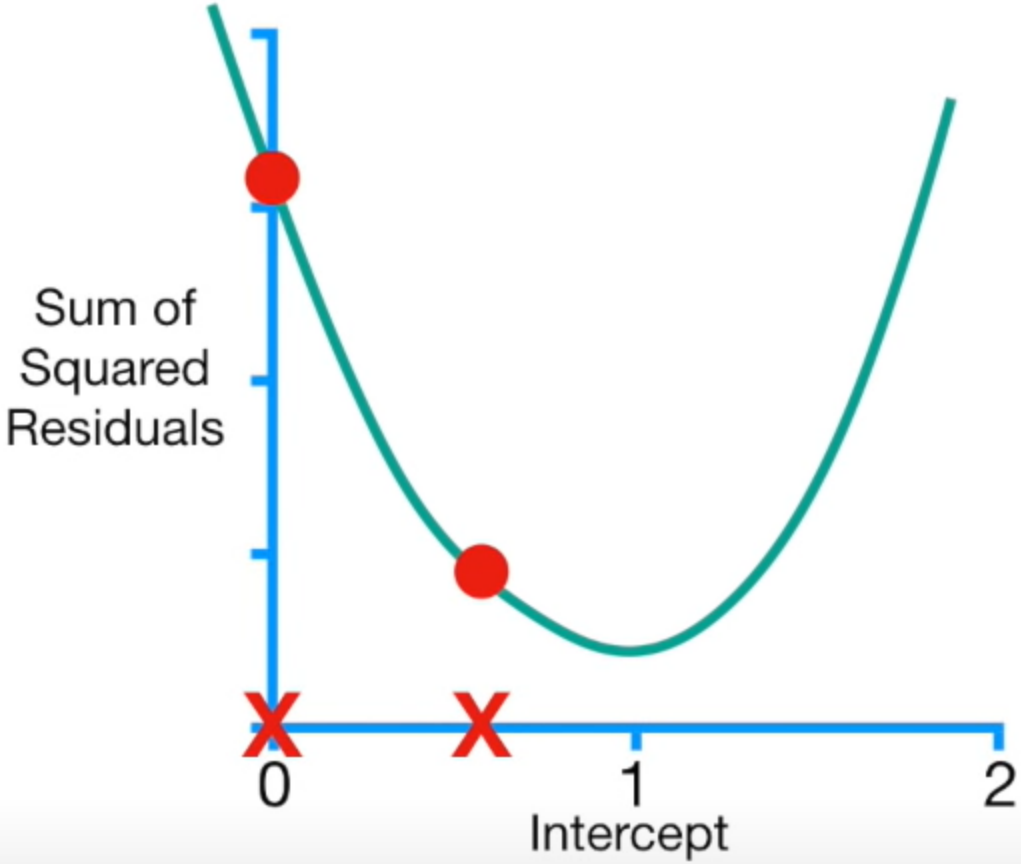
In one big step, we moved much closer to the optimal value for the **Intercept**.



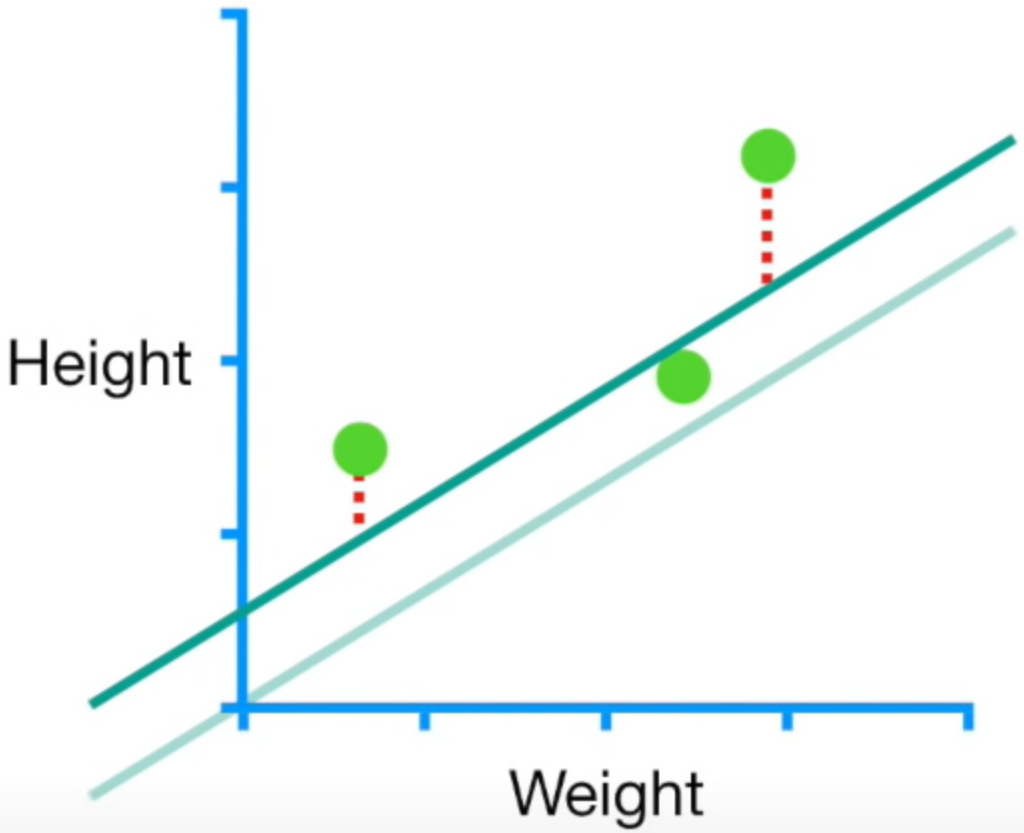


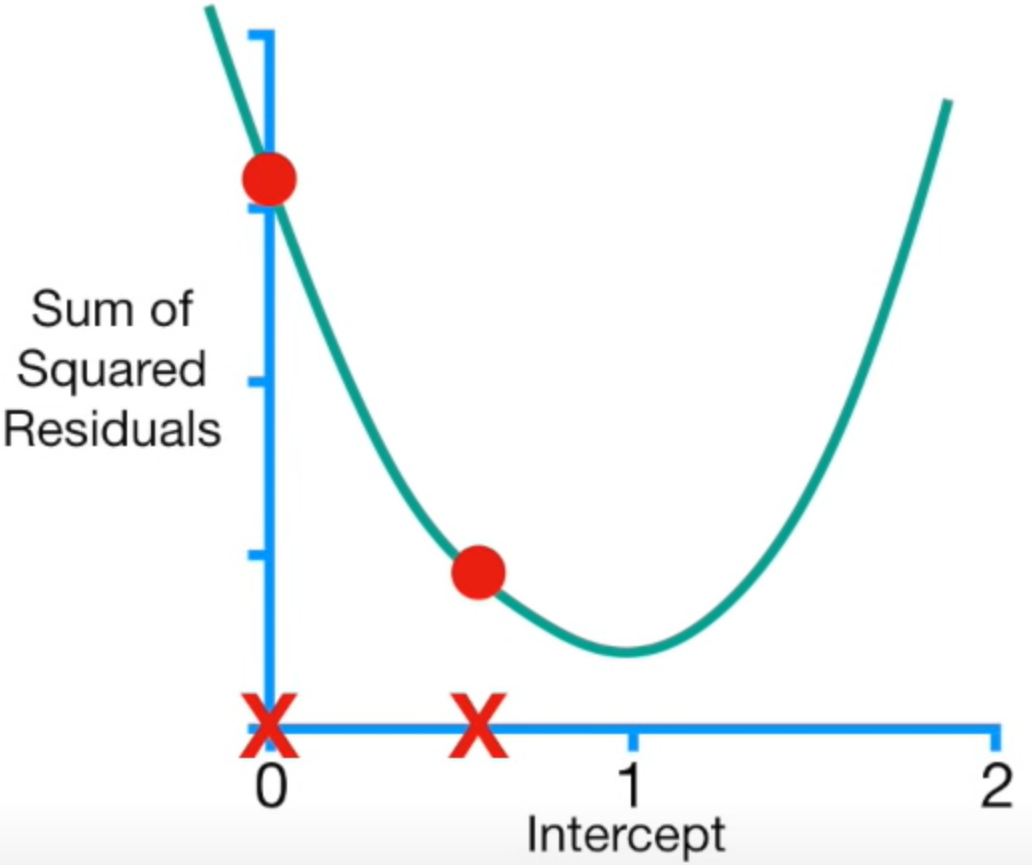
Going back to the original data and the original line, with the **Intercept = 0**…





…we can see how much the residuals shrink when the **Intercept** = **0.57**.





Now let's take another step
closer to the optimal value
for the **Intercept**.

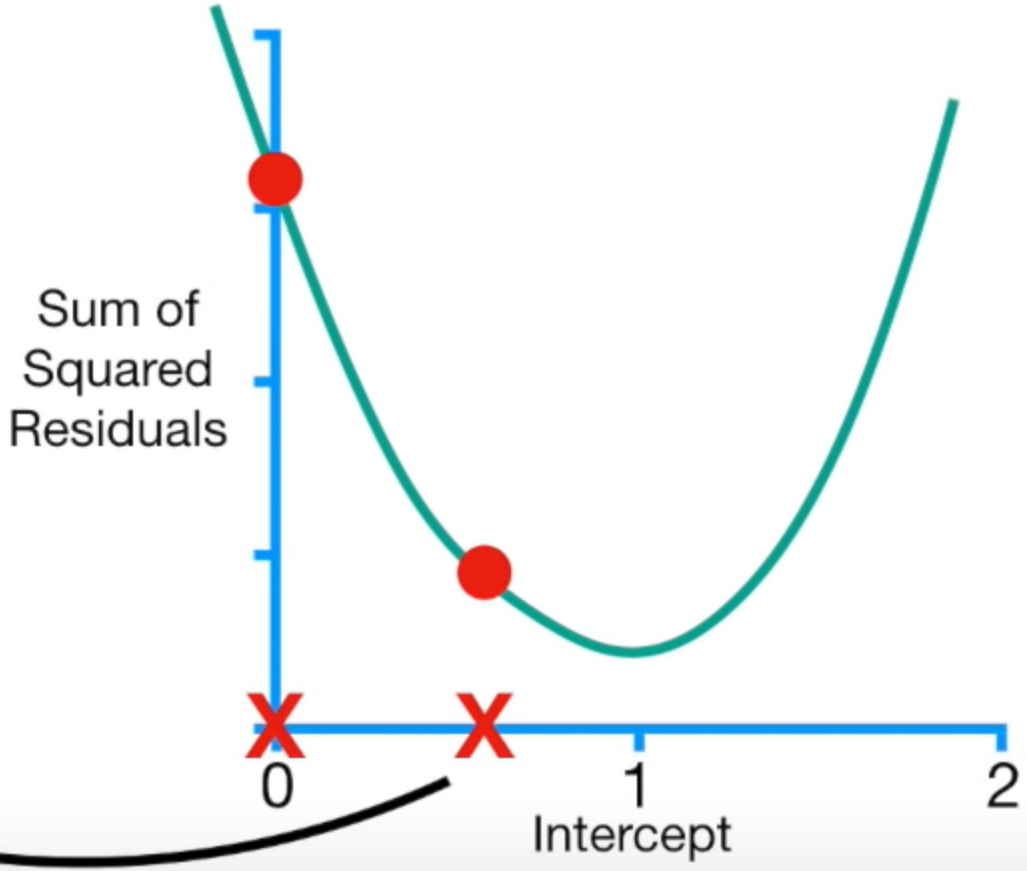
$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

$$-2(\mathbf{1.4} - (intercept + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (intercept + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (intercept + 0.64 \times \mathbf{2.9}))$$

To take another step, we go back to the derivative and plug in the **New Intercept** (**0.57**)…



$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

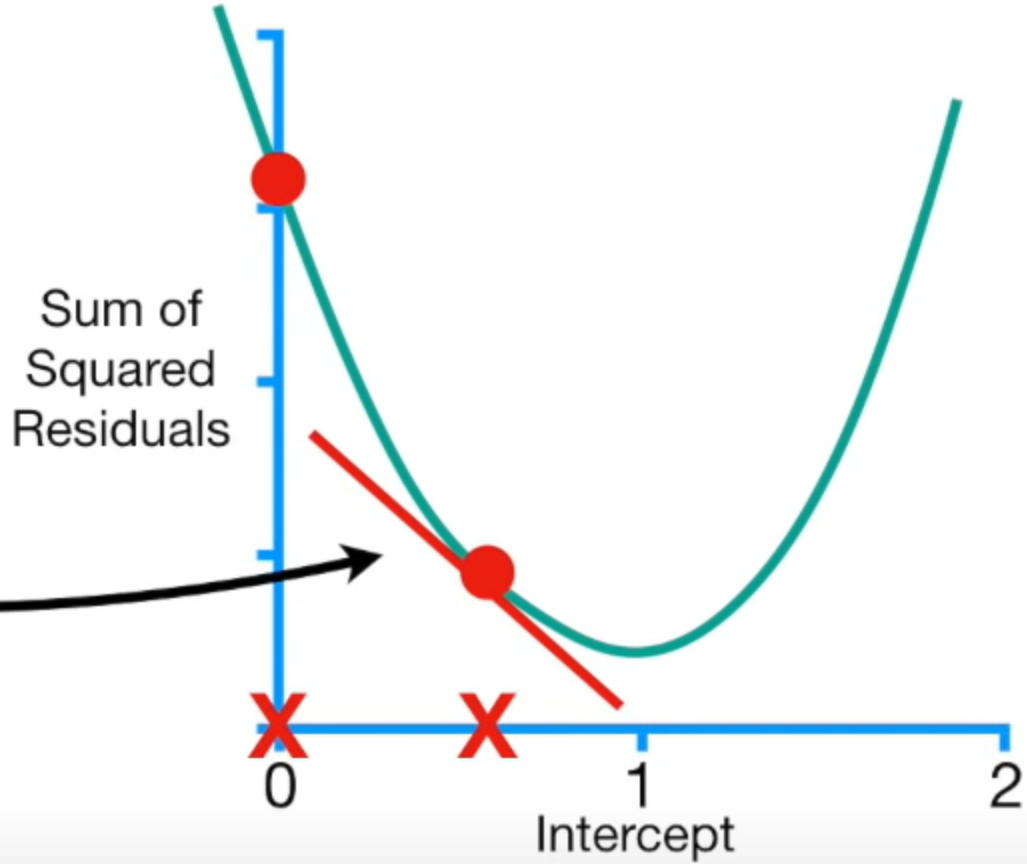
$$-2(\mathbf{1.4} - (0.57 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0.57 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0.57 + 0.64 \times \mathbf{2.9}))$$

$$= \boxed{-2.3}$$

…and that tells us the slope of the curve = **-2.3**.



$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

$$-2(\mathbf{1.4} - (0.57 + 0.64 \times \mathbf{0.5})$$

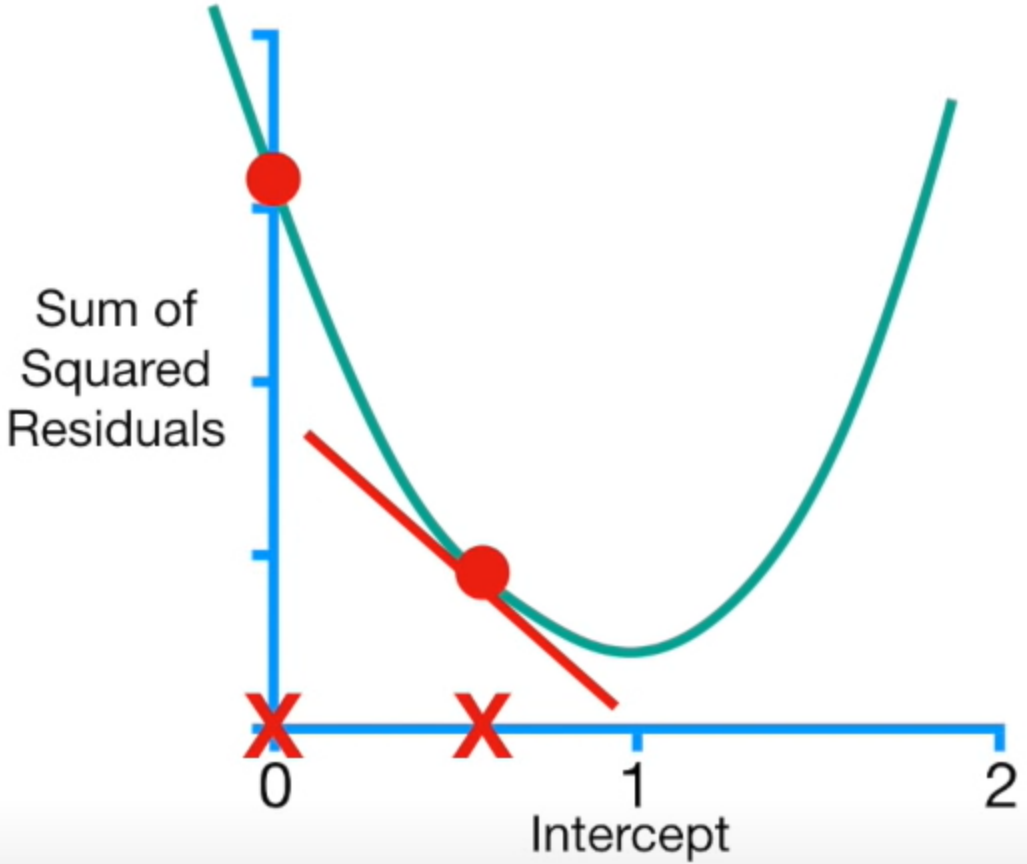
$$+ -2(\mathbf{1.9} - (0.57 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0.57 + 0.64 \times \mathbf{2.9}))$$

$$= -2.3$$

Step Size = **Slope** × **Learning Rate**

Now let's calculate the **Step Size**…



$$\frac{d}{d\,intercept}$$ Sum of squared residuals =

$$-2(\mathbf{1.4} - (0.57 + 0.64 \times \mathbf{0.5})$$

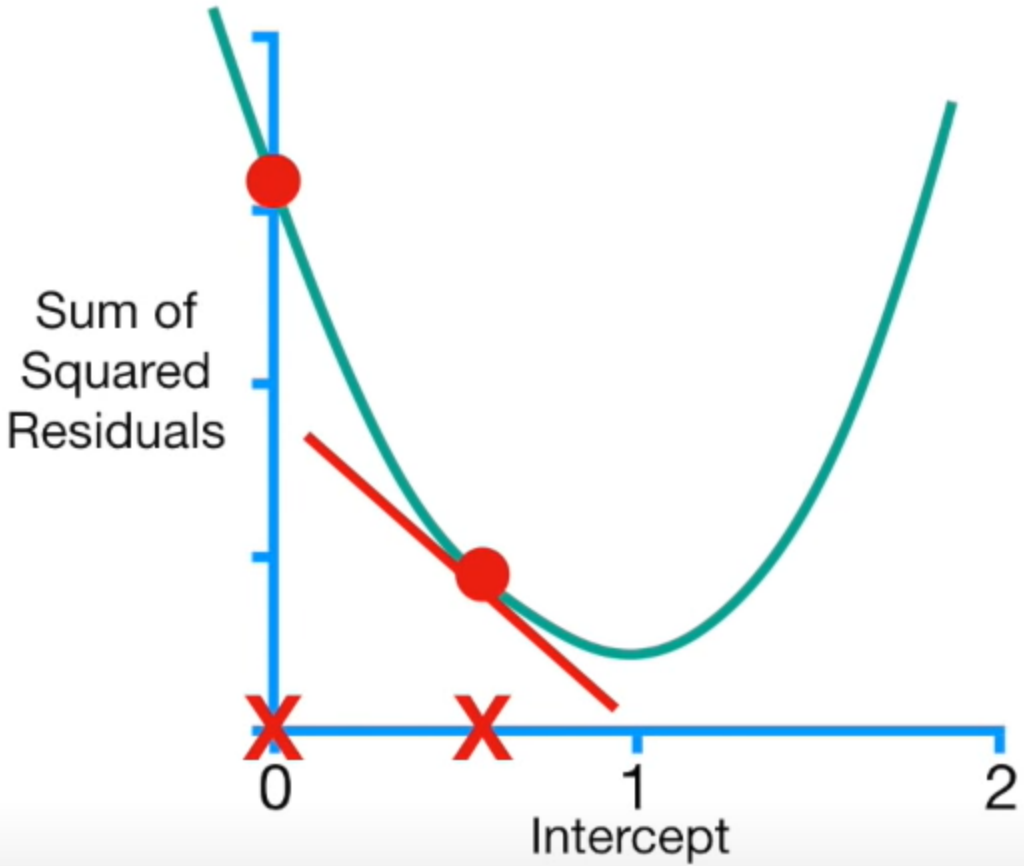
$$+ -2(\mathbf{1.9} - (0.57 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0.57 + 0.64 \times \mathbf{2.9}))$$

$$= \boxed{-2.3}$$

Step Size = -2.3 × **Learning Rate**

…by plugging in **-2.3** for the **Slope**…



$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

$$-2(\mathbf{1.4} - (0.57 + 0.64 \times \mathbf{0.5})$$

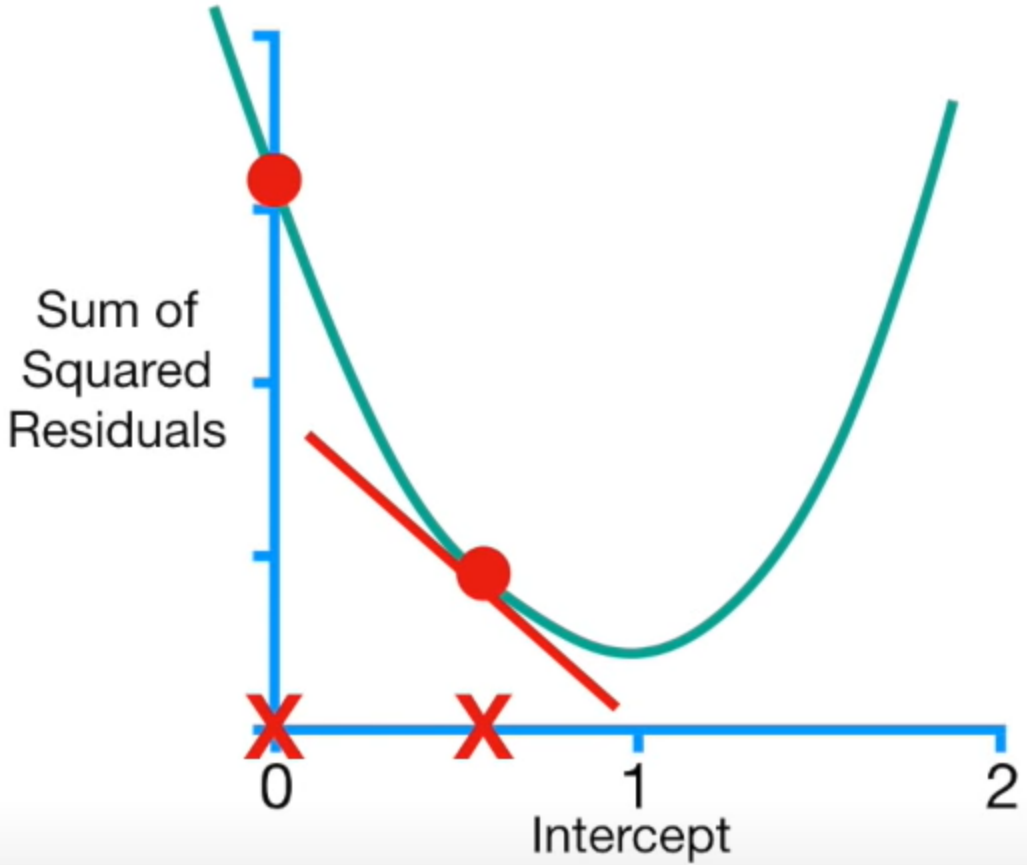
$$+ -2(\mathbf{1.9} - (0.57 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0.57 + 0.64 \times \mathbf{2.9}))$$

$$= -2.3$$

Step Size = -2.3 x 0.1

…and **0.1** for the **Learning Rate**.



$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

$$-2(\mathbf{1.4} - (0.57 + 0.64 \times \mathbf{0.5})$$

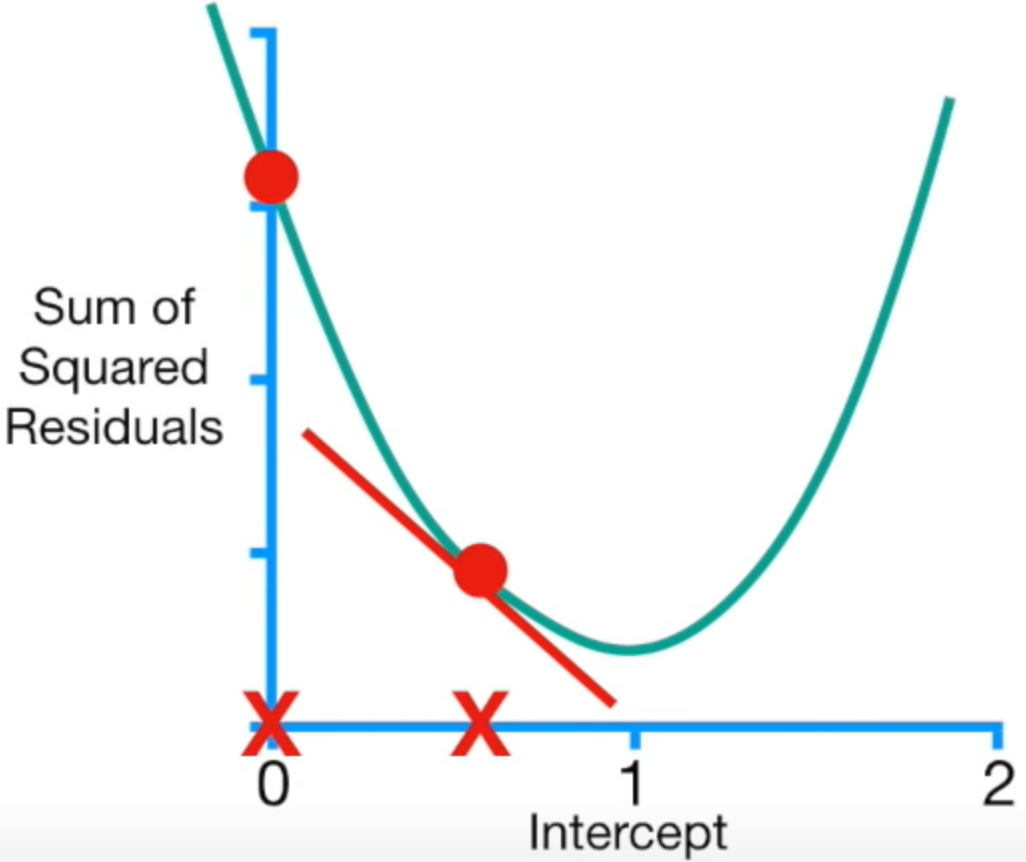
$$+ -2(\mathbf{1.9} - (0.57 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0.57 + 0.64 \times \mathbf{2.9}))$$

$$= -2.3$$

Step Size = -2.3 × 0.1 = **-0.23**

Ultimately, the **Step Size** is **-0.23**…



$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

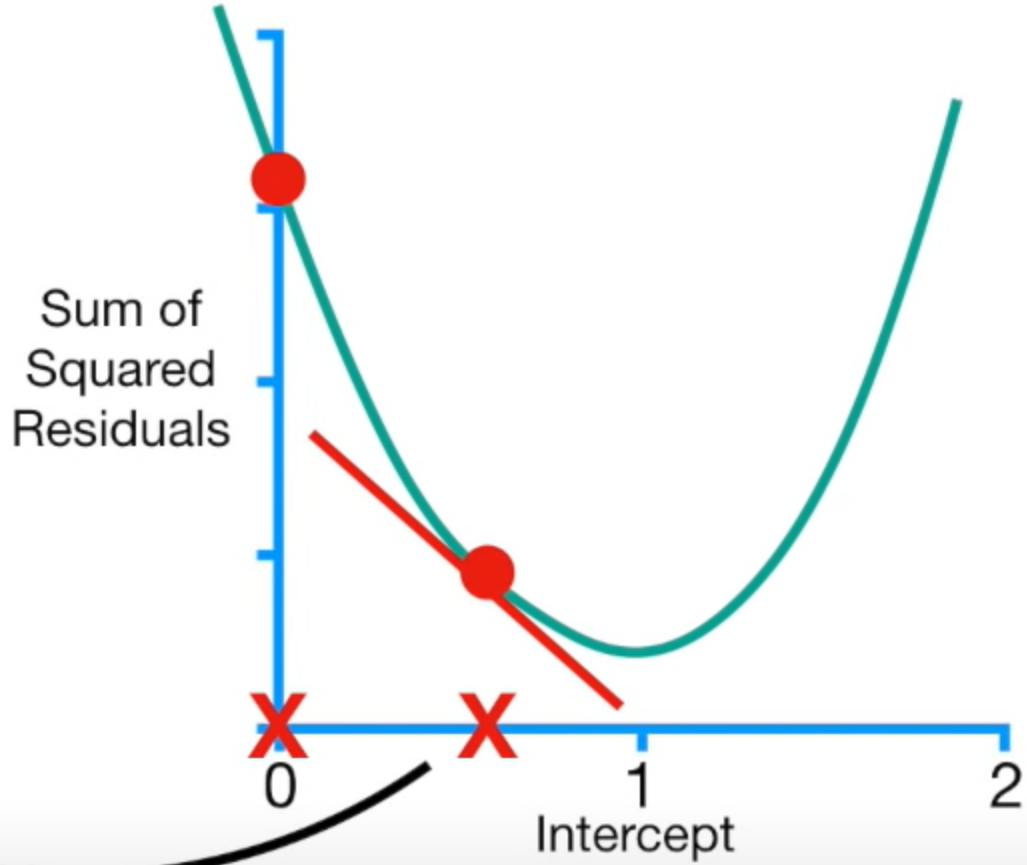
$$-2(\mathbf{1.4} - (0.57 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0.57 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0.57 + 0.64 \times \mathbf{2.9}))$$

$$= -2.3$$

Step Size = -2.3 × 0.1 = **-0.23**



New Intercept = **Old Intercept** - **Step Size**

…and the **New Intercept**…

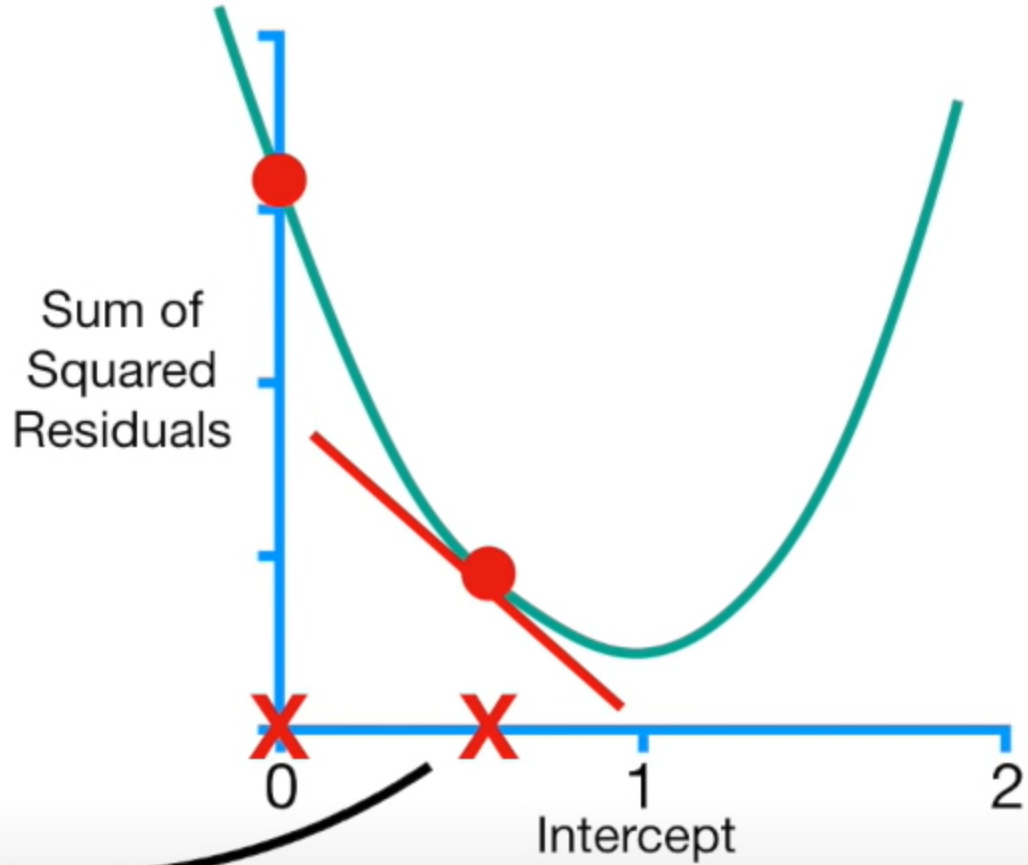
$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

$$-2(\mathbf{1.4} - (0.57 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0.57 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0.57 + 0.64 \times \mathbf{2.9}))$$

$$= -2.3$$



Step Size = -2.3 × 0.1 = **-0.23**

New Intercept = 0.57 - **Step Size**

…and the **New Intercept**…

$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

$$-2(\mathbf{1.4} - (0.57 + 0.64 \times \mathbf{0.5})$$

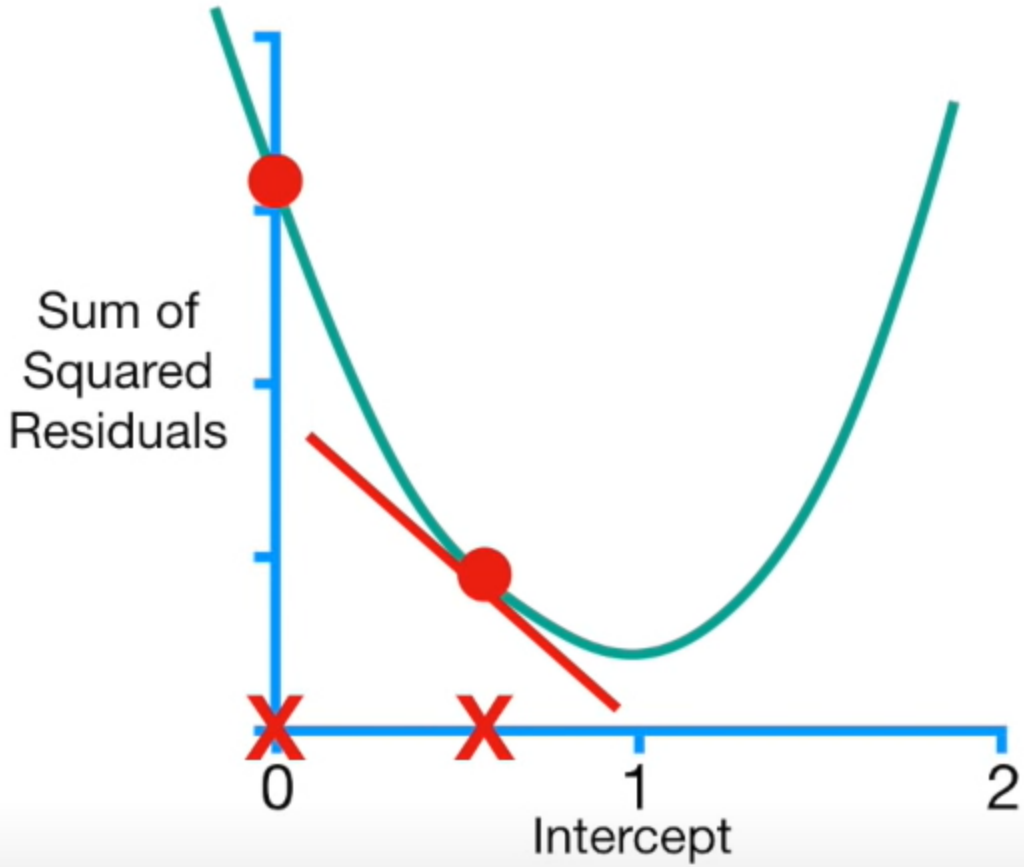
$$+ -2(\mathbf{1.9} - (0.57 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0.57 + 0.64 \times \mathbf{2.9}))$$

$$= -2.3$$

Step Size = -2.3 x 0.1 = **-0.23**

New Intercept = 0.57 - **Step Size**



…and the **New Intercept**…

$\frac{d}{d\ intercept}$ Sum of squared residuals =

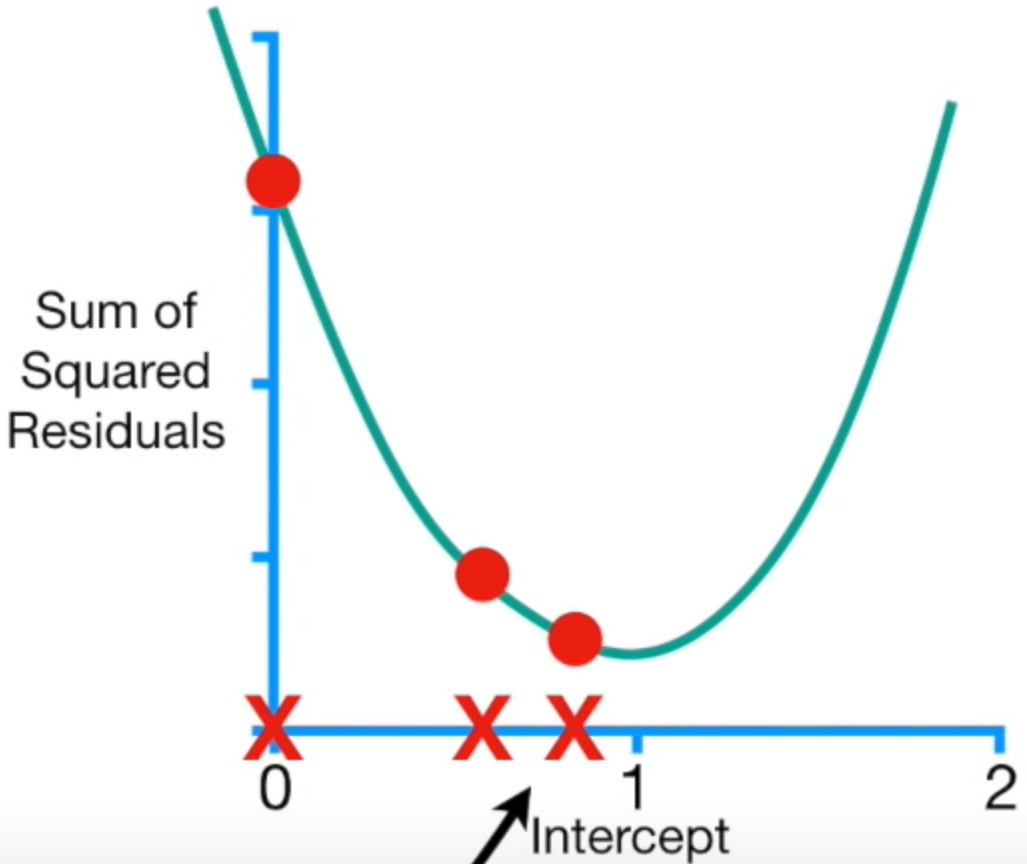
$$-2(\mathbf{1.4} - (0.57 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0.57 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0.57 + 0.64 \times \mathbf{2.9}))$$

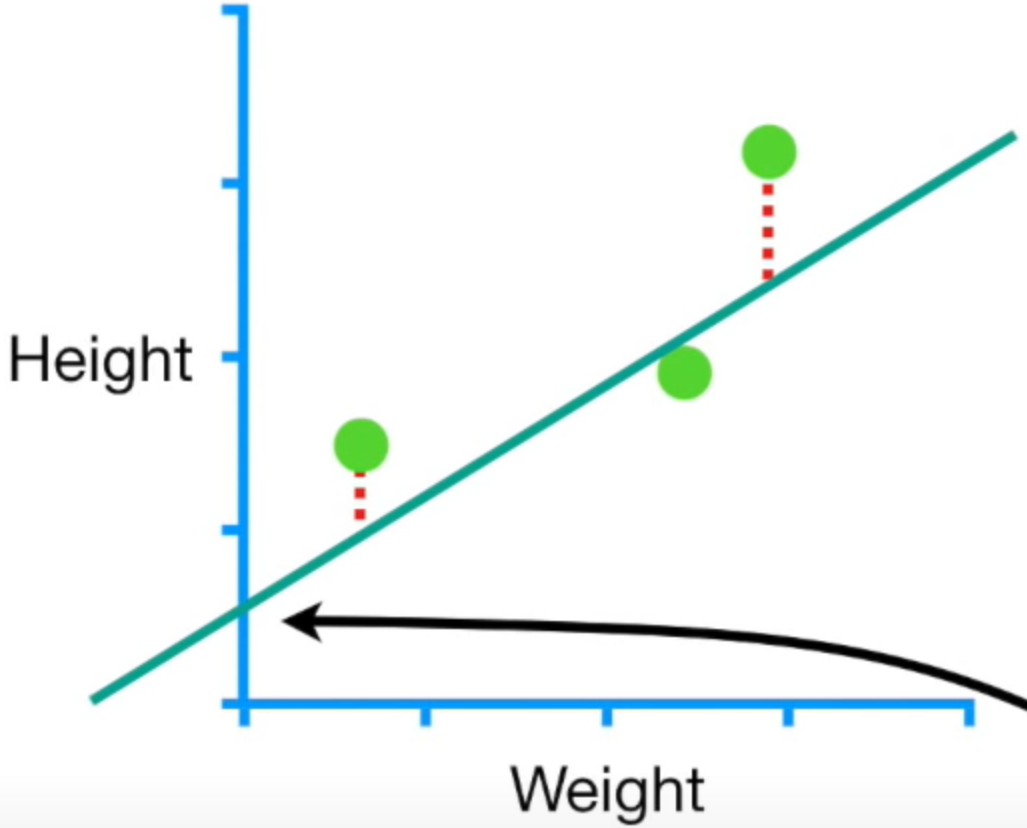
$$= -2.3$$

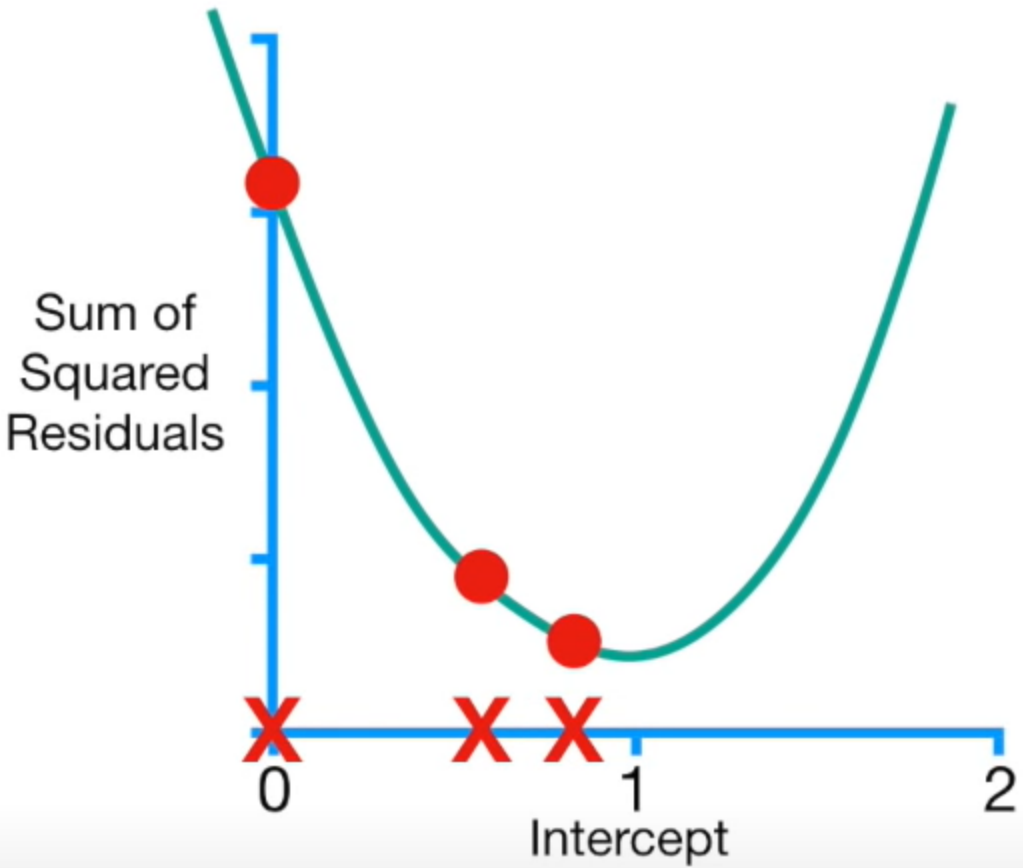
Step Size $= -2.3 \times 0.1 = \mathbf{-0.23}$



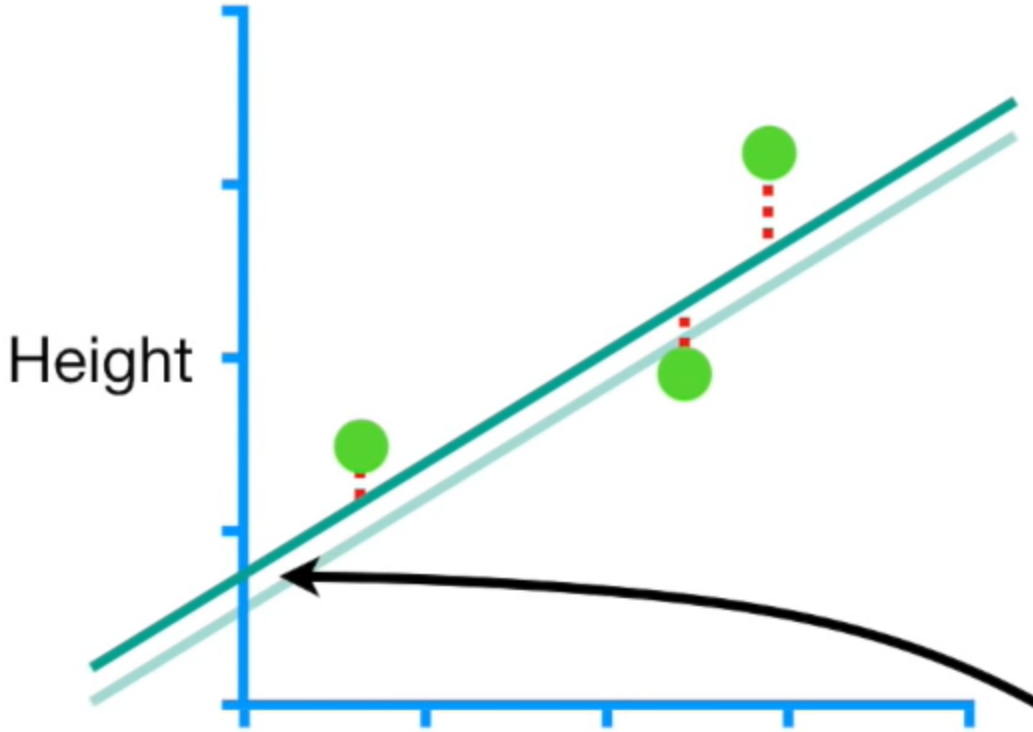
New Intercept $= 0.57 - (-0.23) = \mathbf{0.8}$

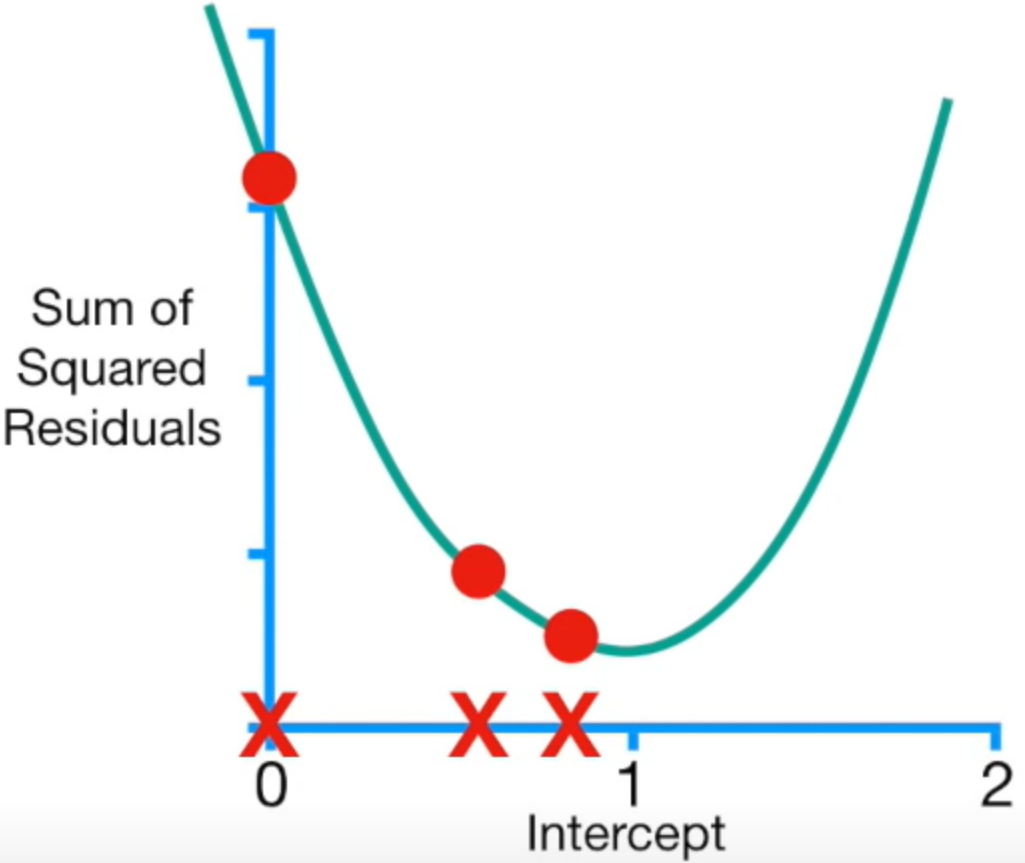
…and the **New Intercept** = **0.8**



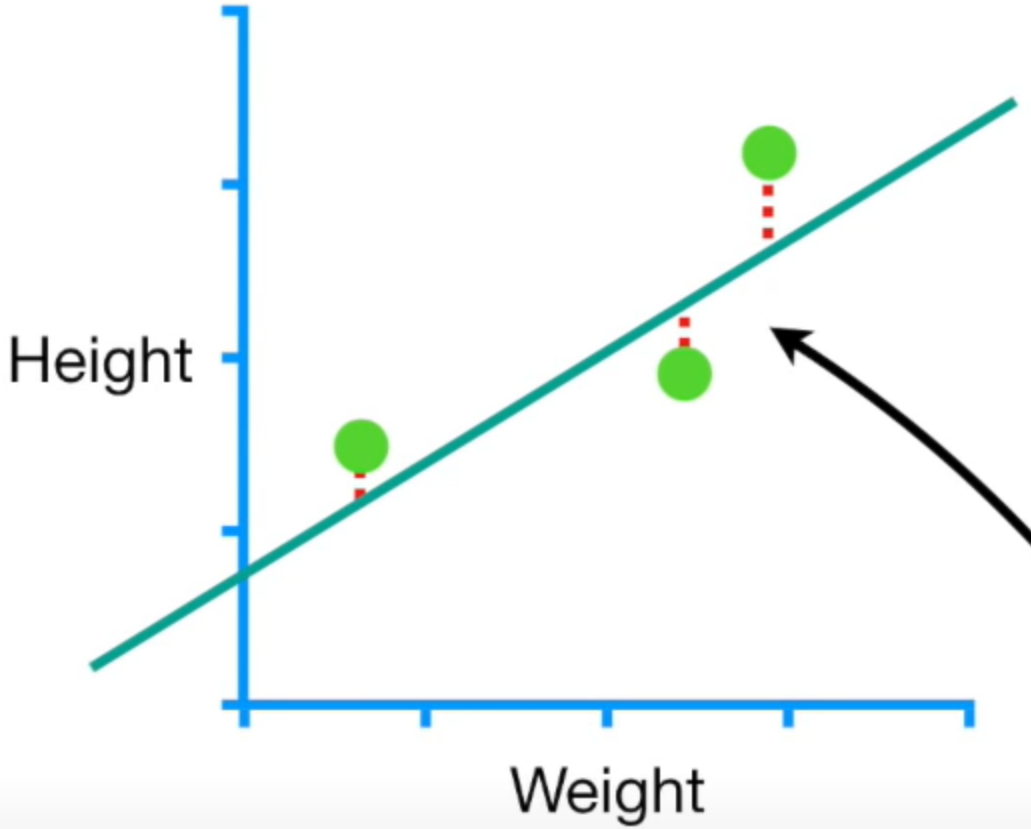


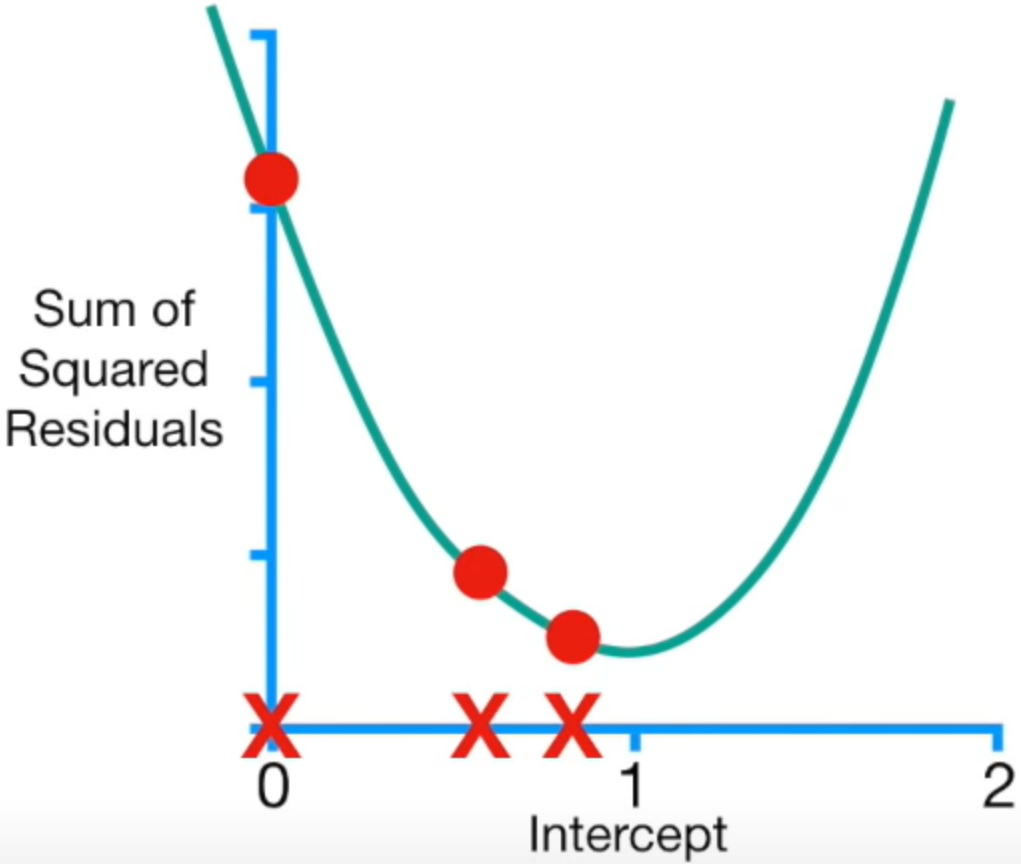
Now we can compare the residuals when the **Intercept = 0.57**…



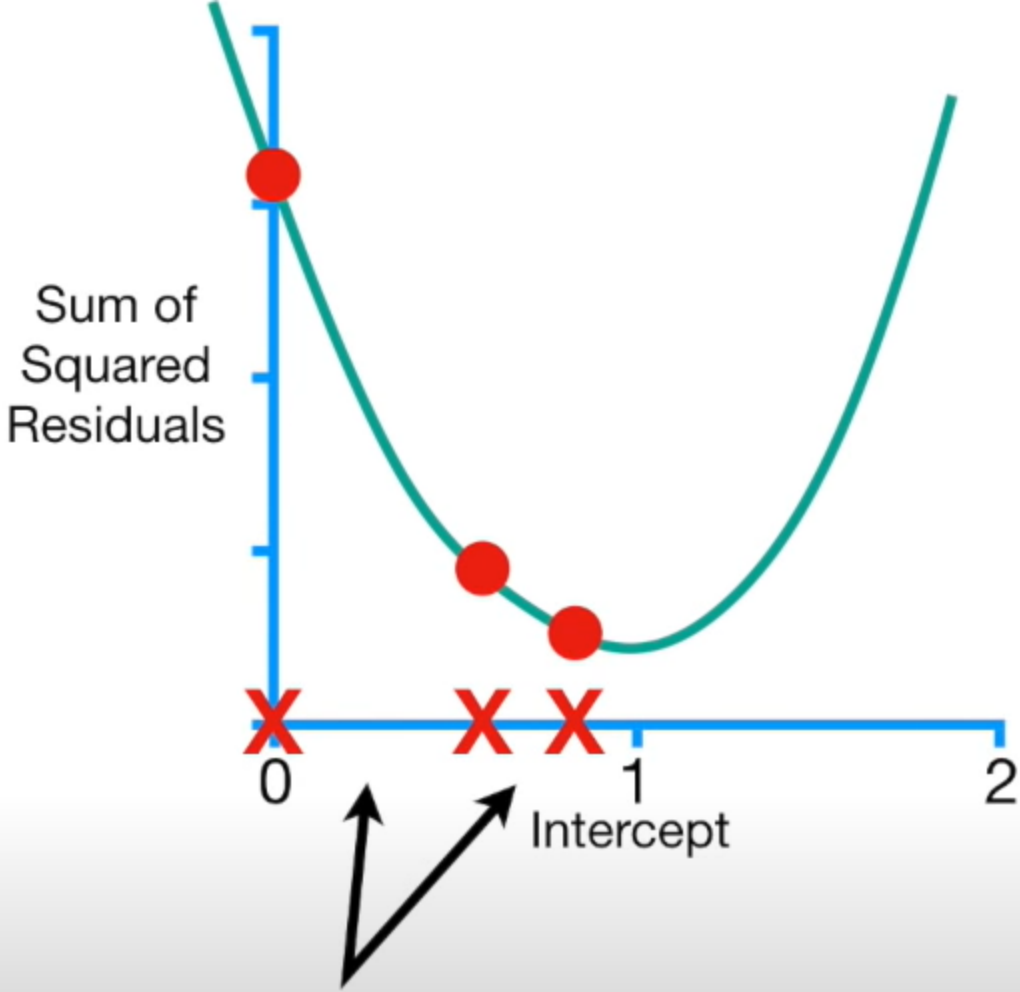


…to when the
Intercept = 0.8





Overall, the Sum of the Squared Residuals is getting smaller.



Notice that the first step was relatively large compared to the second step.

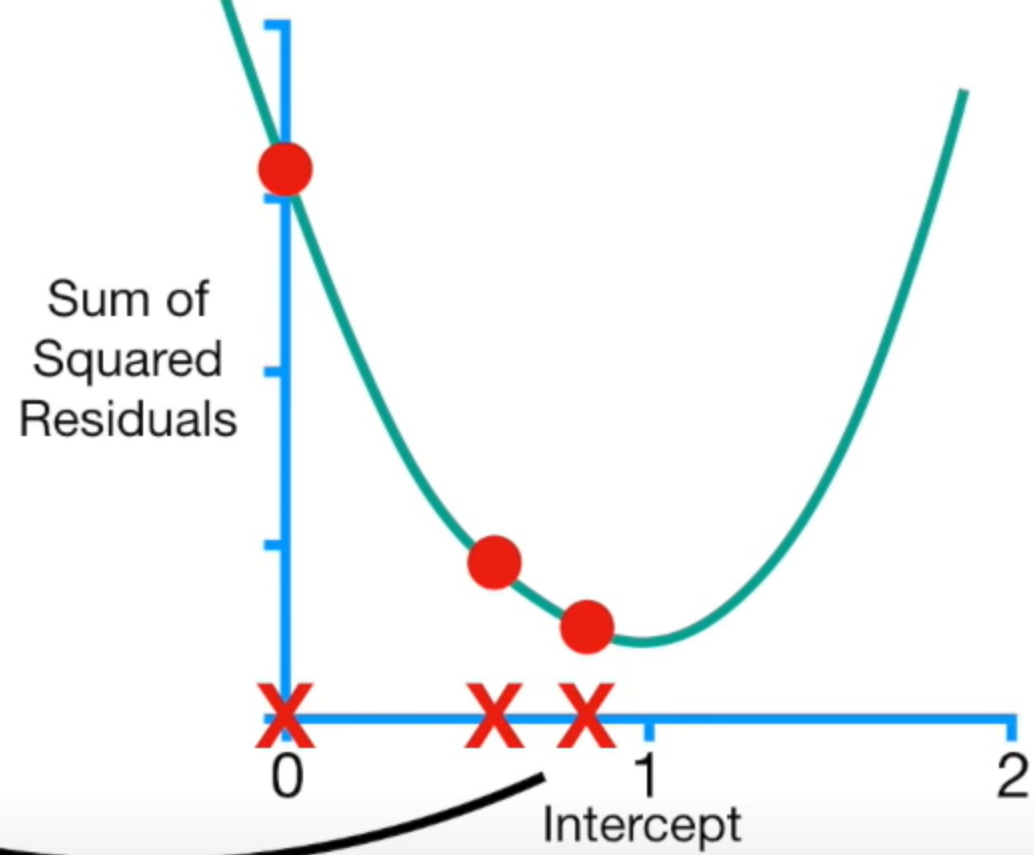
$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

-2(**1.4** - (intercept + 0.64 × **0.5**)

+ -2(**1.9** - (intercept + 0.64 × **2.3**))

+ -2(**3.2** - (intercept + 0.64 × **2.9**))

Now let's calculate the derivative at the **New Intercept** (**0.8**)…



$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

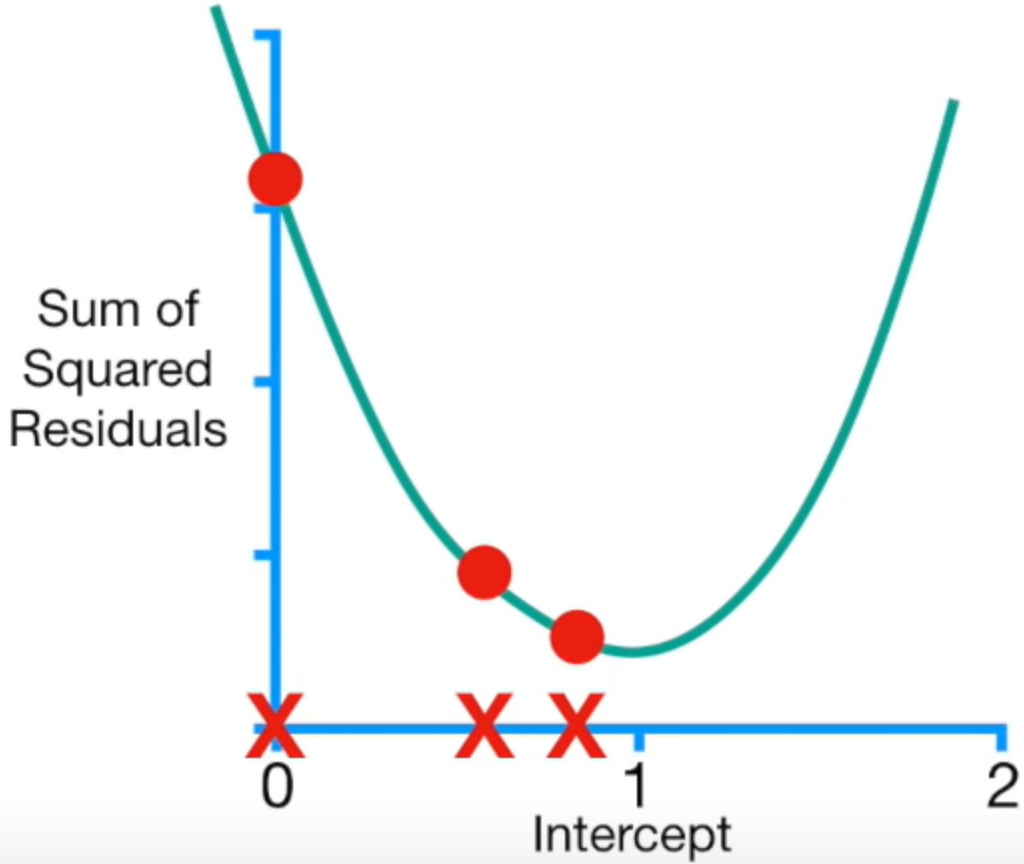
$$-2(\mathbf{1.4} - (0.8 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0.8 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0.8 + 0.64 \times \mathbf{2.9}))$$

$$= \boxed{\mathbf{-0.9}}$$

…and we get **-0.9**.



$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

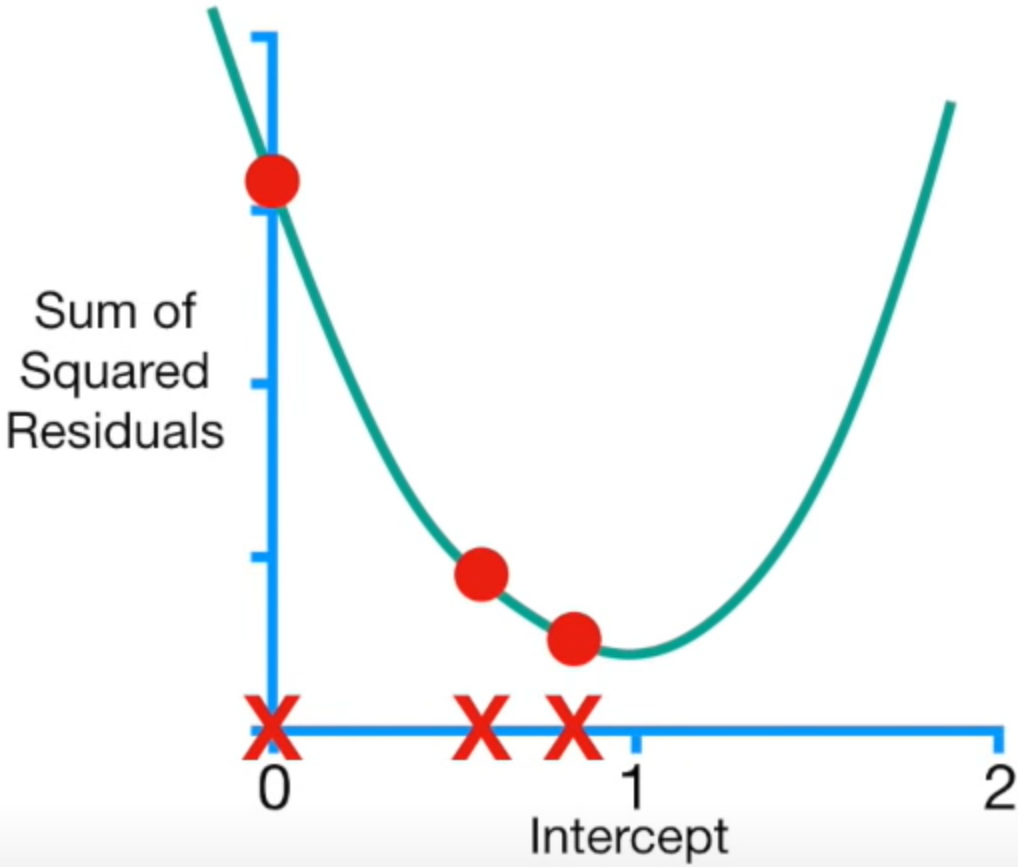
$$-2(\mathbf{1.4} - (0.8 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0.8 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0.8 + 0.64 \times \mathbf{2.9}))$$

$$= \mathbf{-0.9}$$

Step Size = Slope × Learning Rate



The **Step Size**…

$$\frac{d}{d\,intercept}$$ Sum of squared residuals =

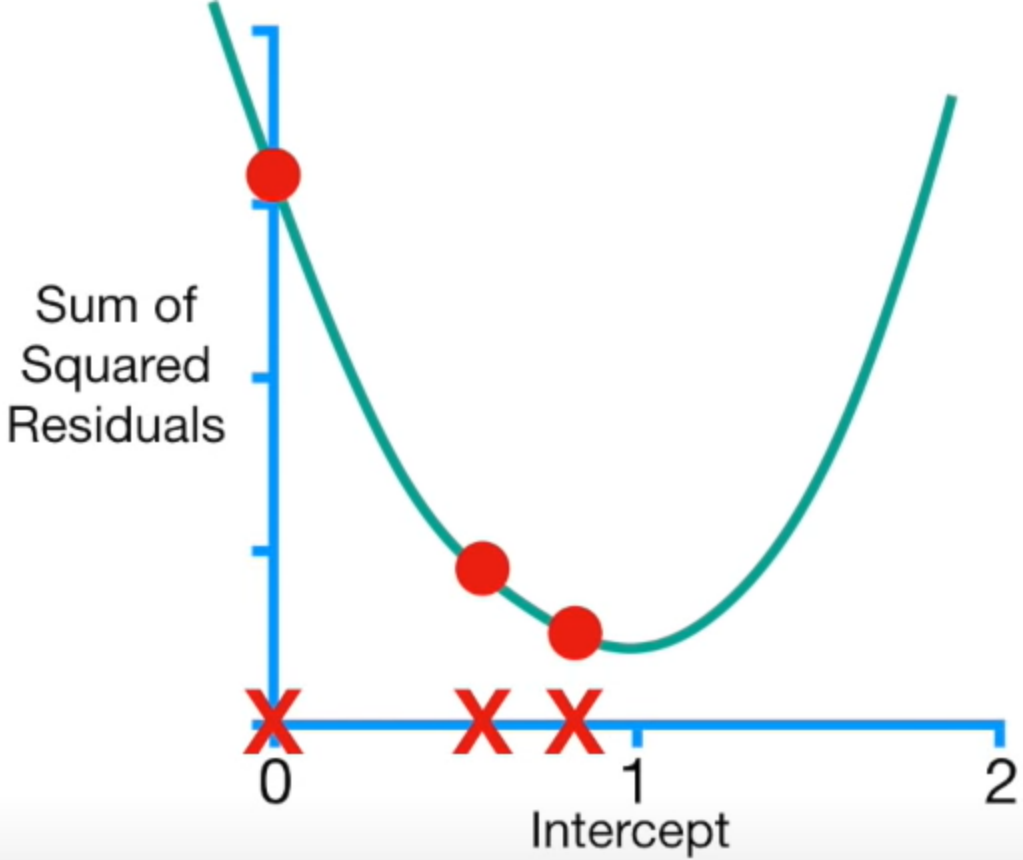
$$-2(\mathbf{1.4} - (0.8 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0.8 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0.8 + 0.64 \times \mathbf{2.9}))$$

$$= \mathbf{-0.9}$$

Step Size = -0.9 × 0.1 = **-0.09**



The **Step Size** = **-0.09**…

$$\frac{d}{d\,intercept}$$ Sum of squared residuals =

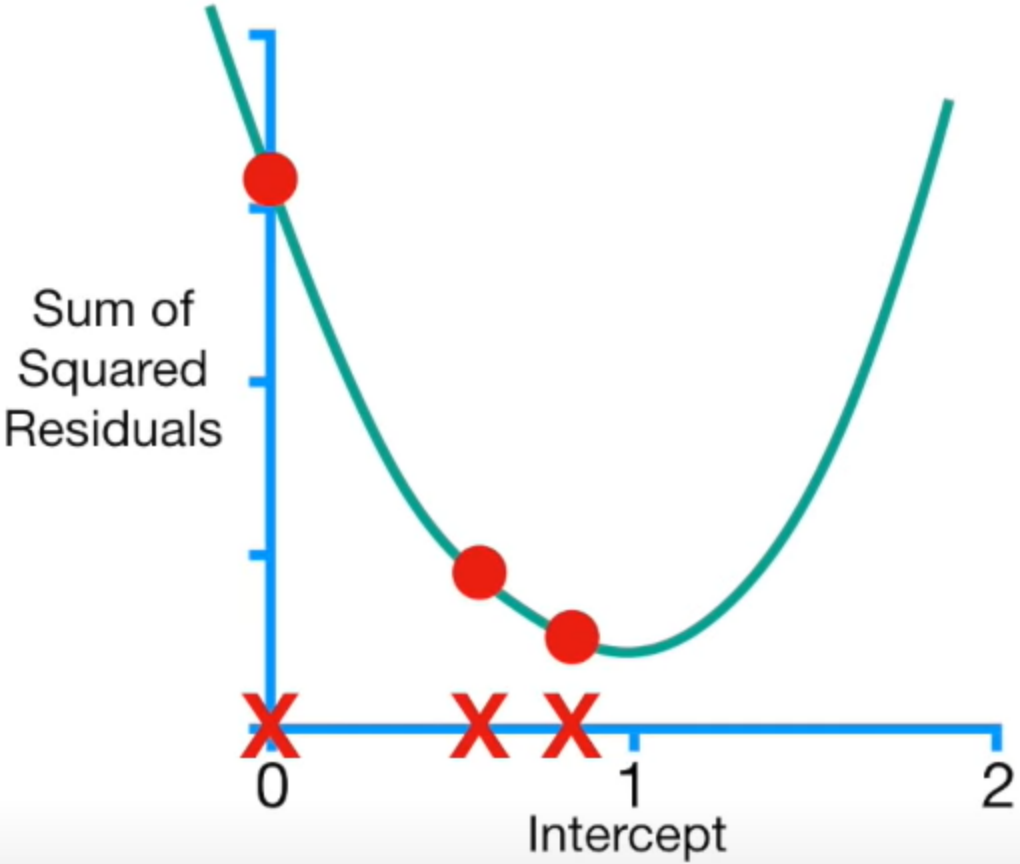
$$-2(\mathbf{1.4} - (0.8 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0.8 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0.8 + 0.64 \times \mathbf{2.9}))$$

$$= \mathbf{-0.9}$$

Step Size $= -0.9 \times 0.1 = \mathbf{-0.09}$



New Intercept = **Old Intercept** - **Step Size**

…and the **New Intercept**…

$$\frac{d}{d\,intercept}\text{Sum of squared residuals} =$$

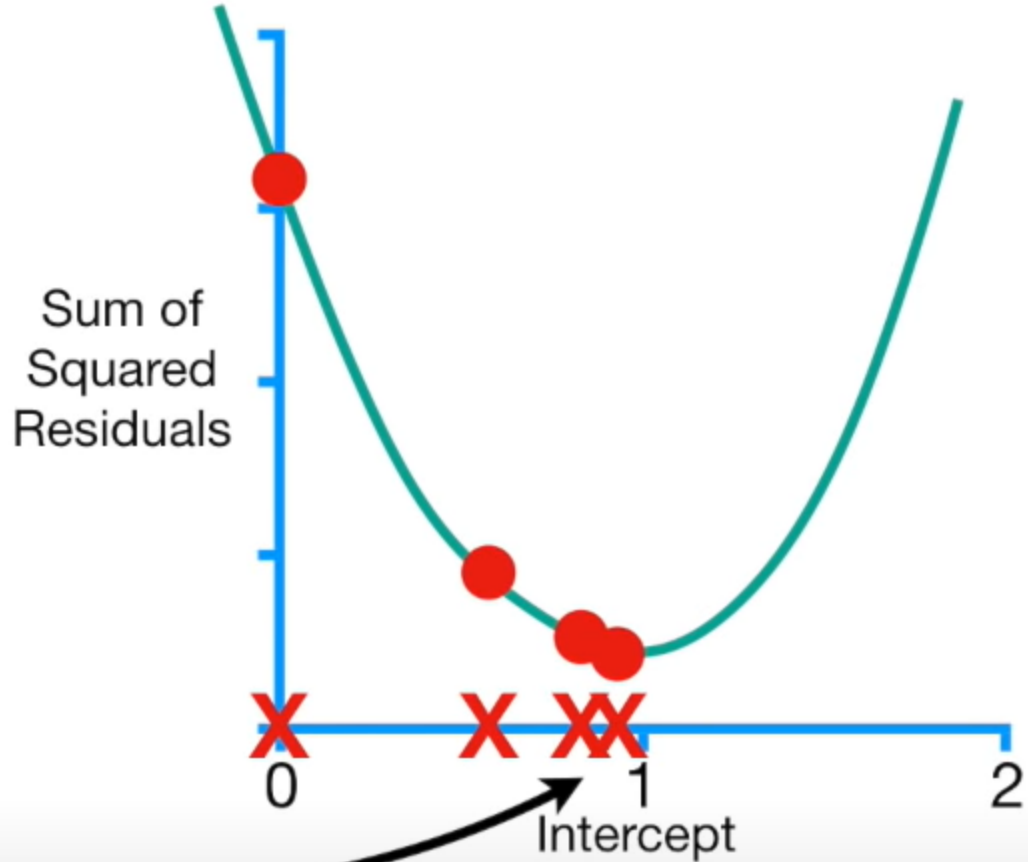
$$-2(\mathbf{1.4} - (0.8 + 0.64 \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (0.8 + 0.64 \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (0.8 + 0.64 \times \mathbf{2.9}))$$

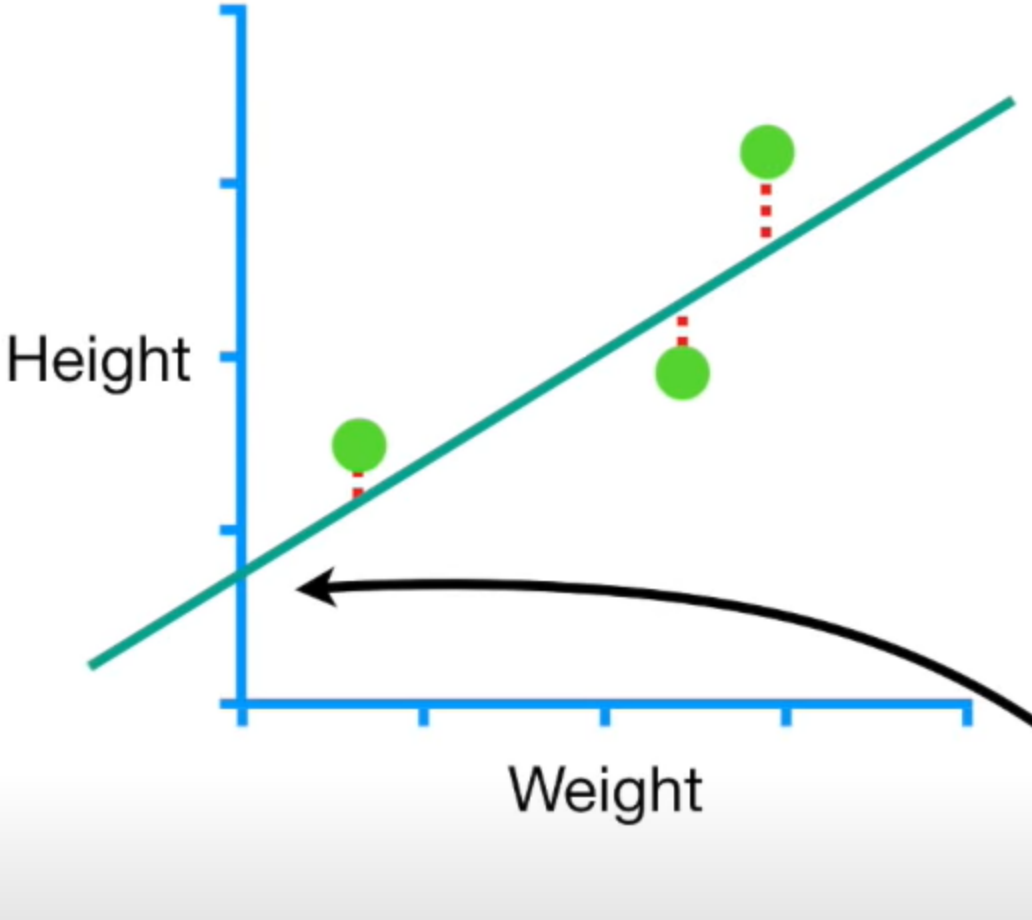
$$= \mathbf{-0.9}$$

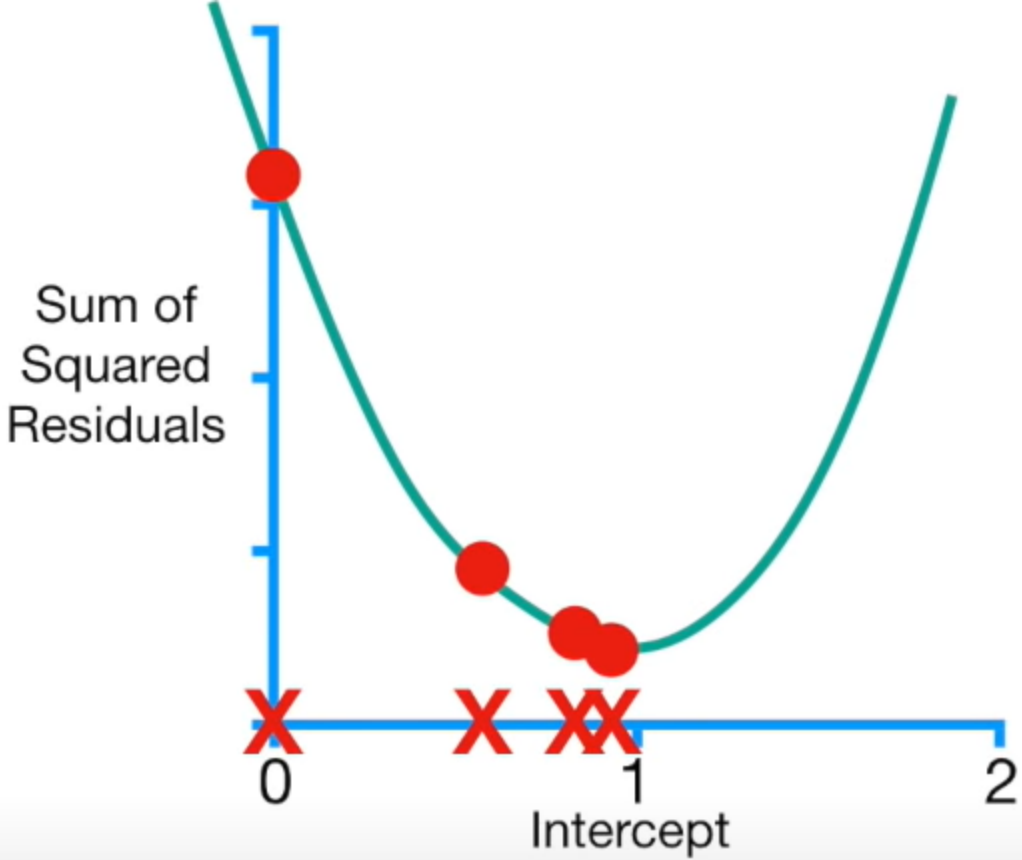
Step Size $= -0.9 \times 0.1 = \mathbf{-0.09}$



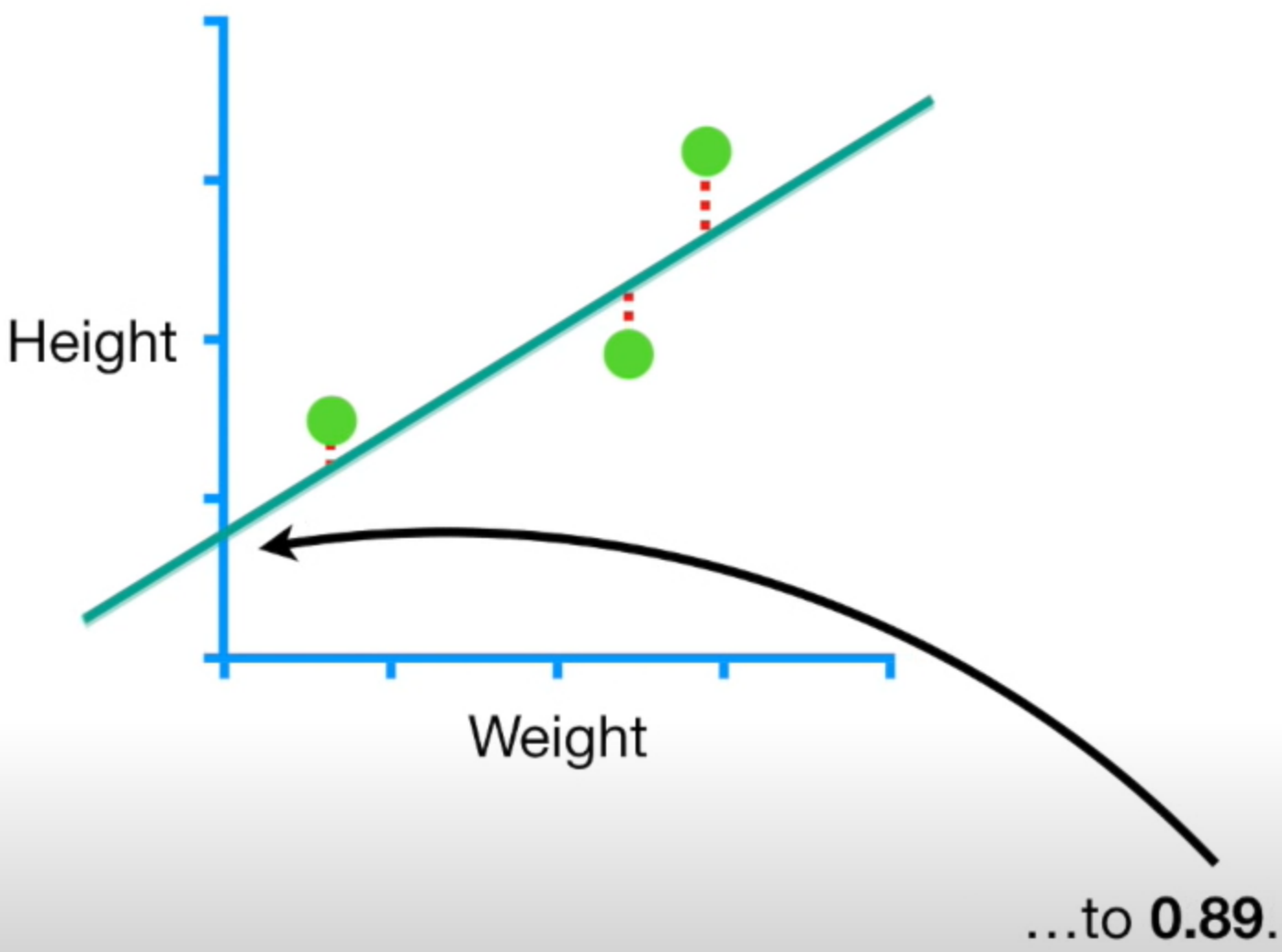
New Intercept $= 0.8 - (-0.09) =$ **0.89**

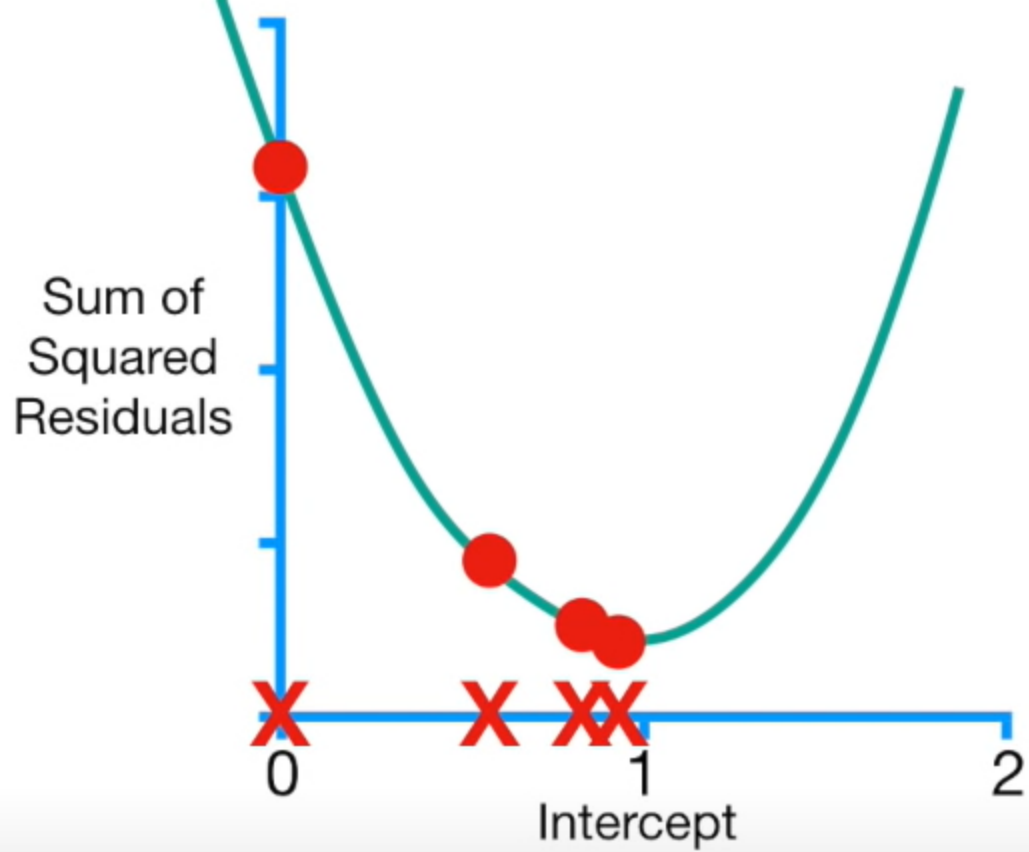
…and the **New Intercept** = **0.89**

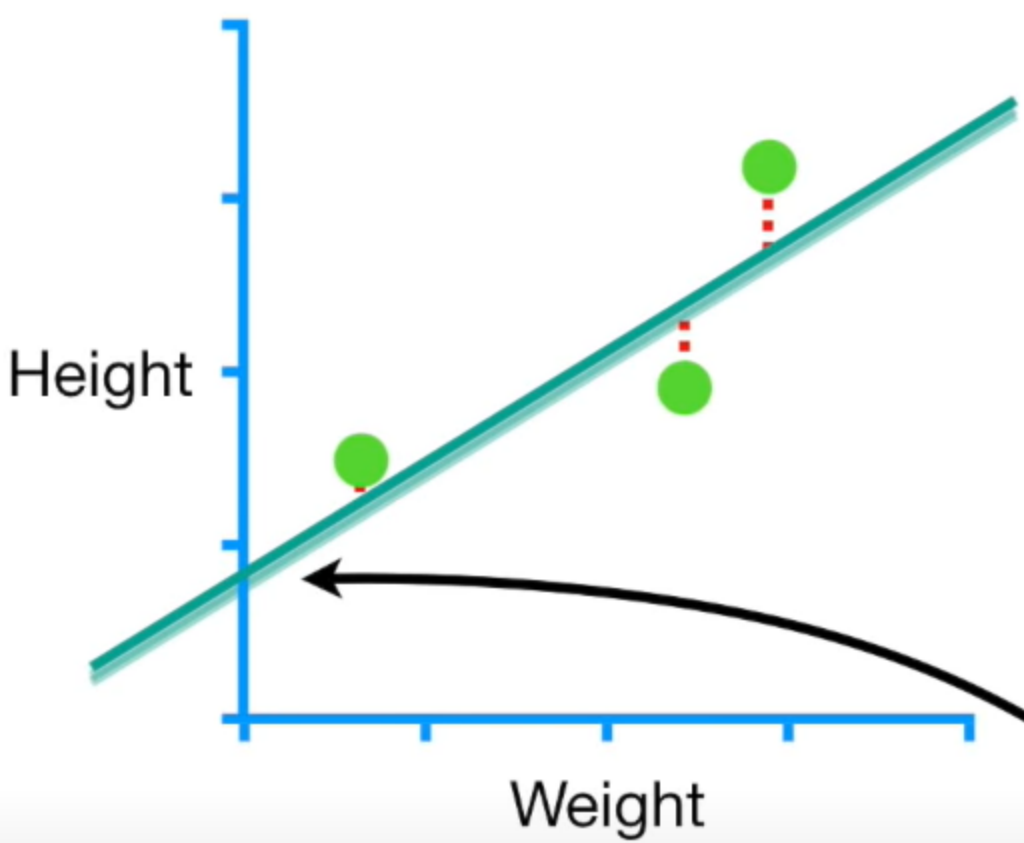


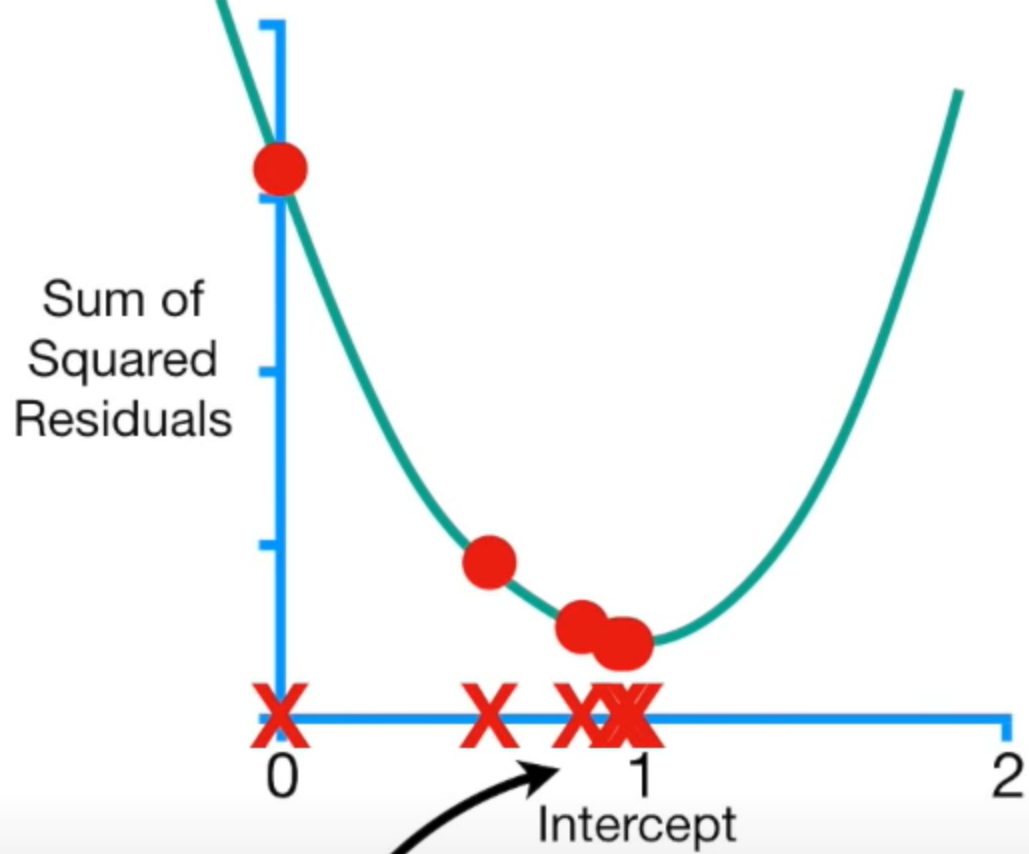


Now we increase the
Intercept from **0.8**…

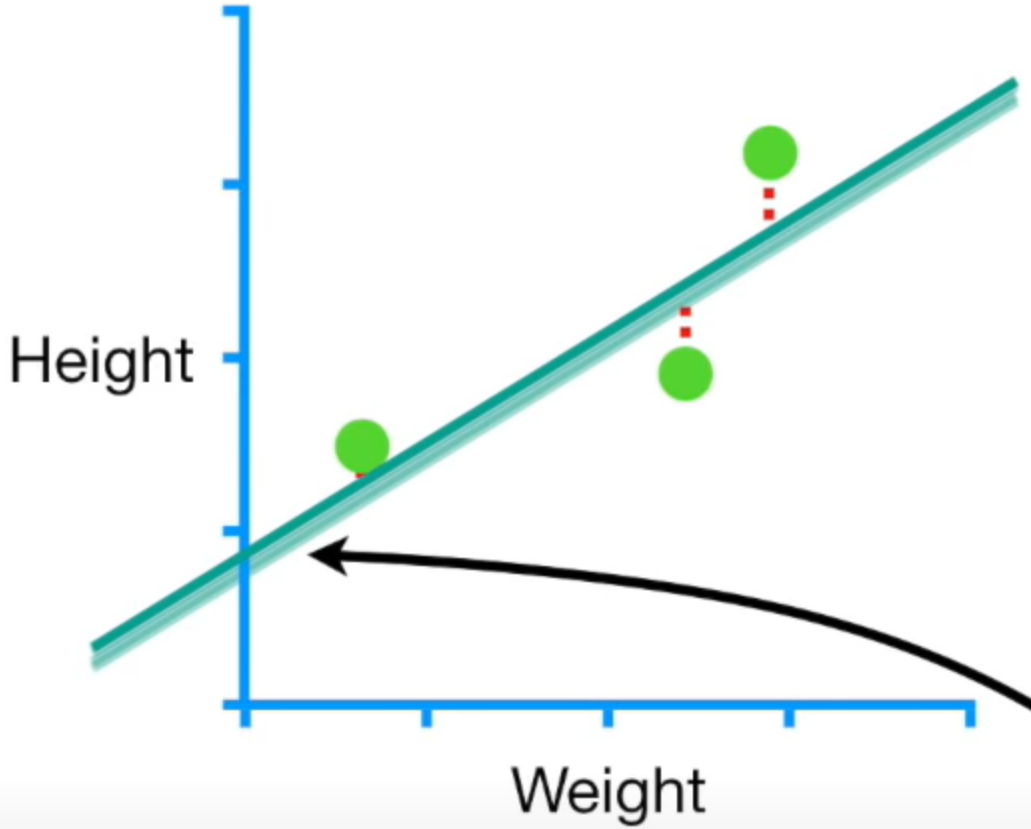

Height
Weight
…to **0.89**.

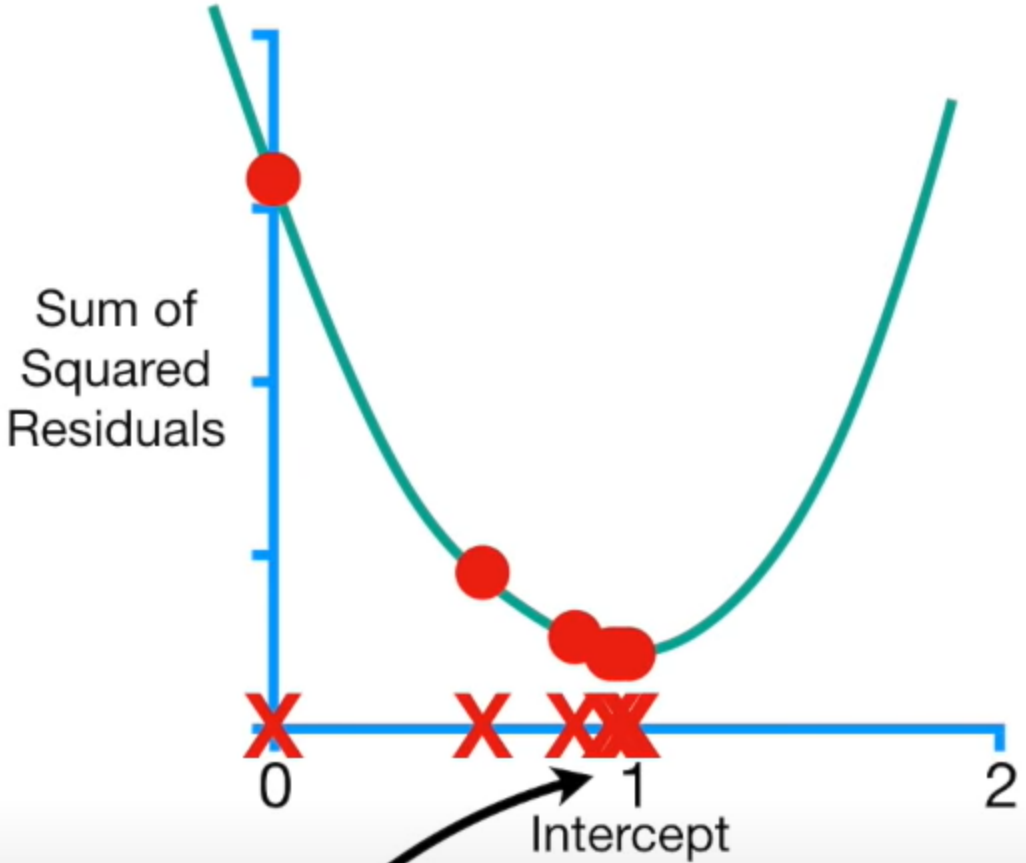

Sum of
Squared
Residuals
0
1
2
Intercept



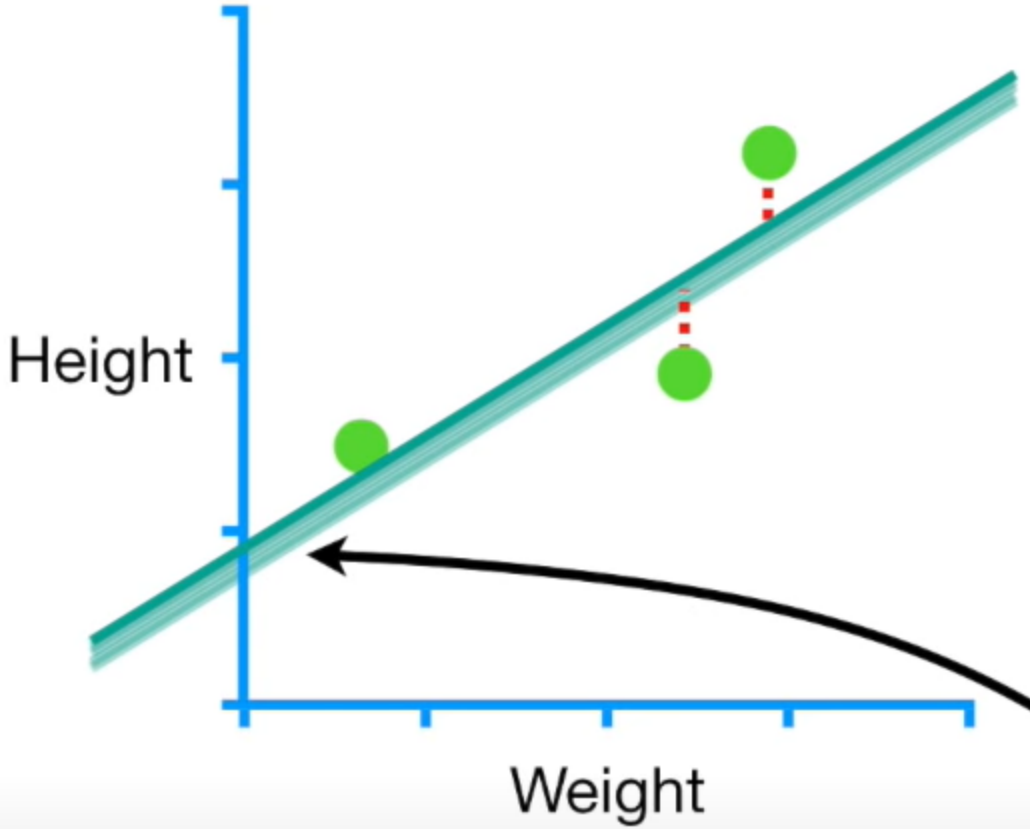


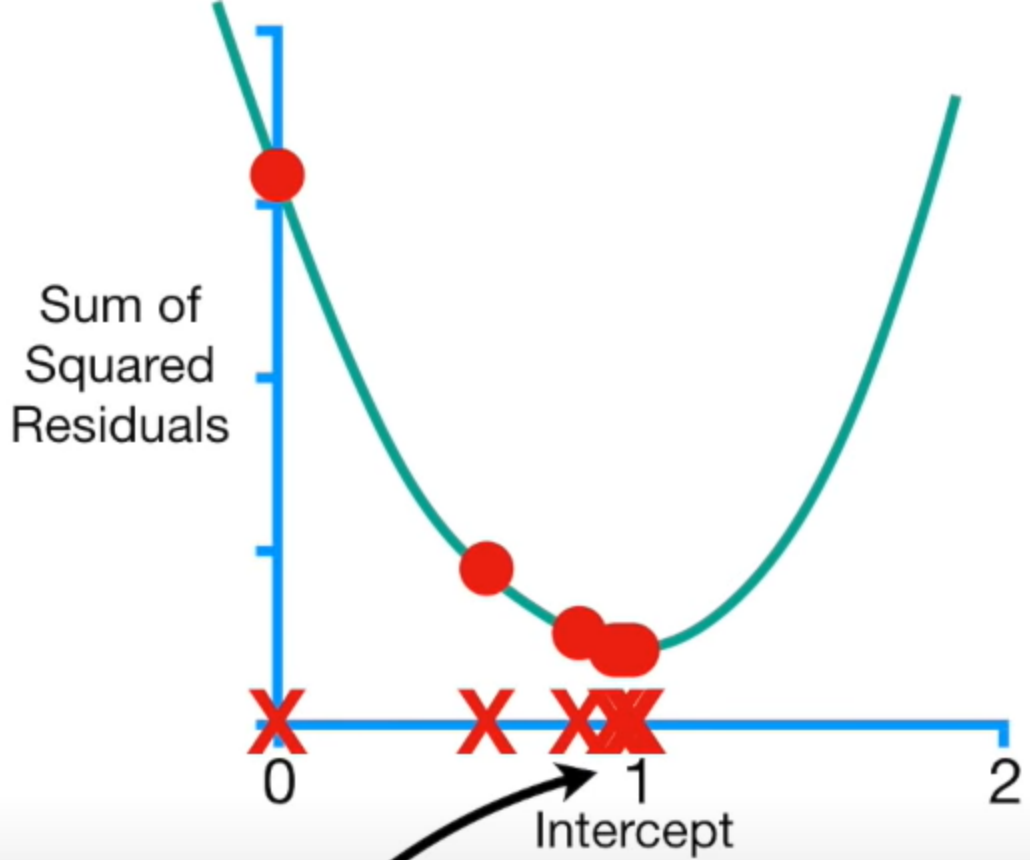
Then we take another step and
the **New Intercept** = **0.92**...



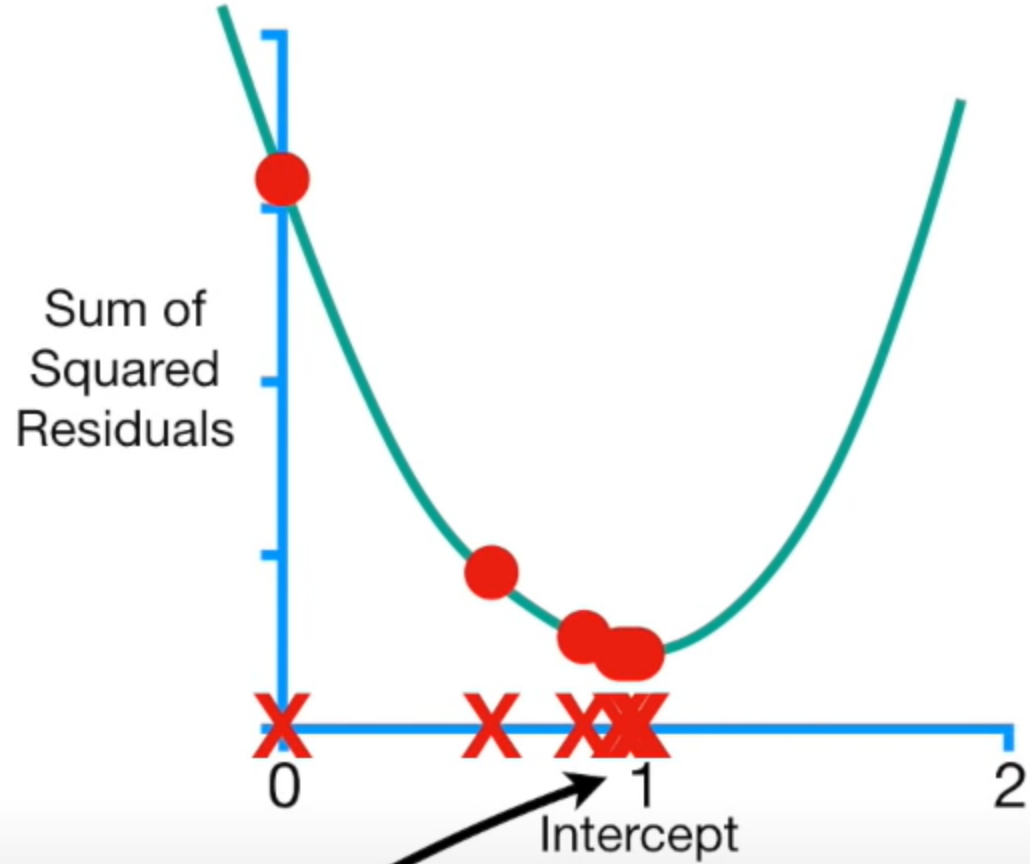


…and then we take another step and the **New Intercept = 0.94**…



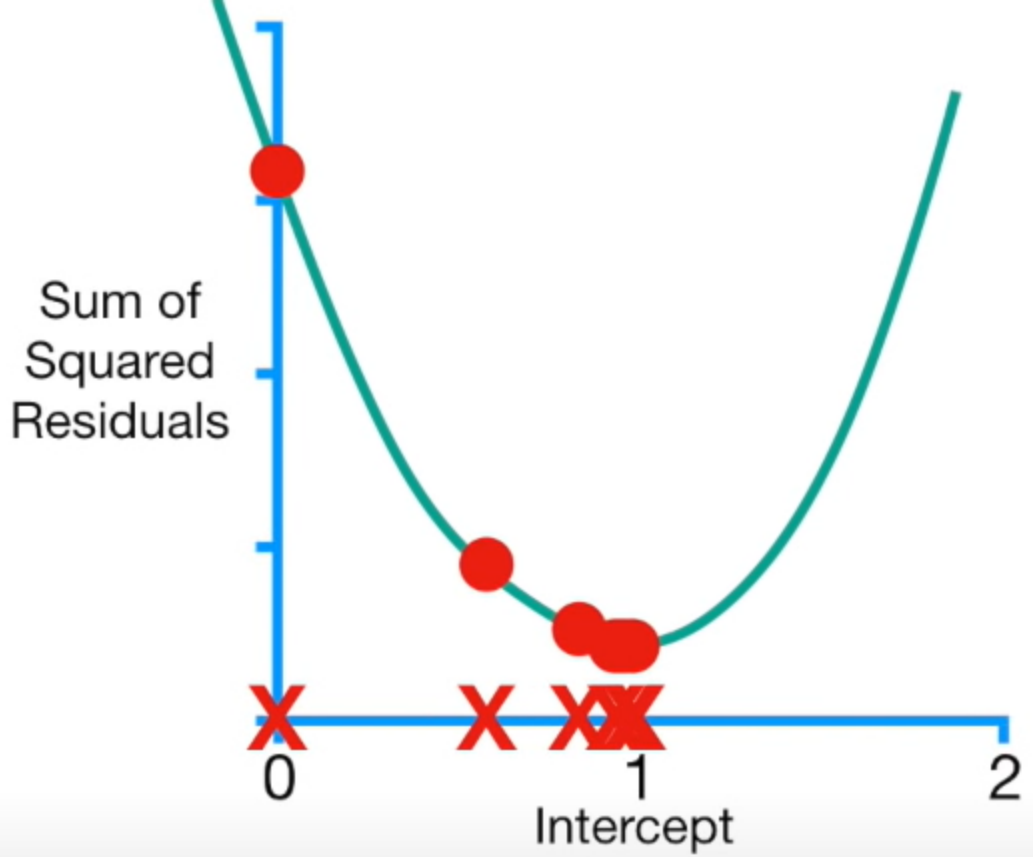


…and then we take another step and the **New Intercept = 0.95**.



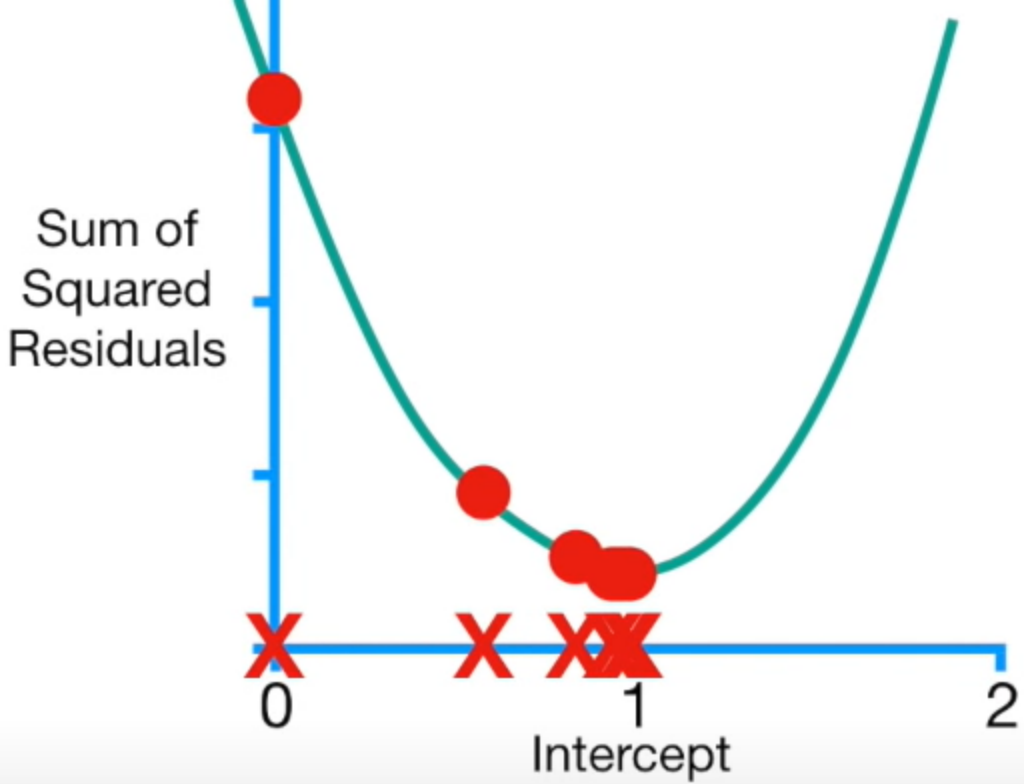
Notice how each step gets smaller and smaller the closer we get to the bottom of the curve.

After 6 steps, the **Gradient Descent** estimate for the **Intercept** is **0.95**.



After 6 steps, the **Gradient Descent** estimate for the **Intercept** is **0.95**.

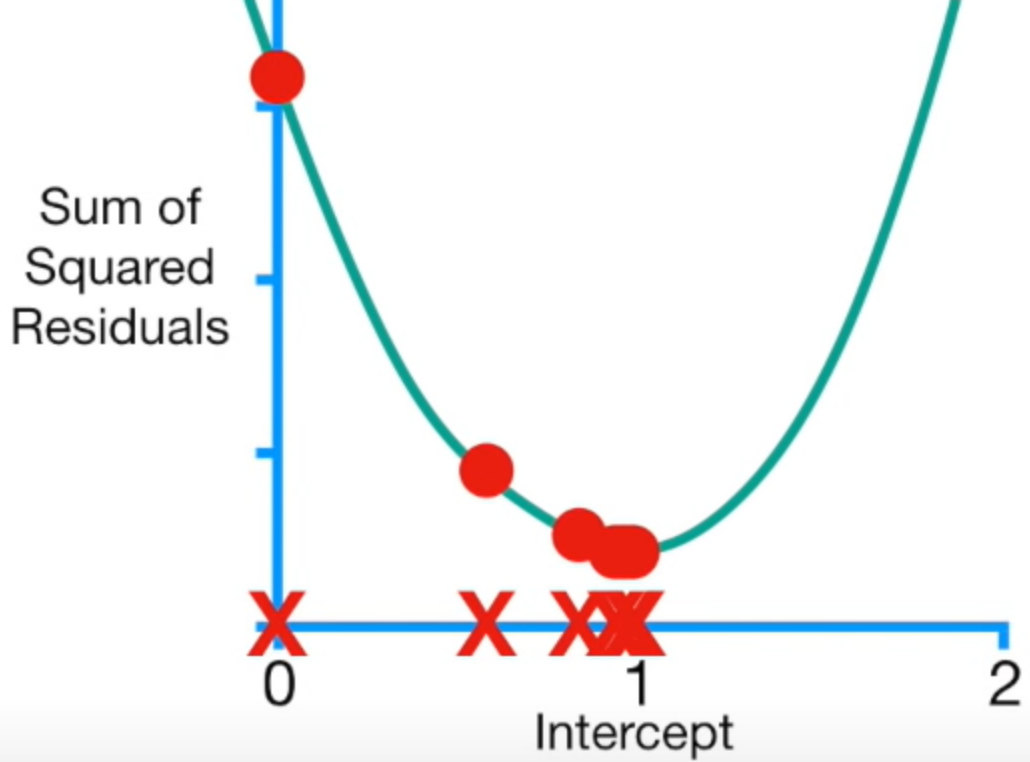
NOTE: The **Least Squares** estimate for the intercept is also **0.95**.



After 6 steps, the **Gradient Descent** estimate for the **Intercept** is **0.95**.

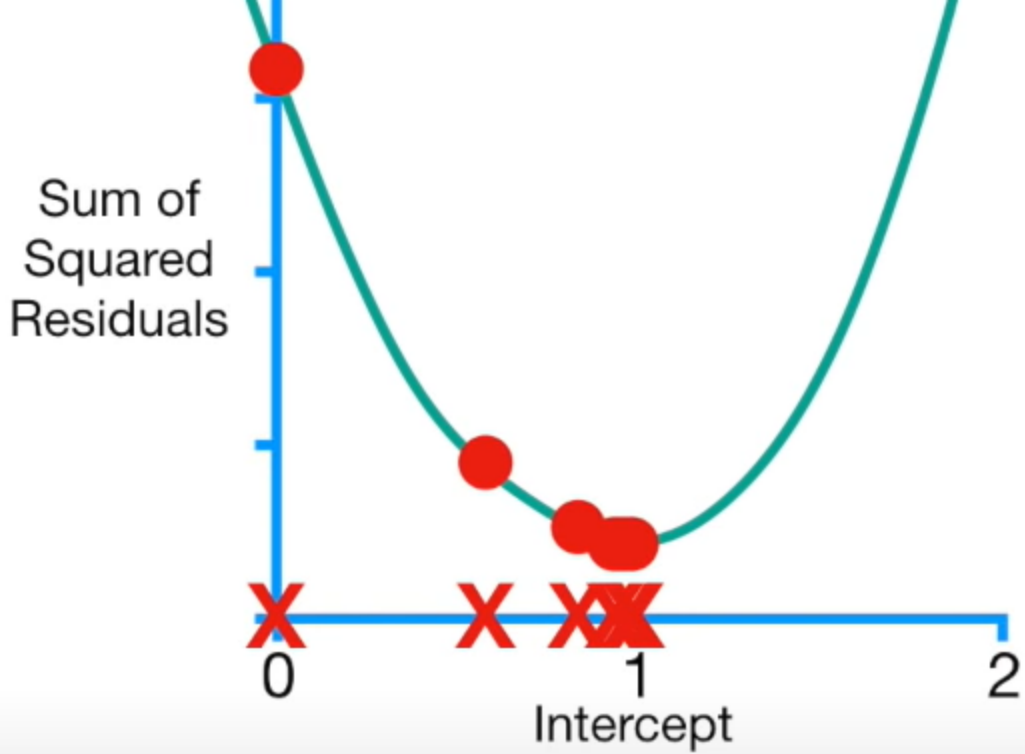
NOTE: The **Least Squares** estimate for the intercept is also **0.95**.

So we know that **Gradient Descent** has done its job, but without comparing its solution to a gold standard, how does **Gradient Descent** know to stop taking steps?



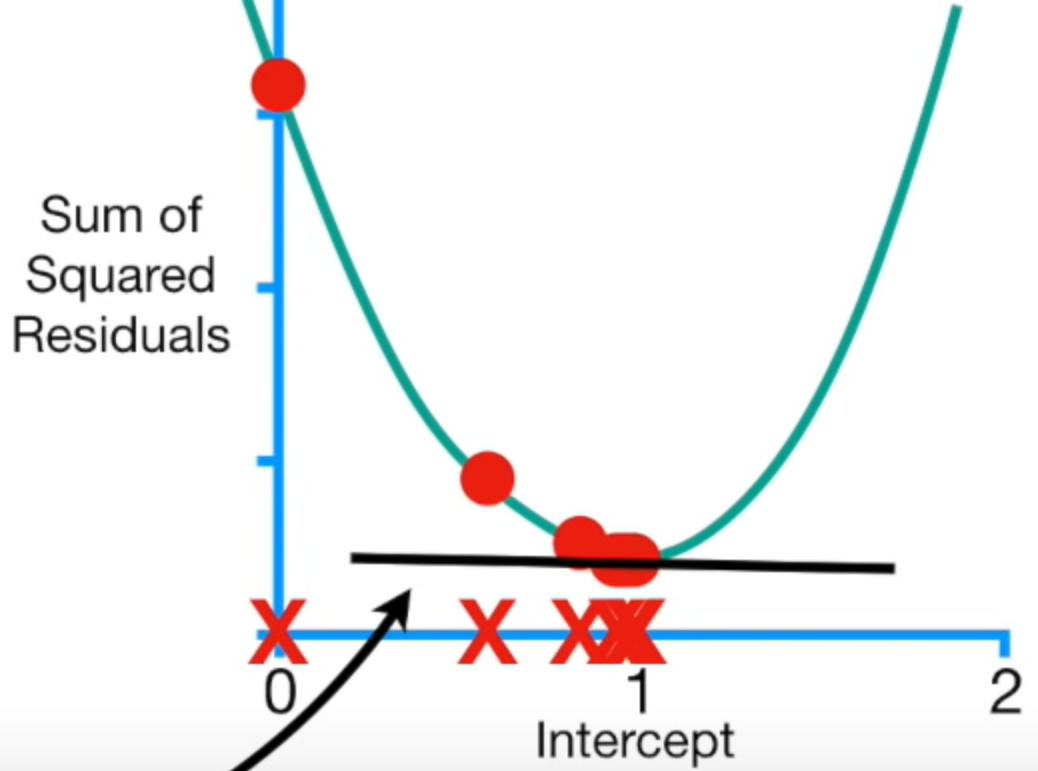
Gradient Descent stops when the **Step Size** is **Very Close To 0**.

Step Size = **Slope** × **Learning Rate**



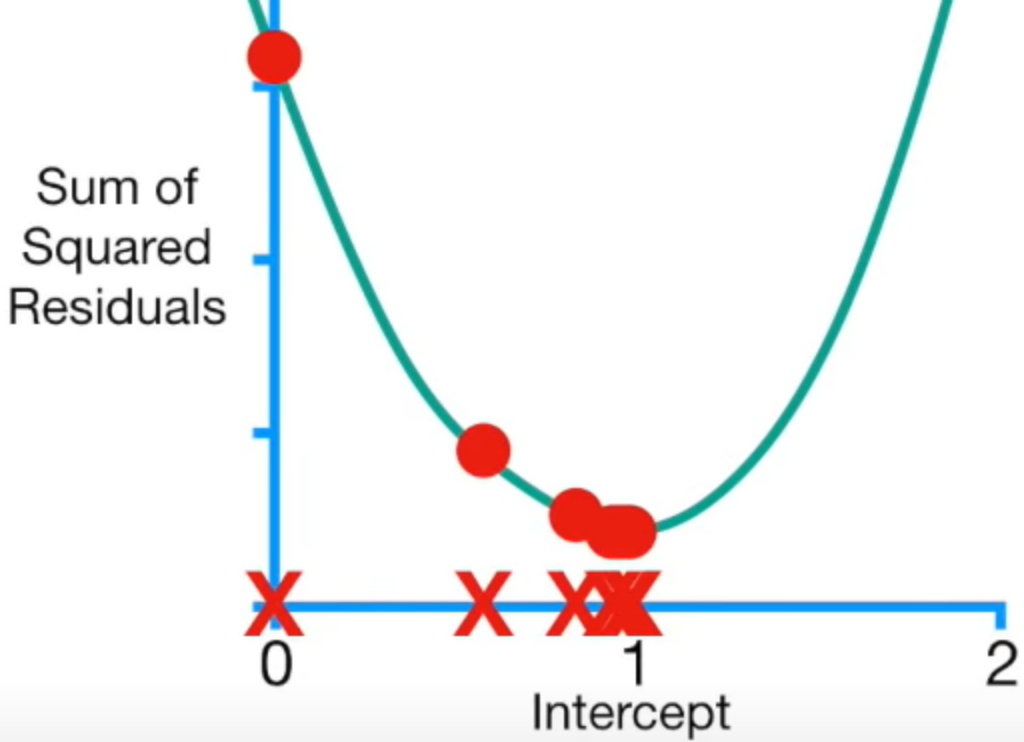
The **Step Size** will be **Very Close to 0** when the **Slope** is very close to **0**.

Step Size = **Slope** × **Learning Rate**



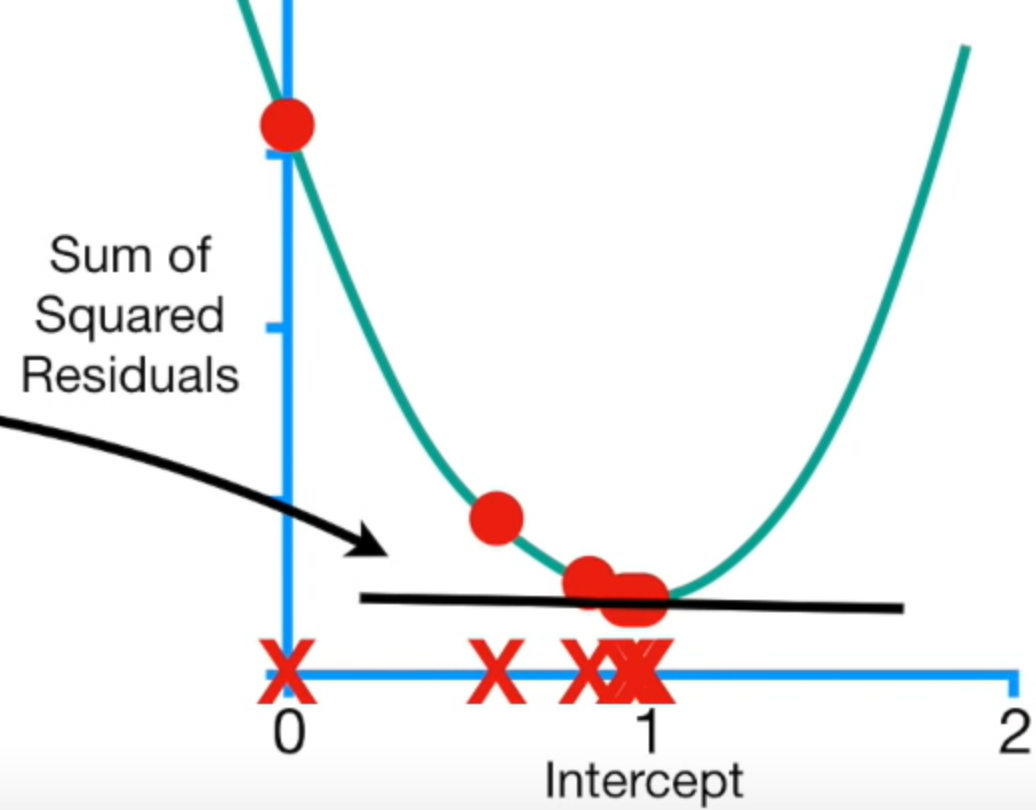
In practice, the
Minimum Step Size = **0.001**
or smaller.

Step Size = **Slope** × **Learning Rate**



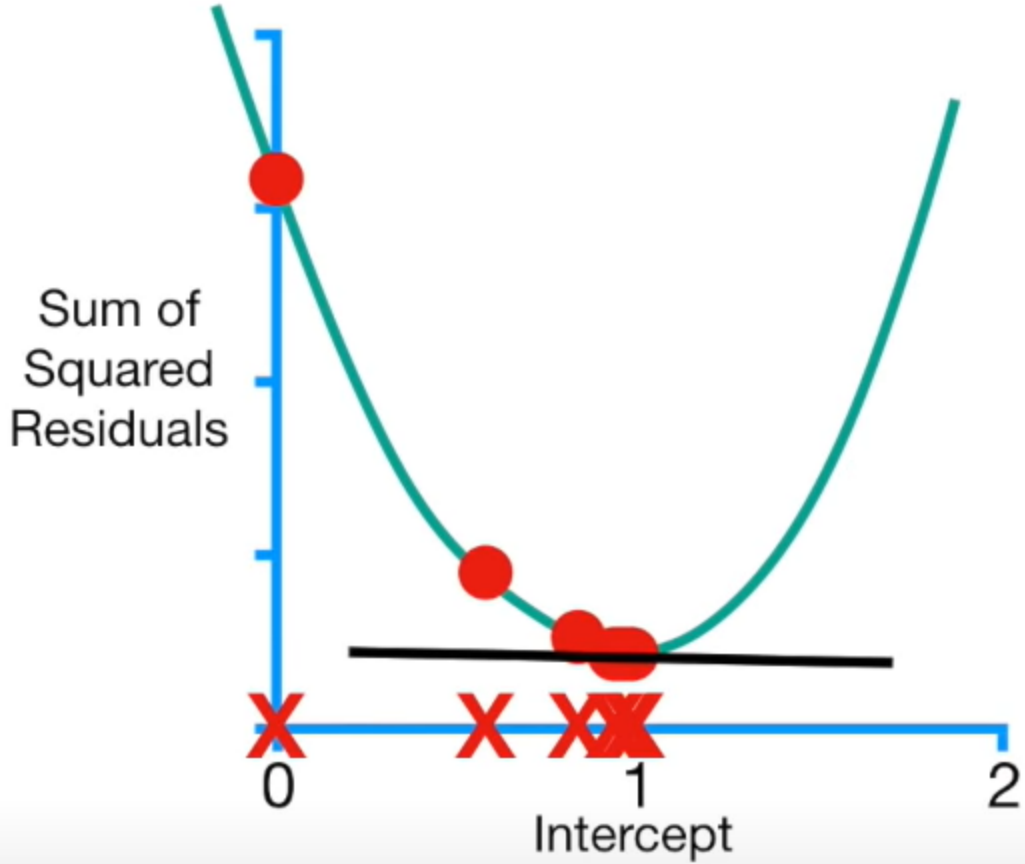
So if this **slope** = **0.009**…

Step Size = **Slope** x **Learning Rate**



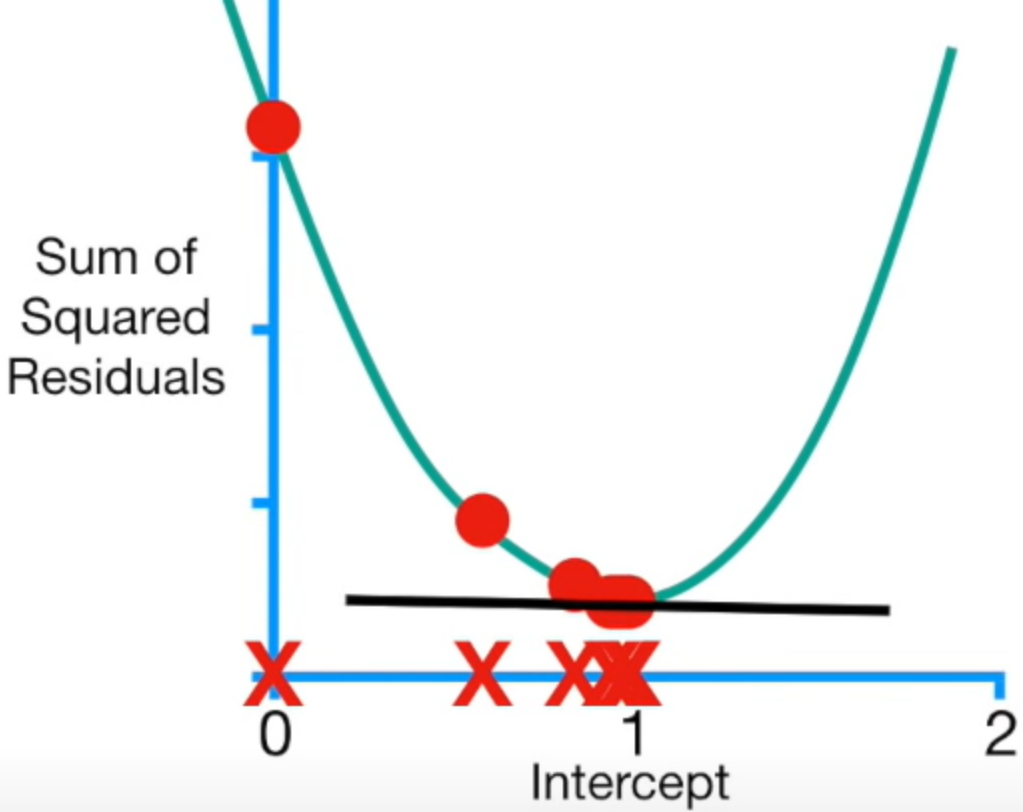
Then we would plug in
0.009 for the **Slope** and **0.1**
for the **Learning Rate**..

Step Size = **Slope** × **Learning Rate**



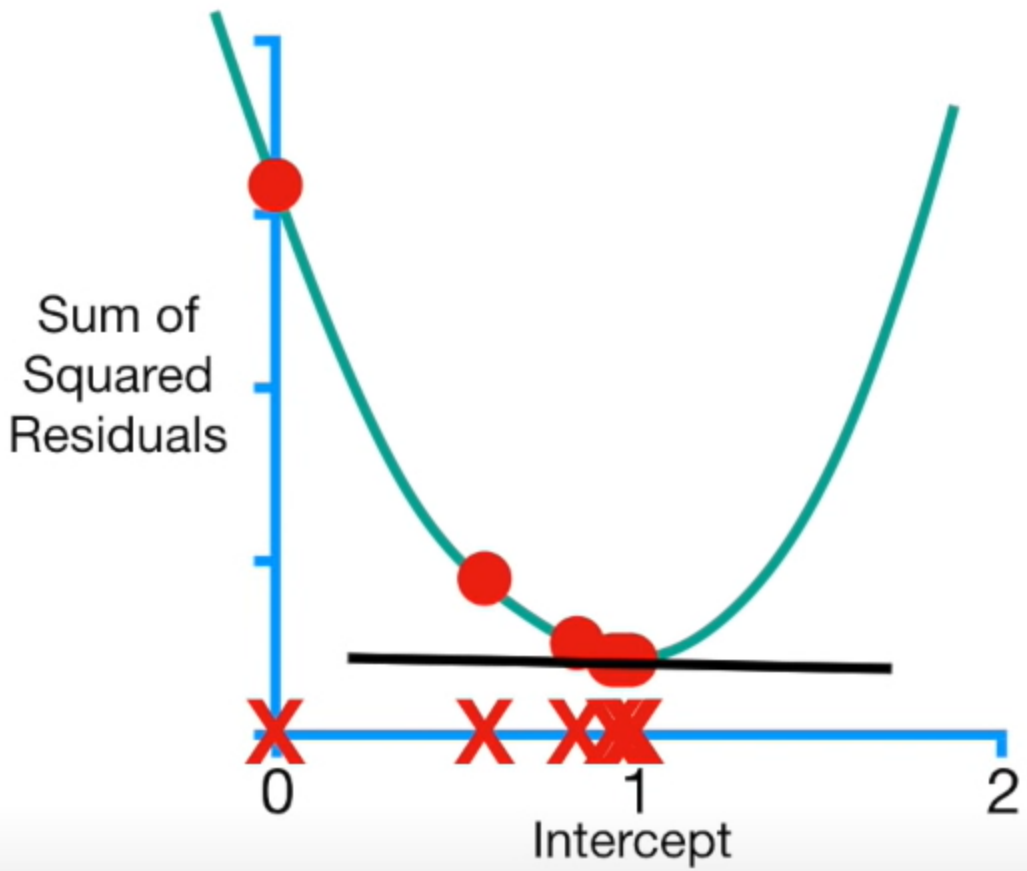
Then we would plug in
0.009 for the **Slope** and **0.1**
for the **Learning Rate**..

Step Size = **0.009** × **0.1**

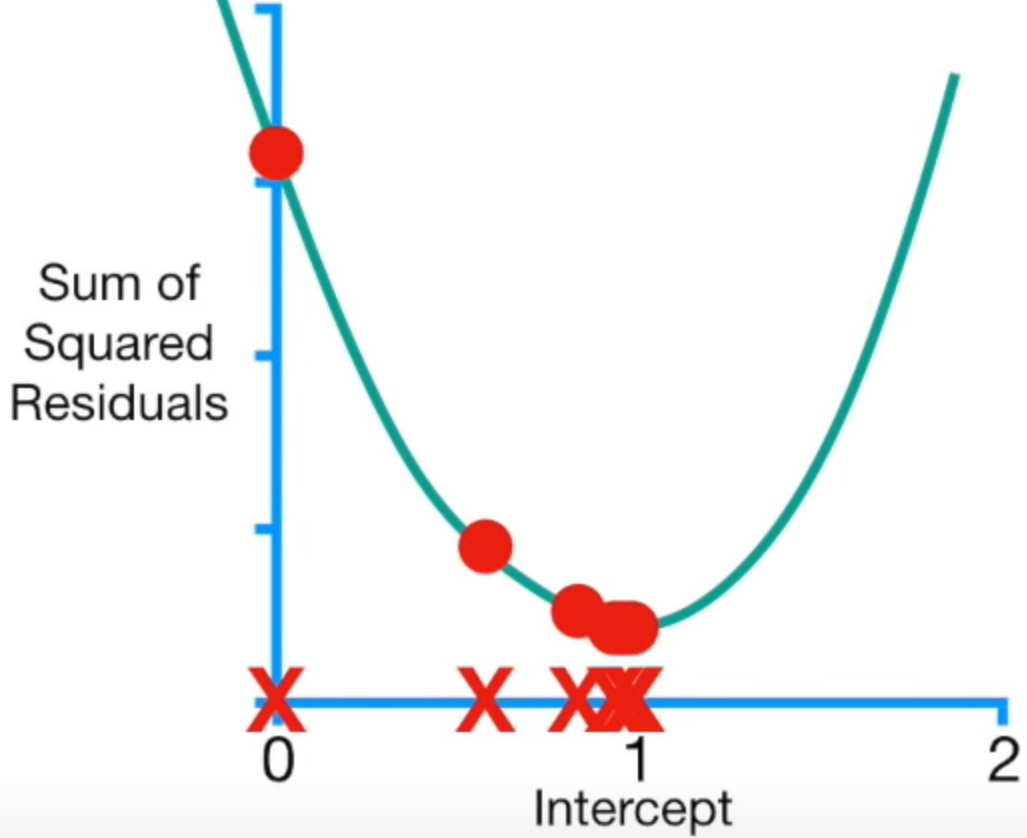


…and get **0.0009**, which is smaller than **0.001**, so **Gradient Descent** would stop.

Step Size = **0.009** × **0.1** = 0.0009

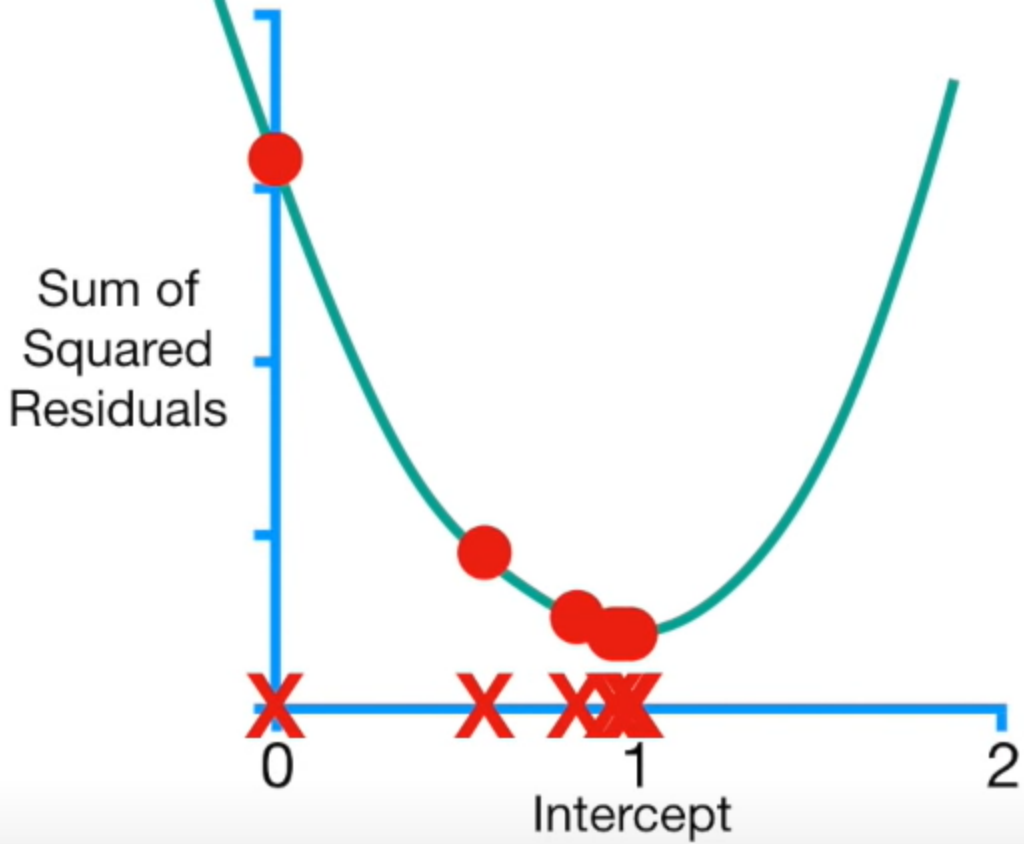


That said, **Gradient Descent** also includes a limit on the number of steps it will take before giving up.

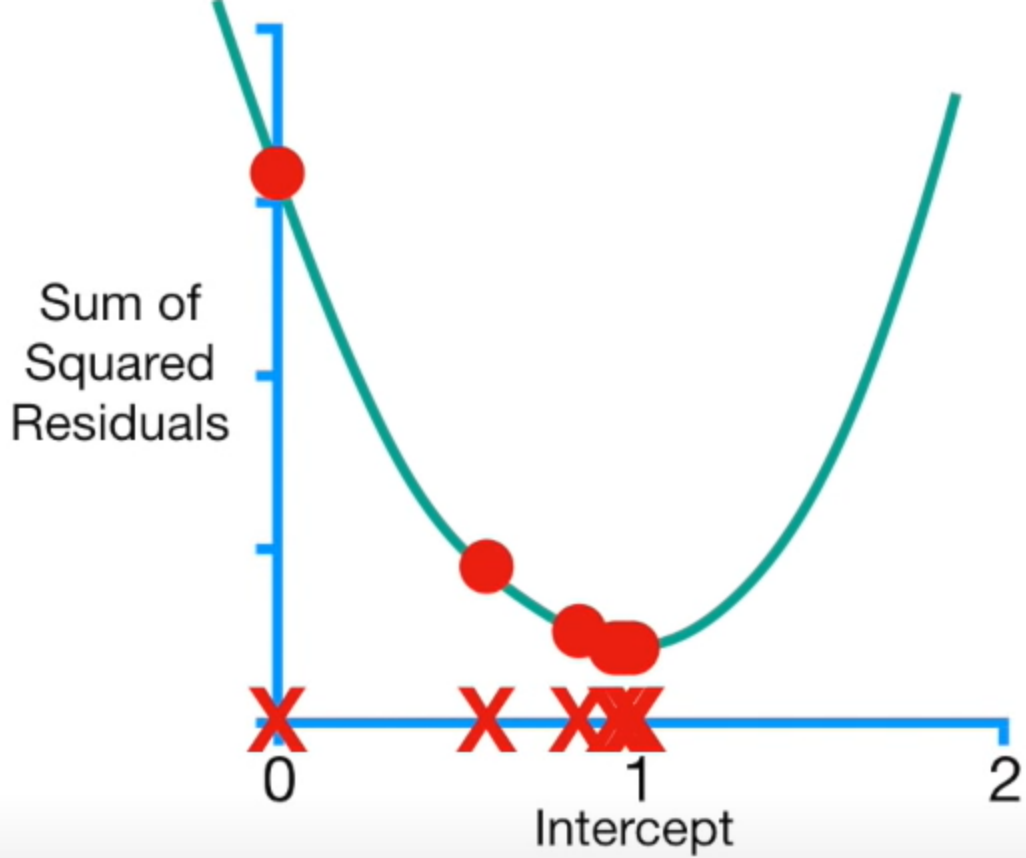


That said, **Gradient Descent** also includes a limit on the number of steps it will take before giving up.

In practice, the **Maximum Number of Steps** = **1,000** or greater.



So, even if the **Step Size** is large, if there have been more than the **Maximum Number of Steps**, **Gradient Descent** will stop.



OK, let's review what we've learned so far…